CAIARARA REDD+ PROJECT

Project title	Caiarara REDD+ Project
Project ID	5390
Crediting period	January 21, 2024 to January 20, 2064 – 40 years
Project lifetime	January 21, 2024 to January 20, 2064 – 40 years
(CCB) GHG accounting period	January 21, 2024 to January 20, 2064 – 40 years
Original issue date	*For pipeline listing, DD-Month-YYYY is the date of submission* *For registration, DD-Month-YYYY is the date on which the project description was completed after the audit was completed*
Most recent issuance date	15-Outubro-2025
Version	01
Standard VCS version	VCS Standard v4.7
CCB Standards Version	CCB Standard v3.1
Project location	Brazil, state of Pará, municipality of Breu Branco.
Project proponent(s)	Biofílica Ambipar Environmental Investments – Plinio Ribeiro (CEO) – email: verra.ambiparenvironment@ambipar.com / address: Av. Angelica, 2346, floor 5 set 53 room 03 – Consolação, São Paulo – SP – Brazil, ZIP code: 01228-200 / telephone: +55 11 3073-0430. Dow Brasil Indústria e Comercio de Produtos Químicos LTDA - Claudia Maria Fonseca Schaeffer – email: projetocaiarara@dow.com / address: Av. Nações Unidas, nº 14.401, andar 8 e 9, B3 – Jatobá, Vila Gertrudes, São Paulo, SP, Brazil / telephone: +55 11 5188-9595.

Validation/verification body	Earthood Services Private Limited "ESPL" – Kaviraj Singh – email: kaviraj.singh@earthood.in / address: 1203-1205,12th Floor, Tower B, Emaar Digital Greens, Sector 61, Gurugram, Haryana, India / telephone: +91 1244204599
CCB status history	First validation attempt.
Gold Level Criteria	Gold Level Climate: to assist communities and biodiversity in adapting to climate change, the Caiarara REDD+ Project has long-term planning for both scopes. In relation to communities, the project aims to promote technical assistance to help families adapt their agriculture through sustainable practices, especially in relation to drought; and educational activities to raise awareness of the impacts of deforestation and training for action on various environmental fronts, such as firefighting techniques. Biodiversity suffers from the decline in species individuals, bordering on extinction. Therefore, the project aims to monitor species in the project area. In addition, with increased asset surveillance to prevent damage to the forest, the project contributes to maintaining the local climate, minimizing global climate change, maintaining habitat connectivity, especially considering a larger spatial scale such as in the Project Area, and protecting ecosystem services such as pollination, seed dispersal, gene flow, soil protection, and hydrographic basins. Gold Level Biodiversity: Four species classified as endangered according to the IUCN Red List were identified in the Caiarara REDD+ Project area: Macaco-Cairara (*Cebus kaapori*) and Acapu (*Vouacapoua americana*), in the "critically endangered" (CR) category; Louro-canela (*Ocotea fragrantissima* Ducke) and Maçaranduba (*Manilkara elata*), in the "endangered" (EN) category. As a result, the project provides for the continuous monitoring of these species to track individuals and assist in the growth of their populations and provides for the preservation of natural habitats by reducing greenhouse gas emissions from deforestation and forest degradation, preventing extinction.
Expected verification schedule	VCS and CCB verifications are expected every two years.
Prepared by	Biofilica Ambipar Environmental Investments S/A

CONTENTS

1 SUMMARY OF PROJECT BENEFITS

1.1 Unique benefits of the project

Estimated outcome or impact by the end of the project's lifetime	Section reference
1) Expected climate benefits: The Caiarara REDD+ Project aims to contribute to climate change mitigation throughout its life cycle. Based on the initial baseline defined, it is expected to avoid the emission of 603,375 tCO2eq over a period of 6 years. In addition, the project provides for the preservation of 1,294 hectares that, in the scenario without the project, would be deforested.	3
2) Benefits to communities: The REDD+ Caiarara Project seeks to promote alternative sources of income for local communities, reducing dependence on activities that lead to deforestation and encouraging sustainable practices that conserve the forest. Actions include, but are not limited to, strengthening local governance, training historically marginalized groups, developing value chains, environmental education workshops, and technical assistance in sustainable agricultural practices. These initiatives strengthen social organization, encourage cooperativism, and expand access to technical knowledge that promotes the responsible use of natural resources. With the active participation of communities in planning and executing activities, the project aims to meet their needs, reduce social vulnerability, combat rural exodus, and foster sustainable development. In addition, strengthening local and regional partnerships seeks to expand the generation of goods and services, promoting economic and social well-being and improving the quality of life of the families involved.	4
3) Benefits to Biodiversity: The activities proposed by the Project aim to maintain the fauna and flora in the Project Area, ensuring the protection and conservation of habitats and local biodiversity, including endemic species and those with some degree of threat according to the IUCN Red List. It also seeks to preserve ecosystems with a high concentration of rare and endemic species, given its location in the Belém Endemism Center.	5

1.2 Standardized benefit metrics

Category	Metric	Estimated by the end of the project's lifetime	Section reference
GHG emissions reductions or carbon dioxide removals	Estimated net removals in the project area, measured against the without-project scenario.	Not applicable	-
	Estimated net reductions in the project area, measured against the without-project scenario	603,375 tCO2eq (6 years)	3
[1] forest cover	For REDD projects[2] :Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario	1,294 hectares (6 years)	3
	For ARR projects[3] : Estimated number of hectares of increased forest cover in the project area measured against the without-project scenario	Not applicable	-
Better land managemen t	[4]Number of hectares of existing forest production land where LULUCF practices are expected to occur as a result of project activities, measured against the without-project scenario	Not applicable	-
	Number of hectares of non-forest land where improved land management practices are expected to occur as a	Not applicable	-

[1] Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters such as minimum forest area, tree height, and canopy cover, and may include mature, secondary, degraded, and humid forests (VCS Program Definitions)

[2] Reducing emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by delaying or stopping the conversion of forests to non-forest lands and/or reduce degradation of forest lands where forest biomass is lost (VCS Program Definitions)

[3] Afforestation, reforestation, and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing, and/or restoring vegetation cover through planting, seeding, and/or human-assisted natural regeneration of woody vegetation (VCS Program Definitions)

[4] Improved forest management (IFM) - Activities that change forest management practices and increase carbon stocks on managed forest lands for wood products such as lumber, pulpwood, and firewood (VCS Program Definitions)

	result of project activities, measured against the without-project scenario		
Training	Total number of community members expected to have improved skills and/or knowledge as a result of the training provided as part of the project activities	Potentially 1,200 people and 240 families.	4
	Number of female community members who should have improved skills and/or knowledge resulting from training as part of the project activities	Potentially 270 adult women	4
Employment	Total number of people expected to be employed in project activities[5] , expressed as number of full-time employees[6]	Not applicable	4
	Number of women expected to be employed as a result of the project activities, expressed as number of full-time employees	Not applicable	4
Livelihood	[7]Total number of people expected to have improved livelihoods or income generated as a result of the project activities	Potentially 1,200 people and 240 families.	4
	Number of women expected to have improved livelihoods or income generated as a result of the project activities	Potentially 270 adult women	4
Health	Total number of people expected to experience improved health services as	Not applicable	-

[5] Employed in project activities means persons working directly on project activities in exchange for compensation (financial or otherwise), including employees, contract workers, subcontracted workers, and community members who are paid to perform project-related work.

[6] Full-time equivalence is calculated as the total number of hours worked (by full-time, part-time, temporary, and/or seasonal personnel) divided by the average number of hours worked in full-time jobs in the country, region, or economic territory (adapted from the United Nations System of National Accounts (1993), points 17.14[15.102];[17.28])

[7] Livelihoods are the capacities, assets (including material and social resources) and activities necessary for a means of living (Krantz, Lasse, 2001. The Sustainable Livelihoods Approach to Poverty Reduction. SIDA). Livelihood benefits may include benefits reported in the employment metrics in this table.

	a result of the project activities, measured against the without-project scenario		
	Number of women for whom health services are expected to improve as a result of the project activities, measured against the without-project scenario	Not applicable	-
Education	Total number of people for whom access to or quality of education is expected to improve as a result of the project activities, measured against the without-project scenario	Not applicable	-
	Number of women and young people expected to have improved access to education or improved quality of education as a result of the project activities, measured against the without-project scenario	Not applicable	-
Water	Total number of people expected to experience improved water quality and/or better access to drinking water as a result of the project activities, measured against the without-project scenario	Not applicable	-
	Number of women who should experience improved water quality and/or improved access to drinking water as a result of the project activities, measured against the without-project scenario	Not applicable	-
Well-being	Total number of community members whose well-being[8] is expected to	Potentially 1,200 people and 240 families.	4

[8] Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics in this table (e.g., Training, Employment, Livelihoods, Health, Education, and Water) and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural importance, etc.

	improve as a result of the project activities		
	Number of women whose well-being should improve as a result of the project activities	Potentially 270 adult women	4
Biodiversity conservation	Expected change in the number of hectares significantly better managed by the project for biodiversity conservation[9] , measured against the without-project scenario	32,625 hectares	5
	Expected number of critically endangered and endangered species worldwide[10] that benefit from reduced threats as a result of the project activities[11] , measured against the without-project scenario	4 species	5

[9] In this context, biodiversity management refers to areas where specific management measures are being implemented within the scope of the project activities with the aim of improving biodiversity conservation, for example, improving the status of endangered species.

[10] According to the IUCN Red List of Threatened Species

[11] In the absence of direct population or occupancy measures, the measurement of reduced threats may be used as evidence of benefit

2 PROJECT DETAILS

2.1 Project objectives, design, and long-term viability

2.1.1 Brief description of the project (VCS, 3.2, 3.6, 3.10, 3.11, 3.13, 3.14; CCB, G1.2)

The Caiarara REDD+ Project is a partnership between Biofilica Ambipar Environmental Investments S/A and Dow Brasil. Its purpose is to promote forest conservation, maintain carbon stocks, and mitigate climate change by reducing greenhouse gas (GHG) emissions from land use change. In addition to climate benefits, the Project also aims to generate social benefits based on sustainable economic development practices and improved well-being for surrounding communities, as well as environmental benefits through actions aimed at conserving local fauna and flora.

The Project Area covers 32,625 ha of Amazon rainforest, located in the municipality of Breu Branco, in the state of Pará. Communities that are directly or indirectly influenced by the Project have been identified, which are geographically close to the Project Area and depend on its ecosystem services. It should also be noted that the project is not covered by a Jurisdictional REDD+ Program.

The region is highly relevant in terms of biodiversity, as it is located in the Belém Endemism Center, a place with a high concentration of rare and endemic species. The predominant vegetation is dense ombrophilous forest with a high density of medium and large trees. It is home to at least 17 species of fauna and flora with some degree of threat according to the IUCN Red List, some of which are endemic to the region.

The history of occupation of the REDD+ Caiarara Project region is directly linked to government strategies for the Amazon between the 1960s and 1970s, which intensified migration due to large projects such as the construction of the Tucuruí Hydroelectric Plant. The municipality of Breu Branco, where the Project is located, developed as a hub to support this migratory flow, growing with activities such as silicon and timber exploration after its emancipation in the 1990s. However, the lack of sufficient jobs led many families to settle in rural areas, developing subsistence agriculture and contributing to the formation of several recent rural communities.

This scenario reflects the 11 communities identified in the Caiarara REDD+ Project region. These communities are not considered traditional, as they are mainly composed of migrant groups that have settled in the area over the last few decades. Their characteristics are mainly linked to subsistence agriculture, and they maintain direct or indirect relationships with the project area, either through the use of natural resources or through their geographical proximity to the properties where the project is located.

The current scenario, prior to the Project, includes illegal practices such as deforestation in the region. These practices, combined with the use of land for slash-and-burn agriculture, timber extraction, and the expansion of extensive cattle ranching and soybean production, do not generate positive forecasts or benefits for the community, the climate, or biodiversity. These land use pressures increase climate, social, and economic risks in the project region.

Given this scenario, it was identified that the main drivers of deforestation and forest degradation in the Caiarara REDD+ Project region are soybean expansion, extensive cattle ranching, and the use of slash-and-burn techniques by local communities. To address these challenges, the Project's activities were designed with a focus on biodiversity conservation and natural resource protection, adopting mitigating and preventive measures such as strengthening property surveillance, monitoring deforestation through satellite imagery, and promoting sustainable alternatives for regional socioeconomic development. These actions include, but are not limited to, strengthening sustainable value chains, biodiversity conservation and monitoring programs in the project area, and environmental education activities aimed at awareness and preservation. In addition, the Project seeks to engage and strengthen local governance by promoting the active involvement of communities and other stakeholders.

Thus, the Caiarara REDD+ Project hopes to improve the well-being of communities and generate exceptional benefits for local biodiversity, in addition to the climate impact of reducing 603,375 tCO_2eq in six years.

2.1.2 Audit history (VCS, 4.1)

Audit type	Period	Program	Name of validation/verification body	Number of years
Validation	January 21, 2024	VCS and CCB	Earthood Services Private Limited	-
Verification	January 21, 2024 – June 30, 2025	VCS and CCB	Earthood Services Private Limited	1.5

2.1.3 Sectoral scope and project type (VCS, 3.2)

Sector	14: Agriculture, forestry, and other land use
AFOLU project category[12]	Reducing Emissions from Deforestation and Degradation (REDD)
Project activity type	Avoided Unplanned Deforestation (AUD)

2.1.4 Project Eligibility (VCS, 3.1, 3.6, 3.8, 3.18, 4.1; CCB Program Rules, 4.2.4, 4.6.4)

The Caiarara REDD+ Project meets the scope of the VCS Program, falling within the following eligibility criteria (Section 2.1 of VCS Standard v4.7):

[12] See Appendix 1 of the VCS Standard

1) It reduces CO_2 emissions into the atmosphere, falling within the Greenhouse Gases stipulated in the Kyoto Protocol.

2) The activities of the REDD+ Caiarara Project follow the parameters established in a methodology eligible under the VCS Program, namely VM0048 (further details in Section 3.1. Application of the Methodology).

3) The implementation of this Project's activities does not violate any laws applicable to the region where it is located (more details in section 2.5.1).

4) It is not located in a jurisdiction covered by a jurisdictional REDD+ program.

5) The activities of the Caiarara REDD+ Project work to prevent emissions from unplanned deforestation and, therefore, do not fall under any of the items excluded from the scope of the VCS Program (presented in Table 1 of item 2.1.3 of the VCS Standard v4.7), which are:

    - Electricity generation activities connected to the grid using hydroelectric power plants.
    - Electricity generation activities connected to the grid using wind, geothermal, or solar photovoltaic (PV) power plants.
    - Waste heat recovery activities for electricity generation in combined cycles, or for heating/cooling through cogeneration or trigeneration.
    - Electricity and/or thermal energy generation activities for industrial use from the combustion of non-renewable biomass, agricultural waste biomass, or forest waste biomass.
    - Activities for the generation of electricity and/or thermal energy using fossil fuels, and activities involving the transition from a fossil fuel with a higher carbon content to one with a lower carbon content.
    - Activities to replace electric lighting with more energy-efficient options, such as replacing incandescent light bulbs with compact fluorescent lamps (CFLs) or light-emitting diodes (LEDs).
    - Activities involving the installation and/or replacement of energy-efficient electricity transmission lines and/or transformers.
    - Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions.

6) According to item 3.8 of *the VCS Standard v4.7*, the start date of an AFOLU project is the date on which the activities that lead to the generation of reductions or removals are implemented. In the Caiarara REDD+ Project, the start date is the date of contracting a new surveillance company with improved monitoring and recording of activities (see details in section 2.1.10).

7) It is established in *VCS Standard v4.7* (item 3.8.2) that AFOLU projects must begin the pipeline listing process within a period of 3 years after the project start date. The start date of the REDD+ Caiarara Project is January 21, 2024, and the listing as “under validation” was carried out on October 14, 2025. Thus, it is understood that the REDD+ Caiarara Project meets this requirement.

8) It is established in *the VCS Standard v4.7* (item 3.8.3) that AFOLU projects with estimated ex-ante emission reductions equal to or less than 20,000 tCO_2per year must complete their validation within 8 years from the project start date. The REDD+ Caiarara Project has an ex-ante estimate of annual average reductions of 99,351 tCO2e, which does not meet this requirement. Item 3.8.4 of *the VCS Standard v4.7* establishes that all other AFOLU projects must complete validation within five years from the project start date. Thus, since the project start date is January 21, 2024, this Project has until January 20, 2029, to complete its validation, which is planned to occur by the end of 2026, with the validation and verification field audit scheduled for December 1st to 5th, 2025, thus meeting this criterion.

9) The Caiarara REDD+ Project was submitted for a 30-day public comment period as "under validation" on the 14th of October 2025, and the field audit will be conducted between December 1st and 5th, 2025.

AFOLU Project Eligibility

This is an eligible project in the AFOLU (Agriculture, Forestry, and Other Land Use) category under the VCS Program through the reduction of emissions from deforestation and degradation (REDD) in the context of avoided unplanned deforestation (AUD) using the VM0048 methodology, eligible under the VCS Program.

According to the VCS Standard v4.7 (Appendix 1), eligible REDD activities are those that reduce greenhouse gas emissions by reducing deforestation (human-induced conversion of forest land to non-forest land) and/or forest degradation. Among the eligible REDD activities is the avoidance of unplanned deforestation, which the Caiarara REDD+ Project falls under.

The project area must meet an internationally accepted definition of forest (FAO, UNFCCC) and must qualify as forest for at least 10 years prior to the project start date. According to data collected from the MapBiomas platform, collection 8[13] , the REDD+ Caiarara Project area has been eligible as forest or wooded savanna (classified according to FAO forest criteria, included in the reference level submitted by Brazil to the UFCCC[14]) for more than 10 years and, therefore, there has been no conversion of the ecosystem for the purpose of generating carbon credits.

CCB Rules Program

The Caiarara REDD+ Project meets the requirements of the CCB Rules Program v3.1 established in this section, in accordance with the following items:

1) According to item 4.2.4, the VCS platform must receive the validation and/or verification report and the validation and/or verification statement within one year of the start of the relevant public comment period. The listing for the start of the public comment period is scheduled for October 14, 2025, and the validation and verification audit for the period of December 1st to 5th, 2025, in accordance with the CCB requirement.

[13] https://brasil.mapbiomas.org/

[14] https://forum.mapbiomas.org/t/area-da-formacao-florestal/163

2) Item 4.6.4 establishes that the public comment period must be completed before the start of the validation/verification body's site visit. If the public comment period ends after the site visit is completed, the validation/verification body shall consider any comments received and may need to return to the project site to do so. As stated in the previous item, the public comment period was planned to end before the start of the validation and verification audit. In the event of any unforeseen circumstances and the public consultation period ending after the audit, the Caiarara REDD+ Project will cooperate in adapting the audit schedule or the auditor's return to the field.

2.1.5 Eligibility of the Transfer Project (VCS, 3.23, Appendix 2)

Not applicable

2.1.6 Project Design (VCS, 3.6)

Indicate whether the project was designed as:

☒ Single site or facility

☐ Multiple sites or instances of project activity (but not a clustered project)

☐ Grouped project

2.1.6.1 Eligibility criteria for grouped projects (VCS, 3.6; CCB, G1.14)

Not applicable. This project is not a grouped project.

2.1.7 Project proponent (VCS, 3.7; CCB, G1.1)

Name of organization	DOW BRASIL INDÚSTRIA E COMÉRCIO DE PRODUTOS QUÍMICOS LTDA
Contact	Claudia Maria Fonseca Schaeffer
Title	Commercial Director of Energy & Climate
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Organization	BIOFILICA AMBIPAR ENVIRONMENTAL INVESTMENTS S/A

Contact	Plínio Ribeiro
Title	Executive Director
Address	Av. Angelica, 2346, floor 5 set 53 room 03 – Consolação, São Paulo, Brazil
Phone	+55
Email	verra.ambiparenviroment@ambipar.com

2.1.8 Other entities involved in the project

Name of organization	STA Technical Solutions and Environmental Engineering Services Ltda.
Role in the project	Responsible for socioeconomic diagnosis, flora diagnosis, and carbon stock estimation
Contact	Raniery Branco
Title	Partner Owner
Address	Rua 28 de Setembro, 1226 – Reduto ZIP Code 66053- 355, Belém/PA – Brazil
Phone	+55 91 3355-4408 & +55 91 9161-2698
Email	raniery@stambiental.com

Organization	Ambiens Resolve Innovation Sustainability and Environment Ltda.
Role in the project	Responsible for fauna diagnosis
Contact	Fábio Bárbara
Title	Partner Owner
Address	Rua Marquês de Sabará, nº 30, Conj. 31 - Real Parque ZIP Code 05.684-020, São Paulo/SP - Brazil

Phone	+55 (11) 97309-7100
Email	fabio@ambiens.com.br

2.1.9 Project Ownership (VCS, 3.2, 3.7, 3.10; CCB, G5.8)

Dow is the legitimate owner and holder of the properties where the Caiarara REDD+ Project is being implemented and developed, as detailed in Section 2.5.5.

To establish responsibility and rights over the Project, as well as the percentage of carbon credits allocated to each party, a contract was signed by the Project proponents. They have the legal right to control and operate the project activities and to generate, register, and trade carbon credits, in accordance with the requirements of VCS Standard v4.7 (section 3.7) and CCB Standards v3.1 (section G5.8) and the definition of "*project* ownership" in VCS Program Definitions v4.5.

It should be noted that in the registrations of the RAAI and RAAII farms, the owner is identified as "Palmyra do Brasil" and that this company is part of the Dow Brazil Group, as mentioned in the contract addendum between the proponents made available to the project validation auditors.

2.1.10 Project start date (VCS, 3.8)

Project start date	January 21, 2024
Justification	The start date of the Caiarara REDD+ Project was set for January 21, 2024, represented by the hiring of a new security company with improved recording of activities, ensuring greater traceability and control of the information collected. Thus, improvements in the security of the Project Area are considered the first activity that led to the generation of GHG emission reductions, as established by the VCS Standard v4.7, since the improvement in surveillance of the area contributed to the deforestation containment actions of the Caiarara REDD+ Project.

2.1.11 Project Benefit Assessment and Credit Period (VCS, 3.9; CCB, G1.9)

Crediting period	The crediting period for the Caiarara REDD+ Project will be for the full 40-year period, considering the VCS v4.7 standard.
Start date of the first credit period or fixed credit period	January 21, 2024 to January 20, 2064.

CCB benefit assessment period	There will be continuous monitoring of climate, community, and biodiversity benefits, subject to verification with the CCB, ideally every two years, throughout the duration of the Project.

2.1.12 Differences in Project Assessment/Credit Periods (CCB, G1.9)

There is no difference between the assessment period and the accounting period for GHG emissions, nor between the assessment periods for community, biodiversity, and climate adaptation benefits. All scopes follow the same project crediting period, as described in Section 2.1.11, beginning on January 21, 2024, and lasting for 40 years. This standardization aims to facilitate integrated monitoring of scopes and ensure consistency between periodic verifications.

2.1.13 Project Scale and Estimated Reductions or Removals (VCS, 3.10)

☒ <300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Calendar year of the crediting period	Estimated reductions or removals (tCO2e)
21-January-2024 to 31-December-2024	93,717
1-January-2025 to 31-December-2025	96,456
1-January-2026 to 31-December-2026	99,193
1-January-2027 to 31-December-2027	101,932
1-January-2028 to 31-December-2028	104,669
1-January-2029 to 31-December-2029	107,408
1-January-2030 to 31-December-2030	110,146
1-January-2031 to 31-December-2031	112,884
1-January-2032 to 31-December-2032	115,621
1-January-2033 to 31-December-2033	118,360
1-January-2034 to 31-December-2034	118,360
1-January-2035 to 31-December-2035	118,360

1-January-2036 to 31-December-2036	118,360
1-January-2037 to 31-December-2037	118,360
1-January-2038 to 31-December-2038	118,360
1-January-2039 to 31-December-2039	118,360
1-January-2040 to 31-December-2040	118,360
1-January-2041 to 31-December-2041	118,359
1-January-2042 to 31-December-2042	118,360
1-January-2043 to 31-December-2043	118,360
1-January-2044 to 31-December-2044	118,360
1-January-2045 to 31-December-2045	118,360
1-January-2046 to 31-December-2046	118,360
1-January-2047 to 31-December-2047	118,360
1-January-2048 to 31-December-2048	118,360
1-January-2049 to 31-December-2049	118,360
1-January-2050 to 31-December-2050	118,359
1-January-2051 to 31-December-2051	118,360
1-January-2052 to 31-December-2052	118,360
1-January-2053 to 31-December-2053	118,360
1-January-2054 to 31-December-2054	118,360
1-January-2055 to 31-December-2055	118,360
1-January-2056 to 31-December-2056	118,360
1-January-2057 to 31-December-2057	118,360
1-January-2058 to 31-December-2058	118,360

1-January-2059 to 31-December-2059	118,359
1-January-2060 to 31-December-2060	118,360
1-January-2061 to 31-December-2061	118,360
1-January-2062 to 31-December-2062	118,360
1-January-2063 to 31-December-2063	118,360
Total estimated ERRs during the first credit period or fixed credit period	4,611,183
Total number of years	40
Average annual ERRs	115,280

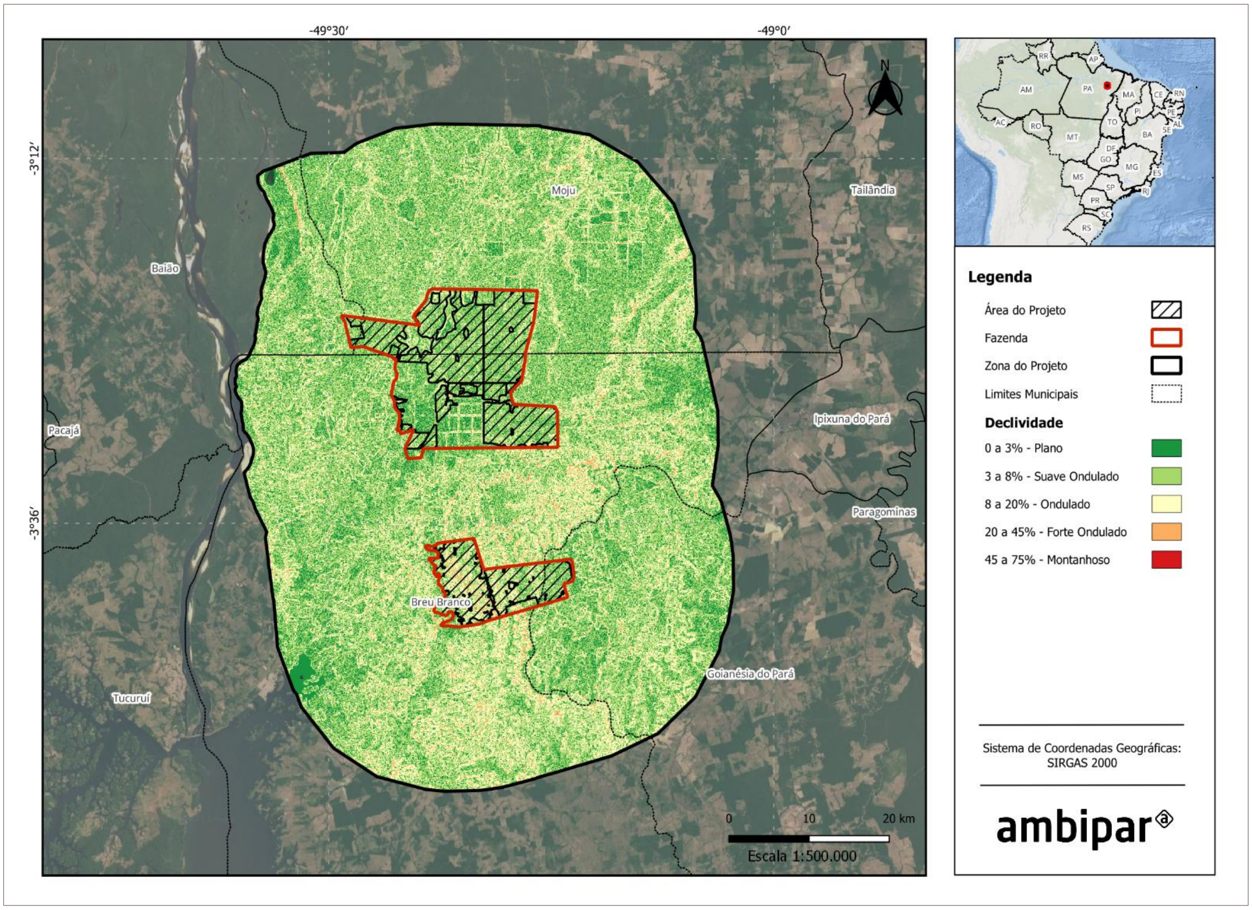
2.1.14 Physical parameters (CCB, G1.3)

Slope

The analysis of land slope in the Project Area, generated from the SRTM digital elevation model, indicates that the area is predominantly gentle, with emphasis on the gentle undulating (3% to 8%) and flat (0% to 3%) classes, representing 49.55% and 21.71% of the area, respectively. These values indicate that more than 70% of the area has a gentle slope, which favors land use and management (Figure 1). Undulating areas (8% to 20%) correspond to 26.57%. The steepest slopes, strongly undulating (20% to 45%) and mountainous (45% to 75%), are insignificant, together accounting for about 2.17% of the area. This indicates that rugged terrain is practically nonexistent in the project area.

In the Project Area, the terrain is predominantly gentle. The flat (0% to 3%) and gently undulating (3% to 8%) classes correspond to 20.60% and 52.35% of the area, respectively, totaling 72.96% of the PA with low slopes. Areas with undulating terrain (8% to 20%) cover 24.78% of the area. Finally, the classes with the steepest slopes, strongly undulating and mountainous from 20% to 75%, are not very representative, together accounting for only 2.26% of the area, which highlights the absence of significant rugged terrain in the Project area.

Figure 1. Slope Map of the Project Zone and REDD+ Caiarara Project Area.



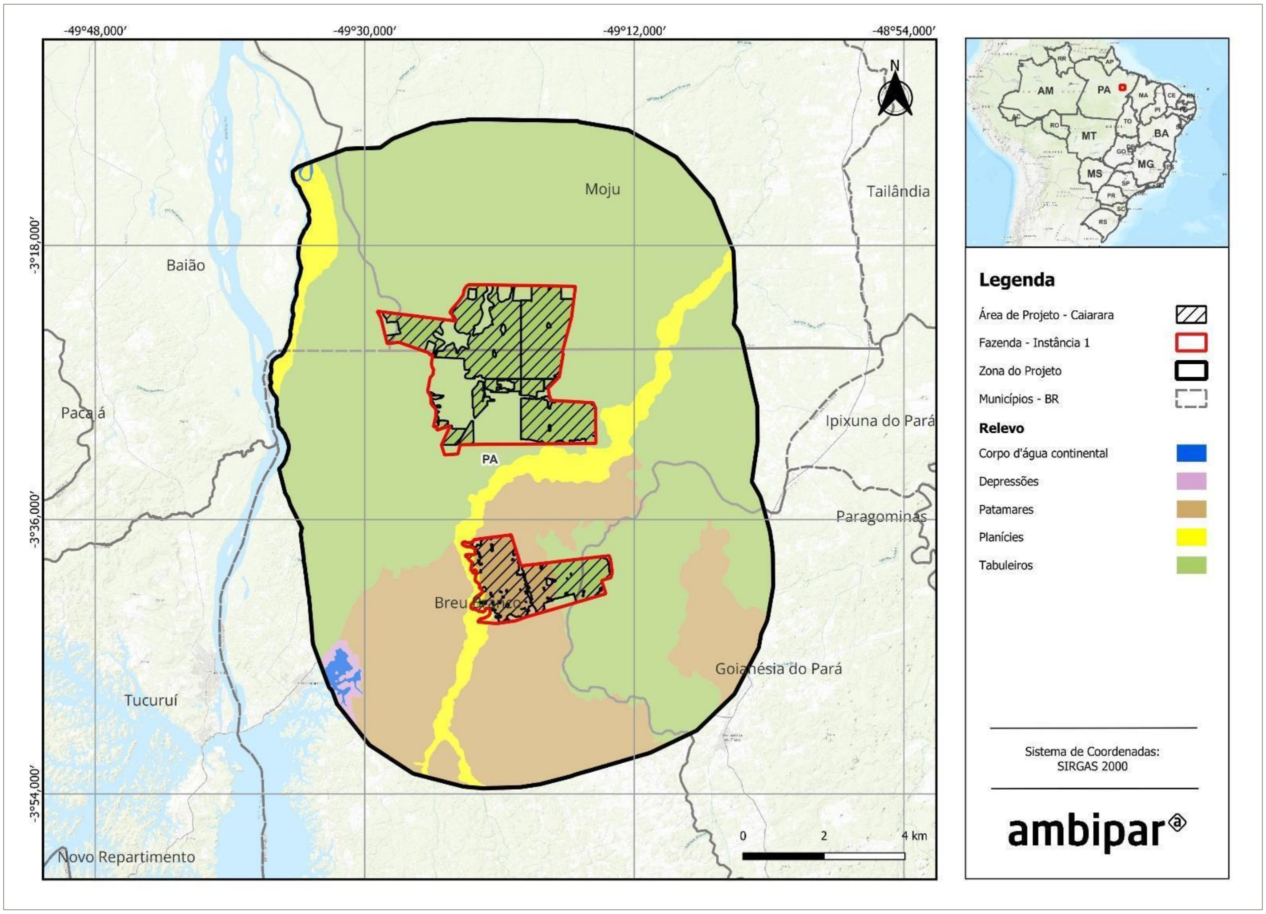
Relief

The Project Area is mainly characterized by tablelands, which occupy 69.43% of the area. These gently undulating landforms are generally associated with sedimentary terrain at medium altitudes, common in more stable regions (IBGE, 2009)[15] . Next are the plateaus, which account for 23.59% of the area and correspond to flat or sloping surfaces located between different altimetric levels. Plains represent 6.19% of the area, indicating areas of almost flat relief, where sedimentary deposition processes predominate. Depressions appear sporadically (0.44%) and indicate areas that are slightly lower than the surrounding area, associated with erosion. Finally, continental water bodies cover 0.36% of the surface.

In the Project Area, the predominant relief is also that of tablelands, which occupy about 79% of the area. The plateaus correspond to 19%, while the plains represent approximately 2% (Figure 2).

[15] IBGE – Brazilian Institute of Geography and Statistics. Technical manual on geomorphology: classification system for the relief of Brazil: relief compartments and landscape units. Rio de Janeiro: IBGE, 2009. 92 p. Available at: https://docs.ufpr.br/~santos/Geomorfologia_Geologia/Manual%20t%C3%A9cnico%20de%20Geomorfologia.pdf Accessed on: July 8, 2025.

Figure 2. Relief map of the Caiarara REDD+ Project Area.

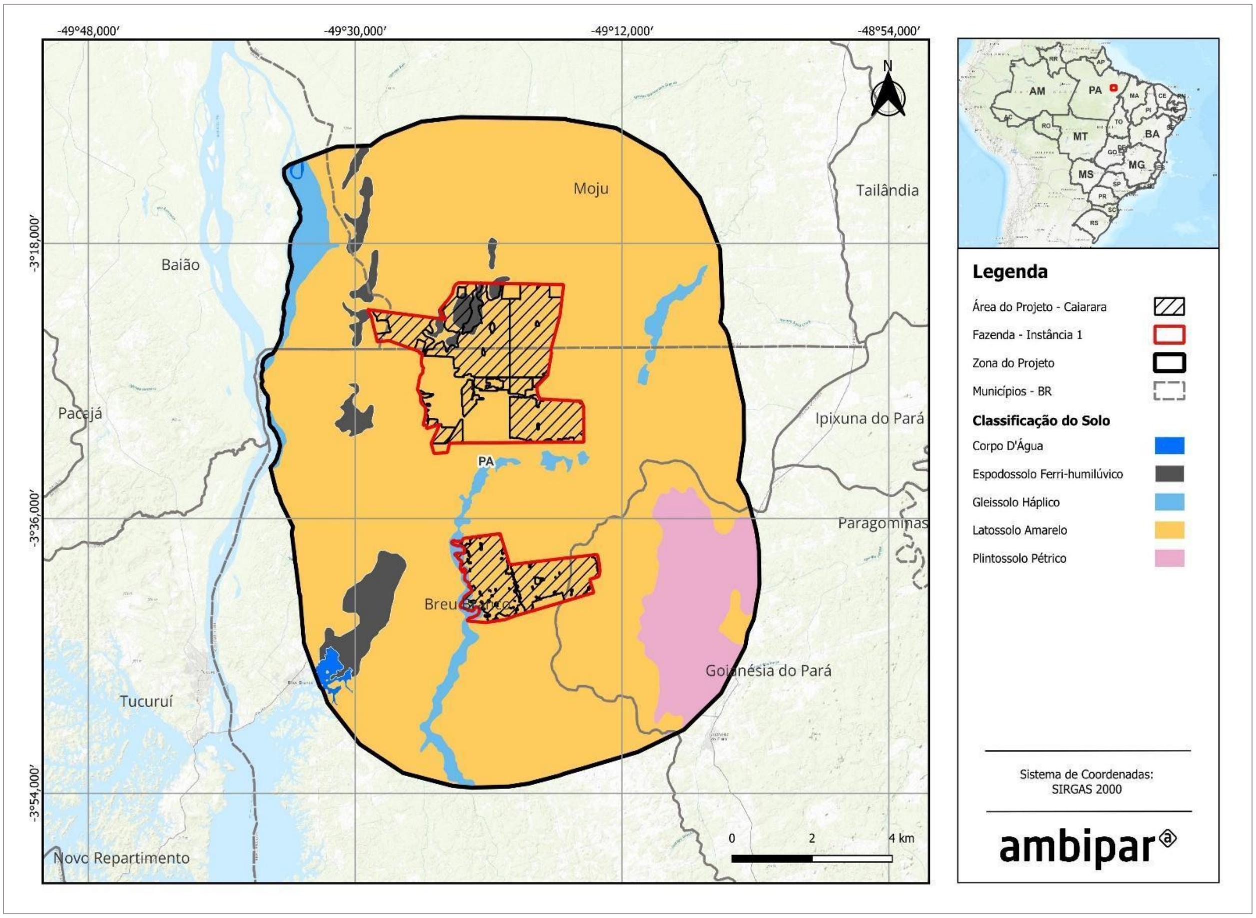


Pedology

The Project Area is largely dominated by Yellow Latosol (Figure 3), which covers approximately 85.29% of the surface. This type of soil is typical of tropical regions, has good depth and drainage, but low natural fertility, and is common in areas with gentle to undulating relief (SANTOS et al., 2018)[16] . Other soils present in smaller proportions include Petric Plinthosol (6.76%), associated with poorly drained areas and subject to the presence of plinthite, which can restrict agricultural use. Eutric Gleysol (3.8%) occurs in permanently or periodically waterlogged areas, while Spodosol Ferri-Humiluvic (3.79%) is generally linked to environments with accumulated organic matter and sand, with low fertility. Continental water bodies account for 0.36% of the area.

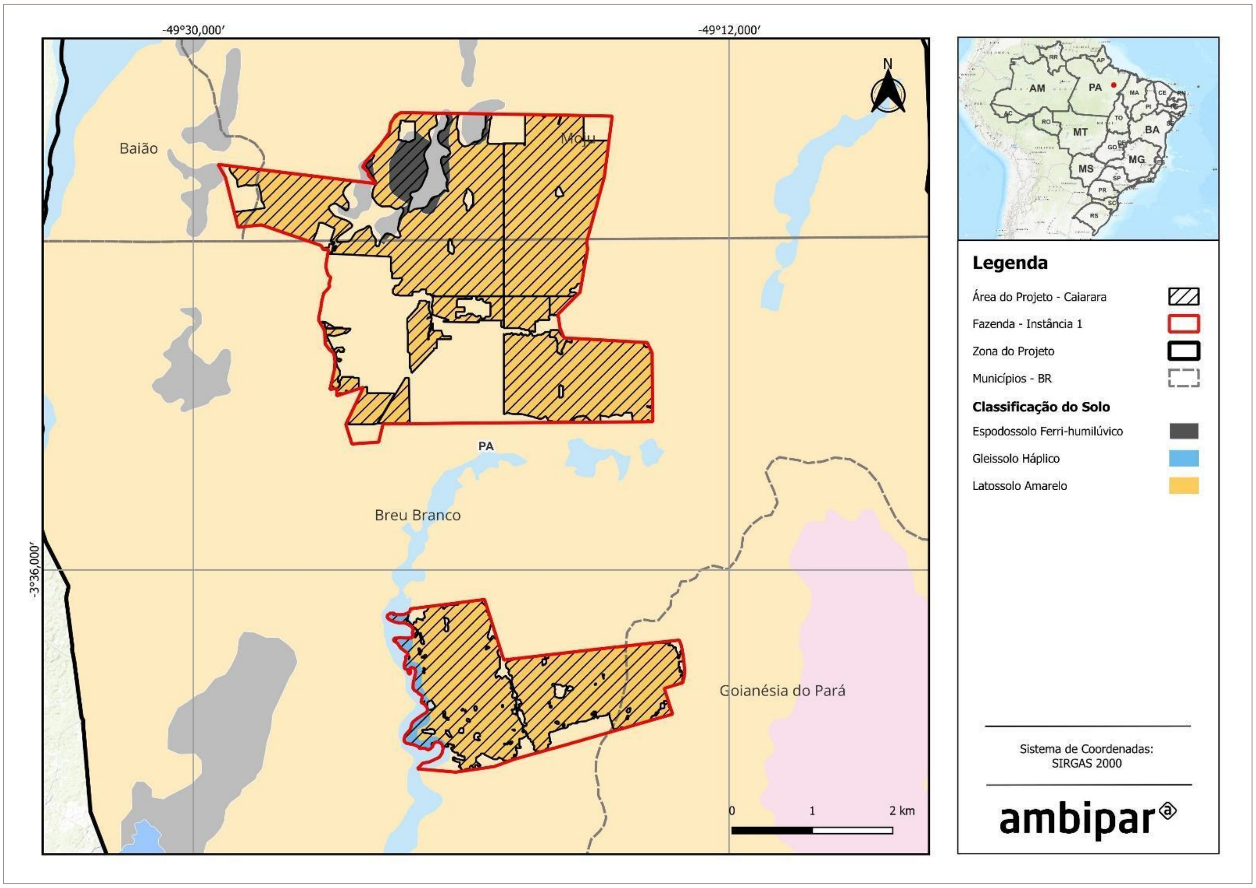
[16] SANTOS, H. G. dos et al. Brazilian Soil Classification System. 5th ed. rev. and expanded. Brasília, DF: Embrapa, 2018.

Figure 3. Pedology map of the Caiarara REDD+ Project Area.



In the Project Area, Yellow Latosol is even more predominant (Figure 4), covering 94.63% of the surface. Spodosol Ferri-Humiluvic represents 3.72% of the area, and Eutric Gleysol 1.65%.

Figure 4. Soil map of the Caiarara REDD+ Project Area.

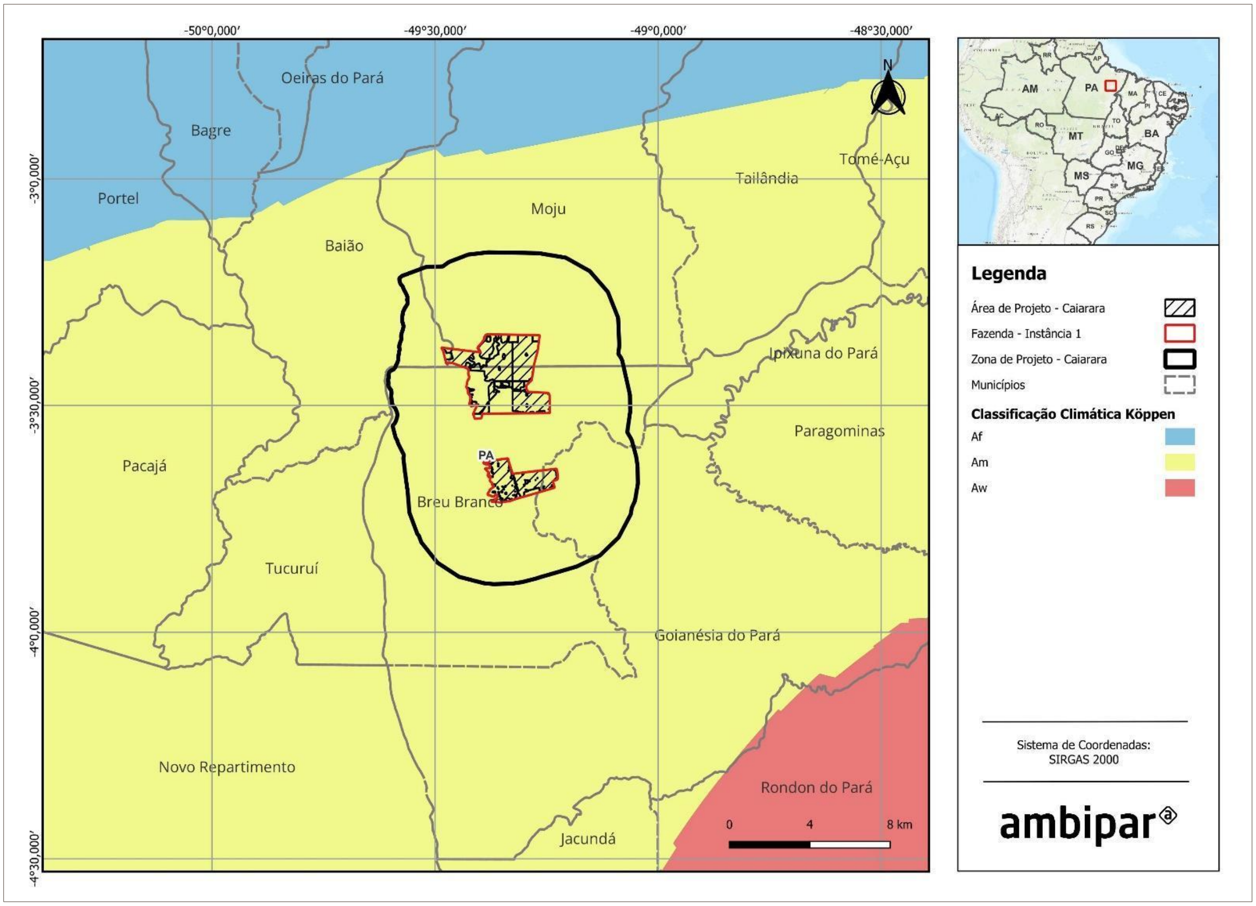


<u>Climate</u>

The project zone and project area are classified as climate type Am (monsoon) according to the Köppen climate classification (1936), as shown in Figure 5. The climate is characterized by abundant rainfall and high relative humidity throughout most of the year. Although there is a slight reduction in rainfall in some months, there is no defined dry season. This climate favors the occurrence of dense, evergreen tropical forests, typical of the Amazon region (ALVARES et al., 2013)[17] .

[17] ALVARES, C. A. et al. Köppen's climate classification map for Brazil. Meteorologische Zeitschrift, v. 22, n. 6, p. 711–728, 2013. DOI: 10.1127/0941-2948/2013/0507.

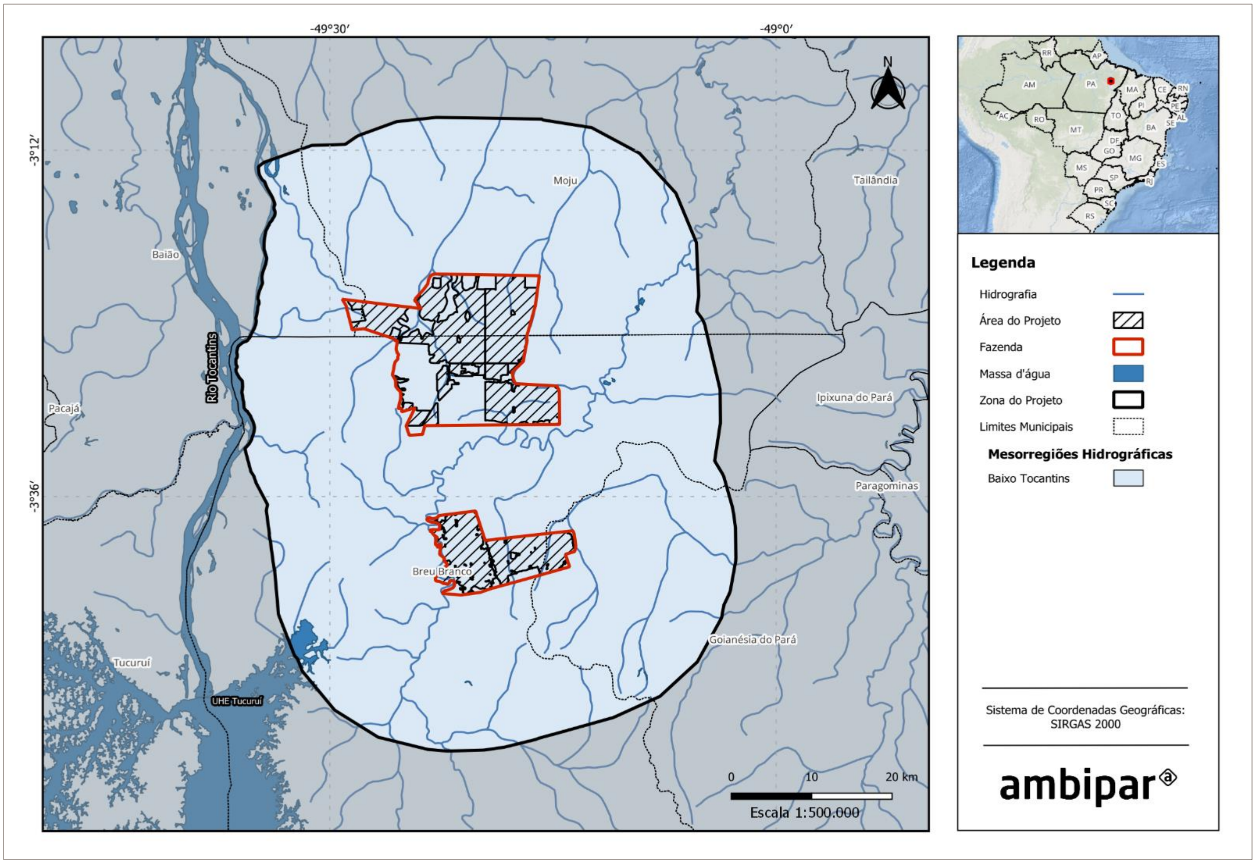
Figure 5. Climate map of the Project Zone and REDD+ Project Area in Caiarara.



Hydrography

The project area is located in the Baixo Tocantins hydrographic mesoregion, according to the IBGE hydrographic division (Figure 6), and is part of the Tocantins–Araguaia macroregion, one of the main hydrographic basins in Brazil, notable for its hydroelectric power generation. It is located near the course of the Tocantins River, one of the main rivers in the North region, and close to the Tucuruí Hydroelectric Plant (UHE Tucuruí), one of the largest in the country.

Figure 6. Hydrography map of the Project Zone and Caiarara REDD+ Project Area

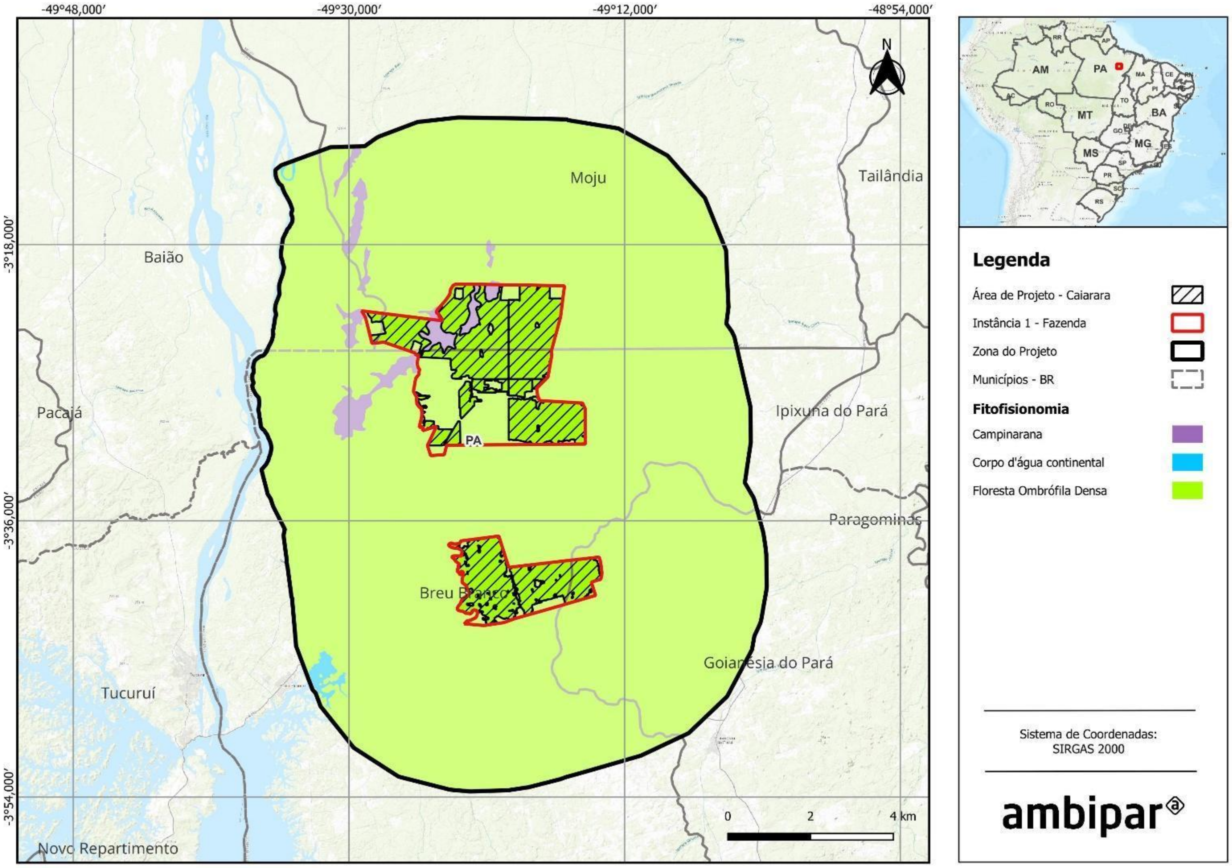


<u>Phytophysiognomy</u>

The project area is predominantly composed of Dense Ombrophilous Forest, which occupies approximately 97.73% of the area. There are also Campinaranas, covering 1.92%, and continental water bodies, covering about 0.35% of the total area. In the project area (PA), the vegetation cover is entirely composed of Dense Ombrophilous Forest, typical of humid regions, with dense vegetation and high species diversity (IBGE, 2012)[18] , as shown in Figure 7.

[18] IBGE – Brazilian Institute of Geography and Statistics. Technical manual on Brazilian vegetation: phytogeographic system, inventory of forest and rural formations, techniques and management of botanical collections: version 2.0. Rio de Janeiro: IBGE, 2012. 271 p. Available at: https://biblioteca.ibge.gov.br/visualizacao/livros/liv63011.pdf. Accessed on: July 8, 2025.

Figure 7. Map of phytophysiognomy in the Project Zone and REDD+ Caiarara Project Area.



2.1.15 Social Parameters (VCS, 3.18; CCB, G1.3)

The Caiarara REDD+ Project is located in the state of Pará, with most of its territory in the municipality of Breu Branco, in addition to small parts of the area in the municipalities of Moju and Goianésia do Pará. In the vicinity of the project area, there are rural communities that use areas surrounding the farms for various activities such as subsistence agriculture, gathering, and other local practices. The Socioeconomic Diagnosis was conducted in the project area by the company STA, involving the communities in the region and farm workers.

- **Surrounding communities**

The most recent history of occupation in the region can be traced back to the government's strategy for the Amazon in the 1960s and 1970s, which intensified the migratory process to the region, almost tripling the population between the 1970s and 1980s (Cavalcante, 2005[19]). Large projects, such as the Tucuruí

[19] CAVALCANTE, F. C. The migratory process in the Amazon linked to labor mobility – the case of the Tucuruí Hydroelectric Plant. Proceedings of the 10th Meeting of Latin American Geographers – March 20-26, 2005 – University of São Paulo.

Hydroelectric Plant, were responsible for attracting a large number of migrants, all in search of work on the construction sites. Most of these migrants came from the states of Maranhão, Pará, and Goiás, with Maranhão accounting for 32.06% of the total number of migrants, followed by Pará with 22.08% (Cavalcante, 2005 *apud* Rocha, 1999[20]).

The creation of the municipality of Breu Branco is directly linked to the construction of the Tucuruí Hydroelectric Plant, as the city was founded to house families who lived on the banks of the Tocantins River, in areas that were submerged by the hydroelectric plant. After its emancipation in the 1990s, Breu Branco underwent a period of development, especially with the exploitation of silicon and timber, investments in infrastructure, and sanitation. The city continues to grow due to its proximity to the hydroelectric plant, the Trans-Amazonian highway, and the presence of mineral exploration companies, which has transformed it into a regional hub. With a growth rate of 4.76%, Breu Branco has the highest growth in its integration region, a pace associated with new mineral exploration and steel industry projects in the area (Castro et al., 2010[21]).

However, not all migrants who moved to the region were able to find employment in large projects, which led many to move to rural areas and develop subsistence agriculture. This new dynamic of work in the countryside contributed to the emergence and expansion of several rural communities, such as those that now surround the Caiarara REDD+ Project. In general, it can be seen that the communities are recently settled, with approximately half of them existing for between 30 and 40 years, while the other half have been established for less than 27 years[22] .

The productive structure of the communities surrounding the project is heavily based on subsistence agriculture, which is the main productive and economic activity, complemented by other practices such as extractivism, fishing, livestock, and charcoal production. In addition to these activities, families' income generation also depends significantly on social benefits, retirement benefits, and pensions. Most families live on a monthly income of one to two minimum wages.

Within family farming, cassava stands out as the main product grown for flour production, followed by crops such as corn, beans, rice, watermelon, cocoa, cashews, and pitaya. However, in many communities, difficulties related to soil and climate, as well as a lack of mechanization and inputs, raise concerns about the long-term continuity of agriculture. The shifting cultivation model, which includes cutting, clearing, and burning practices, is still widely used, with fire playing an important role in soil fertilization.

Charcoal production and cattle raising are also important activities for some families, although they face challenges such as a lack of formal organization for marketing and difficulties imposed by an increasingly dry climate. The absence of basic technical assistance and adequate infrastructure exacerbates these difficulties, leading some families to consider migration in search of better living and productive conditions.

With regard to basic sanitation, the communities reflect the reality of the municipalities in the region, with low rates of access to essential services. Access to drinking water is precarious, with many families

[20] Ibid.

[21] CASTRO, E, R et al. Socioeconomic study of municipalities in the Tucuruí region, Pará. NAEA Paper 258, March 2010.

[22] Final report Socioeconomic Diagnosis – STA.

depending on wells, springs, or cisterns, although some communities have partial access to the water distribution network. Water quality is a critical issue, with reports that a significant portion of the population suffers from diseases related to contaminated water, such as diarrhea, especially among children and young people.

The sewage situation is also alarming. There are no public sewage systems and, for most communities, less than half of the houses have adequate bathrooms or toilets. When they do exist, sewage is discharged into rudimentary or masonry pits built by the families themselves. Solid waste tends to be burned on the lots themselves, as collection services are limited to a few locations.

Although basic health services are available in most communities, access to more complex medical care requires travel to urban centers. In addition, families often use medicinal plants and herbs to treat everyday illnesses.

With regard to migration, despite the challenges faced, migratory movements in recent decades have been limited. However, there are signs that adverse conditions, such as soil deterioration and lack of infrastructure, are driving families, especially younger ones, to leave in search of better opportunities for education, work, and quality of life. Some communities, particularly those most vulnerable in terms of access to water and productive infrastructure, report difficulties in remaining in the region, with an increase in population outflow.

In terms of education, it was observed that there are no daycare centers available for children in the communities studied, which may represent a significant gap for families. In terms of primary education, some communities have schools that offer education up to the 4th grade, while others have the complete primary education cycle. For secondary education, students need to travel to the city of Breu Branco, relying on school transportation guaranteed by public policies that, according to reports, serve all communities. Access to regular school meals is available in most locations, with the exception of a few that reported a lack of this benefit. With regard to literacy, there is a rate of over 90% for individuals up to 34 years of age, but in older age groups, especially among black and brown people, this rate drops significantly, reaching less than 50% in some regions.

- Farm workers

With regard to farm workers, the survey reveals a predominance of males, with 90% of respondents belonging to this gender, although there is also a 10% representation of females. The age distribution of workers is varied, with the 31-40 age group being the most representative (41.46%), followed by the 41-59 age group (30.49%) and, finally, the youngest, between 19 and 30 years old, who represent 27.44% of the sample. Only one person over the age of 60 was interviewed. This age diversity points to a predominantly adult and experienced workforce, but also with a significant presence of younger workers.

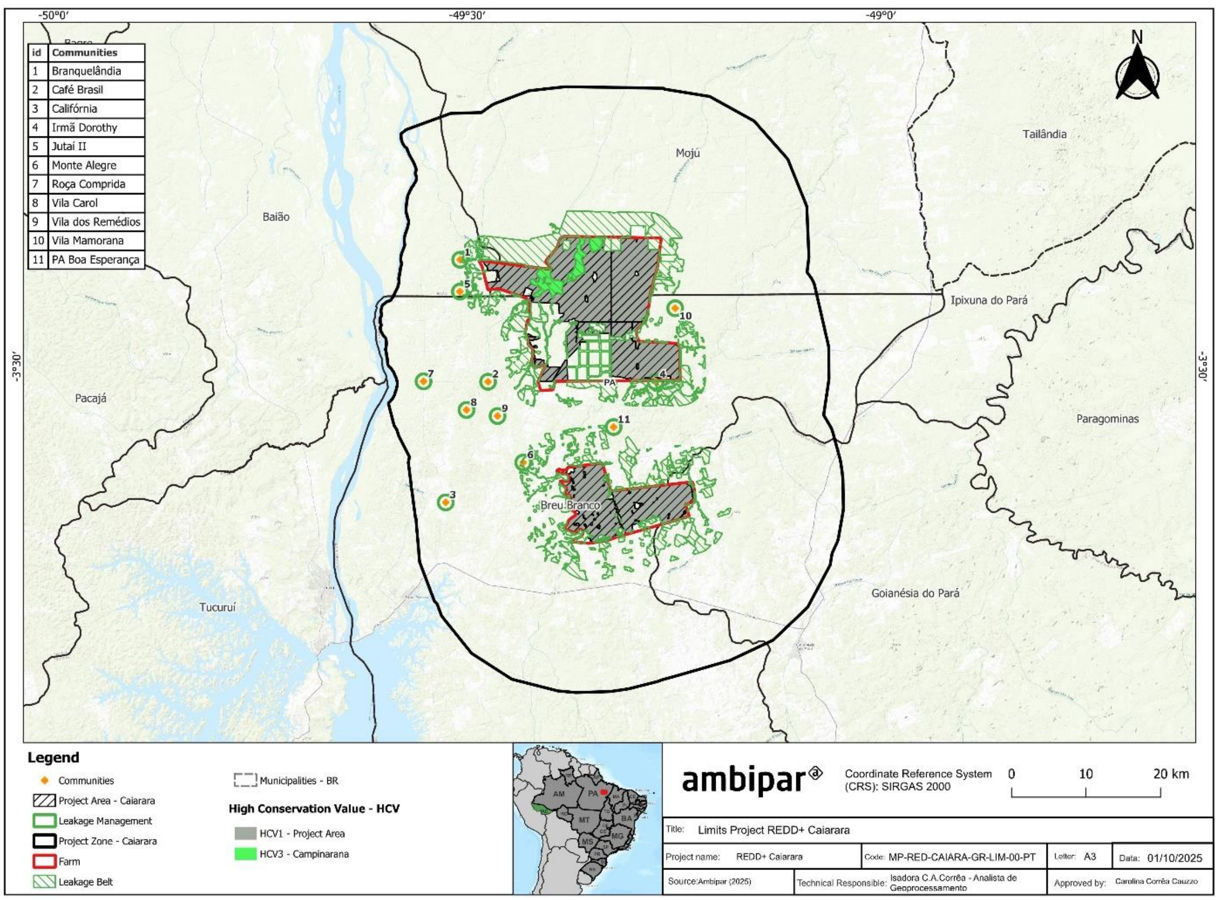
With regard to education, most respondents have completed or incomplete high school education (48%), followed by those with elementary school education (31%). A considerable portion, 18%, have completed or incomplete higher education, while 3% of respondents are illiterate. This educational distribution shows a diversity of educational levels, reflecting different degrees of training and professional opportunities among workers.

In terms of income, most respondents (40.24%) reported monthly family earnings between 1 and 2 minimum wages, with a similar group (39.63%) reporting income between 2 and 4 minimum wages. This points to a predominance of low- to middle-income families. However, 6.71% of respondents said they lived on less than one minimum wage, revealing the presence of families in a situation of financial vulnerability. When asked about the adequacy of their income, 32% considered it more than sufficient, while 41% said it was sufficient to cover expenses. However, 23% pointed out that their income often results in small debts, and 4% reported that it is insufficient to support their families.

As for housing, most workers live in the municipality of Breu Branco (46%) or in surrounding communities (45%). Before working on the farm, 41% already lived in neighboring communities, and 40% came from nearby municipalities such as Breu Branco, Moju, and Tucuruí, demonstrating the geographic mobility of a considerable portion of workers in search of opportunities in the region.

2.1.16 Map of the Project Area and Project Location (VCS, 3.11, 3.18; CCB, G1.4-7, G1.13, CM1.2, B1.2)

Figure 8. Map of the Project location.



2.1.17 Project Activities and Theory of Change (VCS, 3.6; CCB, G1.8)

The Theory of Change is a planning model used by social entrepreneurs and organizations to describe the desired impact caused by an intervention or program in a given area. It brings together attributes for evaluating, measuring, and monitoring the impact of a program, working from a long-term results perspective, analyzed from a sequence of intermediate results.

To construct the Theory of Change for the Caiarara REDD+ Project, a survey was initially conducted of the main problems faced in the project area and its surroundings, considering the information gathered by Socioeconomic and Environmental Diagnostics, and the root causes of each problem were identified. Thus, it was possible to outline activities that would influence changes in search of a solution for each root cause.

After validation between proponents and stakeholders, the project activities were defined: Strengthening asset surveillance; Monitoring deforestation and hot spots; Biodiversity conservation and monitoring; Strengthening local governance; Developing and strengthening value chains; Improving stakeholder well-being; Promoting educational activities; and Strengthening inclusion, diversity, and equity. These activities have a common goal, which is the main objective of the project: to promote joint actions aimed at reducing greenhouse gas emissions resulting from unplanned deforestation and forest degradation (REDD+).

This set of interconnected actions allows for the generation of financial resources, mainly from the sale of REDD+ credits registered with the VCS (*Verified Carbon Standard*), associated with social development and the conservation of natural resources, and ultimately seeking to ensure adequate financing for the fulfillment of the above objectives, as well as allowing for their maintenance throughout the entire life cycle of the Caiarara REDD+ Project. The Project activities are described below:

- **Strengthening asset surveillance**

Asset surveillance plays a crucial role in protecting natural resources and conserving biodiversity. In tropical forest areas, such as the project area, strengthening surveillance is essential to prevent damage to the ecosystem, ensure the integrity of forest carbon stocks, and promote the preservation of environmental services that sustain local and global communities. This initiative also contributes directly to mitigating climate change by preventing greenhouse gas (GHG) emissions associated with deforestation and other illegal practices.

In the project area, it is possible to identify the entry of unauthorized people who engage in illegal hunting and fishing, as well as timber extraction for charcoal production. These activities threaten local biodiversity, compromise ecological balance, ecosystem services, and forest carbon stocks, increasing the area's vulnerability to external pressures.

The activity of strengthening asset surveillance involves intensifying the patrols already carried out in the project area, with the addition of new technologies to ensure continuous, traceable, and effective surveillance. These measures aim to increase the frequency of patrols and standardize the recording of daily activities, providing greater control over the area. In addition, surveillance also involves monitoring and recording forest fires, maintaining fences and firebreaks, and responding to possible invasions, among other occurrences.

As a result, a reduction in invasions, illegal hunting and fishing practices, as well as unauthorized timber extraction is expected, directly contributing to the conservation of forest resources and ecosystem services in the region. This improved surveillance will help preserve forest carbon stocks, avoiding greenhouse gas (GHG) emissions associated with deforestation and other illegal activities. In addition, the protection of wildlife species targeted by illegal hunting and fishing will be strengthened, promoting the conservation of local biodiversity. In terms of relevance, this activity is directly linked to containing anthropogenic pressure that leads to deforestation, playing a crucial role in preventing GHG emissions, protecting fauna and flora, and maintaining the ecological balance of the Project Area, preventing damage caused by illegal activities.

- **Monitoring deforestation and hot spots**

Continuous remote monitoring of forest areas is essential to ensure their conservation and the protection of natural resources. Monitoring technologies, such as the use of satellite imagery, play a key role in identifying deforestation and hot spots, enabling rapid and accurate action to mitigate damage. In addition to contributing directly to the reduction of greenhouse gas (GHG) emissions, this activity is crucial for the preservation of biodiversity and ecological balance in areas of high environmental vulnerability.

As mentioned earlier, with the entry of unauthorized people and agricultural expansion in the region, forest fires and deforestation can be common and can affect the project area. Therefore, the activity of Monitoring deforestation and hot spots complements the activity of Strengthening asset surveillance by providing remote and continuous monitoring of the Project Area. Using satellite monitoring technologies such as MapBiomas Alerta, regular reports are generated that record the occurrence of deforestation and hot spots, providing an accurate view of the dynamics of deforestation and forest fires in the region. This data not only allows for the assessment of the impacts generated by these occurrences but also provides valuable information to guide patrols and field interventions, making asset surveillance more effective.

With active monitoring, it is expected that there will be a greater capacity to prevent and mitigate deforestation, avoiding forest degradation and the consequent emission of greenhouse gases (GHG). In addition, the early identification of possible fire and deforestation hotspots allows for the adoption of rapid corrective measures, minimizing negative impacts on the local ecosystem. This activity is directly linked to the containment of anthropic pressure that leads to deforestation and is essential for the prevention of GHG emissions and forest conservation, reinforcing the project's commitment to the protection of biodiversity and natural resources in the region.

- **Biodiversity conservation and monitoring**

The biodiversity conservation and monitoring activity aims to periodically monitor the fauna and flora species present in the project area, ensuring the preservation of local ecosystems. This activity includes regular monitoring of species, providing specific data on population dynamics and behavior, especially those that are endemic and threatened. In addition, there is coordination for the possible involvement of stakeholders throughout the project, promoting local engagement and awareness of the importance of biodiversity. Continuous monitoring will allow for a deeper understanding of existing species and provide a robust database to support decision-making, especially with regard to the management of invasive species and the preservation of endangered species.

In the long term, the activity contributes to the balance of the ecosystem, empowering stakeholders to become directly involved in biodiversity protection and creating a network of local knowledge. The relevance of this activity lies in monitoring species that are essential to the region, including those threatened with extinction, as well as encouraging community engagement in conservation issues, reinforcing the importance of biodiversity as a vital resource for the environmental sustainability of the Project Area.

- **Strengthening local governance**

The activity to strengthen local governance seeks to consolidate stakeholder participation through regular meetings with community employees and leaders, as well as defining effective communication strategies between the parties involved in the project. The implementation and consolidation of a specific communication plan and channel ensure a continuous flow of information and provide a space for mapping suggestions and complaints. As a result, effective and continuous communication between stakeholders and the project proponent is expected, strengthening community engagement both in relation to basic rights and services and to the activities proposed by the project.

Strengthening local governance, which will be conducted in partnership with local community leaders and trained facilitators, also contributes to the efficient management of activities, empowering communities and workers and promoting social capital by improving the social organization of the territory and relations between different actors. The relevance of this activity lies in its ability to promote stakeholder engagement with the project, ensuring clear and fluid communication between proponents and those involved, which is critical to the success of local initiatives.

- **Development and strengthening of value chains**

The activity of developing and strengthening value chains aims to map existing productive activities in communities and identify new opportunities that can be implemented in a sustainable manner. Through access to technical training and specialized assistance, in partnership with local community leaders and trained facilitators, community members will be prepared to adopt productive systems designed for both the market and their own consumption.

As a result, sustainable production models are expected to be implemented, which will contribute to improving the quality of local agriculture, diversifying activities, and strengthening human capital. These processes will lead to important impacts, such as the sustainable strengthening of agricultural practices, food security, diversification of family income, and a reduction in hunting for food. In addition, this activity also promotes adaptation to climate change, as sustainable and diversified practices enable communities to better face environmental challenges.

The relevance of this activity lies in offering sustainable production alternatives that prevent deforestation and hunting, as well as integrating new environmentally responsible practices into the daily lives of communities, ensuring greater resilience and long-term sustainability.

- **Improved well-being of stakeholders**

The socioeconomic assessment identified habits related to basic sanitation and food hygiene that may be harming the health of communities. In view of this, in partnership with local community leaders and trained facilitators, the activity aims to guide changes in the cultural habits of people around the project.

In the short term, socioeconomic assessments and meetings with communities are held to map risks to well-being, in addition to workshops on topics such as health and working conditions. In the medium term, initiatives include promoting knowledge about food safety and proper consumption practices. In the long term, improvements in health and eating habits are expected to be consolidated, ensuring a positive and lasting change in the quality of life of the communities.

As a result, these actions will contribute to better health conditions and greater awareness of dietary practices. By integrating health, education, and sustainability, the project promotes not only the well-being of communities, but also the construction of a more balanced and resilient future.

- **Promotion of educational activities**

The promotion of educational activities focuses on workshops and seminars on environmental, social, and economic issues relevant to stakeholders, which will be conducted in partnership with local community leaders and trained facilitators. In addition, ongoing meetings and dialogues will be promoted to share information about the Caiarara REDD+ Project, its objectives, and the importance of preserving the forest ecosystem. It also includes training for Palmyra employees, ensuring they have the necessary knowledge to support the project's activities and their daily work. The expected results include increased knowledge about the project and its activities, greater awareness of the importance of the forest and the reduction of harmful practices such as deforestation and hunting, as well as technical training for employees.

In the long term, this activity seeks to promote changes in cultural habits, encouraging sustainable practices within communities, while raising the level of education and qualifications of both employees and the communities involved. As a result, greater engagement with the Caiarara REDD+ Project is expected, strengthening environmental preservation and promoting a greater understanding of the importance of the standing forest and local biodiversity.

- **Strengthening inclusion, diversity, and equity**

The activity of Strengthening inclusion, diversity, and equity, which will be conducted in partnership with local community leaders and trained facilitators, seeks to encourage education through workshops and lectures on topics such as mental and physical health and income alternatives, among others, in addition to offering an active voice in project activities. As a result, it is hoped that historically marginalized groups will become actively involved in the project's activities and gain greater respect and value in society and greater competitiveness in the region's labor market.

In the long term, this activity can influence the improvement of health, greater appreciation of the role of women and other groups in the community and in the workplace, as well as enabling alternative income for affected individuals. Thus, it is possible to have the active participation of this portion of the stakeholders in the Project's activities, give greater value to these groups from the community's perspective, and promote opportunities for personal and professional life changes.

The following table was developed using Appendix 2 of the official VCS/CCB template and provides a summary description of the activities and their respective outputs, outcomes, impacts, and relevance to Climate, Community, and Biodiversity, which will contribute to achieving the Project's intended benefits.

Table 1. Caiarara REDD+ Project activities and their respective outputs, outcomes, impacts, and relevance to the Project.

Climate	Biodiversity	Community	Activity	Expected climate, community and/or biodiversity			Relevance to project objectives
				Output (short term)	Results (medium term)	Impacts (long term)	
			Strengthening asset surveillance	* More frequent patrols using technology in the project area. * Defined and standardized procedure for recording daily activities.	* Reduction in invasions in the project area. * Reduction in illegal hunting and fishing. * Reduction in timber removed from the area.	* Conservation of forest resources and preservation of ecosystem services in the Project Area; * Conservation of forest carbon stocks. * Avoidance of GHG emissions. * Protection of wildlife species targeted by illegal hunting and fishing that inhabit the project area.	* Avoid GHG emissions. * Avoid illegal deforestation, hunting, and fishing by third parties. * Conservation of fauna and flora in the Project Area.
			Monitoring of deforestation and hot spots	* Remote monitoring of the Project Area. * Reports on deforestation and hot spots. * Monitoring and assessment of the impacts caused by deforestation of hot spots.	* Understanding the dynamics of deforestation and hot spots to conduct effective asset surveillance. * Provision of inputs for building and improving field interventions.	* Mitigation and prevention of deforestation. * Prevent and avoid GHG emissions from deforestation and forest degradation in the project area.	* Identifying potential deforestation and forest fire hotspots in the Project Area. * Avoid deforestation, fires, and GHG emissions from deforestation and forest degradation.
			Biodiversity conservation and monitoring	* Periodic monitoring of fauna and flora species in the project area. * Coordination for possible involvement of stakeholders in biodiversity monitoring throughout the project.	* Knowledge of local fauna and flora species. * Specific data on the population dynamics and behavior of existing species, including endemic and endangered species. * Local environmental engagement and awareness of biodiversity and its importance.	* Involvement and training of stakeholders in local biodiversity. * Robust database on local biodiversity. * Balanced ecosystem for resident biodiversity. * Consolidated knowledge about invasive species for sound decision-making.	* Monitor existing species, including endemic and endangered species in the Project Area. * Engagement of stakeholders on the issue of biodiversity protection and its importance.

			Strengthening local governance	* Holding meetings with employees and community leaders. * Communication strategies defined between stakeholders and proponent. * Implementation and consolidation of communication channels with stakeholders.	* Effective communication between stakeholders and proponent through communication channels. * Collective engagement in demanding basic rights and services * Mapping and resolution of suggestions and complaints from stakeholders. * Engagement with the activities proposed by the Project.	* Efficient management and implementation of activities. * Community empowerment and strengthening of social capital. * Improvement in the social organization of the territory and in relations between stakeholders. * Effective communication between stakeholders and project proponents.	* Engagement of stakeholders with project activities. * Good communication between the proponent and stakeholders.
			Development and strengthening of value chains	* Mapping of activities already carried out and others that can be implemented. * Access to technical assistance. * Community members technically trained through courses and able to conduct sustainable land use practices. * Agriculture and other practices planned for the market and for own consumption.	* Implementation of sustainable production systems and models. * Increased quality agriculture and diversification of production. * Strengthening of human capital and increased local knowledge of sustainable practices.	*Sustainable strengthening of agricultural practices. * Food security. * Diversification and increase in income. * Reduction in hunting for food and family income. *Adaptation to climate change	* Production alternatives to prevent deforestation and hunting. * Sustainable alternatives for activities already present in the daily lives of communities. *New activities to complement the communities' production activities.
			Improved well-being of stakeholders	* Mapping risks to stakeholder well-being through socioeconomic diagnosis and stakeholder meetings. * Workshops on health, infrastructure, and working conditions.	* Knowledge about food safety and proper consumption practices. * Alternatives for easier access to healthcare.	* Improvement of health and eating habits.	* Better health conditions for stakeholders.

Promotion of educational activities	* Workshops on environmental, social, and economic issues. * Meetings and dialogues about the project, its objectives and importance. * Training and capacity building for Palmyra employees.	* Knowledge about the REDD+ Água Azul Project and its activities. * Knowledge about the importance of the forest ecosystem and the importance of not practicing deforestation and hunting. * Training for Palmyra employees	* Changes in cultural habits related to hunting and deforestation in communities. * Higher level of education among employees and communities. * Engagement with the REDD+ Caiarara Project.	* Understanding of the importance of standing forests and fauna and flora species for ecosystems. * Stakeholder understanding and engagement with the Caiarara REDD+ Project. * Improved education and training for stakeholders.
Strengthening inclusion, diversity, and equity	* Encouraging education for historically marginalized groups. * Giving historically marginalized groups an active voice in project activities. * Workshops and lectures on various topics involving historically marginalized groups.	* Engaging historically marginalized groups in project activities. * Increased competitiveness in the market.	* Improved mental and physical health of historically marginalized groups. * Historically marginalized groups with income alternatives outside family activities. * Greater appreciation of women's work and role in the community and by women themselves.	* Active participation of historically marginalized groups in project activities. * Opportunity for young people and women to enter the labor market. * Appreciation of the role of women in the community. * Appreciation of historically marginalized groups in society.

2.1.18 Contributions to Sustainable Development (VCS, 3.17)

The REDD+ Caiarara Project will be implemented on the RAAI and RAAII farms, located in the municipality of Breu Branco (PA), and focuses on the conservation of native vegetation through the strengthening of asset surveillance, sustainable agricultural technical assistance, environmental education, and actions aimed at the well-being of communities and workers, promoting sustainable development in the region. Thus, the project will contribute to the following United Nations Sustainable Development Goals (SDGs):

SDG 1 – Eradicating poverty

1.5 By 2030, build the resilience of the poor and those in vulnerable situations and reduce their exposure and vulnerability to extreme climate-related events and other economic, social and environmental shocks and disasters.

The Caiarara REDD+ Project seeks to contribute to improving the quality of local agriculture, diversifying activities, and strengthening human capital through the development and strengthening of value chains. In addition to food security, this activity also promotes adaptation to extreme climate-related events, since sustainable and diversified practices enable communities to better face environmental challenges, offering sustainable production alternatives that avoid deforestation and hunting, as well as integrating new environmentally responsible practices into the daily lives of communities, ensuring greater resilience and long-term sustainability.

SDG 4 – Quality education

4.4 By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent work, and entrepreneurship.

4.7 By 2030, ensure that all students acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship, and appreciation of cultural diversity and of culture's contribution to sustainable development.

The Caiarara REDD+ Project promotes quality education through training for workers, addressing everyday techniques and occupational safety, improving professional qualifications, and encouraging entrepreneurship. In addition, it conducts environmental education workshops focused on forest conservation and the preservation of endangered species, training both workers and local communities. Technical assistance for sustainable agricultural production will also be offered, contributing to community empowerment, promoting sustainable development, gender equality, and global citizenship.

SDG 6 – Drinking water and sanitation

6.6 By 2020, protect and restore water-related ecosystems, including mountains, forests, wetlands, rivers, aquifers, and lakes.

The main objective of the REDD+ Caiarara Project is to preserve forest cover and carbon stocks, preventing illegal deforestation and forest degradation. An essential activity of the project is to strengthen surveillance on farms, ensuring the protection of vegetation in the project area. Forest conservation is crucial for the provision of water-related ecosystem services, as forests play an important role in regulating the water cycle, influencing factors such as precipitation, water availability and purification, as well as protecting soils, lakes and rivers.

SDG 12 – Sustainable consumption and production

12.8 By 2030, ensure that people everywhere have relevant information and awareness for sustainable development and lifestyles in harmony with nature.

The activity of promoting educational actions aims to raise awareness about the importance of forests and reduce harmful practices such as deforestation and hunting. The expected results are to promote changes in cultural habits, encouraging sustainable practices in communities, while raising the level of education and qualification, strengthening environmental preservation, and promoting a greater understanding of the importance of standing forests and local biodiversity.

SDG 13 – Action against global climate change

13.2 Integrate climate change measures into national policies, strategies, and planning.

By focusing on reducing greenhouse gas emissions from deforestation and forest degradation, the REDD+ Caiarara Project promotes sustainable forest management, which is essential for tackling climate change. Through the implementation of effective conservation and monitoring practices, the project aligns its activities with national climate change mitigation strategies, such as the National Policy on Climate Change (PNMC), the Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm), and Brazil's Nationally Determined Contributions (NDC) under the Paris Agreement.

15 VIDA TERRESTRE

SDG 15 – Life on land

15.1 By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in accordance with international agreements.

15.5 Take urgent and significant steps to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species.

15.c Strengthen global support for efforts to combat illegal hunting and trafficking of protected species, including by increasing the capacity of local communities to pursue sustainable livelihood opportunities.

The Caiarara REDD+ Project is located in the Belém Endemism Center, a region with a high concentration of endangered and endemic species. The project's activities are focused on protecting the forest, preventing deforestation and, consequently, reducing habitat loss, species exploitation and the impacts of climate change.

To achieve these objectives, the project promotes awareness among stakeholders about the importance of biodiversity for the maintenance and functioning of ecosystem services, the control of environmental degradation, and the promotion of sustainable use of natural resources. In addition, continuous monitoring of fauna and flora, as well as studies on the natural and socioeconomic resources necessary for CCB certification, also contribute to protecting endangered and endemic species in the region, integrating biodiversity and ecosystem values into local and national planning, aligning development with conservation strategies.

Monitoring contributions to sustainable development will be carried out based on the socio-environmental indicators set out in the project's monitoring plan, and the results will be communicated in periodic reviews and through the communication channels defined with stakeholders.

2.1.19 Implementation Schedule (CCB, G1.9)

Date	Milestones in the project development and implementation
October 2023	Contract between project proponents.
November 2023	Workshop with suppliers and proponents to develop the project design.
December 2023 to January 2024	Socioeconomic, environmental, and carbon stock assessments.
January 21, 2024	Project start date (hiring of new security company)

July 2024	Feedback event with stakeholders
August 2024	Hiring of the Validation/Verification body.
December 2024	Consolidation of the Communication Plan and opening of the Project's official Communication Channels.
March 2025	Lecture on water and health with the Vila Mamorana community.
December 2025	Field audit for validation and verification.
Beginning in 2024 until the last year of the Project	Development and monitoring of social and environmental management activities.
	Continuous monitoring of deforestation and emissions, with annual data analysis and compilation.
Forecast every 2 years	GHG and CCB verifications
January 21, 2024 to January 20, 2064	GHG emissions accounting period

2.1.20 Risks to the Project (CCB, G1.10)

Using the "AFOLU Non-Permanence Risk Tool v4.2," we verified the likely natural and human-induced risks to climate benefits reported in the REDD+ Caiarara Project's "VERRA Project Hub." The Non-Permanence Risk analysis using the aforementioned tool generated a buffer of 17%.

In conjunction with the tool, probable risks to the expected benefits for the climate, community, and biodiversity during the project's lifetime were identified, as well as their respective mitigation measures, as described in the table below.

Table 2. Risks to the project with their potential impacts and mitigation actions.

Identified risk	Potential impact of the risk on climate, community, and/or biodiversity benefits	Necessary and planned actions to mitigate the risk
Forest fires	Forest fires occur naturally but are more frequently induced by humans. Cutting and burning activities are common among community members in the region.	The properties involved in the project are equipped to respond quickly to forest fires. Both properties have water trucks, employees trained in firefighting, and PPE

	Forest fires in the region can spread beyond the project area and cause forest degradation. In addition, this practice pollutes the air, directly affecting the health of people in the region, as well as contributing to a drier and hotter climate, intensifying the impacts of climate change.	for the activity. Fire brigade training is conducted annually for farm employees. In addition, there are also preventive measures to stop forest fires from happening, like building firebreaks in key spots in the forest. To influence the reduction of cutting and burning practices in the region, the project will focus on environmental education to raise awareness of the impacts on the region's biodiversity and the well-being of the community itself.
Drought	Communities in the region may suffer from a lack of water for agricultural irrigation due to reduced rainfall caused by longer periods of drought, which is influenced by climate change. The adversities faced by agriculture directly impact the well-being of communities that use agricultural products for subsistence and as a source of income.	The project will work with relevant agencies and institutions to provide technical assistance and workshops to help farmers adapt their practices to climate change. In addition, all project activities aim to conserve the forest and its biodiversity, contributing to reducing the impacts of climate change.
Entry of third parties into the project area in search of extractive products or to engage in illegal hunting and fishing activities.	Illegal entry by community members to extract forest products may cause deforestation or forest degradation, raising the possibility of timber extraction. Entry may also impact the biodiversity of the project area, as illegal hunting and fishing are still common in the region, even with signs along the perimeter of the properties prohibiting such activities.	To prevent third parties from entering the project area, property surveillance will be reinforced. In addition, educational activities will be carried out for communities surrounding the project with the aim of raising awareness among community members about the negative impacts of deforestation and predatory hunting and fishing activities on the region's biodiversity.
Lack of interest from stakeholders	The lack of interest in participating in Project activities directly impacts the benefits that would be generated at	Strengthening and encouraging greater involvement of all stakeholders in decision-making processes related to

in participating in Project activities	the community and farm worker levels. Furthermore, the lack of interest on the part of public agencies may compromise potential partnerships for the development of Project activities.	Project activities, such as capacity building and technical assistance to encourage sustainable and adaptive agricultural production practices and models. Another important measure is the project communication plan, which involves communication channels tailored to the reality of the stakeholders and procedures for addressing complaints.
Failure to manage financial resources	Difficulty in implementing the activities proposed by the Caiarara REDD+ Project, in addition to the difficulty of allocating resources for conservation and sustainable development in the region.	Biofilica Ambipar Environmental Investments S/A has a financial team responsible for managing project resources.
Market Risk - Difficulty in marketing verified carbon credits	Difficulty in marketing the carbon credits generated by the Project could directly impact the proposed activities, consequently affecting the three scopes of the Project – climate, community, and biodiversity – as it would hinder investment in these activities.	There is an ongoing effort to identify new financing, business, and activity opportunities, such as partnerships and donations for direct use in project activities (not necessarily linked to the sale of credits). Additionally, the focus is on consolidating and expanding the commercial network to promote the project.

The Caiarara REDD+ Project has an Adaptive Management Plan that focuses on identifying, preventing, and/or mitigating risks through the adaptability of the implementation of actions and activities, considering the monitoring of indicators, evaluation of lessons learned, and the incorporation of these into future decision-making. The document is justified by the need to evaluate the implementation of a project, considering actions and/or reactions to adversities and unforeseeable scenarios, and ensuring good management, quality of results, actions directly linked to reality, and financial health. In addition to the Adaptive Management Plan, the project has a Management, Financial, and Monitoring Plan, which aims to demonstrate that the REDD+ Caiarara project has a solid governance structure, financial capacity, and effective monitoring mechanisms to ensure environmental and operational integrity throughout its useful life.

2.1.21 Permanence of Benefits (CCB, G1.11)

The activities designed for the REDD+ Caiarara Project have been carefully planned to achieve positive results in the short, medium, and long term, ensuring that the benefits for the climate, communities, and biodiversity become self-sustaining in the long term. With the aim of maintaining and enhancing these benefits during the 40-year life of the Project and beyond, actions are focused on strengthening local governance capacity, improving human capital, and promoting decisions that favor the sustainability of natural resources.

These actions aim to empower communities to determine their own sustainable paths, ensuring that the benefits for the climate, biodiversity, and communities are perpetuated after the end of the Project. The strategies developed to ensure this permanence are:

Strengthening asset surveillance: strengthening patrol activities around farm perimeters will enable more efficient land management, minimizing invasions, deforestation, and forest degradation. This strengthening will directly result in the maintenance of climate benefits beyond the Project's lifetime.

Strengthening local governance: Empowering community leaders and encouraging social organization aim to increase the capacity for dialogue and resolution of environmental, social, and land issues of collective interest. Awareness and collective engagement are essential to maintaining the social and environmental benefits of the Project in the long term.

Strengthening value chains: Through technical assistance, the Project promotes sustainable agricultural and environmental practices, focusing on production diversification, reduction of burning, and decreased dependence on illegal activities such as logging and predatory hunting. These actions seek to ensure food security and generate economic autonomy, ensuring that families remain in the region and expand their income opportunities.

Biodiversity conservation and monitoring: The Project also includes environmental education and awareness campaigns, as well as a continuous biodiversity monitoring program. These activities seek to transform the relationship between communities and the environment, promoting knowledge and protection of biodiversity beyond the Project's lifetime.

Thus, the Caiarara REDD+ Project seeks to promote sustainable development in the region, mitigating pressure on biodiversity and reducing greenhouse gas emissions. The results of the planned activities will be absorbed by local actors throughout the Project's lifetime, ensuring that climate, community, and biodiversity benefits extend for many years after its completion.

2.1.22 Financial Sustainability (CCB, G1.12)

Biofilica Ambipar Environmental Investments S/A established, together with the proponent of the Caiarara REDD+ Project, a robust partnership instrument that aims, among other objectives, to enable long-term conservation investments in the RAAI and RAAII farms through the commercialization of environmental assets.

Considering current carbon market assumptions and the potential for generating GHG emission reductions, the financial flow of the REDD+ Caiarara Project presents very attractive results. In this model, the proponent and developer expect to recover the investment made in the Project when the commercialization of GHG emission reductions begins.

Other information related to the financial analysis of the REDD+ Caiarara Project and the proponent's financial health statements is considered commercially sensitive and has been shared with the audit team on a confidential basis.

2.2 Land use scenario and additionality without the project

2.2.1 Conditions Prior to Project Start and Land Use Scenarios without the Project (VCS, 3.13; CCB, G2.1)

The project area is located in the Amazon biome, characterized by tropical rainforests with high biological diversity and a closed canopy structure, with native vegetation of Dense Ombrophilous Forest type, characterized by tropical climatic factors of high temperatures (averages of 25ºC) and high rainfall well distributed throughout the year (0 to 60 dry days), which determines a bioecological situation with virtually no dry season. The project area is located in the Baixo Tocantins hydrographic mesoregion and the Project Area is located near the Tocantins River, one of the main rivers in the North region, and close to the Tucuruí Hydroelectric Plant (UHE Tucuruí), one of the largest in the country.

Some main factors are important to contextualise the conditions prior to the project: land insecurity in the state of Pará, slash-and-burn agriculture carried out by rural communities and small landowners, the production of commodities such as meat and grains by capitalised medium and large landowners, and the demand for charcoal for the steel industry.

Historically, the project area has been occupied by rural properties dedicated to extensive agriculture and livestock farming, with predominant use for pasture and cropland. Part of Palmyra's properties have been used for commercial eucalyptus plantations with important reforestation areas for the sustainable supply of vegetable raw materials, charcoal, and wood chips used in the company's silicon metal furnaces. Today, planted forest plots represent 14.9% of the total area of the properties Reflorestamento Água Azul I and II, exceeding the 80% required by Brazilian law for native forest cover for preservation, despite constant pressure from deforestation and invasions.

The municipalities in the Project Zone have an economy based on agricultural activities, especially the temporary crops of manioc and corn and the permanent crops of palm oil, açaí, bananas, black pepper and cocoa, as well as livestock farming, mainly for beef cattle, and the extraction of forest products such as logs and charcoal. In relation to the reference municipalities where the communities studied are located, in Breu Branco and Goianésia do Pará timber extraction and livestock production are the predominant economic activities, while Moju shows diversity in agricultural production (STA, 2024).

The most recent history of occupation in the region was due to the government's strategy for the Amazon in the 1960s and 1970s, which intensified the migratory process to relocate people for the construction of the Tucuruí Hydroelectric Power Station (CAVALCANTE, 2005) [23]. The municipality of Breu Branco arose to house families who lived on the banks of the Tocantins River in areas that were submerged by the

[23] CAVALCANTE, F. C. O processo migratório na Amazônia vinculado à mobilidade pelo trabalho – o caso da UHE de Tucuruí. Anais do X Encontro de Geógrafos da América Latina – 20 a 26 de março de 2005 – Universidade de São Paulo.

hydroelectric dam. In the 1990s, the municipality began to invest in infrastructure, sanitation and services, and the town continued to grow due to its proximity to the hydroelectric dam and the trans-Amazonian motorway and due to the cluster of silicon mining companies available in the region and logging companies (BARATA, 2019)[24]. However, due to the lack of work, many families migrated to the countryside as a way of staying in the region, with the development of subsistence farming. This new working relationship in the countryside has influenced the emergence and expansion of new rural communities, as in the case of the municipality of Breu Branco, where there are several recent communities.

Subsistence farming is the basis of the productive-economic activities of these communities, with most families earning a minimum wage per month (STA, 2024). The agricultural production method used, called itinerant slash-and-burn agriculture, is consolidated as one of the forms of deforestation in the region. It is defined by a few years of cultivation followed by a few years of resting the land and includes slashing, felling and burning to clear new areas. With the exception of one community, all those interviewed in the diagnosis use fire in productive activities (STA, 2024).

In the 1980s, as a result of tax and credit incentive policies, several pig iron mills, which consume large quantities of charcoal as an input in their production process, set up in the eastern Brazilian Amazon (MONTEIRO, 2004)[25]. The establishment of these companies, mainly in the Carajás region in the south-east of Pará, represents an immense pressure on native vegetation: it is estimated that the accumulated export of pig iron up to 2005 has caused illegal deforestation of more than 800,000 hectares of dense forest (Embrapa, 2006)[26]. According to the same article, studies carried out in the south-east of Pará between 1999 and 2004 showed that the felling of forests, the extraction of timber and the planting of swiddens, followed by the formation of pastures in the settlement projects, are related to the utilisation of wood for the production of charcoal.

According to IMAZON[27], in 2013, 39% of Pará's territory had pending land regularisation, and this same area concentrated 71% of the state's deforestation. By analysing and integrating databases from public bodies and registry offices in 10 municipalities in the state, SIG Fundiário (Geographic Land Information System), a platform developed by the Federal University of Pará in collaboration with the Public Prosecutor's Office and the Pará State Court of Justice, identified in its preliminary results 22.7 million hectares of private land and 18.5 million hectares of extra public land, i.e. land that doesn't actually exist, due to the overlapping of areas registered with notaries, creating private and public "ghost lands" (FIAVORANTE, 2022)[28]. In short, it is clear that land insecurity in Pará is serious: a large part of the territory remains

[24] BARATA, C. J. Dinâmica demográfica e redistribuição populacional na área de influência da usina hidrelétrica de Tucuruí, estado do Pará. In: Simpósio nacional de geografia urbana, XVI., 2019, Vitória – ES. Anais [...] Vitória – 14 a 17 de novembro de 2019.

[25] MONTEIRO, M. de A. A Produção de Carvão Vegetal na Amazônia: Realidades e Alternativas. Papers do NAEA nº 173. Belém, junho de 2004. Disponível em: <http://dx.doi.org/10.18542/papersnaea.v13i1.11555>.

[26] HOMMA, A. K. O., ALVES, R. N. B., MENEZES, A. J. E. A. de, MATOS, G. B. de. Guseiras na Amazônia: perigo para a floresta. Embrapa Amazônia Oriental. Ciência Hoje, v. 39, n. 233, p. 56-59, dez. 2006. Disponível em: <https://www.alice.cnptia.embrapa.br/alice/bitstream/doc/376354/1/Guserahomma.pdf>.

[27] BRITO, B., BAIMA, S., SALLES, J. Pendências fundiárias no Pará. IMAZON, 06 de Agosto de 2013. Disponível em: <https://imazon.org.br/pendencias-fundiarias-no-para/>.

[28] FIAVORANTE, C. As Terras Imaginárias do Pará. Pesquisa FAPESP 279, Maio de 2019. Disponível em: <https://revistapesquisa.fapesp.br/wp-content/uploads/2019/05/076-081_Par%C3%A1_279.pdf>.

unregulated, showing deep flaws in the control of properties, which ends up favouring conflicts and deforestation.

All of these factors are relevant because they explain the dynamics of deforestation in the project region: the low diversity of income generation sources in rural communities, which perpetuates slash-and-burn agriculture, the demand for wood to produce charcoal illegally and the pressure on forests from the advance of livestock and grain production, coupled with the lack of land-title regularisation and the inefficiency of public authorities in monitoring, contribute to the deforestation scenario observed in the region. As a result, if nothing is done, the future trend for the scenario without the project is for the dynamics that cause growing deforestation to continue.

In the scenario prior to the project, the proponent and owner of the land already carried out surveillance actions to contain invasions on the property and social actions with a neighbouring community for local development. However, surveillance is carried out randomly, without management, control or traceability of information, with little effect on curbing illegal activities and fully protecting the forest and biodiversity. As for the social activity carried out, the Ybá Project, which focuses on generating income from andiroba extraction in the forests of the Project Area, covers a few people from just one community (Vila Mamorana), resulting in a limited impact. Therefore, the activities in the business-as-usual scenario are not enough to change the dynamics that lead to deforestation. In other words, the actions being carried out prior to the REDD+ Project are inefficient and insufficient to contain the advance of deforestation and develop the region socio-economically in a sustainable manner.

The baseline scenario is equivalent to the land use conditions and environmental dynamics observed before the project was implemented. Therefore, please refer to Section 3.1.5 (Baseline Scenario) for a full description of the without-project scenarios, agents and drivers of land use change. Among the realistic and credible alternative land use scenarios that would occur within the Project boundaries in the absence of the AFOLU Project activity recorded in the VCS, the following were considered:

<u>(I) Continuation of pre-project land use:</u> Traditional land use practices, such as slash-and-burn agriculture, livestock and grain production and illegal charcoal production, continue to be widely employed by the agents causing deforestation in the project region, as described in Section 2.2.1 and 3.1.5. These activities are driven by the lack of sustainable economic alternatives and the growing demand for land and natural resources, resulting in a continuous cycle of deforestation. The expansion of cattle ranching and the increased risk of soya cultivation entering the region are vectors of environmental degradation, and illegal logging and hunting are not contained due to ineffective enforcement, resulting in the constant degradation of native forests. In this scenario, the activities carried out by the landowner to monitor and invest in the surrounding community will not be enough to contain the action of the main agents and vectors of deforestation, and the rural communities will not have the economic strength to produce and generate income in a sustainable way, thus maintaining the growing deforestation of the forests, perpetuating the conditions of land use prior to the Project.

<u>(II) Improvement of surveillance actions without registration in the VCS:</u> This scenario involves an increase in investment in property surveillance to improve inspection. The other actions already carried out by the owner are maintained: good sustainability practices, such as using up to 20 per cent of the land on the RAAI and RAAII farms to plant FSC-certified eucalyptus for charcoal production and keeping the remaining

areas preserved in accordance with current environmental laws, and the Ybá Project are also maintained. The difference with scenario 1 is the greater investment in surveillance of the properties, with the hiring of a new security company with improved management and control of activities, guaranteeing greater traceability of information. The aim of increasing investment in surveillance is to improve surveillance to prevent the entry of invaders, prevent illegal hunting and fishing and curb deforestation, based on improvements in the management and control of property security. However, this initiative is limited in scope and impact, as it only seeks to protect the forest in the Project Area but does not focus on changing the dynamics that lead to deforestation in the region, so it only treats the symptom but does not address the causes of the problem. The lack of resources to expand these actions to alter the actions of the agents prevents effective containment of the vectors of degradation, so the scenario is insufficient to prevent continued deforestation and develop the socio-economy in a sustainable manner in the region.

(III) REDD+ Project activities not registered in the VCS: The activities needed to protect the forest are implemented by focusing on the regional dynamics that cause deforestation, with the aim of preventing the action of the vectors and agents that generate deforestation, promoting climate, social and biodiversity benefits. According to the diagnoses carried out, actions include strengthening surveillance to prevent illegal deforestation, hunting and fires, as well as satellite monitoring and the conservation and monitoring of local fauna and flora, with a focus on endangered and endemic species. In the social sphere, communities require support such as training to generate sustainable income alternatives, promote the empowerment of historically minoritized groups and improve local governance. However, despite the promising initiatives, without VCS registration this scenario is financially unviable, as there is no specific source of income to guarantee the implementation of conservation and social development activities, limiting their long-term effectiveness.

2.2.2 Justification of the most likely scenario (CCB, G2.1)

Given the scenarios described above, scenario I is the most likely to be perpetuated if the Caiarara REDD+ Project is not implemented, continuing the deforestation and land use dynamics that are currently taking place, because field data, information from local agents and secondary data show a clear relationship between existing social practices and the intensification of pressure on forests. The choice of scenario I as the most likely is based on the data collected by the Socio-Economic Diagnosis carried out in 2024, which revealed the predominant use of practices such as slash-and-burn agriculture, illegal extraction of wood for charcoal and extensive cattle breeding, as well as the communities' interest in expanding these practices and the pressure from the advance of soya on medium and large properties around the communities. Deforestation data from MapBiomas (2022)[29] and IMAZON alerts (2021)[30] corroborate this regional pattern, showing continued forest loss in similar areas.

Without effective activities acting directly to contain the dynamics that lead to deforestation, to contain the agents and vectors, such as the expansion of livestock and soya, the illegal exploitation of resources and

[29] MapBiomas, 2023. Relatório Anual de Desmatamento 2022 - São Paulo, Brasil. - 125 páginas.

[30] IMAZON 2021. Boletim do desmatamento da Amazônia Legal (abril 2021) SAD. Disponível em: <https://imazon.org.br/publicacoes/boletim-do-desmatamento-da-amazonia-legal-abril-2021-sad/>.

the inefficiency of inspection, the forests will continue to be severely pressurized. The proponent's current actions of monitoring properties randomly and without management and investing in just one community in a moderate and restricted way, are not enough to contain forest degradation in the Project region and change the current conditions that contribute to deforestation. The social vulnerability of local communities, coupled with the lack of alternatives for sustainable production practices and public authorities unable to mitigate this context, points to a worsening scenario, with the risk of progressive extinction of forest areas if no measures are taken. The selection of the most likely scenario followed the steps described in tool VT0001 and methodology VM0048, with no methodological adaptations or deviations, as described in Section ***Erro! A origem da referência não foi encontrada.***

2.2.3 Community and Biodiversity Additionality (CCB, G2.2)

Without the Caiarara REDD+ Project, illegal deforestation, land invasions, and predatory practices such as illegal hunting and timber removal would continue to advance, aggravating environmental and social degradation in the region. Slash-and-burn agriculture, combined with growing demand for timber and extensive cattle ranching, would increase pressure on forests, leading to habitat destruction, biodiversity loss, and the extinction of local endemic and threatened species. The lack of effective enforcement and institutional weakness would contribute to the perpetuation of these processes. Without the project, illegal activities would not benefit local communities or biodiversity; on the contrary, they would intensify environmental degradation and social vulnerability.

With the implementation of the Caiarara REDD+ Project, the project area will benefit from a set of actions aimed at forest conservation, sustainable economic development, and the protection of biodiversity and natural resources. The project activities, supported by investments from the trade of carbon credits, will directly contribute to reducing deforestation rates, conserving forests, and mitigating pressures on natural resources. By coordinating technical assistance for sustainable agricultural and extractive practices, training and strengthening local governance, the project will offer viable alternatives for communities, generating socioeconomic and environmental benefits that would otherwise not be achieved, ensuring the reduction of deforestation, the preservation of local ecosystems, and improved quality of life for local actors.

The positive impacts of the project are due, in short, to its proposed activities, generating the following net benefits for surrounding rural communities, farm workers, and biodiversity, which would not occur in the absence of the project:

For Communities and Farm Workers:

- Training and capacity building on sustainable agricultural and extractive practices, offered by project partners;
- Strengthening of social organization and community governance practices;
- Improved communication and collaboration between workers and communities;
- Development of skills, knowledge, and human capacities related to income generation and sustainable use of resources;
- Strengthening of agricultural and forestry products value chains;
- Promotion of environmental awareness and retention of families and workers on their land.

- Empowering youth and women in communities and among workers.
- Strengthening and valuing historically marginalized groups.
- Improvement in the well-being of stakeholders.

For Biodiversity:

- Significant reduction in deforestation rates and protection against habitat loss;
- Promotion of the conservation of local fauna and flora, with a focus on species of high conservation value (HCSV), including endangered, rare, and endemic species;
- Ensuring genetic diversity and mitigating extinction risks;
- Implementation of environmental education programs to engage stakeholders in biodiversity protection.

The activities planned in the project would not be implemented without the incentives provided by the REDD+ mechanism, as they face significant barriers. From a financial standpoint, there are no sufficient regular sources of income to sustain conservation and local productive inclusion actions. Institutionally, communities lack technical and organizational support to carry out structured conservation and sustainable use activities. The absence of robust public programs and the difficulty of accessing credit aggravate this scenario, making the autonomous implementation of these actions unfeasible.

There are no government or private programs in the region capable of enabling continuous forest conservation actions focused on benefits generation for communities and biodiversity. Thus, the benefits described above depend directly on the implementation of the Caiarara REDD+ Project and its financial viability through the trade of carbon credits.

Although there are legal provisions aimed at protecting the environment and the rights of communities, such as the Forest Code (Law 12.651/2012), which establishes that 80% of the forested areas of a rural property in the Legal Amazon must be maintained with native vegetation cover, its application in the region is deficient. For example, data from MapBiomas indicates that only 5% of the deforestation that took place in the Amazon in 2019 was authorized and, according to IMAZON (2020)[31], only 30% of a total of 38,573 hectares of forest exploited by logging between August 2017 and July 2018 had authorization from the competent body, while 70% was carried out without proper authorization. According to the Federal Court of Auditors (2024)[32], a number of factors are responsible for the lack of law enforcement in the Amazon: insufficient budgetary resources to implement the actions, delays in implementing the monitoring system, disaggregation of actions between the various agencies responsible, lack of efficient central coordination and staff shortages, which jeopardize the government's ability to apply policies to control illegal deforestation in the Amazon.

[31] Cardoso, D., & Souza Jr., C. Sistema de Monitoramento da Exploração Madeireira (Simex): Estado do Pará 2017-2018 (p. 38). 2020. Belém: Imazon. Disponível em: <https://imazon.org.br/publicacoes/sistema-de-monitoramento-da-exploracao-madeireira-simex-estado-do-para-2017-2018>.

[32] Tribunal de Contas da União. Controle do Desmatamento Ilegal na Amazônia. Lista de Alto Risco da Administração Pública Federal, 2ª edição, 2024. Disponível em: <https://sites.tcu.gov.br/listadealtorisco/controle_do_desmatamento_ilegal_na_amazonia.html>.

Therefore, the existence of these laws does not guarantee the implementation of the actions envisaged by the project. The lack of effective government oversight, coupled with the absence of robust institutional support, makes the Caiarara REDD+ Project essential for the implementation of conservation and sustainable development activities that would not take place in the absence of the project, due to significant financial and institutional barriers.

2.2.4 Benefits to be used as compensation (CCB, G2.2)

As described in Section 2.2.3, additionality to the community and biodiversity scope has been demonstrated. It should be noted that the benefits generated will not be used in other compensation programs; the Caiarara REDD+ Project aims to produce only compensation related to Reducing Emissions from Deforestation, as described in Section 3 – Climate.

2.3 Safeguards and Stakeholder Engagement

2.3.1 Identification of stakeholders (VCS, 3.18, 3.19; CCB G1.5)

To identify stakeholders, STA Soluções Técnicas e Serviços de Engenharia Ambiental Ltda (see section 2.1.8) was hired to conduct a socioeconomic assessment in the project area, property, and surrounding areas.

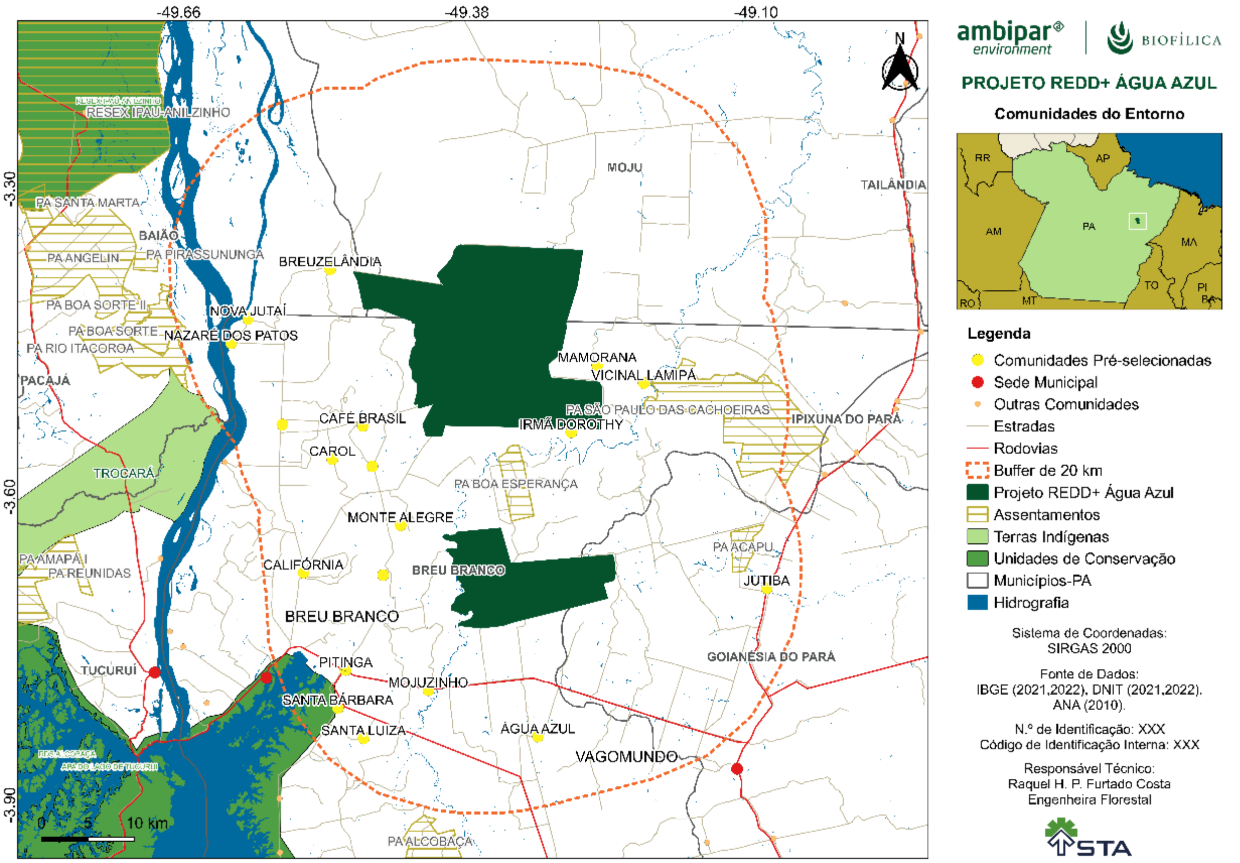
The results of the Socioeconomic Diagnosis carried out by STA were used to compile the information and build processes for section 4, as well as to help build the activities of the Caiarara REDD+ Project.

The identification was conducted by assessing communities around the project area, since there are no communities within the property, public agencies, other local actors, and landowners visited. In addition to the two mentioned above, employees working on the property, employees and contractors of the company proposing this project were considered stakeholders.

2.3.1.1 Identification of communities or population centers

To identify community groups, a 20-kilometer radius was established around the project area to identify communities or population centers that could have some kind of relationship with the project area. Initially, communities with records in secondary data from the IBGE database and previous studies of the region were mapped, resulting in an initial map (Figure 9).

Figure 9. Location of rural communities identified for the pre-field phase. Source: Prepared by STA.



From December 3 to 13, 2023, the technical team began field research based on the initial mapping, working directly with rural communities, traveling around the project area and establishing an initial dialogue with communities within a buffer zone defined as a 20 km radius from the project area, totaling 25 communities (localities, camps, settlements).

During the pre-field activity, it was found that the number of communities located in the area was greater than had been initially mapped, totaling 30 communities/localities identified. Of the total, it was possible to identify and engage in dialogue in the pre-field with 21 communities and map only 11 communities for the socioeconomic diagnosis, which were identified according to the research criteria. The table below presents a summary of the communities identified before the pre-fieldwork, those that were contacted during the pre-fieldwork, and those that were included in the socioeconomic diagnosis research universe.

Table 3. Number of rural communities identified. Prepared by: STA, 2024.

No	Communities	Databases for pre-fieldwork	Date of pre-field visit	Relationship with the project area?	Implementation of the DRP in the community	Date of survey
1	Acampamento Tigre	Yes	No inhabitants	No	Not applicable	-
2	Acapu	No	12/07/2023	No	Not applicable	-
3	Água Azul	Yes	12/05/2023	No	Not applicable	-
4	Branquelândia	No	12/03/2023	Yes	Yes	12/16/2023
5	Breuzelândia	Yes	None	No	Not applicable	-
6	Café Brasil	Yes	12/03/2023	Yes	Yes	01/14/2024
7	California	Yes	12/03/2023	Yes	Yes	12/08/2023
8	Irmã Dorothy (1)	Yes	12/06/2023	Yes	Yes	12/17/2023
9	Jutaí II	No	03	Yes	Yes	12/14/2023
10	Jutiba	Yes	Within a PA	No	Not applicable	-
11	Km-22	Yes	Within a PA	No	Not applicable	-
12	Mamorana	Yes	12/06/2023	Yes	Yes	12/16/2023
13	Mojuzinho	Yes	07	No	Not applicable	-
14	Nazaré dos Patos	Yes	12/04/2023	No	Not applicable	-
15	Nova Jutaí	Yes	12/04/2023	No	Not applicable	-
16	PA Acapu (2)	Yes	Out of area	No	Not applicable	-
17	PA Boa Esperança (2)	Yes	12/05/2023	Yes	Yes	12/15/2023
18	PA Santa Barbara (2)	Yes	09	No	Not applicable	-

No	Communities	Databases for pre-fieldwork	Date of pre-field visit	Relationship with the project area?	Implementation of the DRP in the community	Date of survey
19	PA São Paulo das Cachoeiras (2)	Yes	12/07/2023	No	Not applicable	-
20	Pitinga	Yes	Within a PA	No	Not applicable	-
21	Roça Comprida	Yes	12/10/2023	Yes	Yes	12/16/2023
22	Santa Luiza	Yes	Outside the area	No	Not applicable	-
23	São José II	Yes	12/06/2023	No	Not applicable	-
24	São Pedro	Yes	Outside the area	No	Not applicable	-
25	Vicinal Lamipá	Yes	Outside the area	No	Not applicable	-
26	Vila Carol (3)	Yes	12/03/2023	Yes	Simplified itinerary	Two dates
27	Vila dos Remédios	Yes	03/12/2023	Yes	Yes	01/16/2024
28	Vila Jutuba	No	03/12/2023	No	Not applicable	-
29	Vila Monte Alegre	Yes	03/12/2023	Yes	Yes	12/10/2023
30	Vila São José	Yes	09	No	Not applicable	-

Notes: In (1) Assentamento Irmã Dorothy; (2) Settlement Projects; and (3) Vila Carol Community, there were two attempts to carry out the diagnosis, scheduled with community leaders, but on the appointed days, there was no audience for the diagnosis, and a simplified script was used.

During the initial identification, the technical team scheduled the best date with community leaders to conduct the diagnosis with residents. The photo below is an example of the pre-fieldwork moments, when the research team sought out the community representative or leader to explain the Caiarara REDD+ Project and the socioeconomic diagnosis that would be carried out. The community leader was responsible for mobilizing and inviting residents.

Figure 10. STA research team making initial contact with the leadership of the Vila dos Remédios community. Source: Socioeconomic assessment by STA.

The socioeconomic assessment of all 11 identified communities was conducted in two periods, between December 8 and 17, 2023, and between January 14 and 16, 2024. For the research, a participatory methodology for data and information collection was used, known as Rapid Participatory Rural Appraisal (RPRA), using approaches and methods to empower local communities to share, increase, and analyze knowledge about their own lives and plan for change (CHAMBERS, 1981[33] ; CHAMBERS & GUIJT, 1995[34]). The approach was carried out using research tools such as diagrams (production flows), mapping (of resources and social aspects), matrices (production, history, diseases), interviews (key people, men and women, groups, young people and the elderly), and calendars (production, extraction, cultural). As an example, in the figures below, team members explain the research topics and dynamics to the Monte Alegre community in a collective community space.

[33] CHAMBERS, R. Rapid rural appraisal: rationale and repertoire. Public Administration And Development, Vol. 1, 95-106 (1981). Available at: https://www.researchgate.net/publication/304775751_Rapid_Rural_Appraisal_Rationale_and_Repertoire.

[34] CHAMBERS, R.; GUIJT, I. DRP: después de cinco años, em qué estamos ahora? Revista Bosques, Arboles y Comunidades Rurales, Quito: FAO, n. 26, p. 4-14, 1995. Available at: https://core.ac.uk/download/pdf/19916089.pdf.

Figure 11. Explanation of the socioeconomic diagnosis in the Monte Alegre community on December 10, 2023. Source: Socioeconomic diagnosis by STA.

Figure 12. Identification of environmental protection practices carried out by community residents using a matrix to facilitate information gathering and cross-reference data with visual support during the application of the tools. Source: Socioeconomic diagnosis by STA.

Figure 13. Explanation of the Caiarara REDD+ Project for the Roça Cumprida Community. Source: Socioeconomic diagnosis by STA.

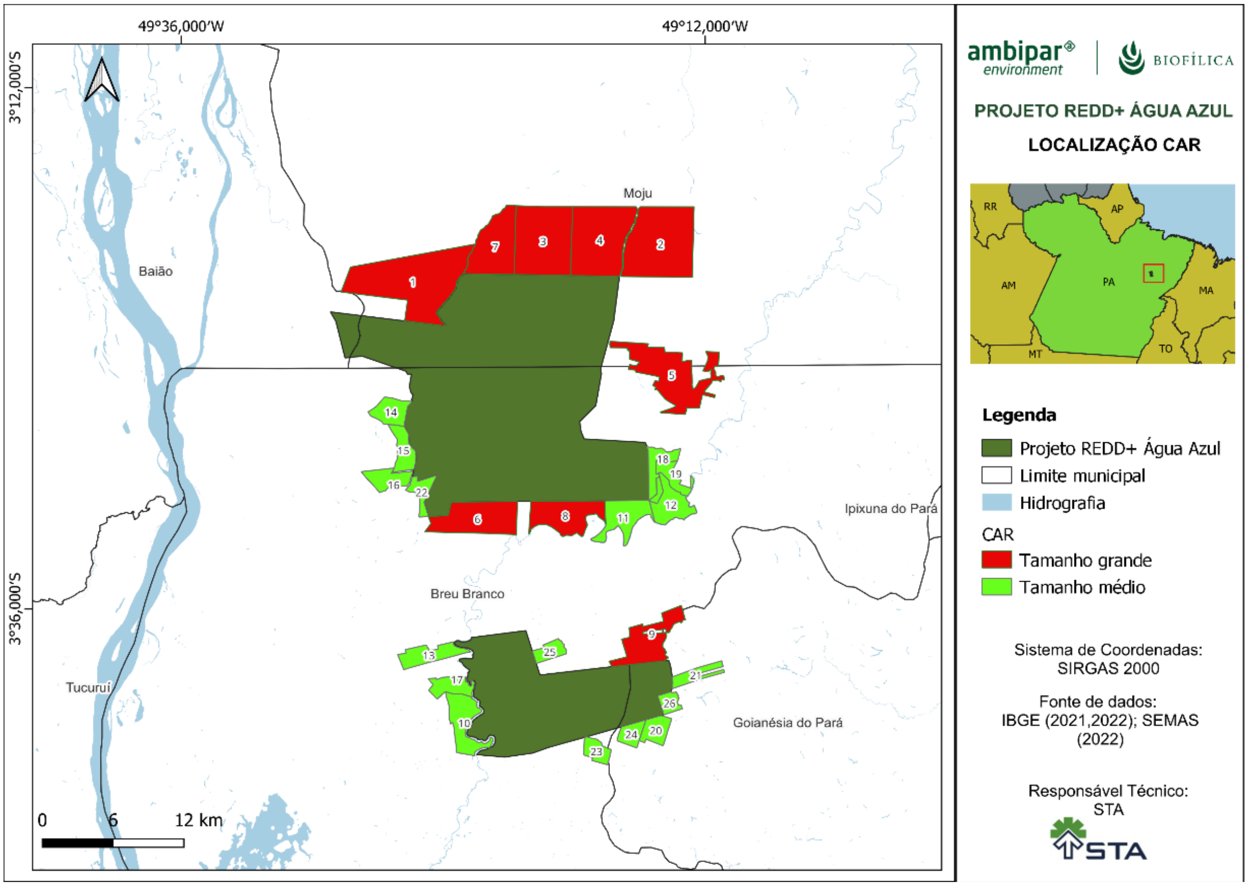
In the case of Vila Carol, the field team was unsuccessful. Two attempts were made to carry out the diagnosis with an appointment scheduled with the community leadership, but both were canceled at the last minute by message or phone call from the leadership to a member of the STA team. On a third attempt, the team traveled to the community, but few people attended the meeting. As a result, it was only possible to apply a simplified research script, collecting general social data from the community members present.

Figure 14. Application of the simplified script in Vila Carol. Source: Socioeconomic diagnosis by STA.

2.3.1.2 Identification of landowners around the project area

For the survey of rural properties surrounding the project property, still in the pre-field period, rural properties registered in the Rural Environmental Registry (CAR) were identified, totaling 110 records, of which 84 were small properties, 17 were medium-sized properties, and 9 were large properties, involving the municipalities of Baião, Breu Branco, and Moju (where 1 rural module = 70 ha) and the municipality of Goianésia do Pará (where 1 module = 55 ha).

Figure 15. Location of CAR registrations in the vicinity of the REDD+ Caiarara Project properties.



With the properties previously identified, the STA team traveled around the project property to identify areas that could be classified as squatter settlements or rural properties, checking for the existence of dwellings and seeking information from other actors about the existence of individual squatter plots and/or rural properties.

For the surveys, a form was developed with open and closed questions about the objectives of the survey, which are:

- Identification of rural property/rural landowners;
- Identification of the productive activities carried out;
- Among the productive activities, identification of practices that could be characterized as vectors or causes of deforestation;
- Among the productive activities, identification of risks that could cause deforestation;

- Identification of future land use prospects, whether they intended to increase the productive area to such an extent that it would become a driver of deforestation;
- Identification of any sustainable productive practices that could be encouraged in the future with the project.

During the socioeconomic assessment period, it was only possible to contact five rural properties, with which the team was able to communicate only by telephone. Three of these refused to respond to the complete survey, providing only a summary of the use of the property.

2.3.2 Descriptions of stakeholders (VCS, 3.18, 3.19; CCB, G1.6, G1.13)

Based on the socioeconomic diagnosis, it was possible to characterize each stakeholder, including communities and population centers, public agencies, farm workers, and rural landowners, which are specified below.

2.3.2.1 Communities and population centers

The communities or population centers were identified based on a Socioeconomic Diagnosis conducted by STA. The methodology and specifications for identification and diagnostic activities are described in section 2.3.1.1.

In total, 11 communities were identified as stakeholders in the Caiarara REDD+ Project, as shown in the table below.

Table 4. Communities identified as stakeholders in the Caiarara REDD+ Project.

No	Communities	No	Communities
1	Branquelândia	7	PA Boa Esperança (2)
2	Café Brasil	8	Roça Comprida
3	California	9	Vila Carol (3)
4	Irmã Dorothy (1)	10	Vila dos Remédios
5	Jutaí II	11	Vila Monte Alegre
6	Mamorana		

Having the communities surrounding the project as stakeholders makes it possible to mitigate risks to the protection of the area, especially in relation to deforestation and hunting, activities that have historically been present in the region. Thus, it is possible to create more effective conservation strategies, aligning the interests of the communities with the success of the project. Integrating these communities into the planning and implementation processes not only strengthens forest protection but also reduces the likelihood that they will become agents of deforestation, promoting sustainable coexistence and generating direct benefits for local populations.

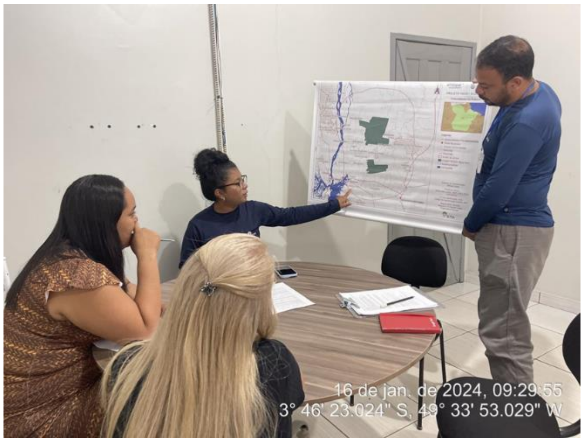
2.3.2.2 Public agencies and other actors

Among the public agencies, we visited and interviewed agencies in the municipality of Breu Branco, the Municipal Secretariat of Agriculture, Fisheries, and Economic Development - Agriculture Sector (SEMAPEC), the Municipal Secretariat for Social Assistance and Development (SEMADS), the Municipal Secretariat for the Environment and Sanitation (SEMASA), the Municipal Secretariat for Education (SEMED) and the Municipal Secretariat for Health (SMS). In addition to the Union of Rural Workers and Family Farmers of Breu Branco (STRAAFBB).

As state public agencies with offices in Breu Branco, visits were made to the Agricultural Defense Agency of the State of Pará (Adepará) and the Technical Assistance and Rural Extension Company of the State of Pará (Emater).

Figure 16. Right, visit to the Municipal Health Secretariat, Breu Branco, January 15, 2024. Left, visit to the Municipal Secretariat for the Environment and Sanitation, Breu Branco, January 17, 2024. Source: Socioeconomic diagnosis by STA.

Figure 17. Right: Municipal Secretariat for Social Assistance and Development, Breu Branco, January 17, 2024. Left: visit to the Breu Branco Rural Workers' Union, December 16, 2024. Source: Socioeconomic diagnosis by STA.



Public agencies were identified in order to expand the collection of information on the local reality, especially on the communities and the productive, economic, and social activities in the area surrounding the project.

2.3.2.3 Employees and third parties on the property

The proponent of the Caiarara REDD+ Project, Dow Brasil, carries out charcoal production and forestry activities on the property where this project is located, but outside the project area. In addition to Dow Brasil's own employees, there are 15 outsourced companies registered as providing services in the development of these productive activities. In total, there are 361 workers.

Therefore, company employees and outsourced workers were considered stakeholders. The following table shows the list of companies and the number of employees considered.

Table 5. Number of employees and outsourced workers considered as stakeholders.

Company	Total number of workers per company
Dow	71
Contractors	290
Total	361

Farm workers were considered stakeholders in the REDD+ project because approximately 50% of them live in communities surrounding the project area. The project can bring benefits that positively impact their daily work, such as improved well-being and training of interest. In addition, by being included as stakeholders, these communities and workers can engage more consciously in forest preservation, contributing to local sustainability and harmonious coexistence between agricultural activities and environmental conservation.

2.3.3 Stakeholder access to project documents (VCS, 3.18, 3.19; CCB, G3.1)

To ensure ongoing communication, consultation, availability of documents, and dissemination of information between proponents and stakeholders, the Caiarara REDD+ Project Communication Plan was established, ensuring that contributions are considered to adapt project management in an efficient and participatory manner throughout its implementation (more details on the Communication Plan in Section 2.3.11).

Access to REDD+ Caiarara Project documents by stakeholders will be provided through three main means: virtual, written, and oral, as they become available during the project's lifetime, following the validated Communication Plan. The main objective is to ensure access to relevant documents and information by all stakeholders.

Virtual communication: the project design document and/or links to access it, information relevant to stakeholders, monitoring reports, and other relevant documents will be available virtually via WhatsApp and on the websites of Biofilica Ambipar Environmental Investments S/A and Verra. News and updates about the Project will also be posted on the social media accounts of Biofilica Ambipar Environmental Investments S/A (LinkedIn: https://www.linkedin.com/company/ambiparenvironment-br/mycompany/verification/ and Instagram: https://www.instagram.com/ambiparenvironment.br/), as well as through a WhatsApp group with stakeholders.

Written communication: complete versions of the Project Design Document (PDD), Monitoring Reports, and other relevant documents will be made available in print at the headquarters of the Reflorestamento Água Azul I (RAAI) and Reflorestamento Água Azul II (RAAII) farms for consultation by all interested parties.

Oral/face-to-face communication: news, updates, and information about Project documents will be shared through meetings, events, and other face-to-face gatherings with stakeholders during the Project activities throughout its cycle.

The complete project documents will be updated in the three communication channels mentioned above as new documents and new versions of these documents are generated throughout the project cycle, and summarized versions in accessible language will be made available to all stakeholders.

In addition, the 30-day public consultation period on Verra will be widely publicized to stakeholders, who will have access to the draft Project Design Document, where they will be able to submit comments, criticisms, and contributions to the project. The proponents will collect, organize, and respond to the contributions received during the period and incorporate them where appropriate.

2.3.4 Disclosure of Project Summary Documents (VCS, 3.18, 3.19; CCB, G3.1)

As described in Section 2.3.3 above, summary documents will be disclosed and made available in the same manner as the complete Project documents: summaries of the Project Design Document (PDD), Monitoring Report and other important documents of the Caiarara REDD+ Project will be made available and disclosed to all interested parties **virtually** on the official website of Biofilica Ambipar Environmental Investments S/A and on the Verra registration platform website and via WhatsApp; and **in writing** through printed summaries at the farm headquarters. Summary information on monitoring results and Project news will be reported **virtually** via WhatsApp and **orally** through meetings, lectures, and other face-to-face meetings with surrounding communities, Dow workers, partners, local institutions, and other stakeholders throughout the Project cycle, in appropriate language to ensure accessibility for audiences with low levels of education or language barriers.

2.3.5 Informative meetings with stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

Meetings, gatherings, and workshops between the proponents of the Caiarara REDD+ Project and stakeholders to discuss the carbon project initiatives will take place throughout all phases of the design, development, implementation, and monitoring of activities.

Whenever necessary, informational meetings will be held between the Project proponents and local actors, depending on local demand and availability. Information on the location and date of meetings will be disseminated in advance through the communication channels established by the project (Section 2.3.11), which includes WhatsApp messages (validated by surrounding communities and workers as the preferred means of communication), the project's institutional email, and in-person visits to announce the meetings in the case of communities with limited internet access.

The informational meetings will always be conducted in accessible language, considering possible language or literacy barriers, in order to reach all stakeholders. Suggestions and criticisms received from stakeholders during the meetings will be recorded in a unified spreadsheet, analyzed, and responded to within defined time frames depending on the complexity of each issue. The project communication team will be responsible for managing these requests, with support from relevant experts, if necessary. When relevant and appropriate, the contributions received will be incorporated into the project's methodologies and activities (see Section 2.3.11 for more details).

The first contact between stakeholders and the Caiarara REDD+ Project initiatives was during the Socioeconomic Diagnosis conducted in the region by STA in early 2024 (see Sections 2.3.1 and 2.3.10). At that time, it was possible to explain the general concepts of a carbon project, provide specific information about the Caiarara REDD+ Project, and establish communication with the communities and workers.

Furthermore, between July 11 and 14, 2024, the Biofilica Ambipar Environmental Investments S/A team, together with STA, held a "feedback event with stakeholders" for the REDD+ Caiarara Project (see Section 2.3.10). The in-person event aimed to present the results of the socioeconomic assessment and provide information on the stages and initiatives of the REDD+ Caiarara Project. In addition, a group questionnaire was conducted with participants to ask about their preferred communication channels and collect additional information for the socioeconomic assessment for workers and surrounding communities.

There was a setback with the Boa Esperança Settlement, and the meeting could not be held in person, but an online meeting was organized on August 2, 2024.

The table below contains information on each meeting held, including the date, stakeholders, and number of participants.

Table 6. Informational meetings held.

Community/ Institution	Date/Time	Number of participants
Palmyra Workers	July 11 at 8:30 a.m.	17
Palmyra Workers	July 11 at 2:30 p.m.	53
Palmyra Workers	July 12 at 2:30 p.m.	23
Monte Alegre	July 13 at 8 a.m.	0
California		
Vila Carol		
Roça Comprida	July 13 at 11 a.m.	10
Vila dos Remédios		
Café Brasil		
Branquelândia	July 13 at 4 p.m.	9
Jutaí II		
Vila Mamorana	July 14 at 9 a.m.	11
Irmã Dorothy		
PA Boa Esperança	August 2 at 2pm	3

2.3.6 Project Risks and No Net Harm (VCS, 3.18, 3.19)

	Identified risk(s)	Potential impact of the risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventive measures taken

Natural and man-made risks to the well-being of stakeholders	No risks identified	-	The project activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. As such, no natural or human-induced risks related to the implementation of project activities have been identified.
Risks to stakeholder participation	Low community involvement in project activities	Communities may not feel identified with and/or interested in the proposals and may not participate in the actions.	Consolidation of mechanisms that promote transparency and encourage the participation of all parties involved in the decision-making processes of project activities, as well as the implementation of communication tools to improve social relations.
	Low engagement of farm workers in project activities	Employees may not feel comfortable with the project and may not identify with the activities.	Promote activities that provide information about the project and its benefits, as well as value the participation of workers and their knowledge of the project area. Also consider actions that strengthen good practices and relations with workers through courses/training on their labor rights, occupational safety, and understanding of their duties and position. In addition, establish efficient communication and feedback mechanisms.
Working conditions	No risks identified		No risks have been identified for existing and future working conditions in relation to project activities.
Safety of women and girls	No risks identified	-	No risks to the safety of women and girls were identified in the implementation of project activities, as these are linked to the intensification of surveillance activities on properties, biodiversity monitoring, and education and technical assistance actions for the community.
Safety of minority and marginalized groups, including children	No risks identified	-	No risks to the safety of minority and marginalized groups, including children, were identified in the implementation of project activities, since these are linked to the intensification of surveillance activities on properties, biodiversity monitoring, and education and technical assistance for the community.

Pollutants (air, noise, water discharges, waste generation, and release of hazardous materials, pesticides, and chemical fertilizers)	No risk identified	-	The project does not intend to use products that have pollution-related impacts.
Discrimination	No risk identified	-	The REDD+ Caiarara Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which guarantee the promotion of an environment of mutual respect by prohibiting discrimination and harassment (including sexual harassment) and unprofessional behavior.
Sexual harassment	No risk identified	-	The REDD+ Caiarara Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which guarantee the promotion of an environment of mutual respect by prohibiting discrimination and harassment (including sexual harassment) and unprofessional behavior.
Equal pay for equal work	No risk identified	-	The project activities do not provide for the payment of salaries; however, if salaries are offered in the future, there will be no distinction in pay for similar positions based on gender, color, religion, or other factors.
Gender equality at work	No risk identified	-	The project activities may be carried out by anyone, regardless of gender.
Forced labor[35]	No risk identified	-	The REDD+ Caiarara Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which ensure compliance with all applicable laws on forced labor in their activities.

[35] The identified risks and proportionate mitigation or prevention measures for forced labor, child labor, and human trafficking must include employees and workers hired by third parties.

Child labor	No risk identified	-	The REDD+ Caiarara Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which guarantee compliance with all applicable laws on child labor in their activities.
Human trafficking	No risk identified	-	The REDD+ Caiarara Project follows the codes of conduct of Biofilica Ambipar Environmental Investments S/A and Dow Brasil, which guarantee compliance with all applicable laws on human trafficking in their activities[36] .
Recognition, respect, and promotion of the rights of IPs, LCs, and customary rights holders	No risk identified	-	The properties of the REDD+ Caiarara Project are private areas and do not have areas with customary rights of third parties.
Preserve and protect cultural heritage	No risk identified	-	Not applicable to the project, as no cultural heritage impacted by project activities has been identified, since the communities in the region are not considered traditional.
Protect and preserve property rights, customary rights, or protect legal or customary rights of tenure/access to land, property, and resources, including collective and/or conflicting rights	No risk identified	-	The properties of the Caiarara REDD+ Project are private areas with no disputes over land or resources.
Impacts on biodiversity and ecosystems	No risk identified	-	The project activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. As such, the project activities aim to

36 Available at: <; https://corporate.dow.com/content/dam/corp/documents/legal/473-00001-11-dow-code-of-conduct.pdf>; and < https://ambipar.com/site2020/wp-content/uploads/2023/09/Codigo-de-Conduta-e-Compliance-2023.pdf>

			have positive impacts on biodiversity and ecosystem conservation.
Soil degradation and erosion	No risk identified	-	The project activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. As such, the project activities do not pose a risk of soil degradation and erosion.
Water consumption and stress	No risk identified	-	The project activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. As such, the project activities will contribute to the conservation of water stocks.
Habitats (and areas necessary for habitat connectivity) for rare, threatened, and endangered species	No risk identified	-	The project activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. As such, the project activities will contribute to the conservation of these habitats.
Areas necessary for habitat connectivity	No risk identified	-	The project activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. As such, no risk has been identified for the degradation of areas necessary for habitat connectivity.
Invasive species	No risk identified	-	The Project does not foresee the use of invasive species in its activities.
Ecosystem conversion	No risk identified	-	The project activities are focused on conserving the forest in the project area, involving actions to protect biodiversity and improve the well-being of stakeholders. Therefore, no risk of ecosystem conversion has been identified.
Loss of access to areas by community members with the definition of the Project Area and increased surveillance and	Loss of access to areas for extractive products in general, subsistence	Illegal and uncontrolled entry into native forest areas of the project to search for extractive products in general, food such as game animals and	Prioritize the involvement of the most affected communities in alternative activities, such as the promotion of diversified and sustainable agricultural production practices and models, and the development and strengthening of value chains for agricultural or forest products, which

monitoring of the forest area.	hunting and fishing.	fish, in order to ensure the food subsistence of some families.	aim to guarantee income and food security while minimizing possible negative impacts and reducing pressure for deforestation.

2.3.7 Community costs, risks, and benefits (CCB, G3.2)

As described above, a socioeconomic assessment was carried out involving the communities in the region and workers on the farms participating in the Caiarara REDD+ Project, in which the local scenario and context were analyzed through interviews (sections 2.3.1 and 2.3.2). During this process, the possible positive and negative impacts that the Project may have on communities were identified, based on the social impact assessment method developed by Richards and Panfil (2011)[37] .

Based on this information, the activities within the social scope of the Project were designed for the social and economic development of local actors, aiming to protect natural resources and reduce deforestation in the region. To validate the proposed activities and present the Project again, a feedback event was held with stakeholders, where the potential costs, risks, and benefits relevant to the communities were presented transparently through posters and slides in accessible language to ensure that the communities understood the information. It was clarified from the outset that participation in the Project is voluntary, and communities were advised that the decision to participate or not can be reviewed throughout the Project cycle without penalty or restriction. In addition, throughout the implementation of the Project, new relevant and appropriate information on the potential costs, risks, and benefits for the communities will be provided at follow-up meetings, consultations, and meetings on the development of activities.

In accordance with the Communication Plan established in Section 2.3.11, suggestions and criticisms from stakeholders received during the meetings will be recorded in a unified spreadsheet, analyzed, and responded to within defined time frames, depending on the complexity of each issue. The management of these demands will be the responsibility of the project communication team, with support from relevant experts, if necessary. When relevant and appropriate, the contributions received will be incorporated into the project's methodologies and activities. Stakeholders also have the opportunity to make suggestions and criticisms via institutional email, WhatsApp messages, and physical forms through suggestion boxes, which will be installed at strategic points to facilitate access to communication channels for communities and workers.

2.3.8 Information to stakeholders on the validation and verification process (VCS, 3.18.6, 3.19; CCB, G3.3)

Palmyra workers, potentially participating communities, and other Project stakeholders will be informed about validation and verification through the communication channels already presented above (more

[37] Richards, M. and Panfil, S.N. 2011. Social and Biodiversity Impact Assessment (SBIA) Manual for REDD+ Projects: Part 1 – Core Guidance for Project Proponents. Climate, Community & Biodiversity Alliance, Forest Trends, Fauna & Flora International, and Rainforest Alliance. Washington, DC.

details in Section 2.3.11), available prior to the event, including institutional email and WhatsApp messages, as well as face-to-face meetings and gatherings, will serve as means of direct exchange of information on the validation and verification process.

Virtual channels, such as the website and social media of Biofilica Ambipar Environmental Investments S/A and Dow Brazil's communication channels will also be used to inform stakeholders and the general public.

Information on the validation and verification process was disclosed during the Rapid Participatory Rural Diagnosis interviews (part of the socioeconomic diagnosis) and the stakeholder feedback event held between July 11 and 14, 2024 (see Section 2.3.10 for more details). On both occasions, stakeholders were informed about the Project's audit process and the presence of auditors.

Doubts and suggestions about the Project's validation and verification process can be submitted via WhatsApp, institutional email, and physical forms available in suggestion boxes, which will be installed at strategic locations to facilitate access to communication channels for communities and workers. The requests will be recorded in a unified spreadsheet, analyzed, and responded to within defined time frames, depending on the complexity of each issue.

2.3.9 Information on site visits and opportunities to communicate with the auditor (VCS, 3.18.6; CCB, G3.3)

The auditor's visit to the Project site will be communicated to the surrounding communities, Palmyra workers, proponents, partners, local institutions, and other stakeholders of the Caiarara REDD+ Project through emails, WhatsApp messages, poster distribution, phone calls, in person, and other means of communication available in a timely manner before the event.

At the "feedback to stakeholders" event held in July 2024, the next stages of the Project were mentioned, explaining the validation and verification process, the audit event, the auditor's visit, the possibility of direct communication, and the expected date.

In addition, the field audit event and the auditor's visit were reinforced by sending official letters to relevant local institutions in the state of Pará, among others that are directly and indirectly involved in the forest conservation and environment sector, such as unions, non-governmental organizations (NGOs), educational institutions, and state and federal government agencies, which also provided updated information on the status of the project, the communication channels used, and an invitation to participate in Verra's public consultation period.

Prior to the opening of the public comment period, a digital banner will be sent via WhatsApp to surrounding communities and workers to inform them about the public consultation, the date of the audit, and the possibility of communicating with the auditor. During the field audit, local residents, workers, and other local stakeholders will have the opportunity to communicate with the auditor in person and independently, without the presence of the project proponents, in order to ensure the independence of communication and a direct channel between the auditor and stakeholders.

2.3.10 Stakeholder consultations (VCS, 3.18; CCB, G3.4)

Two events were held by the Caiarara REDD+ Project to consult local stakeholders (communities surrounding the Project Area and farm workers): the first was the socioeconomic diagnosis meetings and interviews, held between December 2023 and January 2024; and the second was the “feedback to stakeholders” event held in July 2024.

The consultations were conducted in language accessible to all, taking into account possible language or literacy barriers. All information collected was recorded on forms, reports, attendance lists, and/or photos, organized and filed.

Date of stakeholder consultation	December 8, 2023 to January 16, 2024
Stakeholder engagement process	The stakeholder engagement process for the socioeconomic diagnosis followed two different paths, as there are two groups of stakeholders in the project (communities and workers). The entire process was carried out by STA. With the communities, engagement took place in the field, at the time of their identification, where appointments were made with the leaders to set the best date for conducting the diagnosis with the residents. The community leadership was responsible for mobilizing and inviting residents, but the team made itself available to assist with mobilization. The participatory tools of the Rural/Rapid Participatory Diagnosis (DRP) were applied with the communities in two periods, the first from December 8 to 17, 2023, and the second from January 14 to 16, 2024. As for engagement with farm workers, days and times were planned for conducting the diagnosis in conjunction with the managers of Palmyra's own employee groups and outsourced workers. The survey of workers was conducted on December 8, 11, 12, 13, and 14, 2023. All meetings were held adapting the discourse to the reality of the participants, seeking greater understanding of the concepts and details of the project and the questionnaire. In addition, diagrams and dynamics were used to assist in conducting the meeting. To record the data, STA filled out field forms with the number of attendees and the questionnaire with the stakeholders, which were scanned, organized, and filed. At the end of the activities, a report was prepared containing all the information collected, its analysis, and interpretation.

Consultation results	During the socioeconomic diagnosis, it was possible to inform local stakeholders about the concepts and cycle of the Caiarara REDD+ Project, collect information about the reality, and record the demands and suggestions of each stakeholder. As a result, a technical report on the socioeconomic diagnosis was prepared, which is extremely important for the design of the project. The characteristics and demands of the stakeholders were considered in the development of the theory of change and the project activities (see Section 2.1.17).
Contribution of stakeholders	Since this action was the first contact with stakeholders, there was no discussion or decision-making regarding project details, only contributions on the socioeconomic characteristics of stakeholders to contextualize the local reality. For more information on stakeholder contributions, see Sections 2.1.15, 2.1.17, and 4.1.

Date of stakeholder consultation	July 11, 2024 to July 14, 2024
Stakeholder engagement process	For the "feedback to stakeholders" event, the meetings were planned by STA. The mobilization of communities and farm workers began three months before the meetings, in April 2024. Community leaders were contacted via WhatsApp and informed about the meeting, its importance, and the need to spread the word to other community members. During this initial contact, the team also set up a schedule of meetings in each community according to everyone's availability. For farm workers, the schedule was agreed upon with their managers. In May and June 2024, the team contacted community leaders again to confirm the date and time of the meetings. Finally, of the 11 communities identified in the Socioeconomic Diagnosis, 09 confirmed their attendance, while the other two did not proceed with the meeting because a festive event called "Cavalgada" took place on that date. On July 11 and 12, three meetings were held with farm workers, and on July 13 and 14, four meetings were held with the communities.

	When the team was on site, the Boa Esperança PA canceled the meeting hours before the scheduled time, but on August 2, 2024, the team and some community members held the meeting online, without any losses. More details on the schedule and number of participants are in section 2.3.5. The eight meetings addressed REDD+ project concepts, the Caiarara REDD+ Project cycle, information on the public consultation and audit, communication channels, and project activities. At the end of the presentation, the team opened the floor for questions, suggestions, and criticism, and a chat took place in most meetings. In addition, an attendance list was passed around and a questionnaire with specific questions was administered to complement the Socioeconomic Diagnosis, validate communication channels, suggest names for the project, indicate activities that participants would like the project to implement, and record their impressions of the event.
Results of the consultation	During the chat and question sessions, farm workers clarified some doubts about the concepts of REDD+ and talked about the importance of the project for the environment and for the development of actions for well-being within the company. In the meetings with community members, the questions and conversations focused on the activities and what the project could offer them. Most of the requests were for technical assistance in agriculture. All demands were noted and taken into account in the design of project activities (Section 2.1.17).
Contribution of stakeholders	Feedback from stakeholders was essential for establishing communication between proponents and stakeholders, validating the results of the Socioeconomic Diagnosis, communicating information about the project cycle, and discussing and adapting project activities to reality. Based on the questionnaire, stakeholders were able to validate communication channels, suggest names for the project, and indicate activities they would like the project to implement. One contribution was the discussion on the need for technical assistance in agricultural production in surrounding communities, which was considered and included in the project activities.

Comments received during Verra's public comment period will be considered in a future version, as the period had not yet begun when this version of the document was submitted.

2.3.11 Continuous Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

As already mentioned in Section 2.3.3, the Caiarara REDD+ Project has established a robust Communication Plan to ensure ongoing communication and consultation with communities and other stakeholders throughout its implementation. The objective is to ensure that perceptions, suggestions, and concerns are considered to adapt project management in an efficient and participatory manner. The structure of this plan is described below:

Communication Channels: Communication will be carried out through various channels, including institutional email, WhatsApp messages, face-to-face meetings, physical forms, and periodic meetings. In addition, suggestion boxes will be installed at strategic points to facilitate access to communication channels for communities and workers. Meetings and lectures will also serve as a means of direct information exchange and collection of contributions.

Social Mobilization Strategies: To maintain engagement, social mobilization will be carried out for events such as meetings, training sessions, educational activities, and community gatherings. Mobilization will be carried out using different means, such as WhatsApp, emails, and direct contact, adapting to the characteristics and needs of the parties involved in each event.

Communication and Demand Registration Procedures: Demands and suggestions will be received through the available communication channels and recorded in a unified spreadsheet. All entries will be analyzed, and responses will be provided within defined timeframes, depending on the complexity of each issue. The management of these demands will be the responsibility of the project communication team, with support from relevant area specialists, if necessary.

Incorporation of Suggestions and Adaptations into the Project: Suggestions and criticisms received will be evaluated by the Project communication team and, when relevant, incorporated into the project's methodologies and activities. These adaptations will be widely disseminated to stakeholders, particularly participating communities and property workers, through established communication channels, for ongoing monitoring and validation, ensuring transparency and reinforcing the project's commitment to a participatory approach.

Conflict Management: The project will adopt a peaceful and negotiated approach to dealing with conflicts, prioritizing dialogue and collective well-being. The means of communication preferred by the parties involved will be used, ensuring a quick and effective process for resolving issues.

Adaptive Management: Recognizing that needs and contexts may evolve over time, the communication and consultation plan will be continuously evaluated and adjusted to improve its effectiveness. The goal is to ensure that project actions remain aligned with stakeholder expectations and demands.

This plan seeks to promote a collaborative relationship, in which effective communication contributes to the success of the project and to strengthening the participation of communities and other parties involved.

2.3.12 Stakeholder Consultation Channels (CCB, G3.5)

The activities of the Caiarara REDD+ Project were planned and implemented considering the interests, characteristics, and needs of the surrounding communities and workers involved. This was made possible through interviews and workshops conducted during the Socioeconomic Diagnosis and at the meeting to validate the proposed activities with stakeholders ("feedback").

As described in Section 2.3.10, consultations and participatory processes were conducted directly with communities and workers, with the support of local leaders and officials, who acted as legitimate representatives and facilitators of dialogue, formally recognized by the communities themselves during the participatory diagnosis. In addition, a feedback event was held to ensure that more information was shared and communities were consulted, promoting greater understanding of the Project by communities and other stakeholders.

During the meetings, the content was presented in simple language adapted to the local context, using visual and oral resources, such as posters, whenever necessary. All consultations were documented through attendance lists, reports, and photographic records, filed physically and/or in digital format.

The proposed communication channels (described in Section 2.3.11) — including face-to-face meetings, WhatsApp messages, suggestion boxes, and institutional emails — are accessible to stakeholders and have been approved by communities and workers as effective tools for consultation and interaction, ensuring accessibility and transparency.

Communication and consultation will continue throughout the implementation of the Project, using the channels described and adapting them as necessary to ensure that information is adequately shared and that the voices of stakeholders are heard and respected.

2.3.13 Stakeholder participation in decision-making and implementation (VCS, 3.18, 3.19; CCB, G3.6)

Involvement in all phases, from design, implementation, monitoring, and evaluation of the Project, is strongly encouraged and occurs through available communication channels (Sections 2.3.11 and 2.3.12) and informational meetings, in which all stakeholders have the opportunity to participate.

As shown in Section 2.3.10, participatory events were held with stakeholders, taking into account cultural and gender aspects, using accessible language, appropriate schedules, and encouraging the participation of women, youth, and other minority groups, while respecting local social dynamics. The participation of communities in these events during the design and development stage of the Project contributed to incorporating their perceptions and suggestions for activities for the project, according to the local context and needs.

The processes related to decision-making and implementation, as well as the various activities related to the Project, are open to the participation of communities and workers. In addition, the project will seek the equal participation of historically marginalized groups in decision-making and in the implementation of Project activities.

The contributions of stakeholders that are incorporated will be widely disseminated, mainly to participating communities and workers on the properties, through established communication channels, for continuous monitoring and validation throughout the Project's life cycle, ensuring transparency and a participatory approach.

2.3.14 Anti-Discrimination Guarantee (VCS 3.19; CCB, G3.7)

Dow Brazil has a solid policy of respect and responsibility, as established in its code of conduct, which ensures the prevention of any form of discrimination and harassment, including sexual harassment. The company promotes an environment of mutual respect, prohibiting discrimination based on gender, race, religion, sexual orientation or other habits, as well as any unprofessional behavior. The Respect and Responsibility Policy reinforce Dow Brazil's commitment to creating an inclusive and safe workplace, and this commitment will be strictly enforced in all activities of the Caiarara REDD+ Project, ensuring that no discrimination or harassment occurs during its implementation.

Dow Brazil also has a global Ethics and Compliance Office ("Office") that reinforces the company's long-standing commitment to ethical conduct. The Office communicates company standards, provides guidance on issues related to ethical conduct, and oversees action mechanisms. It is available to answer questions or address concerns and has the authority to take immediate action in the event of conduct that is inconsistent with its Code of Conduct. In addition, Dow Brazil has the *Dow EthicsLine* for anyone who wants to ask questions about Dow's policy, seek guidance on specific situations, or report possible violations of its Code of Conduct. All those involved, including local employees and community representatives, will have access to the reporting channel, with guidance in accessible language and a guarantee of confidentiality and non-retaliation. Use of the channel will be reinforced during activities with workers and surrounding communities.

Similarly, Ambipar has policies that represent all companies in the Ambipar Group. The Code of Conduct and Compliance serve as a fundamental tool to guide the conduct of all parties involved in the adoption of good practices in relationships and in the conduct of business, guiding attitudes of respect for diversity and combating any form of prejudice. In addition, the *Diversity and Inclusion Policy* guides respect for human dignity and the preservation of human rights, observing the parameters, precepts, and principles established in the *Universal Declaration of Human Rights* and *the UN Guiding Principles on Business and Human Rights.* The project adopts an inclusive approach in its relationship with local communities, promoting respect for gender, cultural, generational, and ethnic diversity, with special attention to potentially vulnerable groups. To date, no incidents of harassment or discrimination have been reported within the scope of the REDD+ Caiarara Project, and all activities are implemented with specific safeguards to ensure that no form of harassment, including sexual harassment, or discrimination occurs as a result of the project's actions.

The aforementioned policies are directly applied in the REDD+ Caiarara Project through contractual clauses, periodic training for all teams, and a formal requirement for partners and service providers to adhere to ethical guidelines. Dow sensitizes direct and indirect employees on site through information banners distributed on the farms, by sending emails and through campaigns and talks promoted by employee diversity networks, addressing respect for diversity, ethical conduct, and harassment prevention, reinforcing the principles of the Code of Ethics in the context of the implementation of the activities. In this way, the project proponents ensure that they will follow all guidelines applied in their codes of ethics, policies, and contracts established throughout the project.

2.3.15 Feedback and Complaint Redress Procedure (VCS, 3.18.4; CCB, G3.8)

Dow Brazil, together with Ambipar Environment, has established an accessible procedure for conflict resolution, which describes the processes and means of communication for conflict management and the feedback procedure (as described in the Communication Plan, Section 2.3.11). This procedure will be implemented throughout the life of the Caiarara REDD+ Project. It includes receiving questions, complaints, suggestions, and requests from stakeholders through the available communication channels, which must be recorded on a standard form, analyzed, and resolved within a predetermined time frame with the parties involved.

Development Process	The development of the feedback and conflict management procedure was based on the socioeconomic diagnosis carried out between December 2023 and January 2024, and the "feedback with stakeholders" event held from July 11 to 14, 2024, in the Project Area (more details in Section 2.3.10). In both activities, the objective was to collect information, provide information about the REDD+ Project, its status, and next steps, answer questions, and gather contributions. At the “feedback with stakeholders” event, a presentation was made on the proposed communication channels, and a questionnaire was administered asking about the preferred channel of workers on the properties and in the surrounding communities, which in this case was the WhatsApp messaging app. The project proponents incorporated the contributions received into the feedback and conflict management procedure.
Complaint redress procedure	Upon becoming aware of demands related to the Caiarara REDD+ Project, the proponents will follow a process that includes receiving the demands through the defined communication channels, followed by analysis of the information by the responsible team. Next, an attempt will be made to respond and resolve the issue within a reasonable time frame, using, whenever possible, the traditional conflict resolution methods adopted by local actors. In cases where it is not possible to reach an initial resolution informally, mediation will be conducted. If mediation is not effective, arbitration will be adopted as a last resort, applicable in extreme situations that require neutral judgment. The deadline for response and resolution at each stage will depend on the level of complexity of the complaint. Upon receiving a complaint, the person responsible for Project communications will record the information in the unified spreadsheet, analyze it, and provide a response within a predetermined time frame. If the

	form is filled out directly by the complainant, it may be placed in the suggestion box and collected by the person responsible for communications, or it may be sent via WhatsApp or Project's email. More serious requests will be forwarded to the project leadership and, if necessary, to the proponent's management for evaluation. Feedback on requests will be provided through the previously established communication channels, as described in Section 2.3.11, depending on the channel through which the request was received (suggestion box, WhatsApp, or project email). Any interested party may submit comments, suggestions, and complaints at any time, ensuring that the process is accessible and transparent.

2.3.16 Accessibility of the Feedback and Complaint Redress Procedure (VCS, 3.19; CCB, G3.8)

As mentioned in section 2.3.15 above, the Caiarara REDD+ Project will adopt the receipt of requests, complaints, and the conflict management and feedback procedure (Communication Plan, Sections 2.3.11 and 2.3.15) through the communication channels described. These channels have been validated by stakeholders and ensure accessibility to communities and workers, regardless of their technological or geographical access conditions.

All requests will be recorded in a unified spreadsheet for formalization, monitoring, and further analysis. Completed forms will be reviewed and responded to within a predetermined time frame with the interested party. Specific requests will be directed to the person responsible for the corresponding area, while more complex issues will be forwarded to project leadership and, if necessary, to Dow Brazil's management for resolution. To preserve the service history and allow for follow-up on these requests, the records will be summarized in a spreadsheet shared with those responsible for providing responses.

Individual responses to requests will be sent by different means, together with their history, as described below:

- **Suggestion box (QR Code):** Responses will be made available in documents accessible at the locations where the boxes are installed, such as folders containing the records and responses, or through posters and informational materials at support points.
- **Other Communication Channels:** Responses may be sent and discussed during meetings, lectures, by email, WhatsApp, phone calls, letters, or any other means that stakeholders prefer to ensure comfort and accessibility.

The communication channels have been designed to be accessible to all interested groups, respecting local needs and limitations. In all channels for receiving feedback, complaints, and reports, records may be made anonymously, except when identification is necessary to respond to requests more efficiently.

The proponents shall ensure that all records of complaints, questions, suggestions, compliments, and requests, as well as their respective developments and decisions, are recorded and disclosed in a clear and accessible manner. These records shall be available both in physical files (such as folders with forms) and in digital files. In addition to individual responses, the records shall be made available to the public in consolidated form at meetings and face-to-face gatherings and during the course of the Project activities.

The procedure (Communication Plan) will be disseminated through informative messages on the WhatsApp group, in-person meetings and gatherings, and during the Project activities, through conversations, posters, and RQ codes for access to the form, throughout the entire Project life cycle.

2.3.17 Worker Training (VCS, 3.19; CCB, G3.9)

The REDD+ Caiarara Project offers training aimed at both farm workers and participating communities. The training is aligned with the activities proposed by the project, aiming to develop local skills and knowledge that facilitate active participation in the implementation of actions. The themes and formats of the training will be defined based on the demands identified in the socioeconomic diagnosis and local vocations, with a focus on practices that can be replicated by the communities themselves.

The training will be offered at accessible times and with methodologies adapted to different levels of education, with special attention to the participation of women, young people, and people with little previous experience. To avoid loss of knowledge due to turnover, training will be documented in support materials (handouts, videos, protocols) and local multipliers will be trained to ensure that the knowledge acquired is not lost, promoting the retention of technical capacity, the continuous strengthening of the communities and workers involved, and the continuity of the practices learned.

Below are detailed descriptions of the planned training activities.

Strengthening Asset Surveillance: Farm workers will receive specific training on the use of equipment and technologies necessary for asset surveillance. They will also be trained in reporting and surveillance practices, ensuring an efficient and timely response to potential threats to the integrity of the area protected by the project.

Development and Strengthening of Value Chains: Participating communities will have access to technical assistance through training and capacity building with the aim of adapting to climate change and increasing sustainable agricultural production. These activities include integrated management practices and agroforestry techniques that aim not only at environmental preservation but also at generating income from sustainable production chains.

Promotion of Educational Actions: Training in education will cover topics such as wildlife conservation, sustainable use of natural resources, and health-related issues. These and other training courses will be tailored to the needs and demands of communities and workers.

Improvement in the well-being of stakeholders: Workshops on topics such as health, infrastructure, and working conditions will be offered to communities and farm workers. These and other training courses aim to promote better health and well-being among stakeholders, seeking healthier habits and reducing the impacts identified in the socioeconomic assessment.

Strengthening historically marginalized groups: Workshops and meetings will be organized to strengthen historically marginalized groups within communities and farm workers, focusing on technical training, leadership, and participation in productive and environmental activities. These learning spaces will be designed to foster empowerment, ensuring that historically underrepresented groups have greater participation in the project implementation and decision-making.

The actions offered by the Caiarara REDD+ Project are fundamental to ensuring the active engagement of workers and communities, promoting the development of essential skills for the implementation and success of the project. By aligning training with proposed activities and ensuring the continuity of knowledge, the project not only strengthens environmental protection but also improves living conditions in the communities involved. This investment in human capital contributes to the longevity of the environmental, social, and economic benefits generated by the project, creating a positive cycle of sustainable development.

2.3.18 Community employment opportunities (VCS, 3.19.13; CCB, G3.10)

Dow Brazil offers equal employment opportunities to surrounding communities, including management positions if the requirements for the position are met, and provides specific training to prepare local candidates for these roles.

All job positions generated locally by the Project follow a recruitment and selection process that Dow Brazil will carry out according to the need and availability of positions.

No criteria based on race, gender, sexual orientation, color, religion, age, ethnic origin, physical or mental disability, or social class are used. All stages of the selection process, as well as the hiring of professionals, will be based on the criteria established in the job descriptions. The project will develop active inclusion strategies, such as prioritizing female candidates for administrative and operational positions, encouraging the hiring of young people and people with disabilities, and periodically monitoring diversity in hiring.

It is worth noting that Dow Brazil has residents from local communities on its permanent and temporary staff, and the Project will work to reinforce the actions already taken in this regard. Currently, approximately 50% of Palmyra do Brasil's own or outsourced staff is made up of people from surrounding communities.

2.3.19 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

No substantial risk to worker safety has been identified to date that could arise from the implementation of the Project. However, Dow Brazil takes a rigorous and preventive approach to ensuring a safe and healthy work environment. To this end, three key programs have been implemented: the Risk Management Program (PGR), the Occupational Health Medical Control Program (PCMSO), and the Technical Report on Environmental Working Conditions (LTCAT). These programs aim to anticipate, track, and diagnose early any occupational health problems that may arise due to the execution of activities.

Each of these programs supports workers through detailed operating procedures, specific work instructions, environmental procedures, and disease and injury prevention and control strategies. Information is disseminated and communicated on an ongoing basis to ensure that all workers are aware of potential risks, even if they are minimal, and of best safety practices in the workplace.

In addition, the project strictly follows internal safety regulations and complies with occupational health and safety standards defined by federal and state governments. Aware that work processes and the work environment can potentially impact workers' health, the project adopts a series of measures aimed at promoting and maintaining the physical and social well-being of employees. These measures include the prevention of occupational diseases and protection against risks that may arise from adverse health factors.

Risk assessments will be carried out frequently to ensure that the work environment remains safe and in compliance with established guidelines. If any occupational safety risks are identified in the future, they will be included in the project's risk management system, and mitigation and monitoring actions will be implemented immediately to ensure that workers are properly protected and informed.

Through these programs and ongoing risk assessment, the project seeks to ensure that all workers are in a work environment that matches their physical and psychological abilities, promoting a work experience that minimizes risks and preserves their health and safety.

2.4 Management Capacity

2.4.1 Project Governance Structures (CCB, G4.1)

The management of the Caiarara REDD+ Project will be the responsibility of Biofilica Ambipar Environmental Investments S/A and Dow Brasil. The obligations and commitments of the parties are described below:

Responsibilities Biofilica Ambipar Environmental Investments S/A: overall coordination of socioeconomic and environmental diagnostics (DSEA), baseline studies and carbon stock; construction of the PDD (Project Design Description); remote monitoring of forest cover and implementation/coordination of additional actions aimed at reducing/mitigating greenhouse gas (GHG) emissions; conducting validation/verification audits; dissemination of the Project; management and communication with VERRA and co-management of the Project throughout its duration.

Dow Brazil responsibilities: investments necessary for project implementation and validation, project co-management, as well as development of all activities related to environmental and social scopes and infrastructure and logistics support for Biofilica Ambipar Environmental Investments S/A and other professionals involved in the project. In addition, it shall provide all necessary support for the audit processes, development of disclosure materials, and other commercial processes.

During the development of the project, two other companies were involved in conducting diagnostic studies (mentioned in Section 2.1.8). Their responsibilities are described below:

Soluções Técnicas e Serviços de Engenharia Ambiental – STA: development of environmental studies to characterize the region's flora, development of a carbon stock estimation study, and development of a socioeconomic diagnosis of the Caiarara REDD+ Project.

Ambiens Resolve Inovação Sustentabilidade e Meio Ambiente Ltda: development of environmental studies to characterize the fauna of the REDD+ Caiarara Project region.

As presented, the REDD+ Caiarara Project is supported by human resources who have assisted in its development and implementation.

2.4.2 Technical skills required (VCS, 3.19; CCB, G4.2)

The main technical skills required for the implementation of the REDD+ Caiarara Project are knowledge of the methodology to prevent unplanned deforestation, knowledge of the development and management of forest conservation projects in the Amazon biome, experience in the implementation, development, and assistance of community programs, implementation of effective land tenure security, and asset surveillance. All parties involved in the implementation and management of the project have the technical skills necessary for the successful completion of the REDD+ Caiarara Project.

The Ambipar Group operates throughout Brazil, offering integrated solutions for the entire business chain. Rapidly expanding worldwide, Ambipar complies with social and environmental responsibility rules, valuing ethics and prompt response to its customers' demands. As a crucial arm of the Group's strategies, Ambipar Environment focuses on environmental solutions, offering a complete platform for the recovery of industrial waste, post-consumer waste, recycling, co-processing, reverse manufacturing, and other services related to waste management, all focused on the circular economy and environmental, social, and governance principles. (ESG – Environmental, Social and Governance). More recently, the maintenance and preservation of native forests has been added to the Group's Environment area.

As part of the Ambipar Group, within the Environment pillar and aligned with its objectives, Biofilica Ambipar Environmental Investments S/A is a Brazilian company that promotes the management of forest areas in the Amazon and Atlantic Forest biomes. The company has a specialized team and is a benchmark in the development of forest conservation projects, ensuring the quality and effectiveness of the REDD+ activities carried out. Biofilica Ambipar Environmental Investments S/A aims to make environmental conservation an economically attractive activity for forest owners, communities, and investors. Its goal is to reduce deforestation and carbon emissions into the atmosphere, conserve biodiversity and water resources, and promote social inclusion and the development of communities living in the Amazon biome through the trade of credits for environmental services, the development and financing of scientific research activities, and the development of sustainable business chains.

Dow is a global materials science company with operations in more than 160 countries and active in various sectors, such as packaging, infrastructure, consumer goods, and transportation. In Brazil, the company also operates in silicon production, with an industrial unit in Breu Branco, Pará. This plant uses advanced technologies and responsible practices, contributing significantly to the global supply chain and promoting regional economic development.

In addition, Dow plays an important social role in the city where it operates, generating jobs, offering professional growth opportunities, and investing in training for local residents. The company also demonstrates a strong commitment to mitigating environmental impacts, strictly adhering to its code of ethics and prioritizing practices that reduce the effects of its operations on the environment. This stance contributes to strengthening the local economy and ensures operations aligned with values of social responsibility and sustainability.

In the context of the Caiarara REDD+ Project, Dow plays a crucial role as a proponent, bringing its solid governance and social experience acquired over years of operation in the region. Its contribution will be fundamental to the success of the project, especially when working in partnership with Ambipar, which has expertise in the development of REDD+ projects. This union between Dow's social governance and Ambipar's technical knowledge results in a solid and strategic partnership, prepared to generate lasting positive impacts on environmental conservation and community strengthening in the Amazon.

2.4.3 Management Team Experience (VCS, 3.19; CCB, G4.2)

Below is the experience of the members of the Caiarara REDD+ Project management team.

Developer: Biofilica Ambipar Environmental Investments S/A

BOARD OF DIRECTORS:

Plínio Ribeiro – Counselor

Plínio Ribeiro holds a degree in business administration from the Institute of Education and Research (INSPER) and a master's degree in public administration and the environment from Columbia University and the Earth Institute (USA). He has participated in several conservation projects in the lower Rio Negro through the Institute for Ecological Research (IPÊ) since 2005 and was one of the producers of Jean Michel Cousteau's documentary "Return to the Amazon." He has worked at Biofilica Ambipar Environmental Investments S/A since 2008, where he has led projects, operations, and business management. He is currently the company's counselor and shareholder.

Cláudio Pádua – Scientific Director

Cláudio Pádua has a degree in Business Administration and Biology, a master's degree in Latin American Studies and a PhD in Ecology from the University of Florida in Gainsville (USA). A retired professor at the University of Brasília, Pádua is currently dean of the School of Conservation and Sustainability and vice president of the Institute for Ecological Research (IPÊ). He is also a senior research associate at the Center for Environmental Studies and Conservation at Columbia University (USA) and international director of conservation at the Wildlife Trust Alliance, as well as a consultant to the Brazilian Biodiversity Fund (FUNBIO) and WWF Brazil. Pádua represents Brazil on the International Advisory Group (IAG) of the G7 Pilot Program. In 2003, together with his wife, Suzana Pádua, he was named a "Hero of the Planet" by Time Magazine for his activities in support of biodiversity conservation. Between 1997 and 2007, he won six conservation awards, three national and three international. Pádua has published two books and more than 30 articles in national and international scientific journals. Since 2008, he has directed the scientific involvement and production of Biofilica Ambipar Environmental Investments S/A as scientific director and consultant.

Soraya Dias Pires – Head of Carbon Solutions

Soraya Dias holds a degree in agronomy from the University of São Paulo (USP), two MBAs from USP, one in agribusiness and the other in strategic management, and an MBA from the Getulio Vargas Foundation (FGV) in business administration. She has over 15 years of experience in agro-industrial business management and development, with a career developed in large multinationals in the sugar-energy sector: Adecoagro, BP Biocombustíveis, and BP Bunge Bioenergia. She has experience in agro-industries in activities related to agro-industrial planning and control, agricultural operations, and negotiations. At Biofilica Ambipar Environmental Investments S/A, she serves as Director of Operations and New Business.

OPERATIONS & NEW BUSINESS TEAM

Caio Gallego – Intelligence Manager

Caio Gallego is a Forest Engineer graduated from ESALQ-USP with an MBA in Business Sustainability from Fundação Getúlio Vargas. He is a specialist in geoprocessing and remote sensing focused on environmental conservation, mapping, and analysis of land use changes. He has knowledge focused on Sustainable Forest Management, environmental modeling, and the use of alternative GIS for forestry and agribusiness. He has advanced knowledge in the use of GIS software and analysis of changes in land use and coverage, such as ArcGIS, QuantumGIS, and DinâmicaEGO. He has experience in forest conservation and restoration, climate change, environmental modeling, and environmental project management. He works as an operations manager in projects developed for the carbon market.

Marcio Sales – Spatial Statistics Specialist

Márcio Sales is a statistician with a degree from the Federal University of Pará, a master's degree in geography from the University of California, Santa Barbara, and a PhD in Production Ecology and Resource Conservation from Wageningen University in the Netherlands. He specializes in data analysis and conducts research in geostatistical modeling of processes distributed in space and time. He works at Biofilica Ambipar Environmental Investments S/A producing GHG emission projections from future deforestation for project baselines and monitoring deforestation by satellite.

Paula Conde – Administrative Manager and Operational Support

Paula Conde holds a degree in Business Administration from São Luís - PUC and a postgraduate degree in Accounting and Financial Management from FAAP. She has extensive experience, most of it at one of the largest media and education groups in Latin America, Editora Abril, where she worked in Financial Control and Reporting, Treasury, Financial and Accounting Reconciliation, Accounts Payable and Receivable, and Royalties. At Biofilica Ambipar Environmental Investments S/A, she is responsible for administrative and financial activities and logistical support for the team and projects.

Luisa Cotrim – New Business Strategy Specialist

Economist graduated from UNICAMP, with experience in carbon markets, corporate decarbonization and market intelligence. She leads new business development at Ambipar Carbon Solutions, responsible for structuring proposals and technical solutions in emissions management and carbon projects, with a growing focus on strategic partnerships and international expansion.

Carla Paulino – New Business Analyst

Carla holds a degree in Agricultural Engineering and Business Administration from ILES/ULBRA Itumbiara, with a specialization in Compliance and Project Management. She has worked in the development and execution of a project for the continuous improvement of origination processes and procedures. She has 16 years of experience in land analysis for partnership and supply contracts. She is currently a New Business Analyst III, focusing on prospecting and customer service for the implementation of Carbon Projects at Biofilica Ambipar Environmental Investments S/A, one of the pioneering companies in Nature-Based Solutions (NBS). She has had the opportunity to develop teams and provide intercompany training.

Maria Claudia Martinelli Trabulsi Alves– Project Coordinator

Maria Cláudia is a Forest Engineer graduated from UNESP - Botucatu, with a master's degree in Energy in Agriculture from the same university and a specialization in Sustainability from UNESP - Rio Claro. She has over 10 years of experience in auditing forestry, agricultural, sustainability, and carbon credit projects.

Gabriella Hita Marangom Cesilio – Project Analyst

Gabriella Hita is a Forest Engineer and holds a degree in Agricultural Sciences, both from the University of São Paulo (USP/ESALQ). She is also a specialist in Environment and Sustainability from the Getúlio Vargas Foundation (FGV) and is pursuing an MBA in Project Management from the University of São Paulo (USP/ESALQ). At Biofilica Ambipar Environmental Investments S/A, she works on the development, implementation, and monitoring of REDD+ projects in accordance with VERRA standards, in addition to contributing to the establishment of procedures and the development of project intelligence.

Carolina Corrêa Cauzzo – Project Analyst

Environmental Manager graduated from the University of São Paulo (USP/ESALQ), with an inter-university exchange at the Institut Agro Montpellier, in France, where she studied Sustainable Management of Natural Resources and Agroecology. She has experience with forest restoration projects in the Atlantic Forest, socio-environmental studies in rural settlements, and geoprocessing tools. At Ambipar, she works on the development, implementation, monitoring, and support of REDD+ projects, supplier diagnostic assessments, communication with stakeholders, and project intelligence development.

Fábio Souza – Geotechnology Coordinator

Geographer, master's degree in natural resource use and conservation, specializing in climate analysis, big data technologist and analytical intelligence, he has worked in environmental diagnostics, geotechnology service development, and geoprocessing team management. Specialist in PostGIS database modeling, webgis platform development, business intelligence, and geoprocessing process automation. Currently part of the operational team at Biofilica Ambipar Environmental Investments S/A as geotechnology coordinator.

SALES & MARKETING

Laion Pazian – Sales Manager

Laion Pazian holds a degree in economics from USP (ESALQ Campus) and an MBA in Commercial Management from Fundação Getúlio Vargas. He manages the carbon credit sales team, serves key national accounts, and supports relevant international sales. He leads part of the commercial strategies and manages the carbon credit portfolio of Biofilica Ambipar Environmental Investments S/A, reporting results to the company's Executive Board.

Ricardo Cordeiro – Marketing and Communications Coordinator

Ricardo Cordeiro has a degree in digital marketing and an MBA in Marketing, Branding & Growth from PUC in Rio Grande do Sul. He has over 16 years of experience in art direction (online and offline) and project management. A specialist in digital marketing strategies, SEO, Google Adwords, and Social Ads, throughout his career he has worked in digital agencies, trade, and live marketing. At Biofilica Ambipar Environmental

Investments S/A, he leads the marketing team, planning and executing actions focused on Branding and Growth.

ADVOCACY & ENGAGEMENT

Gabriel Trivelino – International Relations Specialist

Gabriel Trivelino is an International Relations Specialist at Biofilica Ambipar Environmental Investments S/A, working on the company's relations with the public sector and representative organizations. With a degree in Political Science from the University of Brasília (UnB) and an MBA in Government Relations from the Getúlio Vargas Foundation (FGV), he has over 10 years of professional experience representing interest groups in specific legislation or regulations in decision-making bodies such as the Brazilian Executive Branch and the country's Legislative Branch.

PROPONENT: DOW BRASIL

Vinicius Sambuc – Commercial Manager of Energy & Climate

Vinicius Sambuc holds a degree in Metallurgical Engineering from the Federal University of Minas Gerais (UFMG) and a specialization in Production Engineering from Uninter and an MBA in Project Management from Fundação Getúlio Vargas (FGV). He has worked at Dow since 2008 and has experience in operations, continuous improvement, and project management. He assumed his current position in 2022 and in 2024 was invited to be the Manager of the REDD+ Caiarara Project.

Adryane Hermann – Senior Product Manager

Adryane holds a degree in Foreign Trade from Universidade São Judas Tadeu (USJT/SP), a Master's degree in Supply Chain Management from BSP- Business School São Paulo, and is currently studying Law at UNIP-Universidade Paulista. She has worked at Dow since 2001 and has extensive experience in product management, projects, and supply chain. She assumed her current position in December 2021 and was invited to co-lead the REDD+ Caiarara Project with Vinicius Sambuc.

Alexandre Forman Prata – Site Operations Leader

Alexandre Forman Prata has a diverse background, including law, mining, and an MBA in Business Management. Since 2001, he has contributed significantly to DOW, where he has accumulated extensive experience in several areas. His professional career spans Supply Chain, Logistics, Forestry Operations, Coal, and Quartz Mining.

Francisco Neto – Regulation and Compliance Specialist

Francisco Neto holds a degree in Forest Engineering from the Federal Rural University of Amazonia (UFRA), with a specialization in Environmental Management from the Federal University of Pará and a specialization in occupational safety engineering (in progress). He has been with Dow since 2008, initially working in forest operations management and currently in the regulation and compliance area, focusing on environmental licensing in coal, forest, and mining production. In 2024, he was invited to join the team of managers for the REDD+ Caiarara Project.

2.4.4 Project Management and Team Development Partnerships (VCS, 3.19; CCB, G4.2)

The REDD+ Caiarara Project has all the necessary partnerships for the construction and implementation of forest asset conservation activities. Currently, the partner institutions mentioned in Section 2.1.8 are responsible for preparing the Socioeconomic and Environmental Diagnostics and Carbon Stock, which comprise the Project Design Document, through service provision contracts. When other initiatives throughout the development of the Project require new technical knowledge and partners, those responsible for the REDD+ Caiarara Project will coordinate partnerships with government, non-governmental, and private sector organizations in order to enable the generation of net positive impacts on society and biodiversity.

2.4.5 Financial health of the implementing organization(s) (CCB, G4.3)

Biofilica Ambipar Environmental Investments S/A is a multinational company that operates in several segments, offering comprehensive services and products focused on environmental management. In full global expansion, Ambipar strictly adheres to compliance and social and environmental responsibility standards, always prioritizing ethics and efficient service to its customers' demands. One of the company's main focuses is the generation and trade of carbon credits through nature-based solutions (NBS). This business area has been significantly strengthened with the acquisition of Biofílica, a Brazilian company with over 15 years of experience in conservation projects through carbon initiatives such as REDD+, ARR, and ALM. The diversification of its business lines and the support of solid investors contribute to Ambipar's financial stability.

Dow Brasil, with more than 64 years of operations in the country, is a multinational company that combines chemistry, biology, and physics to develop innovative technologies that drive progress in various sectors, such as agriculture, transportation, electronics, health, pulp and paper, among others. In the context of this project, the participating headquarters is Dow Brasil, located in Breu Branco (Pará), which is part of the Dow Group and operates in the metallic silicon and natural resources production sector.

Based on this, it can be stated that the project developers and proponents have sufficient financial strength to guarantee the necessary financial support throughout the entire project implementation period, ensuring the commitment of resources with long-term investment capacity and reinvestment of revenues in the REDD+ Project. The documents proving this financial health are classified as Commercially Sensitive Information and were made available to the audit team on a confidential basis.

2.4.6 Prevention of corruption and other unethical behavior (VCS, 3.19; CCB, G4.3)

Biofilica Ambipar Environmental Investments S/A maintains a legal team responsible for implementing and monitoring the good practices established in the "Ambipar Group Code of Conduct and Compliance," including, in particular, the "Anti-Corruption and Anti-Money Laundering Policy," which is in full compliance with the Brazilian Anti-Corruption Law (Law No. 12,846/13), Law No. 9613/98 (Money Laundering Law), as well as International Anti-Corruption Conventions (UN, OECD, Transnational Convention on Combating Corruption in the International Public Service, Merida Convention, Palermo Convention, Convention against Organized Crime); as well as observing the precepts contained in the FCPA (Foreign Corrupt Practices Act) and the UK Bribery Act.

Therefore, Biofilica Ambipar Environmental Investments S/A clearly expresses its position against acts of corruption and/or conduct that seek personal gain to the detriment of the company, society, or the government, strictly complying with national and international anti-corruption principles, as well as the main guidelines and positions of the Ambipar Group regarding the fight against money laundering, terrorist financing, and all forms of corrupt conduct, such as bribery (active and passive corruption), embezzlement, and granting of undue advantages (administrative impropriety), and curbing acts by its employees that impede investigation and inspection activities.

It should also be noted that Biofilica Ambipar Environmental Investments S/A undergoes annual financial audits to ensure that its resources are used responsibly and free from corruption.

In order to promote compliance with these policies, Ambipar has an online reporting channel that can be used by interested parties in the Program for any complaints regarding: conflicts of interest; corruption involving public officials; fraud/theft or misappropriation of funds; favoritism toward suppliers or customers; damage to company property; misuse of resources; leakage/misuse of information; violation of laws or non-compliance with policies/procedures, among others. Complaints should be made, whenever possible, through the channel on the Company's website, at https://ambipar.com/denuncias/ or by email tocanaldeetica@ambipar.com.

All Ambipar Group policies are available on its website: https://ambipar.com/politicas-estatuto/

Dow Brazil is committed to maintaining the highest ethical and legal standards in its relationships around the world. This includes relationships with governments, government authorities, and other companies, as well as with institutions representing communities. Corruption in any form, such as kickbacks, bribes, nepotism, favors, fraud, favoritism, extortion, money laundering, among others, is not tolerated. Dow Brazil also has an Anti-Bribery and Anti-Corruption Policy included in its Code of Conduct, which has a solid global risk-based Anti-Corruption *Due* Diligence process established to conduct *due diligence* on third parties based on their risk profile. The results of *due diligence* are used to establish risk mitigation strategies, including, if necessary, the termination of the business relationship.

Dow Brazil has the *Dow EthicsLine* for anyone who wants to ask questions about Dow's policy, seek guidance on specific situations, and/or report possible violations of its Code of Conduct. Its Ethics and Compliance Office maintains a confidential, secure, and reliable helpline for reporting ethical concerns and allows anonymous reporting when legally permitted. The *Dow EthicsLine* is operated by a third-party company, whose calls are answered by a trained communications specialist who documents the questions or concerns and forwards the report to the Ethics and Compliance Office for further review and handling. The channel for contacting the *Dow EthicsLine* is through the website www.dowethicsline.com, where you can file a report online or by contacting the Dow Ethics and Compliance Office at +1-989-636-2544 or ethics@dow.com.

All Dow Brazil policies are available at https://corporate.dow.com/en-us/about-dow/corporate-governance/living-our-values.html.

2.4.7 Commercially sensitive information (VCS, 3.5.2 – 3.5.4; CCB Regulation, 3.5.13 – 3.5.14)

No sensitive information was excluded from the report.

2.5 Legal status and property rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6. 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

Compliance with laws, statutes, and other regulatory bodies significant to the Caiarara REDD+ Project is related to REDD+ activities in Brazil. With regard to these activities, there is a history of initiatives, despite the construction and negotiation of this concept through agreements and meetings under the United Nations Framework Convention on Climate Change (UNFCCC).

In December 2015, the National Strategy for REDD+ in Brazil (ENREDD+) was established by MMA Ordinance No. 370, a document that formalizes to Brazilian society and the signatory countries of the UNFCCC how the Brazilian government has structured its efforts, contributing to climate change mitigation through the control of deforestation and forest degradation, the promotion of forest recovery, and the promotion of sustainable development. In this context, in Brazil, Decree No. 11,548/2023, which replaced Decree No. 10,144/2019, established the National Commission for REDD+ (CONAREDD+) with the objective of coordinating, monitoring, and reviewing the National Strategy for REDD+ and guiding the development of requirements for access to payments for results from REDD+ policies and actions in the country. Mention is also made of the publication of several subsequent Resolutions that established the internal regulations of CONAREDD+ and technical working groups on REDD+ Measurement, Reporting and Verification, Safeguards and Benefit Sharing.

As for the carbon market in Brazil, on December 12, 2024, Law No. 15,042/2024 was enacted, establishing the Brazilian Greenhouse Gas Emissions Trading System (SBCE), creating the Brazilian Regulated Carbon Market. Among other points, greenhouse gas emission limits were established, based on which it will be possible to trade Brazilian Emission Quotas (CBE), in addition to clarifying aspects of REDD+ projects and programs that generate carbon credits based on reducing emissions from deforestation, degradation, and increasing carbon stocks in native vegetation.

Below is a list and detailed description of the main relevant legislation and regulations at the federal and state levels. In addition, a brief analysis was made of the international climate agreements that have been guiding the creation and development of REDD+ initiatives around the world.

<u>Theme: Brazilian Federal Constitution of 1988</u>

Defines and regulates how laws should be created in the country, how the three branches of government (Executive, Legislative, and Judicial) should function, and the agencies that work with them. The Constitution has the following fundamental principles:

I - sovereignty;

II - citizenship;

III - human dignity;

IV - the social values of work and free enterprise;

V - political pluralism.

<u>Applicability to the project</u>: Regulates and guides all national legislation that influences the project.

Theme: Land regulation

Related legislation:

Law No. 601/1850 - Law on Vacant Lands: Provides for vacant lands in the Empire, and those owned by title of sesmaria without fulfilling the legal conditions. as well as by simple title of peaceful possession; and determines that, once the former have been measured and demarcated, they shall be transferred for consideration, either to private companies or for the establishment of colonies of nationals and foreigners, with the Government being authorized to promote foreign colonization in the manner declared.

Law No. 4,504/1964 - Land Statute: Provides for the status of land and other measures. This Law regulates the rights and obligations concerning rural real estate for the purposes of implementing Agrarian Reform and promoting Agricultural Policy.

Law No. 9,393/1996 - ITR Law - Tax on Rural Land Ownership: Provides for the ITR Tax, on the payment of debt represented by Agricultural Debt Securities and makes other provisions.

Law No. 8,629/1993 - Law on Agrarian Reform - regulates and disciplines provisions relating to agrarian reform, as provided for in the Federal Constitution.

Law No. 10,406/2002 - Brazilian Civil Code - On the rights and duties in private life, of persons, property, and legal facts. Divided into two parts, (1) General Part, referring to persons, property, and legal facts, and (2) Special Part, referring to the law of obligations, companies, things, family, and succession.

Applicability to the project: The project complies with the laws listed, which guarantees the right to property, including the Project Area, and compliance with taxes related to rural properties.

Theme: Environment

Related legislation:

Decree No. 58,054/1966: Promulgates the Convention for the Protection of Flora, Fauna, and Scenic Beauty of the Countries of America.

CONAMA Resolution No. 16, dated December 7, 1989 - Establishes the Integrated Program for Environmental Assessment and Control of the Legal Amazon.

Law No. 7,347/1985 - Law on Public Civil Action governing public civil action for liability for damage caused to the environment, consumers, and property and rights of artistic, aesthetic, historical, and tourist value, and providing other measures.

State Law No. 5,887/1995 and subsequent amendments - State Law that provides for the State Environmental Policy and other measures.

State Decree No. 1,523/1996 - Approves the Regulations of the State Environment Fund (FEMA), created by Law No. 5,887 of May 9, 1995.

State Law No. 5,977/1996 - State Law that provides for the protection of wildlife in the State of Pará.

Decree No. 2,661/1998 - Regulates the sole paragraph of Article 27 of Law No. 4,771, of September 15, 1965 (Forest Code), by establishing precautionary rules regarding the use of fire in agricultural, pastoral, and forestry practices, and provides other measures.

Law No. 9,605/1998 - Federal Law on environmental crimes that provides for criminal and administrative penalties for conduct and activities harmful to the environment and makes other provisions.

MMA Normative Instruction No. 02, of May 10, 2001 - Provides for the economic exploitation of forests on rural properties located in the Legal Amazon, including Legal Reserve areas and excepting those under permanent preservation established in current legislation, to be carried out through sustainable multiple-use forest management practices.

Legislative Decree No. 143/2002 - Approves the text of Convention No. 169 of the International Labor Organization concerning Indigenous and Tribal Peoples in Independent Countries.

State Law No. 6,506/2002 – State Law establishing basic guidelines for the implementation of Ecological-Economic Zoning (ZEE) in the State of Pará and other measures.

State Law No. 6,462/2002* Amended by Law No. 7,289, of 2009 – State Law providing for the State Forest Policy and other forms of vegetation.

State Law No. 6,671/2004 – State Law that provides for the State Environmental Policy of Pará and other measures, and amends Article 122 of State Law No. 5,887, of May 9, 1995.

State Law No. 6,745/2005 – State Law establishing the Ecological-Economic Macro zoning of the State of Pará and other provisions.

Decree No. 5,975/2006 - Regulates Articles 12, final part, 15, 16, 19, 20, and 21 of Law No. 4,771, of September 15, 1965, Article 4, item III, of Law No. 6,938, of August 31, 1981, Article 2 of Law No. 10,650, of April 16, 2003, amends and adds provisions to Decrees No. 6,514/08 and No. 3,420/00, and makes other provisions.

CONAMA Resolution No. 378, dated October 19, 2006 - Defines projects that may cause national or regional environmental impact for the purposes of item III, paragraph 1, article 19 of Law No. 4,771, dated September 15, 1965, and makes other provisions.

CONAMA Resolution No. 379, dated October 19, 2006 - Creates and regulates a data and information system on forest management within the scope of the National Environment System (SISNAMA).

State Decree No. 2,141/2006 - Regulates provisions of State Law No. 6,462 of July 4, 2002, which provides for the State Policy on Forests and other Forms of Vegetation and makes other provisions, with the objective of encouraging the recovery of altered and/or degraded areas and the restoration of legal reserves, for energy, timber, fruit, industrial or other purposes, through forest and agroforestry reforestation with native and exotic species, and makes other provisions.

COEMA Resolution No. 54, dated October 24, 2007 (ANNEX 1): Approves the list of endangered flora and fauna species in the State of Pará.

State Decree No. 1,148/2008 and subsequent amendments - Provides for the Rural Environmental Registry (CAR-PA), Legal Reserve area, and other measures.

Law No. 12,187/2009 – Federal Law establishing the National Policy on Climate Change – PNMC and other provisions.

State Law No. 7,389/2010 – State Law defining activities with local environmental impact in the State of Pará and other provisions.

State Law No. 7,381/2010 – State Law that provides for the restoration of vegetation cover and riparian forests in the State of Pará.

State Decree No. 2,099/2010 and subsequent amendments – Provides for the maintenance, restoration, natural regeneration, compensation, and composition of the Legal Reserve area for rural properties in the State of Pará and other measures.

Law No. 12,651/2012 - Brazilian Forest Code establishing general rules on the protection of vegetation, Permanent Preservation Areas and Legal Reserve Areas; forest exploitation, the supply of forest raw materials, the control of the origin of forest products and the control and prevention of forest fires and providing for economic and financial instruments to achieve its objectives.

Law No. 12,727/2012* amends Law No. 12,651/2012 – Federal Law establishing general rules on the protection of vegetation, Permanent Preservation Areas and Legal Reserve Areas; forest exploitation, the supply of forest raw materials, the control of the origin of forest products, and the control and prevention of forest fires, and provides economic and financial instruments to achieve its objectives.

MMA Ordinance No. 370, dated December 2, 2015 - Establishes the National Strategy for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management, and Increase in Forest Carbon Stocks (REDD+) in Brazil-ENREDD+.

State Decree No. 254/2019 - Establishes the Pará Forum on Climate Change and Adaptation (FPMAC).

State Decree No. 59/2019 - Regulates the Pará State Environment Council (COEMA/PA) and provides other measures.

MMA Ordinance No. 544, dated October 26, 2020 - Publishes the Internal Regulations of the National Commission for the Reduction of Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management, and Increase in Forest Carbon Stocks - REDD+.

Law No. 14,119/2021 – Federal Law establishing the National Policy on Payment for Environmental Services; and amends Laws 8,212/1991, 8,629/1993, and 6,015/1973, to bring them into line with the new policy.

MMA Ordinance No. 148, of June 7, 2022 - Recognizes the following official lists: "Official National List of Endangered Flora Species," "Official List of Endangered Brazilian Fauna Species," and "Official List of Extinct Brazilian Fauna Species."

State Law No. 9,781/2022 - Amends State Law No. 9,048, of April 29, 2020, which establishes the State Policy on Climate Change in Pará (PEMC/PA).

State Decree No. 2,596/2022 - Regulates the registration of forestry activities, the State Environmental Information Management System, and the license for the transport of forest products and by-products in the State of Pará.

Decree No. 11,548/2023 - Establishes the National Commission for the Reduction of Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management, and Enhancement of Forest Carbon Stocks - REDD+.

Decree No. 11,550/2023 - Provides for the Interministerial Committee on Climate Change.

Ordinary Law No. 10,750/2024 – Establishes the Amazon Now State Plan (PEAA).

Applicability to the project: The farms where the Caiarara REDD+ Project is located have Permanent Preservation Areas and Legal Reserve Areas in accordance with current legislation. In addition, according to the assessments carried out in the area, it was possible to identify endangered species of Brazilian fauna and flora, leading the project to recognize protection and preservation activities in accordance with the legislation mentioned above.

It is worth noting that Ambipar Group is committed to the environment, as stated in its Code of Conduct and Compliance, which makes it clear that all its activities and those of its employees must strictly comply with environmental legislation and standards, seeking to optimize natural resources.

Dow, in turn, is committed to complying with environmental laws, standards, and Dow requirements in all regions where it operates. In cases where local or national laws, codes, or regulations impose additional requirements beyond Dow standards, the affected facilities comply with the law and Dow requirements, whichever is more stringent.

In addition to complying with legislation, Dow Brazil adopts voluntary measures and is committed to its environmental performance, with a long-standing commitment to achieving Sustainability Goals, circular economy, and climate protection, among others. To demonstrate its commitment, Dow also publishes Periodic Progress Reports, such as the one found at the link: *2023 INtersections Progress Report.* An important highlight is its adherence to the most recognized environmental/climate reporting format, called CDP - Carbon Disclosure Project, whose reports can be found at the link: *Dow's CDP Scores and Submissions.*

Theme: Combating corruption

Related legislation*:*

Law No. 12,846/2013 – Anti-Corruption Law: Provides for the administrative and civil liability of legal entities for acts against the national or foreign public administration and makes other provisions.

Applicability to the project: Any action taken by the REDD+ Caiarara Project is contrary to acts of corruption or conduct aimed at personal gain, or the benefit of society or the government, and all those involved are required to refrain from such practices.

Theme: International agreements

CITES, of March 3, 1973: "Convention on International Trade in Endangered Species of Wild Fauna and Flora," signed in Washington, D.C., on March 3, 1973, amended in Bonn on June 22, 1979.

FCCC/CP/2005/Misc.1: Reducing emissions from deforestation in developing countries: approaches to stimulate action. Submission from Parties. COP 11, Montreal, 2005.

FCCC/CP/2007/6/add.1: Report of the Conference of the Parties on its thirteenth session, held in Bali from December 3 to 15, 2007. Addendum. Part two: Action taken by the Conference of the Parties at its thirteenth session or “Bali Action Plan”. COP 13, Bali, 2007.

FCCC/CP/2009/Add.1: Report of the Conference of the Parties on its fifteenth session, held in Copenhagen from December 7 to 19, 2009. Addendum. Part Two: Action taken by the Conference of the Parties at its fifteenth session or "Copenhagen Accord." COP 15, Copenhagen, 2009.

FCCC/CP/2010/7/Add. 1: Report of the Conference of the Parties on its sixteenth session, held in Cancun from November 29 to December 10, 2010. Addendum. Part Two: Action taken by the Conference of the Parties at its sixteenth session or "Cancun Agreement." COP 16, Cancun, 2010.

FCCC/CP/2011/9/Add. 1: Report of the Conference of the Parties on its seventeenth session, held in Durban from November 28 to December 11, 2011. Addendum. Part Two: Action taken by the Conference of the Parties at its seventeenth session. COP 17, Durban, 2011.

FCCC/CP/2012/8/Add.1: Report of the Conference of the Parties on its eighteenth session, held in Doha from November 26 to December 8, 2012. Addendum. Part Two: Action taken by the Conference of the Parties at its eighteenth session.

FCCC/CP/2013/Add.1: Warsaw Framework for REDD-plus, held in Warsaw, Poland, from November 11 to 22, 2013, in particular the following decisions:

Decision 9/CP.19: Work programme on results-based finance to progress the full implementation of the activities referred to in decision 1/CP. 16, paragraph 70.

Decision 10/CP.19: Coordination of support for the implementation of activities in relation to mitigation actions in the forest sector by developing countries, including institutional arrangements.

Decision 12/CP.19: The timing and frequency of presentations of the summary of information on how all the safeguards referred to in decision 1/CP.16, appendix I, are being addressed and respected.

Decision 13/CP.19: Guidelines and procedures for the technical assessment of submissions from Parties on proposed forest reference emission levels and/or forest reference levels.

Decision 14/CP.19: Modalities for measuring, reporting and verifying.

Decision 15/CP.19: Addressing the drivers of deforestation and forest degradation.

FCCC/CP/2015/Add.1: Report of the Conference of the Parties on its twenty-first session, held in Paris from November 30 to December 13, 2015. Addendum. Part two: Action taken by the Conference of the Parties at its twenty-first session.

FCCC/CP/2015 Paris Agreement: Global, legally-binding agreement that sets out a global framework to avoid dangerous climate change by limiting global warming to well below 2°C and pursuing efforts to limit it to 1.5°C. Entry into force on November 4, 2016.

FCCC/CP/2016 Decisions adopted by the Conference of the Parties (COP): Especially decisions 1 (preparation into force of the Paris Agreement), 3 (Warsaw International Mechanism for Loss and Damage associated with Climate Change Impacts), 6 (National adaptation plans) and 7 (Long-term climate finance)..

FCCC/CP/2017, FCCC/CP/2018, FCCC/CP/2019 Decisions adopted by the COP: Especially decision 1 reporting on developments of the implementation of the Paris Agreement.

Article 6 of the Paris Agreement (2021): Decision 1/CP.21 mandated the SBSTA to operationalize the provisions of this Article through recommending a set of decisions to the COP serving as the meeting of the Parties to the Paris Agreement at its first session. At COP26, the Parties to the Paris Agreement at its third session (CMA 3) adopted three main decisions related to Article 6: decision 2 (on Article 6.2), decision 3 (on Article 6.4) and decision 4 (on Article 6.8).

Glasgow Leaders' Declaration on Forests and Land Use (2021): Signatories (including Brazil) promise to reverse and end deforestation by 2030. Brazilian Nationally Determined Contribution (NDC): First Brazilian NDC submitted in September 2015 to the UN Framework Convention on Climate Change for mitigation, adaptation and means of implementation, in a manner consistent with the purpose of the Paris Agreement.

Brazilian Nationally Determined Contribution (NDC): First Brazilian NDC submitted in September 2015 to the UN Framework Convention on Climate Change for mitigation, adaptation and means of implementation, in a manner consistent with the purpose of contributions to achieve the ultimate objective of the Convention, pursuant to Decision 1/CP.20, paragraph 9. The updated Brazilian NDC was presented at COP26 on December 8, 2022. Brazil's first NDC was submitted in September 2015 to the UN Framework Convention on Climate Change for mitigation, adaptation, and means of implementation, consistent with the purpose of contributions to achieve the ultimate objective of the Convention, pursuant to Decision 1/CP.20, paragraph 9. The updated Brazilian NDC was presented at COP26 on December 8, 2022).

Applicability to the project: The Caiarara REDD+ Project is in line with international agreements aimed at achieving objectives related to climate change, combating deforestation, protecting biodiversity, sustainable land use, and safeguards, such as the Paris Agreement and other reports from the Conferences of the Parties (COPs).

2.5.2 Relevant laws and regulations related to workers' rights (VCS, 3.18.2; CCB, G3.11)

The Caiarara REDD+ Project ensures that all employees of Biofilica Ambipar Environmental Investments S/A, Dow Brasil, and service providers are legally hired in compliance with Brazilian labor laws. In addition, international agreements ratified by Brazil and issues related to worker welfare are respected.

Compliance with labor standards and laws applied by Biofilica Ambipar Environmental Investments S/A is verified annually through an audit, due to the fact that it is a S.A. company.

In turn, Dow Brazil recognizes and respects all applicable labor and employment laws, including those addressing freedom of association, privacy, and equal employment opportunities. As a publicly traded company, Dow issues detailed financial statements and other reports on best governance practices, including labor issues.

After hiring and before the employee begins work, training and capacity building on technical procedures are provided, and empowerment regarding their rights and applicable laws is promoted. In addition, employees are encouraged to join the institution responsible for defending their rights, the unions specific to their area of work.

The Caiarara REDD+ Project is committed to complying with current labor laws, providing quality of life to its employees. The laws and regulations that protect workers' rights in Brazil, as well as international agreements ratified by Brazil on labor issues, are listed below.

Federal legislation and regulations

Ambipar Environment, in accordance with the Federal Constitution, with special attention to item IV, art. 3, art. 5 and art. 7, as well as current legislation and the Ambipar Group's compendium of policies and statutes, is against any type of discriminatory act based on the individual characteristics of each person. Thus, we value equality and diversity among all Ambipar Environment employees and stakeholders.

In addition to the above regarding attention and respect for the provisions of the Constitution and the policies and statutes already in force at Ambipar Group, we emphasize special respect for legislation and infra-legal instruments concerning the issue of equal treatment, such as:

- Decree-Law No. 5,452/1943, which approves the Consolidation of Labor Laws;
- Law No. 14,611/2023, which provides for equal pay and remuneration criteria between women and men; and amends the Consolidation of Labor Laws, approved by Decree-Law No. 5,452, of May 1, 1943;
- Law No. 7,716/1989, which defines crimes resulting from discrimination or prejudice based on race, color, ethnicity, religion, or national origin;
- Law No. 12,288/2010, which establishes the Statute of Racial Equality;
- Law No. 10,741/2003, which provides for the Statute of the Elderly;
- Law No. 13,146/2015, which establishes the Brazilian Law for the Inclusion of Persons with Disabilities;
- Law No. 8,069/1990, which provides for the Statute of Children and Adolescents;
- Law No. 11,788/2008, which provides for student internships, and Law No. 10,097/2000, which provides for the Apprentice Program; and
- Law No. 9,029/1995, which provides for the prohibition of requiring pregnancy and sterilization certificates, and other discriminatory practices, for admission purposes.

We emphasize that any violation of the above legislation and/or policies and statutes already in force by the Ambipar Group may result in the termination of the employment contract or the termination of the business relationship, without excluding any applicable civil, administrative, and criminal measures.

International agreements ratified by Brazil

International Labor Organization Convention No. 87 of 1940 - Provides for freedom of association.

International Labor Organization Convention No. 98 of 1949, ratified by Brazil on November 18, 1952 - Provides for the right to organize and collective bargaining.

International Labor Organization Convention No. 29 of 1930, ratified by Brazil on April 25, 1957 - Provides for the abolition of forced labor.

International Labor Organization Convention No. 100 of 1951, ratified by Brazil on April 25, 1957 - Provides for equal remuneration between men and women.

International Labor Organization Convention No. 111 of 1958, ratified by Brazil on March 1, 1965 - Provides for discrimination in employment and occupation.

International Labor Organization Convention No. 97 of 1949, ratified by Brazil on June 18, 1965 - Provides for migrant workers.

International Labor Organization Convention No. 105, ratified by Brazil on June 18, 1965 - Provides for the abolition of forced labor.

International Labor Organization Convention No. 143 of 1975 - Provides for immigration carried out under abusive conditions and the promotion of equal opportunities for migrant workers.

International Labor Organization Convention No. 142 of 1975, ratified by Brazil on November 24, 1981 - Provides for the development of human resources.

International Labor Organization Convention No. 131 of 1970, ratified by Brazil on May 4, 1983 - Provides for the establishment of minimum wages, especially in developing countries.

International Labor Organization Convention No. 155 of 1981, ratified by Brazil on May 18, 1992 - Provides for the safety and health of workers.

International Labor Organization Convention No. 182, ratified by Brazil on February 2, 2000 - Provides for the prohibition of the worst forms of child labor and immediate action for their elimination.

International Labor Organization Convention No. 138 of 1973, ratified by Brazil on June 28, 2001 - Provides for the minimum age for admission.

International Labor Organization Convention No. 169 of 1989, ratified by Brazil on July 25, 2002 - Provides for indigenous and tribal rights.

In addition to international legislation and agreements, the REDD+ Caiarara Project is based on well-structured policies that guide the Program's management practices. For example, the Ambipar Group's Code of Conduct and Compliance, which aims to guide all parties involved in the adoption of good practices in relationships and business conduct, based on ethical and moral principles. Thus, the document provides guidelines on topics such as hiring, compensation and benefits, training, labor and union relations, modern slavery, moral harassment, and workplace quality, among others. Another example is the Dow Brazil Code of Conduct, which summarizes many of the ethical principles and policies created to address issues of equal employment opportunity and respect in the workplace, inclusion, diversity, labor practices, and human rights, as well as health and safety in the workplace. More information on ethics and workers' rights can be found in section 2.4.6.

2.5.3 Human Rights (VCS, 3.19)

The Caiarara REDD+ Project implementation area does not include any regions where traditional communities reside, as it is implemented on private properties for which Dow Brazil has signed a long-term agreement with Biofilica Ambipar Environmental Investments S/A for the conservation of surplus vegetation areas within its properties. Although there are no Indigenous Peoples or traditional communities present, the principles of the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention No. 169 on Indigenous and Tribal Peoples are recognized and voluntarily incorporated as ethical guidelines. Should relevant IPs, LCs or customary rights arise in the future, the Project will adopt processes based on Free, Prior and Informed Consent (FPIC) in accordance with the aforementioned international instruments.

The Caiarara REDD+ Project is guided by the policies and codes of conduct of the Ambipar Group and the Dow Group, as described in Section 2.3.14, and is committed to the stakeholders located in its surroundings, acting on the following premises:

- Encourage respect for people, their traditions, and values;
- Adopt and maintain a transparent process for defining social actions;
- Respect the interests and needs of the communities in which it operates;
- Collaborate with knowledge and technical information that may bring public benefits;
- Preserve and guarantee the political freedom and freedom of expression of employees;
- Guide the relationship between representatives of the REDD+ Caiarara Project and authorities, politicians, and public officials through transparent, professional, and correct conduct. Any form of pressure or request from public officials that does not correspond to this definition must be immediately rejected and reported to the management of Biofilica Ambipar Environmental Investments S/A;
- Guide relationships based on respect for laws and conventions dealing with fundamental human rights and the protection of sustainability. In all their relationships, representatives of the REDD+ Caiarara Project must observe ethical, health, safety, and human rights standards and respect social and environmental responsibility.

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

Based on mapping using government databases and the socioeconomic assessment carried out in the project implementation region in 2024, which identified all residents within a 20 km buffer zone around the Project area, it was concluded that the Caiarara REDD+ Project Area does not include any areas inhabited by traditional communities or indigenous peoples, nor were any traditional or indigenous communities identified around the area or dependent on it (more details in sections 2.1.15, 2.3.1, 2.3.2, and 4.1.1).

However, the Project recognizes that it can play a role in preserving and protecting the rich cultural heritage of the Amazon. In this context, the project aims to implement activities aligned with the socio-environmental and economic objectives focused on the identified stakeholders, such as activities to strengthen local governance, develop and strengthen value chains, improve the well-being of stakeholders, promote education and strengthen inclusion, diversity, and equity (more details in section 2.1.17), with a view to forest conservation and sustainable development, thereby enhancing the value of local cultural heritage.

In addition, the Project is committed to activating protocols and competent bodies if cultural elements are identified in the future.

2.5.5 Statutory and Customary Property Rights (VCS, 3.18, 3.19; CCB, G5.1)

The company Palmyra do Brasil, belonging to the Dow Group, is the legitimate owner of the properties known as RAAI (Reflorestamento Água Azul I, certificate no. 921) and RAAII (Reflorestamento Água Azul II, certificate no. 922) and holds the right to use and possess the farms and their resources in the Caiarara REDD+ Project Area. Palmyra do Brasil has all the necessary documentation for the properties proving its ownership, and there are no disputes over titles or rights in the project area. In addition, the Right to Use the Project Area is strictly respected, following the criteria established by *the VCS Standard v4.7* (pages 25-26).

There are no communities residing in the Project Area or depending on it for their survival and/or reproduction of customs and traditions. Therefore, there are no overlapping or conflicting rights to ownership, use, access, and management of land, territories, and resources within the properties that are part of the Project Area. According to the socioeconomic assessment (STA, 2024) carried out in the Caiarara REDD+ Project region (more details in sections 2.1.15 and 4.1.1), 25 communities were identified in the vicinity of the Project Area within a 20 km buffer zone. It can be concluded from the 11 that were studied by the assessment that none have any relationship with the land and resources of the Project Area or the RAAI and RAAII properties.

It is crucial to emphasize that property rights are constitutionally guaranteed and considered a fundamental right in the Brazilian legal system. According to the Civil Code, the owner is the person who has the power to use, enjoy, and dispose of the property, as well as the right to recover it from anyone who possesses or holds it unjustly. Documents proving ownership and formal contracts between the parties will be shared with the audit team on a confidential basis.

2.5.6 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

The Caiarara REDD+ Project recognizes and respects all property rights, complying with significant statutory and regulatory requirements, as well as having the necessary approvals from state and local authorities.

Dow Brasil owns the rights to use and exploit the properties involved in the Project, as well as obtaining the right to access the natural resources existing therein under the terms of the Brazilian Federal Constitution and the Civil Code, by virtue of being the owner of the properties where the REDD+ Caiarara Project is located. All rights have been secured through the execution of a contract, in accordance with the relevant legal requirements, with the free, prior, and informed consent of all identified stakeholders. The documents proving ownership and the contracts formalized between the parties will be shared with the audit team on a confidential basis.

As described in Section 2.5.5 above, there are no communities residing in the Project Area and surrounding areas that depend on it for their survival and/or reproduction of customs and traditions, and therefore there are no customary and/or informal rights. Furthermore, there was no need to support or strengthen third-party rights.

2.5.7 Free, Prior, and Informed Consent (VCS, 3.18; CCB, G5.2)

Free, Prior, and Informed Consent will be carried out throughout the Project's life cycle, always with an approach of dialogue and consent between the parties involved. The proponents ensure that all actors who may be impacted in any way by the Caiarara REDD+ Project have been consulted. It is hereby declared that there are no traditional communities, indigenous peoples, or customary rights holders in the Project Area or in the zone of direct influence, as verified by field checks (STA, 2024) and government databases.

The table below summarizes the outcome of the FPIC process as part of the stakeholder consultation process.

Description of the process for obtaining consent	Consent was obtained from all key stakeholders, especially the landowner, who is the main stakeholder in terms of providing the land and facilitating the implementation of the project, who is fully informed and in agreement about their participation, role, and responsibilities within the project. This was achieved by the proponent through several rounds of discussions and negotiations with the aim of providing a general explanation of the project and its objectives, timeline, conditions, projections, and shared benefits. This step took place in parallel with all the aforementioned documentary analysis on ownership and cross-referencing with all public boundaries to mitigate any risk of conflicts or false information. The proponent and landowner's consent was formalized through a contract between the project proponents. Workshops were held during the field implementation of the socio-environmental diagnostics in order to convey information about the Project to the communities neighboring the Project area and to farm workers. Consultations and questionnaires were also conducted on their opinions regarding the Project and its activities, as described in Section 2.3.10. Project information was presented in an accessible manner and in language adapted to the local reality. Future consultations related to stakeholder consent will continue to be held throughout the Project's life cycle and will be formally documented in accordance with good traceability practices. All information about the Caiarara REDD+ Project can be obtained through Biofilica Ambipar Environmental Investments S/A's virtual channels, such as its website, newsletter, and social media.
Outcome of the FPIC process	The Project is located in a private area (private property) with no residents or traditional communities within its boundaries.

	The Project does not intend to develop any activities on private properties of third parties, traditional and indigenous communities, or the government without being requested to do so. With regard to social and biodiversity activities, it is ensured that no activities will be carried out without the free, prior, and informed consent of the stakeholders. No action related to the Project will result in the relocation of activities that are important to the culture or livelihoods of property rights holders, nor will it aim at the involuntary relocation or removal of their lands or territories. Any proposed removal or relocation must be carried out only after obtaining the free, prior, and informed consent of the appropriate Property Rights Holders.

2.5.8 Benefit Sharing Mechanisms (VCS, 3.18, 3.19;)

The Caiarara REDD+ Project is carried out on duly titled private properties, without overlapping traditional communities, indigenous peoples, or squatters. All areas involved have regular land documentation, and there are no conflicts of use or third-party claims over the land. Although there is no impact on property rights, the project provides for the involvement of neighboring communities through socio-environmental and productive actions, as described in Section 2.3.10. The community engagement plan has been validated with participants and includes initiatives that promote social and environmental benefits in a transparent manner.

2.5.9 Protection of Property Rights (VCS, 3.18, 3.19; CCB, G5.3)

The proponent of the Caiarara REDD+ Project holds the right to use the properties where the Project is located, as they are private properties with all documentation proving land ownership. Additionally, as identified in the socioeconomic assessment, there are no communities or families residing on the property or in the project area. Even though there are no communities present, the Project recognizes and respects the rights of the communities that live in the surrounding area.

As described in section 2.5.7, no Project-related activity will result in the involuntary removal or relocation of Property Rights Holders from their lands or territories, nor will it force them to relocate activities important to their culture or livelihoods. No removals or relocations are planned due to the project, if removal or relocation is proposed in the future, it will only be carried out after obtaining the free, prior, and informed consent of the property rights holders, including provisions for fair and equitable compensation, demonstrating the Project's commitment to respecting the principles and process of FPIC throughout the Project's entire life cycle.

2.5.10 Identification of Illegal Activity (VCS, 3.19; CCB, G5.4)

Deforestation is the main illegal activity that can negatively impact the development of the Caiarara REDD+ Project, as well as the predatory exploitation of fauna and flora. The socioeconomic assessment identified the main causes of this illegal deforestation as members of the communities surrounding the project area who enter the area to remove wood for fence construction and charcoal production, as well as illegal hunting and fishing, even though the perimeter of the forest area is adequately marked to prohibit unauthorized entry and hunting and fishing activities.

The Project seeks to control and combat illegal activities commonly found in the Project region through mitigating and preventive measures such as strengthening property surveillance, as well as encouraging the involvement of other actors and stakeholders, social inclusion, and local socioeconomic development by encouraging alternative economic activities to deforestation and discouraging hunting and predatory fishing.

With the implementation of these measures encouraged by the Project's activities, it is expected to improve the well-being of communities without generating costs for the native forest and local biodiversity. Asset surveillance aims to curb illegal practices such as deforestation, extraction of plant species, and hunting and capture of wild animals by third parties. The mechanisms and procedures for preventing illegal activities are summarized in *Table 6.*

Table 7. Mechanisms and procedures for preventing illegal activities in the Caiarara REDD+ Project

Mechanisms and procedures for preventing illegal activities	
General conditions	- Conduct regular patrols to ensure the protection of Dow Brasil's land assets; - Prevent deforestation, forest fires, or other acts of aggression against the environment; - Prevent illegal logging and trade in timber; - Maintain good relations with surrounding communities and other stakeholders; - Control entry and exit from the RAAI and RAAII farms; - Promote environmental education with local actors who engage in predatory hunting and fishing and encourage the practice of alternative and sustainable economic activities; - Request support from police and enforcement authorities when necessary; - Activate the legal department to take appropriate measures; - Record incidents involving trespassing, property damage, and illegal extraction of forest products; - Incidents involving environmental damage must be reported to the responsible agencies (IBAMA, Environmental Police, etc.);

	- In all situations involving land conflicts, it is necessary to avoid confrontation between the parties, respecting the laws in force in the country.
Forms of registration	- Incident report; - Photographic record of incidents; - Monthly monitoring program; - Report on property security activities.

2.5.11 Ongoing disputes (VCS, 3.18, 3.19; CCB, G5.5)

In the Caiarara REDD+ Project area, there have been no ongoing or unresolved conflicts or disputes over land, territory, and resource rights, nor have any disputes been resolved in the last twenty years, as verified through consultation with local registry offices and land surveys . Thus, the project properties are legally owned and possessed by Dow Brasil, the project proponent. Furthermore, it is emphasized that no project activity that could jeopardize the outcome of a dispute relevant to the project will be initiated in the event of an unresolved conflict.

In the remote event of future disputes over land ownership, the project will implement activities to assist in their resolution and to clarify emerging claims in order to pacify the situation and allow for the consolidation of the developed projects. As described in Section 2.3.15 Feedback and Complaint Redress Procedure, a conflict management procedure will be implemented.

2.5.12 Approvals (CCB, G5.7)

The project proponents obtained recognition and approval for the implementation of the Caiarara REDD+ Project from the stakeholders through face-to-face meetings with communities, farm workers, proponents, and authorities mentioned in Section 2.3.

In January 2024, the first presentation of the project and consultation with the initially selected communities and farm workers was held, a process that gave rise to the report prepared by STA containing the results of the socioeconomic diagnosis (Section 2.1.15 for more details).

Subsequently, a feedback event was held between July 11 and 14, 2024, to present the results of the socioeconomic assessment and information on activities and communication channels to the communities. The meetings were held with the communities selected as stakeholders and with workers from the RAAI and RAAII farms. At the feedback event, participants were able to understand, ask questions, and collaborate on the design and development of the Project as described in Section 2.3.10.

In addition to these meetings and stakeholder participation, the Project will undergo a public consultation event on Verra's registration platform for comments, suggestions, and clarification of questions about the Caiarara REDD+ Project. The importance of stakeholder engagement and collaboration in this process was reinforced by sending formal invitations to project stakeholders through the communication channels established by the project. In addition, letters were sent to relevant municipal, state, and federal institutions containing information about the public consultation, as well as the context of the project and communication channels.

It is worth noting that despite the progress of the National REDD+ Strategy in Brazil (ENREDD+), the passage of Bill No. 572/2020, and the resumption of the Pará Forum on Climate Change and Adaptation (FPMAC), there are still no official REDD+ policies at the national or jurisdictional level. However, the proponents of the Project are always attentive to new information, always present in federal and state government discussion forums in order to contribute to the formulation of these policies and regulations, and are readily available to adapt the Project to the new rules officially established.

2.5.13 Double Counting and Participation in Other GHG Programs (VCS, 3.23; CCB G5.9)

2.5.13.1 No double counting

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

Yes ☐ No☒

2.5.13.2 Enrollment in other GHG programs

Has the project been registered in any other GHG program?

☐Yes ☒ No

2.5.13.3 Projects rejected by other GHG programs

Has the project been rejected by any other GHG program?

Yes ☐ No☒

2.5.14 Double claiming, other forms of credit, and scope 3 emissions (VCS, 3.24)

2.5.14.1 No double claiming with emissions trading programs or binding emission limits

Are the project reductions or removals or project activities also included in an emissions trading program or binding emissions cap? See the *VCS Program Definitions* for definitions of emissions trading program and binding emissions cap.

☐ Yes ☒ No

2.5.14.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or plan to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for the definition of GHG-related environmental credit system.

☐ Yes ☒ No

2.5.14.3 Supply chain emissions (scope 3)

Do the project activities affect the emissions footprint of any product (goods or services) that are part of a supply chain?

☐ Yes ☒ No

2.6 Additional information relevant to the project

2.6.1 Leakage Management (VCS, 3.11, 3.15)

Leakage management is focused on the communities identified for participation in the Project, in the Project Zone, in which a buffer has been made around the 11 communities, these being the Leakage Management Areas. In these areas, activities will be carried out to prevent the occurrence of displacement of illegal activities, so that no leakage is expected due to displacement.

The planned project activities, drawn up using the Theory of Change (Section 2.1.17), were designed precisely to prevent the displacement of illegal hunting, fishing and deforestation activities, by engaging the surrounding rural communities, seeking to offer sustainable production alternatives, encourage new environmentally responsible practices in the daily lives of the communities, raise awareness about the protection of biodiversity and ecosystem services, as well as training and improving the well-being of residents, seeking greater engagement with the Caiarara REDD+ Project. In the long term, the aim is to promote changes in cultural habits, integrate sustainable practices and prevent deforestation and poaching, thus strengthening environmental preservation and fostering a greater understanding of the importance of standing forests, protected local biodiversity and the benefits of a balanced environment.

Leakage management activities are not expected to include agricultural or grazing improvements that increase carbon or GHG stocks compared to baseline scenarios in leakage areas. Therefore, leakage associated with leakage prevention measures is not expected. However, if these activities are implemented during the project period, their emissions will be monitored and accounted for.

2.6.2 More information

Not applicable.

3 CLIMATE

3.1 Methodology Application

3.1.1 Methodology Title and Reference (VCS, 3.1)

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0048	VM0048 – Reduction of emissions from deforestation and forest degradation.	1.0
Methodology	VMD0055	Estimated emission reductions from avoiding unplanned deforestation.	1.1
Tool	VT0001	Tool for demonstrating and assessing additionality in agriculture, forestry, and other land use (AFOLU) project activities under the VCS - VT0001.	3.0
Tool	AFOLU NPRT	AFOLU Non-permanence risk tool	HUB VERRA

3.1.2 Methodology Applicability (VCS, 3.1)

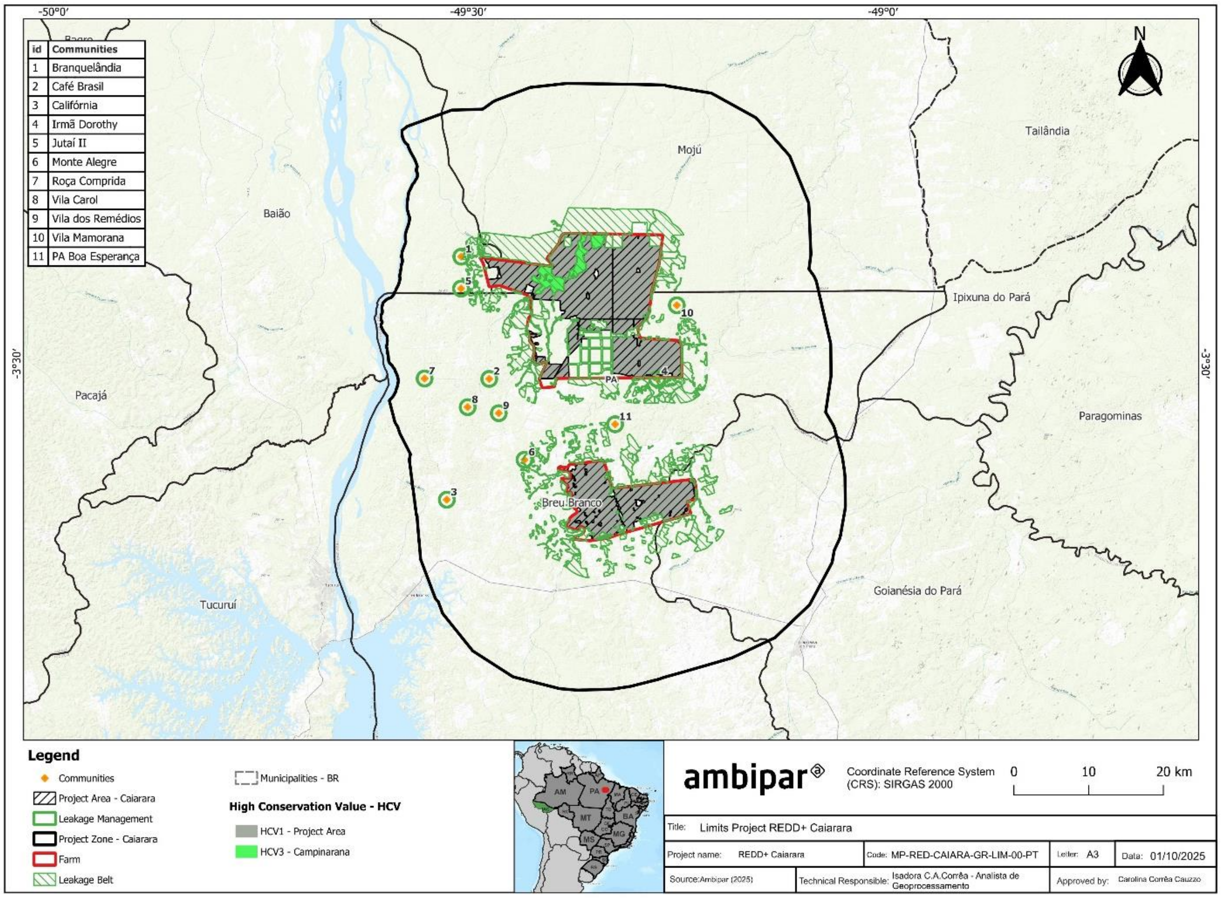
ID/Reference Title	Applicability condition	Justification of compliance
VMD0055	1) The transition in land use in the baseline scenario is from forest land to non-forest land, in accordance with the definition of unplanned deforestation.	The baseline scenario refers to the continuation of land use prior to the Project, where the project area would be subject to significant pressure from unplanned deforestation. Local communities could continue to enter the area, even with the presence of restriction signs, to carry out activities such as timber extraction for charcoal production, as well as illegal hunting and fishing. In addition, growing regional demand for agricultural expansion, especially for soybean cultivation and cattle ranching, would tend to intensify pressure on forest resources in the project area. These conditions characterize the transition from forest to non-forest land, in accordance with the definition of unplanned deforestation.

VMD0055	2) The project involves activities aimed at preventing unplanned deforestation.	The activities of the Caiarara REDD+ project aim to prevent deforestation of the project area by third parties. These include strengthening property surveillance, preventing unauthorized persons from entering the area, as well as containing forest fires, social activities with local communities and farm workers to raise environmental awareness and promote sustainable development, and activities to protect local biodiversity, preventing fauna and flora species from becoming threatened and protecting those that are already threatened.
VMD0055	3) Deforestation agents in the baseline scenario clear land for tree extraction, settlements, roads, unauthorized road expansion or other infrastructure, agricultural production, livestock or aquaculture.	The deforestation drivers identified in the baseline scenario transition land use mainly to the agricultural sector, such as the expansion of soybean production and livestock farming, in addition to deforestation for family farming and timber extraction for charcoal production.

3.1.3 Project Boundary (VCS, 3.12)

Spatial boundaries

Figure 18. Map of Project location.



<u>Jurisdiction Boundary</u>

The jurisdiction of relevance to the project is the Pará State (Brazil).

<u>Avoiding Unplanned Deforestation Project Area (UDef PA)</u>

The limits of the project area were defined considering the existing forest area inside the Fazendas Reflorestamento Água Azul I (RAA I) and Reflorestamento Água Azul II (RAA II) owned by Dow. The total area corresponds to 32,625 hectares. Description of ownership, property rights, and land documents were detailed in Section 2.5.9.

The forest cover estimates were based on PRODES/INPE data and only included land qualifying as forest for a minimum of 10 years prior to the project start date. The Project area has no overlapping boundaries with any other project currently registered at Verra.

<u>Avoiding Unplanned Deforestation Project Leakage Belt (UDef LB)</u>

The avoiding unplanned deforestation project leakage belt (UDef LB) is the forest area to which unplanned deforestation by geographically constrained agents may be displaced and will be monitored.

The spatial extent of the UDef LB was defined by Verra following the criteria described in Section A1.2.2 of Appendix 1 (Module VMD0055).

Land Available for Activity-Shifting Leakage

Areas that are potentially arable, physically accessible, and unprotected land in the country. Determined by Verra.

Leakage Management Zones

These are areas where the project proponent intends to implement activities to minimize the risk of activity-shifting leakage. The leakage management zones comprise a 1km buffer zone around the 11 communities consulted by the Project. The total area corresponds to 3,399 hectares.

Project Activities Region

The project activities region comprises the UDef PA, the UDef LB, and the surrounding deforested areas extending 10 km from the UDef PA.

Carbon pools

The carbon pools considered to the Caiarara REDD+ project are shown below. Methodological details of the estimation of the carbon pools included can be found in the document Forest Carbon Stock Report made available to VVB.

Carbon pools	Included / Excluded	Justification / Explanation of choice
Aboveground biomass	Tree: included	Major carbon pool that will significantly decrease in the baseline scenario in the case of deforestation or forest degradation
	Non-tree: excluded	Conservatively excluded
Belowground biomass	Tree: included	Major carbon pool that will significantly decrease in the baseline scenario in the case of deforestation or forest degradation
	Non-tree: excluded	Conservatively excluded
Dead wood	Included	Significant pool for the forest typology of the UDef PA
Harvested wood products	Excluded	Timber harvest is negligible in the baseline case

Litter	Excluded	Conservatively excluded
Soil organic carbon	Excluded	Conservatively excluded

Sources of GHG Emissions

The GHG emission sources included in or excluded from the boundary of the REDD project activity are shown below:

Source		Gas	Included?	Justification/explanation
Reference	Burning of woody biomass	CO_2	Included	Major emissions source
		CH_4	Included	Included as non-CO_2 emissions from biomass burning in the baseline scenario, according to the methodology
		N_2O	Included	Included as non-CO_2 emissions from biomass burning in the baseline scenario, according to the methodology
		Other	Excluded	Potential emissions are negligible
	Combustion of fossil fuels	CO_2	Excluded	Potential emissions are negligible
		CH_4	Excluded	Potential emissions are negligible
		N_2O	Excluded	Potential emissions are negligible
		Other	Excluded	Potential emissions are negligible
	Use of fertilizers	CO_2	Excluded	Potential emissions are negligible
		CH_4	Excluded	Potential emissions are negligible
		N_2O	Excluded	Potential emissions are negligible
		Other	Excluded	Potential emissions are negligible
Project	Burning of woody biomass	CO_2	Included	Major emissions source
		CH_4	Included	Included as non-CO_2 emissions from biomass burning in the project scenario, according to the methodology

Source		Gas	Included?	Justification/explanation
		N2O	Included	Included as non-CO2 emissions from biomass burning in the project scenario, according to the methodology
		Other	Excluded	Potential emissions are negligible
	Combustion of fossil fuels	CO2	Excluded	Potential emissions are negligible
		CH4	Excluded	Potential emissions are negligible
		N2O	Excluded	Potential emissions are negligible
		Other	Excluded	Potential emissions are negligible
	Use of fertilizers	CO2	Excluded	Potential emissions are negligible
		CH4	Excluded	Potential emissions are negligible
		N2O	Excluded	Potential emissions are negligible
		Other	Excluded	Potential emissions are negligible

3.1.4 Baseline Scenario (VCS, 3.13)

The determination of the most plausible baseline scenario is based on the result of the additionality analysis (Section 3.1.5) and should be consistent with the description of conditions prior to the project start date. The following approach was followed, based on VM0048 v1.0 *Reducing Emissions From Deforestation And Forest Degradation*, to determine the most plausible baseline scenario:

Step 1) Re-use the plausible alternative land use scenarios for the REDD project activity that were listed as a result of Sub-step 1b of Additionality Tool VT0001 (Section 2.2.1).

List of alternative land use scenarios:

I) Continuation of land use prior to the Project*;*

II) Improvement of surveillance actions without registration as a VCS AFOLU Project*;*

III) REDD+ activities without registration as an VCS AFOLU Project.

Step 2b) When the VT0001 investment analysis is used to demonstrate additionality, and if at least one land use scenario generates financial benefits in addition to carbon revenues, select the baseline scenario as follows:

"*When Option I of VT0001 is used, the baseline scenario will be the land use scenario with the lowest costs over the crediting period. Option I can only be applied if the alternative scenarios do not include revenues*."

As none of the alternative scenarios include revenue, option I was used: through simple cost analysis, it was concluded that the proposed VCS AFOLU project does not produce financial benefits beyond the VCS-related revenue. Therefore, the baseline scenario is the scenario with the lowest costs over the long period of obtaining credits, which is scenario I): continuation of land use prior to the Project, since the only costs in this scenario are the payment of the Rural Property Tax (ITR), asset surveillance and the Ybá Project.

According to VMD0055 v1.1 *Estimation Of Emission Reductions From Avoiding Unplanned Deforestation*, once determined, the baseline scenario should be described through a detailed analysis of the agents, immediate causes, drivers and underlying causes of deforestation in the project region. Understanding "who" is deforesting and what drives their decision is necessary in order to understand the risk of leakage through diversion of activity and to design effective measures to deal with deforestation.

The analysis of the agents, vectors and causes of deforestation in the project region was carried out by STA in the Socio-Economic Diagnosis (2024) and was based on secondary data from the region and primary data collected through interviews with local actors.

3.1.4.1 Identification and characterization of deforestation agents

The following identifies the groups of agents that would deforest the region of the project's activities in the absence of its implementation and their relative importance to historical changes in land use and cover. The main groups of deforestation agents identified in the project application region are: I) family farmers, II) rural landowners and III) sawmills, timber and charcoal companies.

I) Family farmers

Family farmers are agents of deforestation insofar as they carry out their farming activities using the traditional slash and burn technique, gradually clearing areas over the years. This agent is directly connected to three vectors of deforestation that will be presented later: slash-and-burn agriculture, cattle ranching and charcoal production.

The rural communities represent the vast majority of the population around the RAAI and RAAII farms, totalling more than 5,000 people, spread over approximately 1,250 families. The majority of this population is made up of families from other states and municipalities who make their living from slash-and-burn agriculture, a traditional technique practised by family farmers in the Amazon, growing mainly manioc, followed by maize and rice, with small livestock.

The rural settlements have collective forest reserve areas in their territories, but residents report that deforestation has caused species of fauna and flora to disappear, some of which have not been seen for years. These communities are suffering from two trends: a) with the gradual increase in the population of families, there will be an advance of slash-and-burn agriculture into arable areas, which may be exhausted by the repeated use of fire and swiddens, and there will be a need to open up new productive areas within the forest reserve areas. As a result, hunting, medicinal and extractive resources can be depleted, causing hardship for the families, who will look for new forest areas to meet their needs for these resources; b) the collective forest areas are targeted by illegal loggers and sawmills who lure families in situations of greater financial and food vulnerability to buy logs and charcoal. This leads to selective logging of timber species and the settlements' timber reserves could disappear. As a result, in the medium and long term, the pressure on the forest areas of the Água Azul I and II reforestation farms could increase.

II) Landowners

Rural landowners are agents of deforestation insofar as they illegally remove forest cover in order to implement agricultural activities, fuel fraud in the CAR, promote land grabbing and speculatively sell plots, reinforcing land insecurity and accelerating deforestation.

Of the five landowners interviewed in the surrounding area, all said they practised cattle farming. In the scenario of expanding production, the owners reported that they would move towards expanding pasture areas with the introduction of cattle. Although the survey did not directly reach the other large and medium-sized rural landowners in the surrounding area, which totalled 26 properties, it is possible to infer that the vast majority carry out cattle ranching activities in the region in areas that have already suffered forest suppression

In addition, cases of CAR registrations were identified under the names of farms that nobody recognised in the region, and others that were not found or were sold. The commercialisation of plots of land in the region to serve as temporary farms for cattle is an existing practice that reinforces the insecurity of land ownership, with high turnover. Revista Pública carried out a study in 2016 which analysed 150,000 CAR registrations in the state of Pará, of which "at least 108,000 showed some overlap with other rural properties" (PÚBLICA, 2023)[38]. Fraud in CAR registrations, according to the research, functions as a kind of "land grabbing", where false owners inform the system of supposed rural properties in forest areas, usually undesignated public areas, in order to simulate a right to the land that they don't have. With these registrations, these supposed owners sell the land to third parties for logging, leaving the land "clean" for cattle and later grain farming.

According to the High Risk List of the Federal Court of Auditors (2024)[39], the land situation in the country is inefficiently managed, favouring land grabbing, which facilitates illegal deforestation, as well as causing social and land conflicts, damaging the environment and directly affecting local communities.

III) Sawmills, lumber mills and charcoal kilns

Sawmills, lumber mills and charcoal factories are historical agents of deforestation in the Amazon, and the project region is no different. Based on secondary data, it was possible to identify illegal logging practices by sawmills and lumber mills located in and around the municipality of Breu Branco. Several seizures have already been made by the competent authorities for illegal timber transport and logging. The local conditions of physical infrastructure and the low capacity of public agents to carry out inspection and control favor the illegal actions of these agents.

38 BARROS, C., BARCELOS, I. As falhas e inconsistências do Cadastro Ambiental Rural. Revista Pública, 01 de agosto de 2016. Disponível em: https://apublica.org/2016/08/as-falhas-e-inconsistencias-do-cadastro-ambiental-rural/

39 Tribunal de Contas da União. Controle do Desmatamento Ilegal na Amazônia. Lista de Alto Risco da Administração Pública Federal, 2ª edição, 2024. Disponível em: <https://sites.tcu.gov.br/listadealtorisco/controle_do_desmatamento_ilegal_na_amazonia.html>.

Characterizing the geographical mobility of agents

An unplanned deforestation project that is avoided by protecting the UDef PA may cause part of the deforestation agents to transfer their activities to another location (i.e. to be "displaced" from the UDef PA). As part of the analysis of deforestation agents, it should be assessed how each group of agents is considered: geographically restricted or geographically mobile.

The initial study of the region (Socioeconomic Diagnosis) was carried out between December 2023 and January 2024, i.e. before the release of VMD0055 v1.1 (21 October 2024), so this point was not included in the scope of the local survey and the data needed to analyze the proportion of agents who have migrated to the project region in the last five years was not collected, making it impossible to carry out such a characterization. Therefore, this issue is a deviation in methodology (more details in Section 3.1.6).

3.1.4.2 Identification of drivers of deforestation

Analyzing the factors that drive the land use decisions of each group of deforestation agents was aimed at identifying the immediate causes of deforestation.

I) Illegal logging and charcoal production

Historically in the municipality of Breu Branco, timber extraction has always been one of the most important activities in the region, with the main species extracted being jarana, acapú, maçaranduba, angelim, melancieira, cipó titica and others (PARATUR, 2012). Illegal logging in the region is not new; logging has been intense and constant for over 40 years. The state, as a supervisory, control and police authority, has been called upon on several occasions to combat this practice in the region.

Several inspection operations have been carried out, for example, in September 2019 in Breu Branco, the Civil Police seized 75m³ of illegal red angelim and maçaranduba wood that was destined for the state of Sergipe (G1 PA, 2019) [40]. In the same year, state agencies IDEFLOR-Bio, Semas and the Environmental Police carried out an operation in the Lago de Tucuruí mosaic region, where they seized three lorries loaded with illegal timber, one with 950 acapu stakes and the other two with logs of various species (Agência Pará, 2019) [41] .

In 2019, IBAMA seized around 600 m³ of logs in the municipality of Breu Branco, and at the time of the operation, the Institute caught several sawmills working illegally (G1 PA, 2019) [42]. In 2020, in a joint action by the Department of the Environment, the Federal Police, the Brazilian Army, the Air Force, the Navy and

[40] G1 PA. Polícia Civil apreende 75 m³ de madeira ilegal em Breu Branco, sudeste do Pará. Belém, 2019. Disponível em: <https://g1.globo.com/pa/para/noticia/2019/09/25/policia-civil-apreende-75-m-de-madeira-ilegal-em-breu-branco-sudeste-do-para.ghtml>.

[41] Margarido, P. Fiscalização na rodovia PA-263 apreende caminhões com madeira ilegal. Agência Pará, 2019. Disponível em: <https://www.agenciapara.com.br/noticia/16644/fiscalizacao-na-rodovia-pa-263-apreende-caminhoes-com-madeira-ilegal>.

[42] G1 PA. Ibama flagra serrarias clandestinas e apreende 600 metros cúbicos de toras de madeira em Breu Branco, no PA. Belém, 2019. Disponivel em: <https://g1.globo.com/pa/para/noticia/2019/08/27/ibama-flagra-serrarias-clandestinas-e-apreende-600-metros-cubicos-de-toras-de-madeira-em-breu-branco-no-pa.ghtml>.

IBAMA, an operation dismantled a gang that sold illegal timber, closing down 10 illegal sawmills in Paragominas, in the southeast of Pará (Agência Pará, 2020) [43] .

In April 2020, the Pará non-governmental organization Instituto do Homem e Meio Ambiente da Amazônia (Imazon) published a study that revealed that 70% of the timber logged in Pará between August 2017 and July 2018 was illegal. The researchers cross-referenced official information from the responsible bodies, such as SEMAS-PA, with satellite images and the results show that of the 38,000 hectares of forest exploited by logging activity in the period surveyed, around 27,000 hectares did not have authorisation from the competent bodies. According to the study's authors, 18,796 hectares of the total logged without authorisation occurred in areas registered with the Rural Environmental Registry (CAR) system (CARDOSO & SOUZA, 2020) [44].

Studies have shown that logging has been profitable in the Amazon because it has several competitive advantages. According to Diniz et al. (2009) [45] , over the last thirty years, beef cattle ranching and logging activities based mainly on log production, with little added value, have been the activities that have shown the greatest competitive advantages for the region, given the market conditions, the opportunity costs of alternative activities, including short and medium-term profitability and even the institutional conditions of easy access to credit and low governance in terms of the fragility of property rights and the associated environmental costs.

Illegal logging tends to continue in the project region. As reported by community leaders interviewed in the socio-economic diagnosis (STA, 2024), hardwood species have practically been exhausted, and there are few protected areas left, such as conservation units, indigenous reserves and private or community areas, so illegal practices enter these areas, putting greater pressure on the forests that still exist.

Another activity that is generally associated with the dynamics of logging is charcoal production. In Breu Branco, most of the charcoal is produced by hand, in kilns with a capacity of approximately 16 m^3 of wood, where the process of burning and cooling the charcoal takes between six and seven days (PARATUR, 2012) [46]. The activity is so important for the region that until mid-2011 there was an Association of Charcoal Burners in the municipality, which had around 170 members and 21 sawmills legalised by IBAMA (PARATUR, 2012).

For family farmers, charcoal production has always taken place locally, mainly for domestic use and with sporadic sales of surplus and has intensified with the rise in the price of cooking gas in recent years. According to the IBGE, in the last five years the municipality of Breu Branco has seen a significant increase

[43] MELLO, A. P. Semas fecha 10 serrarias clandestinas em operação conjunta com órgãos federais. Agência Pará, 2020. Disponivel em: <https://www.agenciapara.com.br/noticia/23969/semas-fecha-10-serrarias-clandestinas-em-operacao-conjunta-com-orgaos-federais>.

[44] Cardoso, D., & Souza Jr., C. Sistema de Monitoramento da Exploração Madeireira (Simex): Estado do Pará 2017-2018 (p. 38). 2020. Belém: Imazon. Disponível em: <https://imazon.org.br/publicacoes/sistema-de-monitoramento-da-exploracao-madeireira-simex-estado-do-para-2017-2018>.

[45] DINIZ et al. Causas do desmatamento da Amazônia: uma aplicação do teste de causalidade de Granger acerca das principais fontes de desmatamento nos municípios da Amazônia Legal brasileira. In: Nova Economia, Belo Horizonte, 19 (1) 121-151, janeiro-abril de 2009. Disponível em: <https://doi.org/10.1590/S0103-63512009000100006>.

[46] PARATUR. Inventário da oferta turística de Breu Branco. Breu Branco, 2012. Disponível em: <https://www.setur.pa.gov.br/sites/default/files/pdf/iot_breu_branco_12-06-12_ok.pdf>.

in the amount of charcoal it produces and has been leading production in the state through investment in planted forests, rising from 32,187 tonnes produced in 2018 to 42,787 tonnes in 2022.

According to Oliveira et al (2023) [47], the production of charcoal from native forests has decreased and there has been a significant increase in production from planted forests, which began to show higher percentages in 2017. Despite this scenario, the production of illegal charcoal from native forests by small local producers and even companies is still constant in the region. The current scenario for 11 charcoal factories located in the Breu Branco region is that the majority are unfit due to the omission of declarations and proceedings linked to slave labour.

Secondary research revealed the existence of a company located in Breu Branco, which has been named by the Federal Public Prosecutor's Office as a defendant in a R$3.8 million lawsuit for using 3,000 cubic metres of illegal timber (REPORTER BRASIL, 2014) [48], and the owner had already been included on the "dirty list" of slave labour. Another charcoal company in the region was fined for slave labour in 2008 for having 180 kilns, 60 of which were illegal (SINAIT, 2012) [49]. At the end of 2011, the Saldo Negro (Black Balance) operation tracked down the 40 largest charcoal factories in Pará and found fraud in 25 charcoal producers - 14 were front companies, selling much more than their production capacity, 11 had no head office (SINAIT, 2012) [50].

The photo below was taken by the field team during research in the communities, and is a common image in the region, of kilns producing charcoal.

[47] OLIVEIRA, M. E. Q et al. Análise da conjuntura da produção de carvão vegetal no Brasil e no Estado do Pará. CONCILIUM, VOL. 23, Nº 21, 2023. DOI:10.53660/CLM-2393-23S17.

[48] OJEDA, I. Carvoarias representam um quinto das inclusões na 'lista suja' do trabalho escravo. Reporter Brasil, 2014. Disponivel em: <https://reporterbrasil.org.br/2014/01/carvoarias-representam-um-quinto-das-inclusoes-na-lista-suja-do-trabalho-escravo/>.

[49] SINAIT. A rota do carvão, do desmatamento e do trabalho escravo na Amazônia. 2012. Disponível em: <https://www.sinait.org.br/noticia/6583/a-rota-do-carvao-do-desmatamento-e-do-trabalho-escravo-na-amazonia>.

[50] Ibid.

Figure 191. Photo of charcoal kilns in rural communities. Source: STA, 2024.

The municipality of Breu Branco more than doubled its charcoal production between 2017 and 2022, from 3,000 tones to 13,822 tones over the period. Communities such as Branquelândia, Califórnia, Vila Mamorana and Vila dos Remédios have charcoal production as one of the main sources of income for their families. Branquelândia stands out as having charcoal as the first product to be commercialised by families.

Thus, with the great influence of steel mills in the region, the price of cooking gas remaining high and the state's difficulty in overseeing it, charcoal production will continue to be strong in the region surrounding the project, compromising the biodiversity of local species and violating fundamental social rights, given that working conditions are inadequate,

II) Slash and burn agriculture

In the Amazon, subsistence farming stands out as one of the forms of land use for agricultural purposes, practiced by farmers on a small scale, by cutting and burning primary or secondary forest (FREITAS et al., 2013) [51]. In a broader definition, slash-and-burn agriculture is any continuous agricultural system in which clearings are opened to be cultivated for shorter periods of time than those intended for fallow (KLEINMAN et al., 1995). In this type of cultivation, the vegetation is cut in the dry season and burned at the beginning of the rainy season (SOTTA et al, 2019). This is followed by the planting of agricultural species for family production and consumption. This model currently covers the entire tropical region of the planet and extends to subtropical forests (PEDROSO JÚNIOR; MURRIETA; ADAMS, 2008). However, slash-and-burn agriculture becomes unsustainable as repeated burns are carried out, reducing the amount of fallow time between crops (DENICH; KANASHIRO; VLEK, 1999), reinforcing the studies by Zarin et al. (2005), which show that a history of successive burns reduces the growth rate of secondary forest in the Amazon basin,

[51] FREITAS, J. L.; SANTOS, E. S.; LIMA E SILVA, R. B.; SILVA, T. L. Comparação e análise de sistemas de uso da terra de agricultores familiares na Amazônia. **Biota Amazônia**, Macapá, v. 3, n. 1, p. 100-108, 2013. Disponivel em http://www.iepa.ap.gov.br/biblioteca/artigo/2015/comparacao-e-analise-amazonia.pdf.

mainly due to a reduction in nutrient cycling stocks. In addition, forests that have already been burned become more susceptible to fire (MALHI et al., 2008).

On the other hand, there are authors who defend the slash and burn system used as a form of soil preparation as a sustainable system in agriculture (SOTTA et al, 2019), since burning vegetation provides nutrients for soil fertilisation, increases pH and reduces the amount of weeds in agricultural cultivation in the long term (KATO; DENICH; VLEK, 1999); and, when practised traditionally in large forested areas, with low population density, low impact technology and long fallow periods, it can be managed in an ecologically sustainable way without drastically compromising soil fertility (KLEINMAN et al., 2005).

From the perspective of the scientific viewpoint in contradictory analyses of slash-and-burn agriculture and the headlines about deforestation rates attributed to agricultural activity, which for the Amazon in particular, Serrão et al (1996) estimate at between 30 and 35%, it is estimated that in the family farming areas that practice slash-and-burn cultivation around the project, if there is no change in agricultural technology, the tendency is for the degradation of small family farmers' arable land to grow steadily, since the need for arable land will remain the same, increasing every year in proportion to the increase in the population of families in rural communities. Thus, the risk of deforestation with the opening up of new agricultural areas and even expansion into forest areas is a constant risk.

The total population mapped in the communities surrounding the project totalled approximately 1,300 families, with each family producing an estimated 1.5 hectares of manioc plantations. This calculates to an average of almost 1,941 hectares per year used for new plantations, which could mean more areas cut down using the slash and burn technique each year if there is no change in the slash and burn technique practised by the families. According to the survey data, the Vila Monte Alegre community, which has 30 families, does not carry out agricultural production, as most of them work for the DOW company or receive some kind of aid, even so, the numbers are still significant in terms of areas for agriculture.

III) Planting grains - commodities

There are studies that prove that the dynamics of land exploitation in the Amazon have followed a mapped path over the last 40 years that involves agricultural activity with temporary and permanent crops, followed by the introduction of extensive livestock farming, which gradually occupies more land and expels small farmers to new areas to then plant monoculture grain crops such as soya. Throughout this trajectory, soya, which until then was considered a high latitude crop, has been transformed, invading the south-east of the country, taming the extensive cerrado areas of central Brazil and today being cultivated at latitudes of 0º in the Amazon and expanding rapidly in the states of the northern region, especially in the state of Pará (COSTA et al, 2017).

In the project region, although there are no official records of soya plantations in the municipality of Breu Branco, there are reports that soya is already in the region and in certain parts of the municipality it is possible to find plantations of the crop, as well as maize. In the municipality of Tailândia, a grain centre is being formed, made up of 12 municipalities, one of which is Breu Branco. The main aim of creating this centre is precisely to expand the areas of grain (soya and maize) in the region.

Some factors contributed to the establishment of soya in the region: low land prices and high tax and financial incentives, creating favorable conditions for the activity. The main grain crops planted in the

Amazon were rice, beans, corn, wheat and soya, but soya was the most profitable relative to investment. Soya is the main agro-product exported by Pará, and the state has entered the list of the top 10 grain exporters in Brazil, the best position since the product was introduced to Pará soil 30 years ago (BOTELHO, 2024). As a result, the search for more land to plant grain crops in the state is growing.

Another vector of future deforestation is the Abaetetuba Private Use Port Terminal (TUP-Abaetetuba/PA), a venture by Cargill Agrícola S.A., whose main objective is to transport grain produced in the states of Pará, Rondônia and Mato Grosso to foreign markets by river. The installation of the port is likely to generate a soya boom and the invasion of indigenous territories and riverside and quilombola communities in the municipality of Abaetetuba and could also expand this trend to other municipalities in the region. The consequence could be a crisis in the subsistence economy based on hunting, fishing and family farming. The growth of property speculation on agricultural land, including the project's area of influence, could stimulate production in the surrounding areas, as speculation about the start of the port could attract farmers who come mainly from the south and Mato Grosso.

The project area could indirectly induce deforestation due to advertising and its proximity to the port area, making it easier to transport goods. Indigenous peoples, quilombolas and artisanal fishermen could be deprived of the natural resources essential to their way of life, triggering pressure on forest resources in denser forest areas, such as the Caiarara REDD+ Project area.

Since October 2021, there has been no movement in the licensing application process with SEMAS and in addition to social and political mobilization in the region with strong pressure on public and legal bodies to contain the TUP project, it is assessed that the installation of the port may suffer a delay in its implementation, however, it remains a strong vector of future deforestation if it materializes in the medium and long term. Associated with the installation of ports, the advance of grain crops, such as soya, could also be boosted, causing even more deforestation in the region of the Caiarara REDD+ Project.

IV) Livestock

Brazil is one of the world's largest beef exporters, with 466,377 tones exported in the first two months of 2024, a 39% increase on the same period in 2023. The position that Brazil has gained since the 2000s is that of a major exporter of the product, in different forms of packaging. The biggest current destinations are China and the United States. China remains the main buyer of Brazilian beef, but its relative share of the country's total exports has fallen from 51.4 per cent in 2023 to 41.6 per cent in 2024, due to a greater diversification of markets. The United States of America is in second place, increasing its imports year on year. In the first two months of 2024, imports from the US totalled 89,103 tones, representing growth of 150% compared to the same period in 2023 (Abrafrigo, 2024).

Because of the preservation of the product, the transport time between the slaughter of the cattle and the consumer center was the determining factor for the spatial organization linked to livestock farming, directly influencing the location of the meat-packing plants. With the improvement of the road network and electrification, the gradual establishment of large-scale meatpacking companies and slaughterhouses in the region became possible. In Pará, the policy of tax legislation favorable to the advancement of livestock farming, which encourages the development of the sector (SEDEME, 2021) [52], led to the installation of several meat-packing plants, which quickly established the dairy industry and then the leather industry. In 1996 there was no leather industry, but by 2004 four companies had been set up and one was under construction, and by 2009 there were 22 meatpacking plants in Pará.

The livestock production chain has several interconnected branches, such as the leather industry and the slaughterhouses and meatpacking plants, which are trying to set up operations where there is more cattle production. The installation of slaughterhouses is a fundamental reference point for the expansion of livestock farming in a region and, for large ranchers, represents the consolidation and modernisation of the cattle production chain.

Secondary research mapped companies in the meatpacking and slaughterhouse sector in Breu Branco and neighbouring municipalities. Eight companies were identified, one in Breu Branco, three in Tucuruí, one in Goianésia do Pará, one in Ipixuna do Pará and two in Tailândia. Based on the data on the number of head of cattle in the period from 2018 to 2022, available in the Pará statistical yearbook (2023), it is possible to see that there has been an increase in the cattle herd in most of the municipalities in the project region. Among the seven municipalities, Goianésia do Pará ranks first in cattle production, followed by Breu Branco and thirdly Ipixuna do Pará, with Baião standing out, which has seen its herd grow considerably over the last five years, demonstrating that the territory has a certain vocation for livestock farming in the region.

The survey found that beef cattle ranching is the main activity of 100 per cent of the medium and large landowners interviewed in the study. This activity is considered to have a tendency to remain in the region and is a vector of deforestation with potential for growth. When it comes to the issue of livestock farming for small family farmers in communities, historically in the Amazon cattle farming has always been associated with the logic of "savings": cattle farming in rural communities is common, and has always functioned as an economic guarantee for "hard times", i.e. for paying off loans, family illnesses, travelling needs, etc. The sale of a single head allows the corresponding money to be recovered three years later; the time needed for the animal to gain weight and acquire market value. The idea of this type of "savings economy" is explained by author Glaúcia Silva:

“Savings activities are highly valued by floodplain dwellers. As we've said, it's a way of describing what isn't perishable, what persists despite the flood and can be transformed into money in case of need, preventing loss and thus allowing accumulation to begin. But this classification is peculiar because it doesn't coincide with its usual definition, which is the withdrawal of money from circulation. So the river dweller doesn't keep the money; he avoids selling a commodity according to a calculation of what is needed or necessary but rather makes a store of value” (SILVA, 2005, p. 296).

During the survey, the communities of Branquelândia, California and the Boa Esperança PA stated that they raise cattle and want to expand production. Based on the data, it can be concluded that beef and pork

[52] ABREU, G. Incentivos fiscais do Governo do Estado contribuem para o desenvolvimento do setor industrial no Pará. Secretaria Estadual de Desenvolvimento Econômico, Mineração e Energia do Pará, 2021. Disponível em: <https://sedeme.pa.gov.br/noticias/incentivos-fiscais-do-governo-do-estado-contribuem-para-o-desenvolvimento-do-setor#:~:text=A%20pol%C3%ADtica%20de%20incentivos%20%C3%A9,10%2C%2015%20ou%2012%20anos.&text=Mesmo%20em%20um%20cen%C3%A1rio%20adverso,de%206.370%20postos%20de%20trabalho>.

farming is a productive activity that generates income and has the prospect of increasing in the region, not only because of the intentions of the landowners interviewed to increase production, but also because of the motivations presented by the rural communities. As such, it is considered that the activity tends to remain and possibly grow in the region, being a vector of deforestation with growth potential.

V) Palm Oil Cultivation

Currently, the state of Pará is the country's largest oil palm producer (98.27%), with a production of around 2.9 million tones on 185,96 thousand hectares. The municipality of Tailândia is the largest producer (942,08 thousand tones), accounting for 32.47% of Pará's production (BARBOSA, 2024). Historically, the first plantings of the crop in the state began in mid-1964/65 and, since then, oil palm has established itself in the state, mainly in the municipalities of Benevides, Santa Bárbara, Moju, Acará, Tomé Açu and Tailândia, and continues to expand, now with an eye on the Lower Tocantins and neighbouring municipalities.

According to studies carried out by EMBRAPA Amazônia Oriental, the areas located in the micro-regions of Almeirim, Portel, Furos de Breves, Arari, Belém, Castanhal, Bragantina, Cametá and Tomé-Açu are suitable for oil palm cultivation, as they have favourable environmental conditions for the development of the species (BRASIL, 1992). A survey carried out by Embrapa technicians found that, of the total area of Pará, 124,804,200 ha, more than 5,500,000 ha are suitable for oil palm cultivation in edaphoclimatic terms (MULLER et al, 2006).

Investment in the crop is only increasing. The expansion of oil palm is carried out both in the companies' own areas and in the integrated modality, the latter of which involves investments in family farming, a process that began in 2012 in the municipality of Moju. Since then, the companies have expanded this modality by injecting resources into the following municipalities: Abaetetuba, Acará, Aurora do Pará, Baião, Bujaru, Cametá, Castanhal, Concórdia do Pará, Garrafão do Norte, Irituia, Mocajuba, Moju, São Domingos do Capim, Tailândia and Tomé-Açu.

Along with this expansion of local palm oil cultivation, there is growing concern about the deforestation caused by this activity. According to Vijay (2016), since 1989 oil palm expansion has been taking place in areas of native forest, causing deforestation and consequently the loss of biodiversity. He also shows that oil palm was responsible for an average of 270,000 ha of forest conversion every year from 2000 to 2011 in the main palm oil exporting countries.

Deforestation of tropical rainforests increases carbon emissions, replacing natural forests with monoculture palm plantations reduces overall plant diversity and eliminates animal species that depend on natural forests (VIJAY, 2016). For Mongabay (2024), the conversion of forests into oil palm plantations is a serious problem throughout the north-east of Pará, because they are entering areas of collective, quilombola and indigenous territories. Communities allege that, in addition to deforestation, oil palm plantations are causing water contamination due to the use of pesticides, the appearance of pests due to monoculture and the loss of biodiversity in general.

There were no oil palm plantations in the project region, but in Breu Branco, where the communities studied are concentrated, the northern side of the municipality borders two of the largest oil palm producers in the state, and the municipality of Baião to the north is receiving investments to expand oil palm plantations in the region. It is therefore assumed that there is a real risk of oil palm plantations moving into Breu Branco, initially in the north of the municipality, given the influence of these three municipalities.

3.1.4.3 Identification of the underlying causes of deforestation

The main underlying causes of deforestation are directly related to the decisions of deforestation agents, shaping the current land use scenario and driving deforestation both directly and indirectly.

I) Land tenure insecurity

As previously mentioned, CAR registrations in Brazil overlap with other properties, in addition to cases identified around the project in which the CAR is registered under the names of non-existent farms or farms that have already been sold, which supports illegal land grabbing practices and fraudulent commercialisation of land, as land grabbers simulate ownership through false CAR registrations, selectively deforest for illegal logging and then convert the area. The future trend is that land tenure insecurity is likely to persist if there are no improvements in the systems for monitoring and regularising land tenure, since the CAR will continue to be used as an instrument for the fictitious legalisation of illegal occupations, maintaining pressure on public, collective and private forests.

II) Socioeconomic dynamics and vulnerability of rural communities

Communities close to the project number around 1,250 families, most of whom live from slash-and-burn agriculture. Charcoal production is also common to meet the demand of illegal charcoal factories, jeopardizing areas of collective reserve forest and nearby private reserves. Due to population growth and the use of fire as a traditional technique, the communities end up expanding into new forest areas.

Family farmers, who don't have access to viable alternatives, reproduce a cycle of continuous deforestation as cultivation areas run out. With population growth in the communities, slash-and-burn agriculture tends to expand, which can lead to a spiral of degradation of agricultural production areas and increased deforestation of native forests. The future trend is that without public technical assistance policies and incentives for sustainable agricultural alternatives, these populations will continue to advance into the forest.

III) Lack of governance and environmental enforcement

Illegal logging has been going on for more than 40 years in the region, with constant operations by IBAMA, the Civil Police and SEMAS, which have seized thousands of cubic meters of wood. Illegal sawmills, logging companies and charcoal factories take advantage of the absence and weakness of the state to operate with little chance of punishment, which encourages predatory exploitation and subsequent agricultural use of the open areas, especially by cattle ranchers and land speculators. The lack of structural investment in inspection and monitoring favors the continuation of deforestation and the future tendency is for weak institutional capacity to maintain the cycle of impunity that encourages the actions of illegal agents.

IV) Policies to encourage agricultural expansion and market pressures

Pará is a major producer of soya and beef, with state policies to encourage the installation of meatpacking plants and the expansion of production, with reports of the advance of the crop in the municipality of Breu

Branco itself. The soya and beef chains put pressure on the occupation of new forest areas and strengthen land speculation in the region, as rural landowners and agricultural entrepreneurs are encouraged to expand their herds and convert areas to grain cultivation. The tendency for these production chains to grow is high, given international demand and logistical investments in the region, which will result in greater pressure on forest areas that are still intact.

The underlying causes identified are deeply interconnected and feed a land use system that favours deforestation, driven by economic pressures, lack of governance, institutional fragility and social vulnerability. The trend of deforestation in the region is one of continuous growth, with severe environmental and social implications.

3.1.4.4 Analysis of the chain of events leading to deforestation

Deforestation in the Caiarara REDD+ Project region stems from a chain of interconnected events in which structural causes, various agents and immediate vectors combine. The process begins with land insecurity, which acts as a gateway to land grabbing and encourages the irregular appropriation of areas. This situation, coupled with low enforcement capacity, creates the conditions for the actions of loggers and sawmills, who illegally exploit the forest, opening branches and clearings, while charcoal companies make use of the waste and firewood to produce charcoal. The wood harvested finances the first stages of conversion.

At the same time, family farmers, faced with a lack of technical assistance and productive alternatives, resort to slash-and-burn to open small fields, a practice that constantly degrades forest edges due to soil exhaustion, as well as often supplying firewood for the charcoal market, reinforcing the cycle.

Once open and degraded, the areas are converted into pastures, attracting medium and large cattle ranchers who introduce cattle as a way of consolidating ownership and generating liquidity. The growth of the herd, combined with the presence of meatpacking plants and regional transport channels, gives economic stability to cattle ranching and encourages new openings. In this context, there are also practices of enticing small family farmers by large grain producers, who offer to buy their plots, and it has already been reported in conversations with communities that this practice is already happening in the region. Faced with economic pressure and the difficulty of maintaining subsistence activities in degraded areas, many families may end up giving up and transferring their land to more capitalized agents, who will then direct the plots towards growing grain on a large scale.

As market and infrastructure conditions improve - especially the prospect of new ports and transport routes - the attractiveness of converting to soya and corn increases, reinforcing the replacement of forests and pastures with monocultures. Furthermore, to the north of Breu Branco, there is a risk of the advance of oil palm, which has already been consolidated in neighboring municipalities.

Based on secondary data and information gathered in the field from local agents, it is possible to find conclusive evidence that explains the relationships between agents, vectors, underlying causes and the pressure of deforestation in the project region. In this way, it can be concluded that the relationships between social variations (sources of income generation in the surrounding communities), the permanence of slash-and-burn agriculture, the motivation for the advance of cattle ranching, the demand for wood for charcoal production and others, added to the inefficiency of public authorities in supervising illegal activities, contribute to the deforestation scenario observed in the region. Considering this evidence, the future trend is for the current scenario of influence from the agents, causes and vectors of growing deforestation to continue.

3.1.5 Additionality (VCS, 3.14)

3.1.5.1 Regulatory Surplus (VCS, 3.14)

Is the project located in an Annex 1 or Non-Annex 1 country of the UNFCCC?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities required by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located within a country not included in Annex 1 and the project activities are required by a law, statute, or other regulatory framework, are those laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

The Brazilian Forest Code (Law 12.651 of 25 May 2012) provides for the protection of native vegetation and establishes rules for the protection of native vegetation in Permanent Preservation Areas, Legal Reserves, restricted use, forest exploitation and related matters, regulating the actions and activities that can take place in each type of area and what the roles of the state and the forest owner are.

Many parts of the Project area are protected by law and must not be legally deforested by the owner, and it is the role of public bodies to assess, monitor and combat illegal deforestation. However, for various reasons, the protection and permanence of these areas is not guaranteed by legal provisions alone, without the presence of REDD+ Project actions and funding, since government bodies at both state and federal level have limited options for ensuring compliance with the laws and regulations created to prevent deforestation. In these cases, the Forest Code has also defined the additionality of the actions taken by landowners to prevent deforestation in Article 41, Section 4: "The maintenance activities of Permanent Preservation Areas, Legal Reserves and restricted use areas are eligible for any payments or incentives for environmental services, constituting additionality for the national and international markets for certified reductions in greenhouse gas emissions."

As already described in Section 2.2.3, enforcement of environmental legislation is not very efficient in Brazil. The high cost of enforcement activities, the small number of employees in environmental protection and control agencies and the vast expanse of territory require many people, institutions and tools capable of monitoring any change in forest cover (Moura, 2016) [53]. According to the Federal Court of Auditors (2024)[54], many factors are linked to compromised government action in the application of policies to control illegal deforestation in the Amazon, such as insufficient budgetary resources, delays in implementing the monitoring system, disaggregation of actions between the various agencies responsible, lack of central coordination and staff shortages, which facilitates land grabbing and favors deforestation. The weak applicability of laws relating to environmental protection is evident in various areas of national policy, from the low rate of land regularization, which is a major factor in deforestation in the Amazon, to the implementation of command-and-control instruments established by the Forest Code, (Law n. 12.651/2012).

At the Breu Branco plant, on the properties covered by the REDD+ Project, Dow has a factory complex that concentrates activities in sustainable management of planted forests, a carbonization plant and the production of silicon metal. The charcoal used in production comes from eucalyptus grown and harvested on the farms following the guidelines of the FSC (Forest Stewardship Council), an international action organization that promotes responsible management of the world's forests through wood certification (DOW, 2020) [55]. Eucalyptus is used to produce charcoal at the carbonization plant, which is part of the Natural Resources Operation, consisting of 120 kilns and two high-tech production gas burners, with 100% of the charcoal coming from FSC-certified forests (DOW, 2021)[56]. The FSC guarantees that forest management is carried out in a responsible manner, respecting strict environmental, social and economic standards. It also ensures that eucalyptus plantations comply with the best sustainability practices, promoting the protection of the rights of workers and local communities, as well as guaranteeing the economic viability of forestry production. This certification reinforces the applicant's commitment to responsible management of natural resources.

3.1.5.2 Additionality Methods (VCS, 3.14)

The additionality of the Caiarara REDD+ Project was verified using the VCS-approved tool "VT0001 – Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities," version 3.0, dated February 1, 2012.

The conditions for applicability of the tool are met since:

- The AFOLU activities are the same or similar to the activities proposed in the Project, within their respective limits, whether or not registered as a VCS AFOLU Project, and do not lead to the violation of any applicable law, even if this law is not enforced; and

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[53] Moura, A.M.M. Governança ambiental no Brasil : instituições, atores e políticas públicas. Brasília: Ipea, 2016. 352 p. ISBN: 978-85-7811-275-2.

[54] Tribunal de Contas da União. Controle do Desmatamento Ilegal na Amazônia. Lista de Alto Risco da Administração Pública Federal, 2ª edição, 2024. Disponível em: <https://sites.tcu.gov.br/listadealtorisco/controle_do_desmatamento_ilegal_na_amazonia.html>.

[55] Dow Inc. INtersections Environmental, Social and Governance Report. 2020. Disponivel em: <https://br.dow.com/content/dam/corp/documents/about/066-00338-11-2020-esg-report.pdf>.

[56] Dow Inc. INtersections Relatório Ambiental, Social e de Governança. 2021. Disponivel em: <https://br.dow.com/documents/about/066-00397-11-2021-esg-report.pdf>.

- The VM0048 baseline methodology provides a step-by-step approach to justify the determination of the most plausible baseline scenario (Section 6.1 - Determination of the Most Plausible Baseline Scenario[57]).

The step-by-step process for analyzing project additionality according to the VT0001 tool is described below.

Step 1. Identification of alternative land use scenarios for the proposed activities of the VCS AFOLU Project.

Sub-step 1a. – Identify alternative land use scenarios that are credible for the proposed VCS AFOLU Project activities.

The alternative land use scenarios identified were described based on the Socioeconomic and Biodiversity Diagnosis carried out in the Project region in 2024, which are based on secondary data from a literature review and primary data resulting from the Participatory Rapid Rural Diagnosis (DRP) with rural communities surrounding the RAAI and RAAII farms; and individual interviews with farm workers. Among the realistic and credible alternative land use scenarios that would occur within the Project boundaries in the absence of the AFOLU Project activity registered in the VCS, the following were considered:

- **(I) Continuation of pre-Project land use (baseline scenario)**

Occupation of the Caiarara REDD+ project area intensified in the 1970s, driven by Brazilian government initiatives and incentives for development in the Amazon region, attracting new ventures and, consequently, migrants from various regions of the country in search of new opportunities, mainly by displacing people for the construction of the Tucuruí Hydroelectric Power Station[58]. The municipality of Breu Branco was created to house families who lived on the banks of the Tocantins River in areas that were submerged by the hydroelectric dam. In this context, the municipality of Breu Branco experienced significant growth in the 1990s, with investments in infrastructure and economic expansion, due to its proximity to the hydroelectric power station and the trans-Amazonian motorway and due to the cluster of silicon mining available in the region and logging companies (BARATA, 2019). However, the demand for jobs exceeded the number of available positions, forcing many migrants to seek alternative livelihoods. Most chose to settle in the rural area of the municipality, working in subsistence agriculture and related activities. This new working relationship in the countryside has influenced the emergence and expansion of new rural communities, as in the case of the municipality of Breu Branco, where there are several recent communities.

Rural communities and small farmers with this historic establishment represent the vast majority of the population surrounding the Dow farms. During the Socioeconomic Diagnosis interviews, it was identified that the most widely used agricultural practices involve the traditional slash-and-burn technique, clearing and burning new areas over the years as demand for land increases. This process is used for temporary crop production and livestock farming, but areas are also cleared for charcoal production, a product

[57] Available at: < https://verra.org/wp-content/uploads/2023/11/VM0048-Reducing-Emissions-from-Deforestation-and-Forest-Degradation-v1.0-1-1.pdf>

[58] CAVALCANTE, F. C. The migratory process in the Amazon linked to labor mobility – the case of the Tucuruí Hydroelectric Plant. Proceedings of the 10th Meeting of Latin American Geographers – March 20-26, 2005 – University of São Paulo.

intended for family use and in high commercial demand in the region, representing a potential influence on local deforestation.

The majority of charcoal production in Breu Branco is handmade in kilns with a capacity of approximately 16 m^3 of wood. However, in the last five years, the municipality of Breu Branco has seen a significant increase in the amount of charcoal it produces, and has been leading production in the state through investment in planted forests, rising from 32,187 tonnes produced in 2018 to 42,787 tonnes in 2022 (IBGE). According to Oliveira et al (2023) [59], the production of charcoal from native forest has decreased and there has been a significant increase in production from planted forest, but even with this scenario, the production of illegal charcoal from native forest is still constant in the region. The municipality of Breu Branco saw a notable increase in charcoal production between 2017 and 2022, from 3,000 tonnes to 13,822 tonnes in the period (IBGE). The communities of Branquelândia, California, Vila Mamorana and Vila dos Remédios have charcoal production as one of their main sources of income, with Branquelândia standing out as having charcoal as the first product commercialized by families (STA, 2024).

Burning practices used in agriculture are a significant problem for forest preservation, as measures taken to prevent fires from spreading to other properties are not effective, putting extensive areas of native forests at risk. In addition, the high demand for charcoal in the region exacerbates this situation, as it increases the risk of invasion of private properties for illegal logging, putting even more pressure on forests and contributing to environmental degradation.

Another relevant risk factor for the region is the expansion of soybean cultivation. Although there are no official records of planted areas in the municipality of Breu Branco at this time, in interviews and consultations with the communities there are reports that soybeans are already present in nearby areas, and in certain locations in the municipality there are already soybean and corn plantations. Soybean cultivation has been expanding rapidly in the state of Pará, driven by factors such as low land prices and high tax incentives. Soybeans, which were initially grown at high latitudes, have adapted to the Amazonian climate and are now one of the main agricultural exports of Pará (COSTA, 2017)[60], which has become one of the 10 largest grain exporters in Brazil, the best position since the product was introduced on Pará soil 30 years ago [61]. As a result, the search for more land to plant grain crops in the state is on the rise and this search for new planting areas poses a direct threat to native forests in the project area, increasing pressure on land and contributing to deforestation.

Beef cattle ranching is another important driver of deforestation in the region. The socioeconomic assessment conducted in the region found that this activity is predominant among 100% of the medium and large rural landowners interviewed, suggesting that the trend is for the activity to continue and even grow. Based on the data on the number of head of cattle in the period from 2018 to 2022, available in the

59 OLIVEIRA, M. E. Q et al. Analysing the charcoal production situation in Brazil and the state of Pará. CONCILIUM, VOL. 23, Nº 21, 2023. DOI:10.53660/CLM-2393-23S17.

60 COSTA et al. Agricultural activity in the state of Pará. Belém, PA : Embrapa Amazônia Oriental, 2017. 174 p.

61 BOTELHO, C. Soy is the main agricultural product exported by Pará. Available at:<https://agenciapara.com.br/noticia/21749/soja-e-o-principal-produto-agro-exportado-pelo-para>. Accessed on: 27.02.2024.

statistical yearbook of Pará (FAPESPA, 2024) [62], it is possible to see that there has been an increase in the cattle herd of most of the municipalities in the project region, with Baião and Goianésia do Pará having a notable increase of around 50 per cent and Breu Branco increasing its cattle herd by 31 per cent, reaching more than 200,000 head of cattle in 2022.

For small family farmers, cattle raising has always been a form of "savings" in the Amazon, serving as an economic resource in times of need (SILVA, 2005)[63]. In the 2010 census, 50% of the communities in the region expressed a desire to expand cattle raising, indicating that the activity has growth prospects. Both cattle and pig farming are seen as productive activities with the potential to increase pressure on forests, contributing to deforestation, this was revealed by IPAM's technical note (2021) [64] which, based on data from INCRA, INPE, MapBiomas and the Brazilian Forestry Service, analyzed that of the areas cleared in public forests that were not earmarked, around 75% became pasture and remained so after ten years of conversion, demonstrating that pasture is the main tool for land occupation in the Amazon, especially in deforestation frontier regions.

The farms are monitored 24 hours a day to ensure the safety of employees and the protection of the property's forests, but due to the large area and the lack of control and management of the monitoring processes and operations, there are still incidents of third parties entering the area illegally. One action is carried out within the Project Area, in the proponent's native forests: the Yba Project, carried out with some residents of the Mamorana Community, which aims to generate income for the families through the extraction of andiroba, a fruit native to the Amazon rainforest, used in the production of cosmetics and medicines. Although important, the action is not enough to curb the advance of deforestation due to its limited and moderate impact.

Given the practices mentioned, such as soybean expansion, cattle ranching, increased demand for charcoal and the use of slash-and-burn techniques, the native forest areas of the RAAI and RAAII farms are under constant threat. Even with the implementation by the owner before the REDD+ Project of monitoring measures and investment in a project with a surrounding rural community, the vastness of the areas makes it difficult to contain invasions and illegal deforestation and the impact on one community is not enough to sustainably develop the region and change the regional dynamics that lead to illegal deforestation. The growing demand for land for agriculture and livestock, combined with commercial pressure for products

[62] FAPESPA – FUNDAÇÃO AMAZÔNIA DE AMPARO A ESTUDOS E PESQUISA. Anuário estatístico do estado do Pará: Tabela de dados. Belém, 2024. Disponivel em: <https://fapespa.pa.gov.br/sistemas/anuario2024/>.

[63] SILVA Gláucia (2005), Sustainability or subordination: ways of life in floodplain communities at the mouth of the Amazon. In: LIMA Deborah (Org). Socio-environmental diversity in the floodplains of the Amazon and Solimões rivers: perspectives for the development of sustainability. Ministry of the Environment/Brazilian Institute of the Environment and Natural Resources/Project for the Management of Natural Resources in the Floodplains Pilot Program for the Protection of Tropical Forests in Brazil, Manaus, 2005. Available at: <https://docplayer.com.br/184334771-Diversidade-socioambientalnas-varzeas-dos-rios-amazonas-e-solimoes-perspectivas-para-o-desenvolvimento-da-sustentabilidade.html> . Accessed on: 03/22/2023.

[64] SALOMÃO, C.S.C., Stabile, M.C.C., Souza, L., Alencar, A., Castro, I., Guyot, C., Moutinho, P. Amazônia em Chamas - desmatamento, fogo e pecuária em terras púbicas: nota técnica nº 8. Brasília, DF: Instituto de Pesquisa Ambiental da Amazônia, 2021. Disponível em: <https://ipam.org.br/wp-content/uploads/2022/05/Amazo%CC%82nia-em-Chamas-8-pecua%CC%81ria-pt.pdf>.

such as charcoal, increases the risk of environmental degradation, compromising forest integrity and jeopardizing local conservation efforts.

In this scenario, deforestation will advance as predicted and demonstrated in Section 3.1.4, totaling 1,294 hectares of deforestation in the project area in the first fixed baseline period. In this scenario, the actions of the main drivers of deforestation will not be impeded, communities will not have the economic strength to produce sustainably and will not have the leadership to curb invasions and land grabbing, thus maintaining the increasing deforestation of the remaining forests in the area, mainly in Permanent Preservation Areas.

- **(II) Improving surveillance actions not registered as an AFOLU VCS Project**

This scenario represents an increase in investment in property surveillance to improve inspection, by hiring a new security company that improves information management and control, in addition to maintaining sustainability actions and good practices already carried out by the owner within all the relevant regulations, norms, standards and laws, without additional investment in forest conservation, communities and biodiversity.

In this scenario, the only change is an increase in investment in property surveillance, aimed solely at containing invasions and illegal activities on the farms, with a 275% increase in cost compared to scenario (I). However, in this scenario there is no implementation of activities beneficial to communities and biodiversity, there is no investment in the surrounding region for sustainable development, due to the increase in cost it would require and the lack of revenue for this, not reaching the regional dynamics that leads to deforestation through the action of agents, vectors and underlying causes of deforestation.

Despite significant initiatives and actions, the scope of these activities is limited by the lack of resources available to expand their impact. Although the proponent involves a community in the region in social projects, the number of people benefiting is still limited, which prevents a broader influence in the areas of the region, failing to have a significant impact on combating deforestation and leaving the forest areas of the property vulnerable to hunting and illegal deforestation. Therefore, in this scenario, the properties remain vulnerable, as the proposed activities are not comprehensive enough to contain all agents and drivers of deforestation.

- **(III) Scenario with REDD+ Project activities not registered with the VCS**

The scenario with REDD+ Project activities in Caiarara aims to preserve the native forests of the RAAI and RAAII farms and promote a positive impact on the Climate, Community, and Biodiversity scopes through activities planned for the short, medium, and long term, covering the 40 years of the project.

To protect the forests, asset surveillance is reinforced by hiring a new security company that improves information management and control to prevent hunting and illegal deforestation by third parties, in addition to acting more quickly in cases of fires or similar events, as in scenario II. But, beyond that, there is the addition of satellite monitoring of the two farms, in which it is possible to verify the pressure of deforestation in the region and monitor hot spots, if any, in the project area.

These two activities also have an impact on the preservation of the region's fauna, as the habitat of species is preserved and the possibility of hunting is reduced. In addition, the biodiversity of fauna and flora will also be monitored, carried out through frequent monitoring campaigns. Existing species will be monitored and the results compared to assess population growth and movement in the project area, and threatened species will be monitored more frequently to ensure their biological success and prevent population decline.

This scenario also includes a series of social actions aimed at strengthening local governance and the sustainable development of communities around the RAAI and RAAII farms. Among the planned initiatives are training activities that seek to develop value chains for local products, in order to promote the generation of alternative income through sustainable activities; and activities to improve the well-being of stakeholders, including the promotion of education and technical training in accordance with community demand.

These efforts also have a specific focus on strengthening historically marginalized groups, recognizing their crucial role in building a sustainable future. Training and leadership opportunities are offered to empower young, women and historically vulnerable groups, encouraging their active participation in local governance and community development initiatives. These actions not only promote gender equality and the empowerment of historically marginalized groups but also strengthen the resilience of communities in the face of environmental and socioeconomic challenges.

In addition to the surrounding communities, farm workers will also benefit from the education and training activities promoted by the Caiarara REDD+ Project. These initiatives aim not only to improve the technical skills and environmental awareness of employees, but also to improve their well-being in their daily work.

With the activities involving the three scopes, the preservation of the properties' native forest is guaranteed, and biodiversity is protected against hunting and illegal deforestation, since the main vectors and agents of deforestation are contained and their impact reduced through alternative income and awareness-raising activities. In addition, species of fauna and flora are protected from hunting and illegal logging. However, in this scenario, these activities are not economically viable, as there is no specific income for forest conservation and the socio-economic development of the surrounding communities. The ideal would be to adopt the measures proposed in this scenario from the revenue from the sale of carbon credits registered with the VCS.

<u>Sub-step 1b. – Consistency of credible land use scenarios with applicable laws and regulations</u>

Scenario (III) is consistent with the laws and regulations applicable to REED+ activities. Section 2.5.1 National and Local Laws provides details on this topic. The practices in scenarios (I) and (II), however, are not in accordance with applicable laws and regulations, as they include illegal deforestation and hunting by third parties, which is widespread in the Project Area due to the lack of systematic application of laws and regulations.

Considering Federal Law 12.651/2012 – Native Vegetation Protection Law, also known as the Forest Code, and Normative Instruction No. 02/2015, on the Authorization for Vegetation Suppression (ASV), which discuss the suppression of native vegetation in rural areas or the cutting of trees in urban areas, it is understood that any suppression is conditional on the issuance of an ASV by the state or federal government agency. Analyzing the Annual Report on Deforestation in Brazil conducted by MapBiomas

(2023)[65], referring to the year 2022, it is noteworthy that only 5% of the areas deforested in the Amazon biome in 2022 had ASV and that this license does not mean that the entire deforested area on the property is within the authorization, which may further reduce the percentage of regularized deforestation.

The document, states that only 7.6% of deforestation alerts for the state of Pará, in the period from 2019 to 2022, underwent some type of authorization or inspection by federal and state agencies, highlighting the difficulty of environmental institutions in controlling and inspecting illegal clearing.

The MapBiomas Annual Deforestation Report (2025) [66] shows that in the state of Pará from 2019 to 2024, around 22,000 hectares of areas with deforestation alerts crossed state or federal authorisations, i.e. only 1.2% of the areas deforested in the state in this period overlapped with a Vegetation Suppression Authorisation. In view of the above, as a result of sub-step 1b, it was shown that only scenario III complies with mandatory laws and regulations, and scenarios I and II do not comply with mandatory laws and regulations due to the systematic lack of enforcement of laws and regulations in the Project region. Therefore, as a result of sub-step 1b: all the scenarios listed, scenarios 2 and 3 are plausible alternative land use scenarios for this VCS AFOLU project activity.

<u>Sub-step 1c. Selection of the baseline scenario</u>

As described in Section 3.1.4 Baseline Scenario: based on data collected in the field from local agents and information obtained through secondary data, it is possible to find conclusive evidence that explains the relationships between agents, vectors, underlying causes, and deforestation pressure in the Project Zone. Thus, it is concluded that the relationships between social variations (social vulnerability of communities), the persistence of slash-and-burn agriculture, the constant demand for timber and hunting, combined with the inefficiency of public authorities in enforcing illegal activities, contribute to the deforestation scenario observed in the region.

Considering this evidence, the future trend is for the current scenario to continue, with the agents, causes, and drivers of deforestation growing steadily (scenario I – Continuation of land use prior to the Project), with native forest areas on the properties threatened and at constant risk of degradation, deforestation, and biodiversity loss, more details in Section 3.1.4.

<u>Step 2. Investment analysis</u>

<u>Sub-step 2a. Determining the appropriate analysis method</u>

Evaluations and comparisons were made of the results of the scenarios described in sub-step 1a), where scenario (i) represents the continuation of land use prior to the project, scenario (ii) represents the improvement of patrimonial surveillance without additional investments in communities and biodiversity, and scenario (iii) represents additional investments in forest conservation, community development and

[65] MapBiomas, 2023. Annual Deforestation Report 2022 - São Paulo, Brazil. - 125 pages.

[66] RAD2024: Relatório Anual do Desmatamento no Brasil 2024 - São Paulo, Brasil - MapBiomas, 2025 - 209 páginas. DOI 10.1088/1748-9326/ac5193. Disponivel em: <https://alerta.mapbiomas.org/wp-content/uploads/sites/17/2025/05/RAD2024_15.05.pdf>.

biodiversity monitoring (REDD+ project activities) within all the relevant regulations, norms, standards and legislation.

The purpose of comparing the scenarios was to assess whether, with the registration of the VCS AFOLU project and the inclusion of complementary and necessary activities to prevent deforestation and increase the social and environmental benefit, what the financial impact would be for the project proponent and what the economic viability would be of adding these complementary expenses, given that the current activities are proving to be insufficient to contain deforestation and social development in the region.

Sub-step 2b. Option I. Simple cost analysis

Option I was chosen for evaluation because in the project area there are no other financial and economic benefits from any other economic activity that are not or will come from the VCS, so in order to be in line with VT0001, the option chosen was simple cost analysis, because there are no other economic resources other than those related to the VCS.

The following table lists the activities that make up each of the scenarios: scenario I with no complementary investments, scenario II with investments only in surveillance, and scenario III with complementary investments in environmental conservation and local social development.

Investments	Scenario I	Scenario II	Scenario III
Property surveillance	X		
Social project (Ybá)	X	X	X
Improved property surveillance		X	X
Initial diagnosis			X
Community engagement and development			X
Biodiversity protection and monitoring			X

Based on the investments described above, in scenario I the landowner would continue to invest in surveillance and the social project in the Project Area; in scenario II, with the improvement in surveillance, the increase in cost would be 115%; and in scenario III, with the aim of containing deforestation, protecting biodiversity, developing the region sustainably and adapting to climate change, investments in social, biodiversity and surveillance activities, the increase in cost would be 69%, compared to scenario II. Therefore, scenario III represents a percentage increase of 264% in annual investment compared to the conditions prior to the project (scenario I), and the activity does not produce financial benefits as there is no economic activity in the project area in this scenario.

Therefore, without the financial benefits of registering the project in the VCS from the sale of carbon credits, there is no economic availability to carry out the complementary activities of conservation, protection and environmental and social development of the surrounding community in scenario III.

Step 3 – Barrier Analysis

The VCS "VT0001 – Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities" requires investment analysis (Step 2) or Barrier Analysis (Step 3). In this case, Investment Analysis, described in Step 2, was chosen.

Step 4 – Analysis of common practice

The fourth stage of the additionality analysis considers the assessment of areas similar to the model proposed by the REDD+ project to identify common practices. For this verification, the geographical delimitation of the Caiarara REDD+ Project Zone was considered.

The similarity analysis applied had basic premises such as property size, land use category, regulatory structure, environmental characteristics and the context of action of the agents and vectors of deforestation. In order to identify possible areas similar to the REDD+ project, a survey was carried out of all the properties within the Project Zone, analyzing the size of the area and evaluating the land ownership situation of the property, considering the regularized areas according to the data available on SIGEF[67] and/or SNCI[68] (Figures 20 e 21).

[67] INCRA - Instituto Nacional de Colonização e Reforma Agrária. Sistema de Gestão Fundiária (SIGEF). Disponível em: https://sigef.incra.gov.br/.

[68] INCRA - Instituto Nacional de Colonização e Reforma Agrária. Sistema Nacional de Certificação de Imóveis Rurais (SNCI). Disponivel em: https://certificacao.incra.gov.br/Certifica/.

Figure 20. Map of private properties and Conservation Units in the Caiarara REDD+ Project Zone and land use categories.

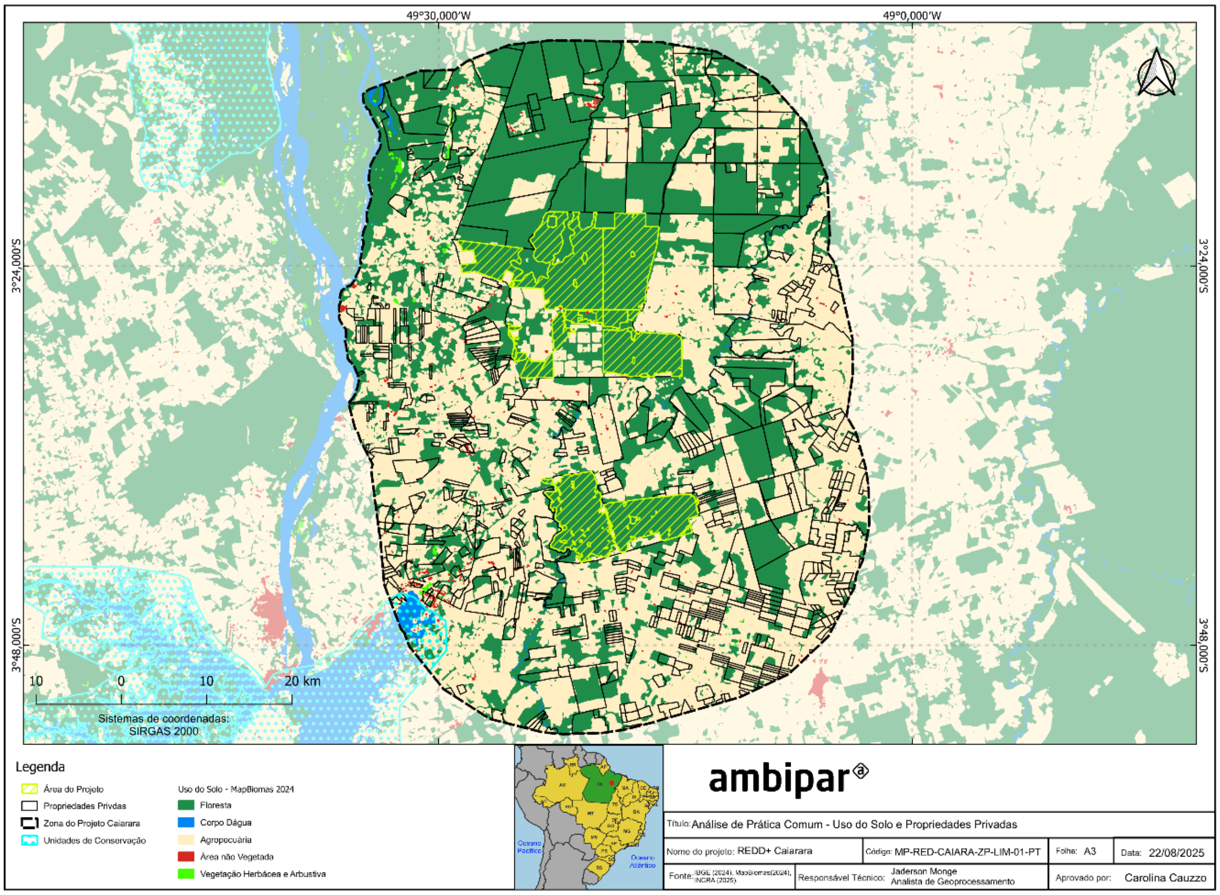
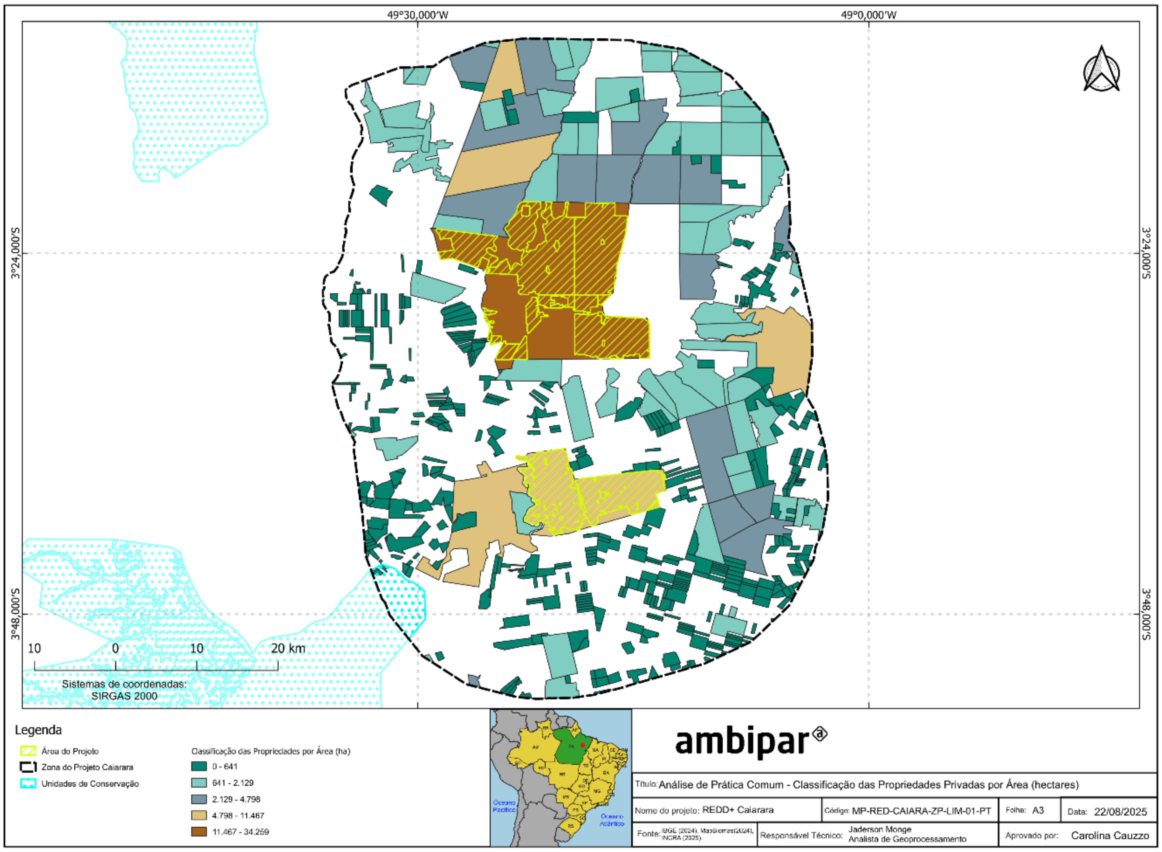


Figure 21. Map of private properties in the Project Zone by size class.



When analyzing and comparing land use with the size of private properties, it can be seen that the majority of properties in the surrounding area have the Agriculture and Livestock class as land use and occupation, and those that have the Forest class do not compare in size with the Project Area. Therefore, no other property in the Project Area is similar in size and forest land use.

A similar activity was identified, the Ybá Project, carried out by the proponent and owner of the farms with the support of partners in the Project Area and in the Vila Mamorana community. This initiative received a total investment of U$300,000.00 from May 2021 to May 2025, which covered demographic and bioactive mapping of commercial interest, training of the association chosen to harvest andiroba seeds in Dow's forests and the installation of a meliponary in the community area[69].

Using the VT0001 guidance for analyzing common practice, it was identified that this initiative takes place in a relevant geographical area, as it is located in the Project Area and in the area of the Mamorana community, a REDD+ Project participant. However, it does not take place on a comparable scale, as it does not include activities focused on monitoring and conserving biodiversity, improving the well-being of surrounding communities, improving property security and monitoring deforestation and hotspots, as well as covering only one community group, presenting a moderate, restricted and smaller impact than the

[69] https://www.portalcop30.org/case/projeto-yba-conservacao-que-transforma-dow/.

proposed VCS AFOLU Project activity. In addition, the Ybá Project's investment comes from the Dow Foundation's Business Impact Fund, a different fund from the REDD+ Project, which comes from the company's own resources, and Ybá does not seek to neutralize the company's carbon emissions, demonstrating the different circumstances in which the similar activity was implemented[70].

Conclusion

It is concluded that the similar activity identified presents essential distinctions in relation to the VCS AFOLU Project model proposed, and the private areas in the region do not compare to the size and land use of the Project Area. Thus, the areas analysed do not represent the "trend" scenario of the Project Area and, as a conclusion of this similarity analysis, it can be seen that there is no common practice for the REDD+ Project in the region analysed. Thus, the proposed VCS AFOLU project activity is not the baseline scenario and is therefore additional.

3.1.6 Methodology deviations (VCS, 3.20)

Characterization of the deforestation agents mobility

According to module VMD0055 v1.1 Estimation of Emission Reductions From Avoiding Unplanned Deforestation, item 5.2.1 Identification and Characterization of Deforestation Agents and item 5.3.4.4 Emissions from Activity Shifting Due to Displacement of Unplanned Deforestation to Areas Outside the UDef LB, as part of the analysis of deforestation agents (Section 3.1.5.1), it should be assessed how each group of agents is considered: geographically restricted or geographically mobile. The initial study of the region was carried out between December 2023 and January 2024, that is, before the launch of VMD0055 v1.1 (October 21, 2024). Therefore, this point was not included in the scope of the local survey.

The data necessary to analyze the proportion of agents that migrated to the project region in the last five years were not collected, making it impossible to perform such characterization. The deviation refers only to measurement procedures and will have no negative impact on the conservation of the quantification of GHG emission reductions, as the Estimated Proportion of Migratory Land Cover Transition Agents in the Baseline (PROPmig) was given a conservative value of 1.0 (Section 3.2.4), as directed by the methodology itself.

3.2 Quantification of estimated GHG emission reductions and removals

3.2.1 Initial Baseline Validity Period (BVP)

The jurisdiction in which the project is located is the State of Pará. The Baseline Validity Period (BVP) for this jurisdiction extends from 2019 to 2024. Considering that the Caiarara REDD+ Project has a start date in 2024, the project opts to extend its baseline for an additional two years, covering the period from 2024 to 2026. This decision allows the project to enter a new cycle of the jurisdictional baseline validity,

[70] https://jatobapr.com.br/archives/cazes/2021/9a65aa93-6614-4c23-8569-316690a81378/files/oH1Hd3gtUEPVjVukILn3loT4nPvg1sOiCjN3mTgr.pdf

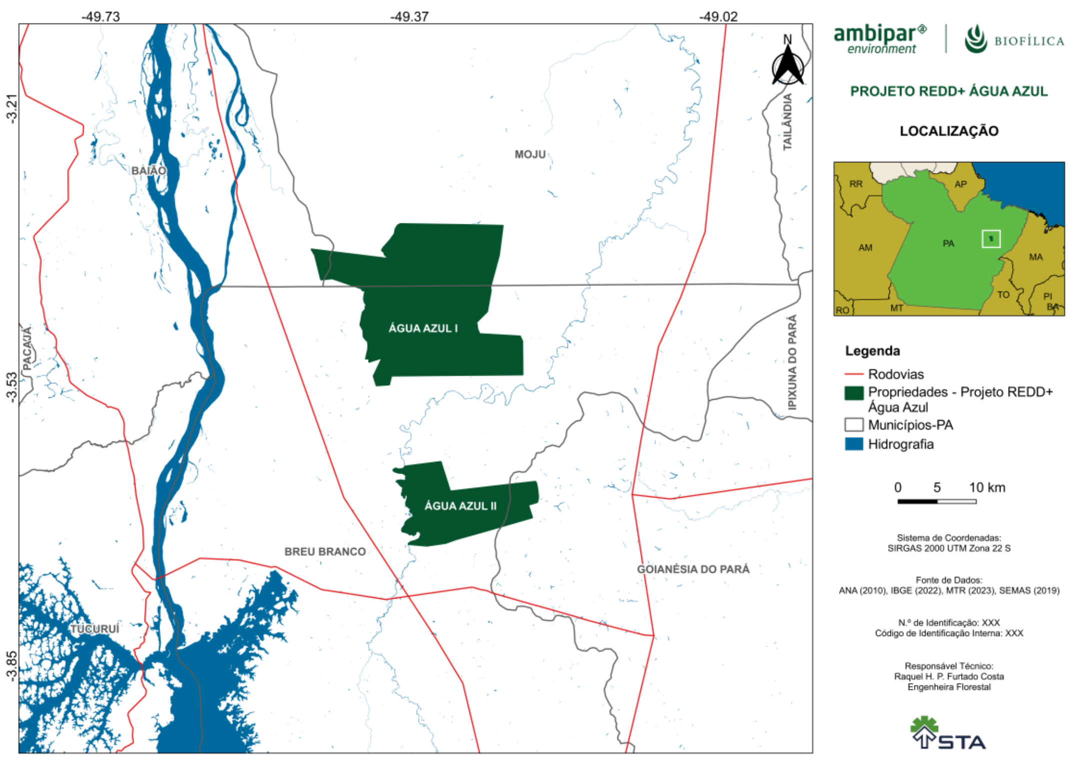
maintaining consistency with the technical and temporal parameters outlined in Section 5.3.1 of module VMD0055.

3.2.2 Baseline emissions (VCS, 3.15)

3.2.2.1 Forest Stratification Map

The area of the Caiarara REDD+ Project is located on the Reflorestamento Água Azul I and Reflorestamento Água Azul II farms belonging to the municipalities of Breu Branco and Moju, State of Pará. Although the farms are spatially adjacent, the areas belonging to the Project are separate, constituting two forest blocks, the RAA I Farm block being the largest of them (Figure 22). The RAA I Farm is located on the border of the municipalities of Moju and Breu Branco, while the RAA II Farm is fully inserted in Breu Branco.

Figure 22: Location of the Caiarara REDD+ Project area.



Based on the shapefiles of the Continuous Database of Vegetation of IBGE[71] in the scale 1:250.000, a classification of the vegetation cover on the farms was carried out. The territory analyzed presents a

71 IBGE – Instituto Brasileiro de Geografia e Estatística. Banco de Dados e Informações Ambientais (BDiA): Mapeamento de Recursos Naturais (MRN): escala 1:250.000 – versão 2023: nota metodológica. Rio de Janeiro: IBGE, Coordenação de Meio Ambiente, 2023. 64 p. Disponível em: . https://biblioteca.ibge.gov.br/index.php/biblioteca-catalogo?view=detalhes&id=2102042. Acessado em 19/08/2025.

combination of planted and native vegetation. Planted Forest is present in part of the area, while native vegetation types predominate, covering the largest extension of the project.

The main vegetation types are:

Dense Ombrophilous Forest, which occupies approximately 80.8% of the total area of the properties.

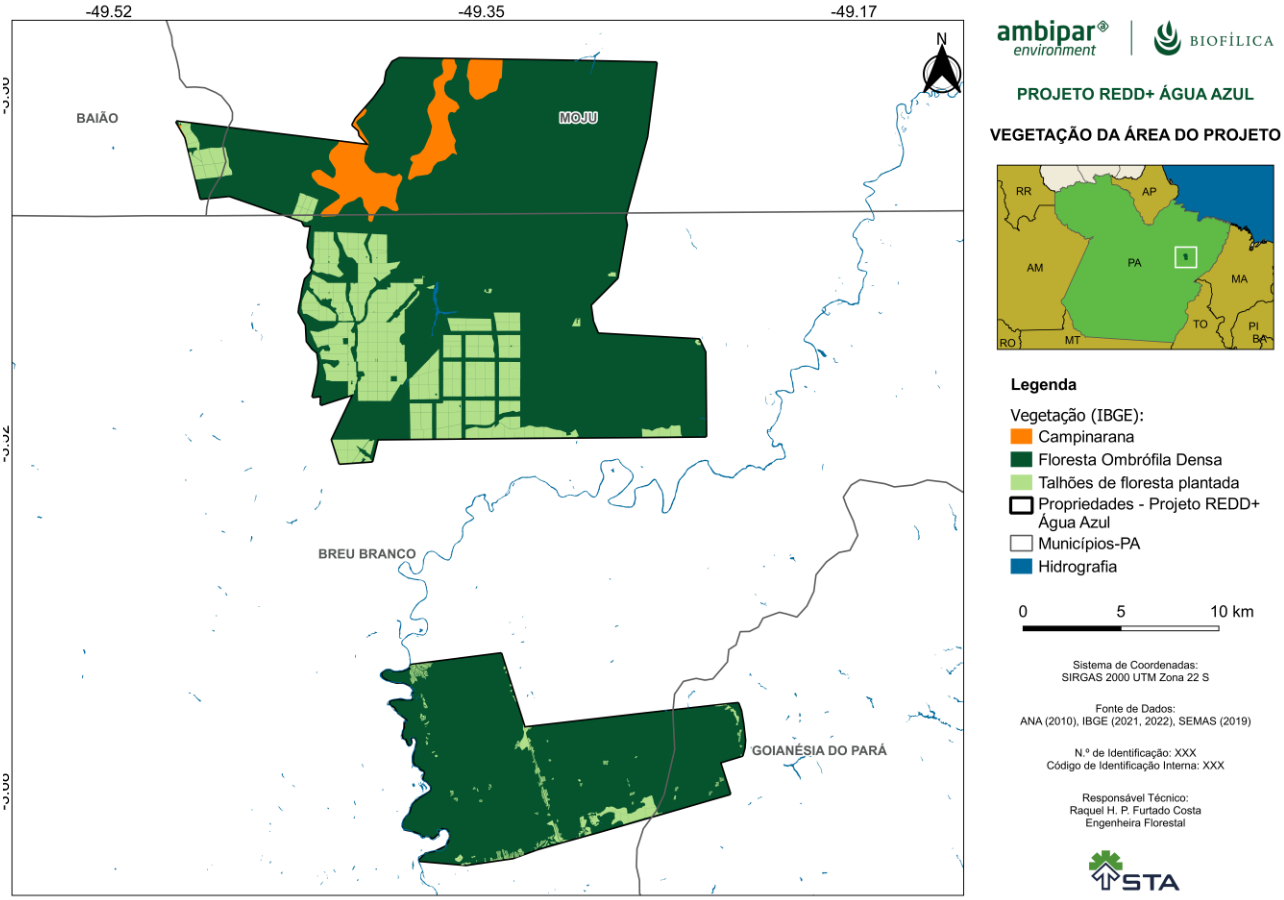
Campinarana, present in about 4.2% of the area of the properties.

This information is detailed in Table 8 and Figure 23.

Table 8: Type of existing native vegetation area of the RAA I and RAA II.

Forest type	Farm Reflorestamento Água Azul I		Farm Reflorestamento Água Azul II		Total	
	Area (ha)	%	Area (ha)	%	Area (ha)	%
Ombrophile dense forest	26,050.40	76.04	10941.83	95.43	36992.24	80.90
Campinarana	1921.39	5.61	0.00	0.00	1921,39	4.20

Figure 23: Vegetation cover of the properties.



<u>Project Area Stratification</u>

Given the above, and that forest plantations and Campinaranas were removed from the REDD+ Project, only one forest stratum was considered for the project area: **dense ombrophile forest.**

3.2.2.2 Estimation of Annual Unplanned Deforestation in the Project Boundaries

In compliance with the requirements established in module VMD0055, v1.1, a formal request for the allocation of the activity data of unplanned deforestation baseline (AD) was submitted through the AD Baseline Allocation Request Form. Following submission, the projected parameters of annual unplanned deforestation area were received for both the Project Area (ADPA-UDef) and the Baseline (ADLB-UDef).

For the determination of baseline emissions associated with unplanned deforestation, the following inputs were used, as specified in the AD Baseline Allocation Report:

- Activity data allocated to the Project Area (UDef PA) and the Leakage Belt (UDef LB);
- Forest stratification map;
- Risk map;

- Boundaries of the UDef PA and UDef LB.

An analysis was carried out to identify and exclude potential overlaps with other Project Areas (PAs) and Baselines (LBs) from VCS AFOLU projects. In cases where overlapping areas were excluded, the forest stratification map was duly adjusted to remove strata existing exclusively in those regions, thereby ensuring the integrity of AD allocation among the remaining strata.

In addition, the boundary of the Leakage Belt was refined in order to identify and, if necessary, exclude non-forest areas or those falling under the categories established in Table 11 of Appendix 1 of VMD0055. Tables 9 and Table 10 present the AD values for the Project Area and the Leakage Belt, respectively.

Table 9. AD values allocated to the Project Area (UDef PA), in hectares

		$AD_{BSL,PA-UDEF,i,t}$ (ha)					
		Stratum i					
BVP	t	Forest					
1	1	216					
1	2	216					
1	3	216					

Table 10: AD values allocated to the Leakage Belt (UDef LB), in hectares

		$AD_{BSL,LB-UDEF,i,t}$ (ha)					
		Stratum ID					
BVP	t	Forest					
1	1	480					
1	2	480					
1	3	480					

3.2.2.3 Estimation of Emissions from Carbon Stocks Changes

Step 1: Estimation of carbon stocks by forest stratum

The estimates of carbon stocks of the different reservoirs were estimated from data from field forest inventories and expansion factors obtained from the literature. The methodology used to carry out the inventory and the calculation of all the estimates presented were based on the VM0048 methodology and its respective modules. The list of modules and tools used in the inventory, as well as their applicability are listed in the tables below.

Table 11: Identification of the modules and tools used in the construction of the Carbon Stock Diagnosis and Monitoring Plan of the Caiarara REDD+ Project.

Module	Title: original Methodology and reference weblink
Methodology	
VM0048	Title: VM0048 -Reducing Emissions From Deforestation And Forest Degradation versão 1.0. https://verra.org/methodologies/vm0048-reducing-emissions-from-deforestation-and-forest-degradation-v1-0/
Modules to measure emissions at baseline	
AUDef	Title: VMD0055 Estimation of Emissions Reductions from Avoiding Unplanned Deforestation version 1.0 https://verra.org/methodologies/vmd0055-estimation-of-emission-reductions-from-avoiding-unplanned-deforestation-v1-0/
Modules to measure the carbon stock of reservoirs	
CP-AB	Title: VMD0001 Estimation of carbon stocks in the above and belowground biomass in live tree and non-tree pools (CP_AB). https://verra.org/methodologies/vmd0001-estimation-of-carbon-stocks-in-the-above-and-belowground-biomass-in-live-tree-and-non-tree-pools-cp-ab-v1-1/
CP-D	**Title:** VMD0002 Estimation of carbon stocks in the dead wood pool (CP-D) https://verra.org/methodologies/vmd0002-estimation-of-carbon-stocks-in-the-dead-wood-pool-cp-d-v1-0/
Modules and Tools for Monitoring	
LK-ME	Title: VMD0011 Estimation of emissions from market-effects (LK-ME), v1.2 https://verra.org/methodologies/vmd0011-estimation-of-emissions-from-market-effects-lk-me-v1-1/

E-BPB	**Title:** VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E–BPB), v1.3 https://verra.org/methodologies/vmd0013-estimation-of-greenhouse-gas-emissions-from-biomass-and-peat-burning-e-bpb-v1-2/
Miscellaneous Modules and CDM Tools	
UDef-AT	Title: VT0007 Unplanned Deforestation Allocation Tool (UDef-AT), v1.0 https://verra.org/methodologies/vt0007-unplanned-deforestation-allocation-udef-a-v1-0/
X-STR	**Title:** VMD0016 Methods for stratification of the project area (X-STR). https://verra.org/methodologies/vmd0016-methods-for-stratification-of-the-project-area-x-str-v1-2/
VT0001	**Title:** VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities, v3.0 https://verra.org/methodologies/vt0001-tool-for-the-demonstration-and-assessment-of-additionality-in-vcs-agriculture-forestry-and-other-land-use-afolu-project-activities-v3-0/
T-BAR	**Title:** VCS AFOLU Non-Permanence Risk Tool (T-BAR) https://verra.org/wp-content/uploads/AFOLU_Non-Permanence_Risk-Tool_v4.0.pdf

Table 12: Applicability conditions for the methodology, modules and tools, as well as the justification for use in the Caiarara REDD+ Project

Applicability condition	Justification
VM0048 - REDD+ Reducing Emissions From Deforestation And Forest Degradation	
● The project area must qualify as forest for at least 10 years prior to the project start date.	● The project area meets this condition, with full forest cover demonstrated in the history of the last 10 years.
● The project area may include forested wetlands (such as floodable forests) as long as they do not grow in peat.	● As shown below, the project area is covered by dense ombrophilous forest; There are no organic soils (or peatlands) in the project area.

● Deforestation and forest degradation at the baseline, in the project area, must fall into one or more of the following categories: Unplanned deforestation (VCS AUDD category); Planned deforestation (VCS APD category); Degradation through the extraction of wood for fuel (production of firewood and charcoal) (category VCS AUDD).	● Deforestation at baseline, in the project area, falls into the category of unplanned deforestation; In addition to the history of invasions that have occurred on farms, there is still the possibility of population advance in the surroundings and in the project area.
● Step 2: Estimation of emissions from changes in carbon stocks	● The baseline will be renewed every 10 years starting in the year 2023.
● All land areas registered under the CDM or any other carbon trading scheme (both voluntary and compliance-driven) must be reported in a transparent manner and excluded from the project area.	● This is the area's first record in a carbon trading program; the area will also be registered with the CCB.
● If the land is not fully converted to an alternative use, and its natural regeneration is allowed (i.e. the land is temporarily depleted), this framework should not be used.	● The deforestation of the forest at baseline is followed by the establishment mainly of pastures in the surrounding area, and planted forests, on the farms in which the two areas that are part of the Project are located, which prevents the growth of the forest.
● Activities to prevent leaks should not include: Farmland that is flooded to increase production (e.g., rice in wetlands); or, intensify livestock production through the use of "feedlots".	● Activities to prevent leaks do not include flooding farmland or creating feedlots.
VMD0055 Estimation of Emissions Reductions from Avoiding Unplanned Deforestation	
● The land use transition in the baseline scenario is from forest land to non-forest land, meeting the definition of unplanned deforestation. ● The project involves activities aimed at preventing unplanned deforestation. ● Deforestation agents in the baseline scenario clear land for tree harvesting, settlements, wheels, unauthorized expansion of roads or other infrastructure, crop production, livestock, or aquaculture.	● There is a history of wood removal before the implementation of the project. So, one of the goals is to avoid unplanned deforestation.

VMD0001 Estimation of carbon stocks in the above and belowground biomass in live tree and non-tree pools (CP-AB)	
● The Module allows ex-ante estimation of carbon stocks in above- and below-ground tree and non-tree biomass in the baseline scenario (pre- and post-deforestation stocks) as well as in the case of the project, and for ex-post estimation of changes in the carbon stocks of above- and below-ground tree biomass in the case of the project. ● The Module applies to all forest types and age classes. The inclusion of the above-ground tree biomass reservoir as part of the project is mandatory.● A biomass aérea não-arbórea deve ser included as part of the project boundary if the following applicability criteria are met: ● Non-arboreal aerial biomass stocks are higher at baseline than in the project scenario; and ● Non-arboreal aerial biomass is considered significant (using the T-SIG Module).	● This Module applies to all forest types and age classes. ● The inclusion of the above-ground tree biomass reservoir as part of the project boundaries is mandatory according to the REDD-MF Module; thus, the biomass of trees with DBH > 10cm will be considered. ● Non-arboreal above-ground biomass will be considered from the biomass of palm trees (with DBH > 10cm); as demonstrated, this biomass is significant, according to the T-GIS tool. ● Subterranean biomass (tree and non-tree) will also be considered in the carbon stock estimates, which is optional according to the requirements of the Methodology, as demonstrated this biomass is significant, according to the T-GIS tool.
VMD0002 Estimation of carbon stocks in the dead wood pool (CP-D)	
● This Module is applicable to all forest types and age classes. ● This Module is applicable if the deadwood reservoir is included as part of the project.	● The Module is applicable for all forest types and age classes. ● This reservoir is included when it represents a significant component of forest biomass.
VMD0011 Estimation of emissions from market-effects (LK-ME)	
● This Module is applicable to calculate the leakage of the market effects on logging so that logging is reduced substantially and permanently. When project activities result in reductions in timber harvesting, it is likely that	● Before the implementation of the project, there was logging in part of the project area. There is also a history of illegal logging, in the project area and in the surrounding area. That is why

production can be shifted to other areas of the country to compensate for the reduction.	monitoring in the area of the leak is mandatory.
VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E– BPB)	
● This Module provides a step-by-step approach to estimating greenhouse gas emissions from biomass and peat burning. ● This Module is applicable to REDD project activities with emissions from biomass burning and REDD-WRC project activities with emissions from biomass and/or peat burning. This Module is also applicable to RWE and ARR-RWE project activities with emissions from peat burning.	● In the reference scenario, fire is used to clear the land, resulting in greenhouse gas emissions. ● When used in the baseline, accounting should occur both in the baseline and in the project scenario and both within the project area and in the leak belt. ● When post-fire incidents occur in areas that coincide with deforested or degraded areas at baseline, the Module should be used to account for greenhouse gas emissions.
VT0007 Unplanned Deforestation Allocation Tool (UDef-AT)	
● The project is stand-alone and meets the conditions of applicability for projects that aim to prevent unplanned deforestation as defined in VMD0055 and is seeking the allocation of jurisdictional data of unplanned deforestation activities, or ● A higher-tier jurisdiction is seeking to allocate its Forest Reference Emisson Level (FREL) to lower-tier projects or jurisdictions with the aim of avoiding unplanned deforestation	● The project. REDD+ Caiarara falls within the scope of the VMD0055.
VMD0016 Methods for stratification of the project area (X-STR)	
● This Module provides guidance on how to stratify the project area into discrete and relatively homogeneous units to improve the accuracy and precision of carbon stock and carbon stock change estimates.	● It was verified that there were no discrete clusters of biomass that differed greatly from the average, only one stratum "Open ombrophilous forests" was considered within the project area. ● The post-deforestation scenario (conversion) will not be stratified; instead, the average carbon stock values for land uses after deforestation were applied,

	according to the guidelines of Modules BL-UP and BL-PL.
VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities	
● AFOLU activities identical or similar to the proposed on-site project activity within the boundaries of the proposed project, executed with or without registration as a VCS AFOLU project, will not result in the violation of any applicable laws, even if the law is not enforced. ● The use of this tool to determine additionality requires the baseline Methodology to provide a stepwise approach that justifies the determination of the most plausible baseline scenario. Project proponents proposing new baseline Methodologies should ensure consistency between the determination of a baseline scenario and the determination of the additionality of a project activity.	● To define additionality to the project, which complies with all relevant laws, statutes and regulatory frameworks. ● The approach to defining the baseline scenario using this tool will not be described in this report, but should be described in the PD.
VCS AFOLU Non-Permanence Risk Tool (T-BAR)	
●This tool provides the procedures for performing the non-permanence risk analysis and determining the buffer required for Agriculture, Forestry and Other Land Use (AFOLU) projects. ●The tool sets out the requirements for project proponents and validation/verification bodies to assess risk and determine the appropriate risk classification.	● The Module is mandatory when using the Methodology REDD-MF. ● The approach to defining risks using this tool will not be described in this report but should be described in the DP.

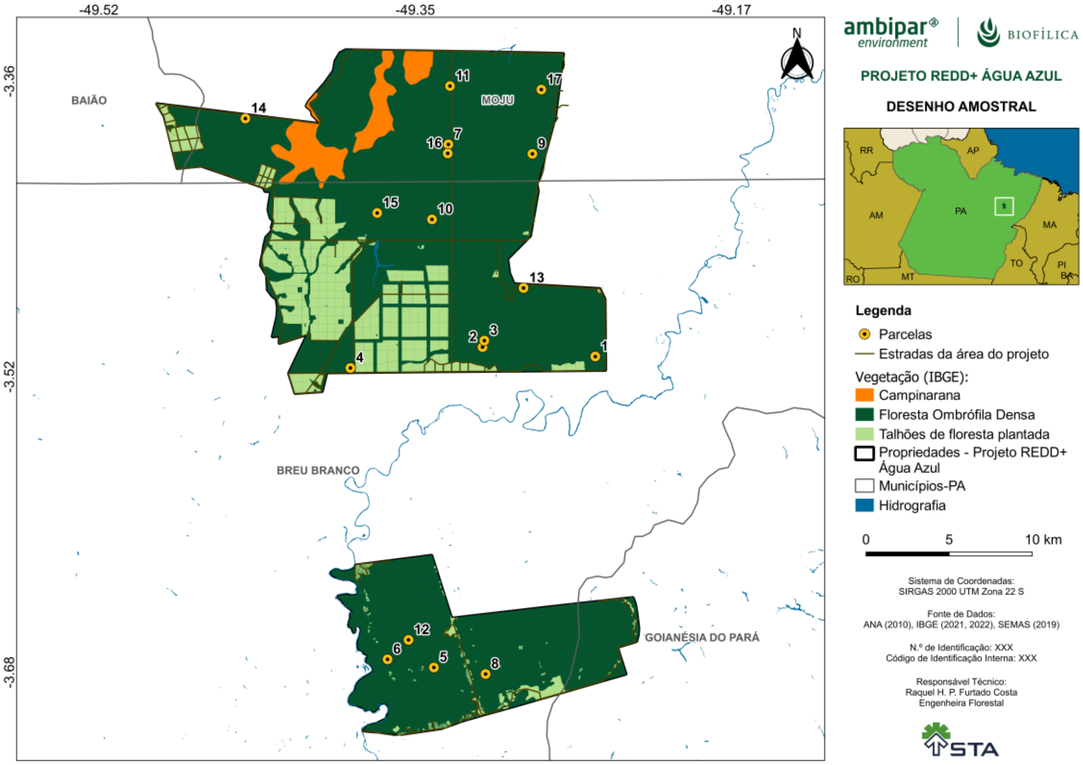
The complete methodology is described in the report "Carbon Stock - Caiarara's REDD+ Project", made available for the audit. Here the main points and results of the inventory will be presented.

<u>Sample design and response design</u>

A total of 17 sample plots of fixed area were randomly allocated in the project area for inventory. All plots have an area of 1 hectare, with dimensions of 100m x 100m each. The number of samples was defined based on a pre-assessment of the variability of above-ground carbon stocks in dense ombrophilous forests, and with the objective of estimating the average stocks with a confidence level of 95% and an admissible error of a maximum of 10%.

The location of the plots in the field is shown in the following figure:

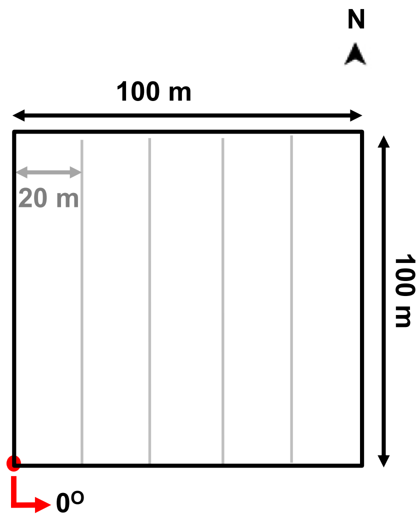
Figure 24: Map of the location of the sampling units (or plots) of the Caiarara REDD+ Project.



<u>Installation of Forest Inventory Plots</u>

Each plot has dimensions 100 m x 100 m (1 hectare); Square and large plots were chosen due to the lower installation cost and to facilitate field operation. Also to facilitate the operationalization in the field and later the audit, each plot was divided into five 20 m strips in the north-south direction (Figure 25). The installation of the lanes aims to assist in the control of the trees measured; Additionally, during the audit, the strips will help locate inventoried trees.

Figure 25: Model of the 1 ha plot (or sampling unit) indicating the position of the azimuth 0° and the position of the strips in the north-south direction.



Inventoried data and field measurement procedures

All trees with DBH ≥ 10 cm were inventoried. It is worth noting that VM0048 does not discuss the botanical identification of individuals; however, an identification based on popular name was carried out for better control in the field. Each individual tree had a circumference at a height of 1.30m (CAP) measured (in centimeters) with a steel tape measure, resistant to humidity and temperature variations, with graduation in millimeters/inches, with a precision class equal to II, which is the highest precision class for steel tapes, according to ABNT NBR 10123) (Figure 26). Later, the NAC was transformed into DBH (DBH = CAP/π). To facilitate the subsequent location of the trees in the field, all individuals received an aluminum plate (or labels) (Figure b) and there was a note of which strip the individual was located. This annotation can be seen in the spreadsheet of data collected in the field and made available to the VVB.

Figure 26: Image of the tape measure used to measure the circumference and/or diameter of the trunks of living trees.

Figure 27: Illustrative images of the diameter measurement (a) and aluminum plates (or labels) on trees (b).

In view of the existence of some individuals with bifurcations below the 1.30m line, the PAC of each branch was measured (only branches with DBH > 10cm) to calculate a single DBH for the tree; for this, an estimated DBH was calculated using the equation as suggested by Loetsch et al. (1973):

Estimated DBH = √(DBH12 + DBH22 + DBH32 + ... DAPn2)

Where, DAP1, DAP2, DAP3, ... DAPn represent the DAP of the branches.

The NAC was normally taken at 1.30 m above the ground (Figure a), however this is a reference value, because in some situations the taking of the NAC at exactly 1.30 m above the ground is prevented, as there are several variables that make it difficult, such as: shape of the trunk base, deformities at the measurement point, terrain topography, etc. In these situations, it is recommended that a new 'Measuring

Point' (POM) be established (Branthomme, 2009; Soares et al., 2011). The Figure illustrates the various possibilities that lead to the establishment of a new POM. Due to some of these situations, the DBH was estimated; In these cases, a note was made in the 'Observations' field of the field worksheet.

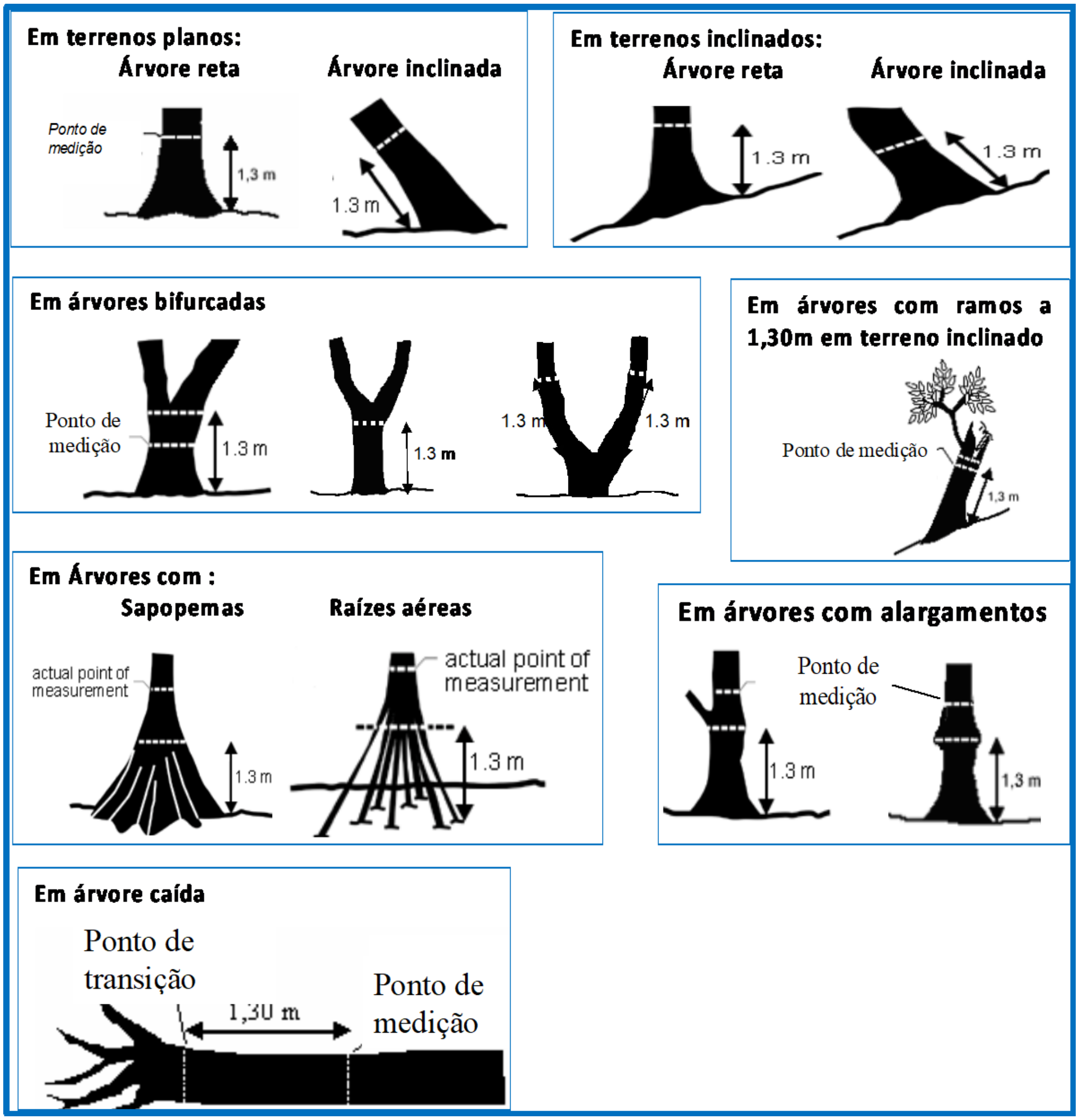


Figure 28: Position for measuring the CAP (Circumference at Chest Height) in the main situations that occur in the field. Source: Modified from Branthomme (2009).

It is worth mentioning that the field team received training before starting the measurements in the field, and special attention was given to the trees that had some impediment to measurement at a height of 1.30m. The images in Figure 11 show some moments of the training carried out in the project area, lasting one day, being the first day of fieldwork.

Figure 29: Training of the team to measure Circumference at Chest Height (PAC).

Measures for data evaluation and quality control (QA/QC)

As measures to ensure the quality of the data, in addition to the plating of all the trees inventoried, the primary data (taken in the field) were recorded in printed spreadsheets and then digitized; The printed spreadsheets were scanned (to be made available in the virtual environment). After digitizing the printed spreadsheets, there was an extensive data check; and before starting the analyses, all trees with annotations in the 'observation' field were inspected and treated according to annotation (for example, if the diameter was estimated, the same value noted in the CAP field was passed on to the DBH field). All data tables were organized as 'Supplementary Tables' and are available virtually, digitally, including those with primary data (these were scanned).

Choice of allometric equation for conversion of DBH to biomass

Seven allometric equations that convert DAP to biomass were evaluated. These equations were selected because they are potentially suitable for the region of the inventory area, and Equation 1 of Higuchi et al (1998) is broken down into two equations according to the DBH of the tree (Table 13). These equations were evaluated to be used in the Caiarara REDD+ Project. The evaluation was carried out through a linear regression analysis between the estimated biomass (y-axis) and the diameter of the trees (x-axis), using the data obtained from the inventory carried out in the Caiarara REDD+ project; The number of trees inventoried constitutes the replications.

Table 13: Allometric equations evaluated to be used in the area of the Caiarara REDD+ Project.

#	Equations
Equations evaluated on trees with DAP ≥ 10,0 cm	
1	Higuchi et al. (1998): For trees in which: 5 > DBH < 20 cm: AGBfresca = 0,6*exp (-1,754 + 2,665 $\log_e$(DBH)) For trees in which: DBH > 20 cm: AGBfresca = 0,6*exp (-0,151 + 2,17 $\log_e$(DBH))
2	Brown (1997): AGB = exp (-2,134 + (2,53 *ln(DBH)))
3	Brown (1997): AGB = 42,69 + (-12,8*DBH) + (1,242*DBH^2)
4	Carvalho et al. (1998): AGB = (1000*0,6) *exp (3,323 + 2,546 $\log_e$ (DBH/100))
5	Araujo et al. (1999): AGBFresca = 0,465*$DBH^{2,202}$/(1-M)
6	Araujo et al. (1999): AGB =0,6(4,06*$DBH^{1,76}$)
7	Chambers et al. (2001): AGB=Exp(0.37+(0.333* ln(DBH)+(0.933*ln(DBH^2))+(0.1222* ln(DBH^3))))

Notes:

In the equations of Higuchi et al (1998), the initial value of 0.6 corresponds to the proportion of dry mass, since the equation was elaborated for fresh mass. This proportion value was also applied to equations number 4, which was also constructed for fresh biomass;

The two equations of Higuchi et al (1998) were used in combination;

DBH is the diameter at the breast height, in centimeters.

As advised by Daniel et al. (2021), if the equation is adequate to the data in the area, as it was to the data from the original study, the results will also show high R2 values and low standard error of the estimate. Thus, the results of the regressions were analyzed, and the equations that presented the highest adjusted

R2, best fit of the residuals and low standard error of the estimate or SEE (the √Mean Standard Error) were considered adequate for use in the Caiarara REDD+ Project.

The equation of Araújo et al (1999), identified in the Table as equation number 6, was the one that presented the highest coefficient of determination and the lowest standard error of the estimate, which is why it was the equation chosen to be used in the Caiarara REDD+ Project.

Calculation of Live Biomass Above Ground

After choosing the equations to be used to estimate the biomass of the trees (Tree), the biomass of each tree was estimated and the sum of the biomass of the plot was established according to Module CP-AB:

$C_{\mathrm{AB}\,(i)} = \sum_{i=n}^{i} \int_i \ (X, Y \ldots)_{(i)}$

$C_{\mathrm{AB}\,(i)}$ = live biomass above the ground (tree) of plot i (Mg. ha-1).

$\int_i \ (X, Y \ldots)$ = estimated biomass for tree "i" from the chosen function;

i = 1, 2, 3 ... n trees inventoried

The means and other statistical descriptors of the $C_{\mathrm{AB-tree}}$. were calculated for the total area of the project, considering the number of 17 plots inventoried. We recall that all values were later transformed to MgC ha-1 using the factor 0.47 and later transformed to MgCO2e ha-1 using the factor 44/12, as explained above.

Calculation of Above-Ground Dead Biomass

This reservoir was considered to correspond to about 12.1% of the biomass of dense lowland rain forests in the Amazon, according to the Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference Report: Land Use Sector, Land Use Change and Forests, 2020.

Thus, the above-ground dead biomass was obtained from the application of an expansion factor of +12.1% on the respective estimates of above-ground living tree biomass. This factor corresponds to the ratio of dead wood:above-ground biomass (tree biomass or CAB found in the Dense Ombrophilous Lowland Forests of the Amazon, originated from Nogueira et al. (2008):

$$C_{DW\,(parcela\ i)} = C_{\mathrm{AB}\,(i)} * \ R_{DW}$$

$C_{DW\,(parcela\ i)} = C_{\mathrm{AB}\,(i)} * \ R_{DW}$ where:

$C_{DW\,(parcela\ i)}$ = dead wood biomass from plot i (Mg. ha-1);

$C_{\mathrm{AB}\,(i)}$ = living above-ground biomass (from trees) of plot i; (Mg. ha-1);

R_{DW} = ratio 'Dead wood biomass: above-ground biomass' = 0.121;

i = 1, 2, 3 ... n inventoried plots.

We recall that all values were later transformed to MgC ha-1 using the factor 0.47 and later transformed to MgCO2e ha-1 using the factor 44/12, as already explained above.

The uncertainty for the dead wood stock considered the range of variation found in dense forests. According to the Report: Land Use Sector, Land Use Change and Forests, the ratio for estimating dead wood biomass in dense forests ranges from 0.085 (found in alluvial forests) to 0.131 (found in submontane forests). To calculate the uncertainty in the dead wood stock, these ratios were applied to each plot and the average interval between the maximum and the minimum was calculated and used as a proxy for the confidence interval in the dead wood stock, according to the guidance of VM0055 (item 2 on page 19).

Calculation of living tree biomass below ground per plot

Below-ground living biomass was also calculated using root:stem expansion factors, as predicted by the CP-AB module, page 17. The CP-AB module and Table 4.4 of the 2006 IPCC Guidelines, Chapter 4, define an average expansion factor of 24% for the inclusion of dry root biomass. This expansion factor was then applied to the result of the biomass density of each plot:

$$C_{BB\ (parcela\ i)} = C_{\mathrm{AB}\ (i)} * R_{\mathrm{R:S}}$$

where:

$C_{BB(parcela\ i)}$ = Below-ground biomass of plot i (Mg. ha^{-1});

$C_{\mathrm{AB}\ (i)}$ = Live above-ground (tree) biomass of the plot i; (Mg. ha^{-1});

$R_{\mathrm{R:S}}$ = root:shoot ratio = 0.24 (ratio of groundwater biomass to above-ground biomass);

i = 1, 2, 3 ... n inventoried plots;

The mean and the other statistical descriptors of the C_BB were calculated for the total area of the project, considering the number of 17 plots inventoried. We recall that all values were later transformed to MgC ha-1 using the factor 0.47 and later transformed to MgCO2e ha-1 using the factor 44/12, as explained above.

The uncertainty for the root stock considered the range of variation found in dense forests. According to the IPCC (Table 4.4 of the 2006 IPCC Guidelines, Chapter 4) the ratio for estimating root biomass in dense forests ranges from 0.22 to 0.33.

Calculation of total biomass per plot

The total biomass of the plot is the result of the sum of the reservoirs analyzed:

$$\text{total Biomass}_{\ plot\ i} = C_{AB\ (plot\ i)} + C_{DW\ (plot\ i)} + C_{BB\ (plot\ i)}$$

Where:

$\text{total biomass}_{plot\ i}$ = Total plot biomass i (Mg ha^{-1});

$C_{AB-(plot\ i)}$ = Tree biomass (com DAP > 10cm) (Mg ha^{-1});

$C_{DW-(plot\ i)}$= Dead wood biomass (Mg ha^{-1});

$C_{BB-(plot\ i)}$= Underground biomass/roots (Mg ha^{-1});

The mean and the other statistical descriptors of the total Biomass were calculated for the total area of the project, considering the number of 17 plots inventoried). We recall that all values were later transformed to MgC ha-1 using the factor 0.47 and later transformed to MgCO2e ha-1 using the factor 44/12, as explained above.

Estimation of the average density of forest carbon in the Project Area and Leakage Belt of the Caiarara project

In total, the biomass originated from the inventory was 442.87 + 38.43 Mg ha-1 (mean + standard deviation), with a coefficient of variation of 8.69% and uncertainty of 5.192% in the tree reservoir. These results show that the total biomass in the project area presents relative errors (=uncertainty) within the expected by VM0048, i.e., less than 10%; Therefore, the average value of total carbon per hectare can be used to estimate the total carbon density in the project area.

Table 14: Statistical descriptors of Biomass, Carbon and CO2e of the dense ombrophilous forests of the Caiarara REDD+ Project area. Results obtained with 90% confidence; n=17; t-tabulated value = 1,746.

Parameters	Trees	Dead wood	Belowground (roots)	Total
Biomass (Mg ha^{-1})				
Average	324,873	39,310	77,970	442,153
Standard deviation	28,243	3,417	6,778	38,438
CV (%)	8,693	8,693	8,693	8,693
Standard error	7,061	4,280[3]	10,234[3]	-
Confidence Interval (90%)[1]	12,328[2]	7,472[4]	17,868[5]	-
Uncertainty (U%)	3,795	19,008	22,917	5,192[6]
Carbon (Mg ha^{-1})				
Average	152,690	18,476	36,646	207,812
Standard deviation	13,274	1,606	3,186	18,066
CV (%)	8,693	8,693	8,693	8,693
Standard error	3,319	2,011[3]	4,810[3]	-

Confidence Interval (90%)[1]	5,794[2]	3,5122[4]	8,398[5]	-
Uncertainty (U%)	3,795	19,008	22,917	5,192[6]
Carbon eq – CO_2e (Mg ha^{-1})				
Average	559,865	67,744	134,368	761,976
Standard deviation	48,672	5,889	11,681	66,242
CV (%)	8,693	8,693	8,693	8,693
Standard error	12,168	7,375[3]	17,636[3]	-
Confidence Interval (90%)[1]	21,245[2]	12,877[4]	30,793[5]	-
Uncertainty (U%)	3,795	19,008	22,917	5,192[6]

[1] Uncertainty corresponds to half the width of a confidence interval of a given level of confidence;
[2] Calculated value normal of the tree inventory (DBH > 10cm), as described in the item;
[3] Value calculated from the proxy value of the Confidence Interval (Confidence Interval divided by t-tabulated);
[4] Value calculated from the minimum (0.085) and maximum (0.131) ratios found in dense forests, as presented in the Brazilian Fourth Communication - Report: Land Use Sector, Land Use Change and Forests, 2020;
[5] Value calculated from the minimum (0.22) and maximum (0.33) ratios found in dense forests, according to the IPCC (Table 4.4 of the 2006 IPCC Guidelines, Chapter 4)
[6] Value calculated to propagate the uncertainties of the three reservoirs analyzed, according to Line 1 of Table 3 of VMD0055, page 19).

Estimation of Non-Forest Carbon Stocks

For post-deforestation stocks, standard values were used for post-deforestation classes present in a buffer of 10km of the project area, adopted by the Ministry of the Environment (MMA) and presented in the Fourth National Communication of Brazil to the United Nations Framework Convention on Climate Change (UNFCCC) (Table 23, page 159):

Table 15: Carbon stock by type of use, likely to occur, after deforestation in the vicinity of the Água Azul REDD+ Project area. Data source: The values in Mg C ha-1 were extracted from the "Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference Report:

Land Use Sector, Land Use Change and Forests, 2020"; and the values in Mg CO2e ha-1 were calculated with the factor 44/12.

Pool / post-deforestation land use class	Pasture Low Degradation	Pasture High Degradation	Agriculture	Planted Forest
Area in the project activities region (ha) [1]	19,453[2]	- [2]	416	0[3]
Total Biomass (MgC ha-1)	10[4]	3,67[4]	7,44[5] (7,14-7,44)	40,43[6] (39,7-43,17)
Aboveground CO2e (MgCO2e ha-1)	14.128[7]	5.170[7]	25.334[8]	96.358[9]
Belowground CO2e (MgCO2e ha-1)	22.605[7]	8.272[7]	1.946[8]	51.885[9]
Dead wood CO2 (MgCO2e ha-1)	-	-	-	-
Aboveground CO2e Uncertainty (%)	19.946[10]	19.946[10]	2.016[11]	4.291[11]
Belowground CO2e uncertainty (%)	19.946[10]	19.946[10]	2.016[11]	4.291[11]
Dead wood uncertainty (%)	-	-	-	-

[1] Forest area converted to each land use class from 2014 to 2024, based on data from the MapBiomas project.

[2] Mapbiomas does not discriminate between high and low degradation pastures, therefore we conservatively used the highest biomass values.

[3] Silviculture (forest plantation) was excluded from this analysis, since no forest has been illegally replaced with silviculture during 2014-2024.

[4] Reference Value for the Amazon (Table 29, page 159 of the Report);

[5] Reference Value for the state of Pará: Average of 2010 (minimum-maximum, between 1994 and 2010) - Table 37, page 170 of the Report;

[6] Reference Value for the state of Pará: Average of 2016 (minimum-maximum, between 2002 and 2010) - Table 36, page 169 of the Report, and specifically for eucalyptus (Table 35, page 168 of the Report); It is recommended to check if the company has this value calculated from inventory in the planted forest;

[7] Calculated from the root:shoot ratio of 1.6 (Table 29, page 159 of the Report);

[8] Calculated from the root:shoot ratio of 0.0713 presented for Soybean by Mazzilli et al. (2014) – Table 29, page 159 of the Report cites Mazzilli et al. (2014) for the quantification of groundwater carbon;

[9]Calculated from the root:shoot ratio of 0.35 (Table 34, page 167 of the Report)

[10] Calculated as a proxy from the Standard error presented by Silva Neto et al. (2012) – Table 29, page 159 of the Report cites Silva Neto et al. (2012) for the quantification of carbon;

[11] Calculated as a proxy based on the amplitude (minimum and maximum, see notes 2 and 3).

The VMD0055 module (pg 18) states that carbon stocks of post-deforestation land use classes following deforestation should be calculated as the area-weighted average stocks of all post-deforestation land use classes present within a 10-km wide area surrounding the UDef PA. We considered carbon stocks estimates of table 13 and Mapbiomas Collection 10 data to identify the area of land cover classified as pastures and agriculture obtain a weighted average estimate. Post-deforestation carbon stock estimate, based on this weighted average, for each pool is provided in the table below:

Table 16: Estimate of post-deforestation carbon stock change and respective uncertainty

Pool[0]	Carbon stocks estimate ($MgCO_2e\ ha^{-1}$)[1]	Uncertainty[2] (CI)	Uncertainty (%)[3]
Aboveground live	14,36	2,76	19,2%
Belowground live	22,17	4,41	19,9%
Total	36,54	5,21	14,2%

[0] The post-deforestation carbon stock of dead wood pool was assumed 0.

[1] Area-weighted averages of carbon stocks estimates by pool, using land cover area estimates from the Mapbiomas project, from table 13 above.

[2] Represents the confidence intervals calculated by propagation of errors of carbon stocks estimates for each pool, using data from Table 13 above.

[3] Uncertainty% for each pool was calculated by area-weighting using combined error propagation equations for $\mathrm{U}(c \times A)$ and $\mathrm{U}(A + B)$. Uncertainty of the total carbon stocks was calculated by error propagation of the uncertainty estimates of each pool, using equation for $\mathrm{U}(A + B)$.

<u>**Step 2: Estimation of emissions from changes in carbon stocks**</u>

Carbon stock changes (emission factors) by pool were obtained by using equation 4 of the VM0055 module:

$$\Delta C_{p,i} = C_{p,i} - C_{p,post,i}$$

Where:

$\Delta C_{p,i}$: Estimated emissions from carbon stock change in pool p in forest stratum i (t CO2e/ha);

$C_{p,i}$: Estimated carbon stock in pool p of forest stratum i (t CO2e/ha);

$C_{p,post,i}$: Estimated carbon stock in post-deforestation pool p in forest stratum i (t CO2e/ha);

p: Carbon pool - aboveground tree biomass (AB_tree), aboveground non-tree biomass (AB_nontree), belowground tree biomass (BB_tree), belowground non-tree biomass (BB_nontree), dead wood (DW), litter (LI), soil organic carbon (SOC), and harvested wood products (WP);

I,M: 1 forest stratum.

Table 17 shows the estimates for each carbon pool p, calculated using data from Table 14 and since we have only one stratum, i = {1}.

Table 17: Estimates of carbon stock changes per pool and total, and respective uncertainties

Stratum i	1	
Stratum name	Dense Forest	Total (W)
ΔCAB_tree,i	148.8	148.8
Std Err (ΔCAB_tree,i)	3.3	3.3
U (ΔCAB_tree,i)	5.8	5.8
U% (ΔCAB_tree,i)	3.9%	3.9%
ΔCBB_tree,i	30.6	30.6
Std Err (ΔCBB_tree,i)	4.8	4.8
U (ΔCBB_tree,i)	8.5	8.5
U% (ΔCBB_tree,i)	27.7	27.7
ΔCDW,i	18.5	18.5
Std Err (ΔCDW,i)	2.0	2.0
U (ΔCDW,i)	3.5	3.5
U% (ΔCDW,i)	19.0%	19.0%
Total WΔC,i	197.8	197.8
Std Err (Total WΔC,i)	6.2	6.2
U (Total WΔC,i)	10.9	10.9
U% (Total WΔC,i)	5.5%	5.5%

Step 3: Assessing uncertainty in carbon stock estimates

The uncertainty reported in Table 17 was calculated using the formula presented in Table 3 of VMD0055 (page 19).

f(A,B,c)	Standard Error of Estimate	Percentage Uncertainty of Estimate
$A - B$	$S(A-B) = \sqrt{S(A)^2 + S(B)^2}$	$U\%(A-B) = 100 \times t_{alpha=10\%} \times \frac{S(A-B)}{A-B}$

Step 4: Estimation of an uncertainty discount factor

The procedure for calculating the uncertainty discount factor in emissions from the change in carbon stock (DFWΔC) was duly carried out according to the requirements established by the VM0048 methodology, module VMD0055, v1.1.

According to the module, the discount factor is:

- If U%(WΔC) ≤ 10%, the discount factor is DFWΔC = 0, i.e., there is no discount.
- If U%(WΔC) > 10%, the discount factor is calculated using the formula:

$$DFW\Delta C = U\%(W\Delta C) \times t_{alpha=66.67\%} / 100 \times t_{alpha=10\%}$$

Where:

DFWΔC = Uncertainty Discount Factor for Changes in Carbon Stock

U%(WΔC) = Percentage uncertainty in emissions due to carbon stock change (%)

talpha=10% = Value of the t-distribution for a two-sided 90% confidence interval

talpha=66.67% = Value of the t-distribution for a one-sided confidence interval of 66.67%

Table 17 shows carbon stocks changes estimates uncertainty is less than 10% and therefore the discount factor was 0 (zero).

Step 5: Conservative estimation of emissions from change in carbon stocks

The conservative estimates of emissions due to carbon stock change in the Project Area (UDef PA) were calculated by adding the different carbon compartments and applying the uncertainty discount factor (DFWΔC). This adjustment was performed separately for each carbon reservoir: aerial biomass, ground/below ground biomass, and dead wood. Stocks associated with soil organic carbon, litter and carbon stocks in wood products were omitted.

For aerial biomass, the following equation was used:

$\Delta C_{CAB\text{-}LI,i} = (\Delta C_{AB\{tree,i} + \Delta C_{AB\{nontree,i)} \times (1 - DF_{W\Delta C})$

where:

$\Delta C_{CAB\text{-}LI,i}$ = Estimated conservative emissions from the change of carbon stock in the UDef PA in the aerial biomass and litter compartments in the forest stratum i (t CO_2e/ha);

$\Delta C_{AB\{tree,}$ = Emissions from carbon stock change in aerial tree biomass;

$\Delta C_{AB\{nontree,i}}$ = Emissions from aerial biomass from non-arboreal woody vegetation;

$DF_{W\Delta C}$ = Uncertainty discount fator.

For below-ground biomass and dead wood, the following equation was applied:

$\Delta C_{CBB\text{-}DW,i} = (\Delta C_{BBtree,i} + \Delta C_{BBnontree,i} + \Delta C_{DW\_i}) \times (1 - DF_{W\Delta C})$

where:

$\Delta C_{CBB\text{-}DW,i}$ = Estimated conservative emissions over 10 years in underground biomass and deadwood compartments;

$\Delta C_{BBtree,i}$= Emissions from underground tree biomass;

$\Delta C_{BBnontree,i}$ = Emissions from groundwater biomass from non-arboreal vegetation

ΔC_{DW_i} = Emissions from dead wood

Emissions from carbon stock change at baseline are calculated similarly, but without applying the uncertainty discount factor.

Aerial biomass:

$\Delta C_{LB,AB\text{-}LI,i} = (\Delta C_{ABtree,i} - C_{WPi}) + \Delta_{CABnontree,i} + \Delta C_{LIi}$

where:

$\Delta C_{LB,AB\text{-}LI,i}$ = Estimated emissions from the change in carbon stock at UDef Baseline in the above-ground biomass and litter compartments in forest stratum i (tCO_2e/ha);

$\Delta C_{ABtree,i}$ = Emissions from the change in carbon stock in the aerial biomass compartment of trees in forest stratum i (tCO_2e/ha);

$\Delta_{CABnontree,i}$ = Emissions from the change in carbon stock in the non-arboreal woody aerial biomass compartment in stratum i (tCO_2e/ha)

$\Delta C_{ABtree,i}$ = Emissions from the change in the carbon stock in the litter compartment in forest stratum i (tCO_2e/ha)

$\Delta C_{LI,i}$ = Emissions from the change in the carbon stock in the litter compartment in forest stratum i (tCO_2e/ha). The value is 0.

C_{WPi} = Carbon stock entering the wood products compartment in forest stratum i (tCO_2e/ha). The value is 0.

i = 1, 2, 3, ..., M: Forest strata

Below-ground biomass and dead wood:

$\Delta C_{LB,BB\text{-}DW,i} = \Delta C_{BBtree,i} + \Delta C_{BBnontree,i} + \Delta C_{DW\_i}$

where:

$\Delta CL_{B,BB\text{-}DW,i}$ = Estimated emissions from change in carbon stock, over a 10-year period, at Baseline UDef in the below-ground biomass and deadwood compartments in forest stratum i (tCO_2e/ha);

$\Delta C_{BB\_tree,i}$ = Emissions from the change in carbon stock in the underground biomass compartment of trees in forest stratum i (tCO_2e/ha);

$\Delta C_{BB\_nontree,i}$ = Emissions from the change in carbon stock in the non-arboreal underground biomass compartment in forest stratum i (tCO_2e/ha);

$\Delta C_{DW,i}$ = Emissions from the change in carbon stock in the deadwood compartment in forest stratum i (tCO_2e/ha);

i = 1, 2, 3, ..., M forest strata

Table 18 presents the emission factors, after discounting for uncertainty, for each pool considered in the Caiarara REDD+ Project.

Table 18: Aggregate emission factors by carbon pool, after discount (tCO2-e ha-1)

		UDef PA			UDef LB		
Stratum ID i	Stratum name	$\Delta C_{AB\text{-}LI,i}$	$\Delta C_{BB\text{-}DW,i}$	$\Delta C_{SOC\text{-}WP,i}$	$\Delta_{CLBAB\text{-}LI,i}$	$\Delta_{CLBBB\text{-}DW,i}$	$\Delta C_{LBSOC\text{-}WP,i}$
1	Forest	545.5	179.9	0.0	545.5	179.9	0.0

<u>**Step 6: Baseline estimate of annual emissions from changes in carbon stocks**</u>

The estimated projected emissions to occur in UDef PA and UDef LB is estimated using the AD values for the Project Area and Leak Belt (Table 9 e Table 10) and the emission factors calculated in the previous step. Emissions from stock changes in the carbon reservoirs of aerial biomass and litter are assumed to be emitted at the time of the land use transition. After the land use transition, emissions from reservoirs of below-ground biomass, dead wood, soil organic carbon, and wood products are assumed to occur gradually over time; those of groundwater biomass and dead wood at an annual rate of 1/10 of the stock change, and those of soil organic carbon and wood products at an annual rate of 1/20 of the stock change.

For a given year t (the year for which emissions are to be estimated), emissions from unplanned deforestation are summed from time t − 10 to t (for below-ground biomass and dead wood) and from time t − 20 to t (for soil organic carbon and wood products).

The equation below was applied to the Project Area:

$$\Delta C_{BSL,PA-UDef,i,t} = \left(AD_{BSL,PA-UDef,i,t} \times \Delta Cc_{\mathrm{AB-LI},i}\right) + \sum_{t-10}^{t}\left(AD_{BSL,PA-UDef,i,t}\right) \times \frac{\Delta Cc_{\mathrm{BB-DW},i}}{10} + \sum_{t-20}^{t}\left(AD_{BSL,PA-UDef,i,t}\right) \times \frac{\Delta Cc_{\mathrm{SOC-WP},i}}{20}$$

where:

$\Delta C_{BSL,LB\text{-}UDef,i,t}$ = Total emissions from the change in baseline carbon stock across all carbon compartments in forest stratum i within UDef LB in year t (tCO_2e)

$AD_{BSL,LB\text{-}UDef,i,t}$ = UDef AD in the baseline scenario allocated to forest stratum i in UDef LB for year t (ha);

$\Delta Cc_{AB\text{-}LI,i}$ = Conservatively estimated emissions from the change in carbon stock in UDef LB in the aerial biomass and litter compartments in forest stratum i (tCO_2e/ha);

$\Delta Cc_{BB\text{-}DW,i}$ = Conservatively estimated emissions from the change in carbon stock, over a period of 10 years, in UDef LB in the groundwater biomass and dead wood compartments in forest stratum i (tCO_2e/ha);

$\Delta Cc_{SOC\text{-}WP,i}$ = Conservatively estimated emissions of the change in carbon stock, over a period of 20 years, in UDef LB in soil organic carbon and wood products compartments in forest stratum i (tCO_2e/ha). The value is 0;

i = 1, 2, 3, ..., M forest strata;

t = 1, 2, 3, ..., t* years since the beginning of the project.

Table 9 and Table 10 presents the AD values for the Project Area and Leakage Belt, respectively.

Table 19: Total emissions from carbon stock changes at baseline within UDef PA

BVP	t	Start	End	Total	$\Delta C_{BSL,PA\text{-}UDef,1,t}$ (tCO_{2e})
1	1	21-Jan-2024	31-Dec-2024	121,535	121,535
1	2	1-Jan-2025	31-Dec-2025	125,416	125,416
1	3	1-Jan-2026	31-Dec-2026	129,297	129,297

Table 20: Total emissions from carbon stock changes at baseline within UDef LB

BVP	t	Start	End	Total	$\Delta C_{BSL,LB\text{-}UDEF,1,t}$ (tCO_{2e})
1	1	21-Jan-2024	31-Dec-2024	270,647	270,647
1	2	1-Jan-2025	31-Dec-2025	279,290	279,290
1	3	1-Jan-2026	31-Dec-2026	287,932	287,932

3.2.2.4 Estimation of Other Baseline GHG Emissions

Baseline emissions from burning fossil fuels and using nitrogen fertilizers were conservatively omitted. Emissions from biomass burning were calculated using the VMD0013 Estimation of Greenhouse Gas Emissions from Biomass and Peat Burning (E-BPB) module. The baseline estimate is determined using the following:

$$E_{biomassburn,i,t} = \sum_{g=1}^{G} \left(\left(\left(A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i} \right) \times 10^{-3} \right) \times GWP_g \right)$$

where:

$E_{biomassburn,i,t}$ = Greenhouse gas emissions due to biomass burning in stratum i in year t of each GHG (CO_2, CH_4, N_2O) (tCO_2e);

$A_{burn,i,t}$ = Burned area in stratum i in year t (ha);

$B_{i,t}$ = Average stock of aerial biomass before burning in stratum i, year t (t m.s. ha^{-1});

$COMF_i$ = Combustion factor for stratum i (dimensionless);

$G_{g,i}$ = Emission factor for stratum i for gas g (kg t^{-1} m.s. burned);

GWP_g = Global warming potential for gas g (tCO_2/t gas g);

g = 1, 2, 3 … G greenhouse gases including carbon dioxide, methane, and nitrous oxide (dimensionless);

i = 1, 2, 3 … M strata (dimensionless)

t = 1, 2, 3 … t* Time elapsed since the start of the project activity (years)

Table 21 and Table 22, respectively, present the parameters used and the baseline estimate of greenhouse gas emissions from biomass burning.

Table 21: Parameters used to calculate emissions due to biomass burning

Gas	COMF	$G_{g,i}$	GWP_g	Source
N_2O	0.36	0.2	298	IPCC 2006
CH_4	0.36	6.8	25	IPCC 2006
CO_2	0.36	1,580	1	IPCC 2006

Table 22: Estimated total emissions from biomass burning

BVP	t	Start	End	Total	$E_{BSL,biomassBurn,1,t}$ (tCO_{2e})
1	1	21-Jan-2024	31-Dec-2024	11,303	11,303
1	2	1-Jan-2025	31-Dec-2025	11,303	11,303
1	3	1-Jan-2026	31-Dec-2026	11,303	11,303

Finally, the equation below was applied to the estimate of other baseline emissions in UDef PA:

$$GHG_{BSL_{PA}-UDef,E,t} = \sum_{t=1}^{t*}\sum_{i=1}^{M}\left(E_{BSL,FC,i,t} + E_{BSL,BiomassBurn,i,t} + N2O_{BSL,direct-N,i,t}\right)$$

where:

$GHG_{BSLPA\text{-}UDef,E,t}$ = Other cumulative baseline GHG emissions in year t resulting from unplanned deforestation within UDef PA since project inception (tCO_2e) ;

$E_{BSL,FC,i,t}$ = Emissions from the burning of fossil fuels in the forest stratum i in UDef PA in year t (tCO_2e). The value is 0;

$E_{BSL,BiomassBurn,i,t}$ = Non-CO_2 emissions due to biomass burning as part of unplanned deforestation activities in forest stratum i in year t (tCO_2e);

$N_2O_{BSL,direct\text{-}N,i,t}$ = Direct N_2O emission as a result of nitrogen application in an alternative land use in forest stratum i in year t (tCO_2e). The value is 0;

i = 1, 2, 3, …, M forest strata

t = 1, 2, 3, …, t* years since the beginning of the project;

Table 23: Estimation of total baseline emissions from other GHG emissions

BVP	t	Start	End	Total	$GHG_{BSLPA-UDef,E,t}$ (tCO_{2e})
1	1	21-Jan-2024	31-Dec-2024	11,303	11,303
1	2	1-Jan-2025	31-Dec-2025	11,303	11,303
1	3	1-Jan-2026	31-Dec-2026	11,303	11,303

3.2.2.5 Estimation of Net Baseline Emissions

Net emissions under baseline conditions for UDef PA are calculated as:

$$\Delta C_{BSL,PA-UDef,t} = \left(\sum_{t=1}^{t*} \sum_{i=1}^{M} \Delta C_{BSL,PA-UDef,i,t} \right) + GHG_{BSL_{PA}-UDef,E,t}$$

where:

$\Delta C_{BSL,PA-UDef,t}$ = Cumulative net GHG emissions since project inception within UDef PA at baseline in year t (tCO_2e);

$GHG_{BSL,PA-UDef,E,t}$ = Other cumulative GHG emissions at baseline in year t, resulting from unplanned deforestation within UDef PA since project inception (tCO_2e);

$\Delta C_{BSL,PA-UDef,i,t}$ = Sum of baseline emissions from the change in carbon stock in all carbon compartments in forest stratum i within UDef PA in year t (tCO_2e);

i = 1, 2, 3, …, M forest strata;

t = 1, 2, 3, …, t* years since the beginning of the project.

Net emissions under baseline conditions for UDef LB are calculated as:

$$\Delta C_{BSL,LB-UDef,t} = \sum_{t=1}^{t*} \sum_{i=1}^{M} \Delta C_{BSL,LB-UDef,i,t}$$

where:

$\Delta C_{BSL,LB-UDef,t}$ = Cumulative net GHG emissions since project inception within UDef LB at baseline in year t (tCO_2e);

$\Delta C_{BSL,LB-UDef,i,t}$ = Cumulative emissions from the change in baseline carbon stock across all carbon compartments in forest stratum i within UDef LB in year t (tCO_2e);

i = 1, 2, 3, …, M forest strata;

t = 1, 2, 3, …, t* years since the beginning of the project.

Table 24: Estimation of net emissions under baseline conditions for UDef PA

BVP	t	Start	End	$\Delta C_{BSL,PA\text{-}Udef,i,t}$ (tCO_{2e})	$GHG_{BSLPA\text{-}UDef,E,t}$ (tCO_{2e})	$\Delta C_{BSL,PA\text{-}Udef,t}$ (tCO_{2e})
1	1	21-Jan-2024	31-Dec-2024	121,535	11,303	132,838
1	2	1-Jan-2025	31-Dec-2025	125,416	11,303	136,719
1	3	1-Jan-2026	31-Dec-2026	129,297	11,303	140,600

Table 25: Estimation of net emissions under baseline conditions for UDef LB

BVP	t	Start	End	Total	$\Delta C_{BSL,LB\text{-}UDEF,1,t}$ (tCO_{2e})
1	1	21-Jan-2024	31-Dec-2024	270,647	270,647
1	2	1-Jan-2025	31-Dec-2025	279,290	279,290
1	3	1-Jan-2026	31-Dec-2026	287,932	287,932

3.2.3 Project Emissions (VCS, 3.15)

3.2.3.1 Ex Ante Estimations of Emission Reductions

In the Project scenario, some unplanned deforestation is inevitable and significant. However, due to the implementation of effective forest cover monitoring, the strengthening of governance in the area through management activities, the actions foreseen by the Project, and greater alignment with local communities, the Project is expected to achieve high levels of effectiveness over its 40-year duration.

Nevertheless, some unplanned deforestation may still occur in the Project Area, depending on the effectiveness of the proposed activities, which cannot be measured ex ante. Ex post measurements, prepared for the Monitoring Report, will be essential to determine the actual emission reductions. To enable ex ante projections, a conservative assumption regarding the effectiveness of the proposed activities was used to define the ex ante Effectiveness ($EA_{EF,t}$). The estimated $EA_{EF,t}$ value is applied to multiply the baseline projections by the factor ($1 - EA_{EF,t}$), and the result is considered the estimated ex ante emissions from unplanned deforestation in the Project scenario. The following equation is used to calculate the actual ex ante Project emissions.

Ex ante Project Emission = $\Delta C_{BSL,PA\text{-}UDef,t} * (1 - EA_{EF,t})$

Where:

$\Delta C_{BSL,PA\text{-}UDef,t}$: Total variation of the carbon stock of the baseline in the year, in the Project Area (tCO2-e);

$EA_{EF,t}$: Ex ante Effectiveness. The value is 17%.

t: 1, 2, 3 ... T, year of the proposed project credit period (dimensionless)

The Project proposed to implement a range of activities designed to prevent invasions and predatory practices, thereby reducing deforestation and greenhouse gas (GHG) emissions. These actions primarily focus on improving property surveillance and enhancing satellite image monitoring. In addition, the Project includes social initiatives aimed at promoting sustainable productive and economic practices among the communities surrounding the Project Area.

Given that the REDD+ Caiarara Project combines robust surveillance and monitoring strategies with targeted interventions addressing key drivers of deforestation, its potential to curb expected deforestation - and, consequently, its effectiveness —is considered relatively high. Based on this, the Project decided to apply a 17% conservative discount to account for the risk of the Project’s ineffectiveness in preventing deforestation. This value was based on the Project’s Non-Permanence Risk Tool. Therefore, the Project is considered to have an effectiveness of 83%.

3.2.4 Leakage emissions (VCS 2.5, 3.2, 3.6, 3.15, 4.3)

3.2.4.1 Ex Ante Estimation of Leakage

Activities that would cause deforestation within the Project Area under the baseline scenario may be shifted outside the project boundaries due to the implementation of the AUD project activity. A greater decrease in carbon stocks within the leakage belt during the project scenario, compared to ex-ante projections, would indicate the displacement of deforestation activities as a result of the Project.

The ex-ante activity-shifting leakage was calculated based on the anticipated combined effectiveness of the proposed leakage prevention measures and Project activities. As explained above, the Project seeks to prevent deforestation through the activities presented in Section 2.1.17.

Although the Project aims to reach 100% of baseline agents, a conservative Leakage Displacement Factor (DLF) was applied. The leakage simulation was carried out by multiplying the estimated baseline carbon stock changes in the Project Area by the DLF, which represents the percentage of deforestation expected to be displaced outside the project boundaries. This factor was set at 15% (default value according to Section 3.15.14 of VCS Standard v4.7) considering the possibility that a portion of the deforestation activities driven by some of the mapped agents would be displaced. The following equation is used to calculate the actual ex ante Leakage emission:

Ex ante Leakage Emission = $\Delta C_{BSL,PA\text{-}UDef,t} * \text{DLF}$

Where:

$\Delta C_{BSL,PA\text{-}UDef,t}$: Total emissions in the baseline within the UDef PA in year t (tCO_2e);

DLF: Displacement Leakage Factor (%).

The value is 15%.

3.2.5 Estimated GHG emission reductions and carbon dioxide removals (VCS, 3.15, 4.1)

The equation below was used to estimate ex ante GHG emission reduction. The calculation results are presented in Table 26.

$NER_{UDef,t}$ =($\Delta C_{BSL,PA\text{-}Udef,i,t}$ + $GHG_{BSLPA\text{-}UDef,E,t}$) – Ex ante Project Emission – Ex ante Leakage emission

Where:

$NER_{UDef,t}$: ex ante GHG emission reduction attributed to the project activity in year t (tCO_{2e});

$\Delta C_{BSL,PA\text{-}Udef,i,t}$: Sum of baseline carbon stock changes in the Project Area in year t (tCO_{2e});

$GHG_{BSLPA\text{-}UDef,E,t}$: Sum of ex-ante other GHG emission in the Project Area in year t (tCO2e);

Table 26: Total net GHG potential emissions reductions.

Period	Baseline emissions from carbon stock change within the UDef PA in year t (tCO_{2e})	Other GHG emissions in the PA (tCO_{2e})	Net GHG emission within the UDef PA in the baseline in year t (tCO_{2e})	Ex ante project emissions (tCO2-e)	Ex ante Leakage emissions (tCO2-e)	Total net GHG potential emissions reductions (tCO_{2e})

BVP	t	Start	End	$\Delta C_{BSL,PA\text{-}Udef,i,t}$	$GHG_{BSLPA\text{-}UDef,E,t}$	$\Delta C_{BSL,PA\text{-}Udef,t}$			$NER_{UDef,t}$
1	1	21-Jan-2024	31-Dec-2024	121,535	11,303	132,838	22,583	19,926	93,717
1	2	1-Jan-2025	31-Dec-2025	125,416	11,303	136,719	23,242	20,508	96,456
1	3	1-Jan-2026	31-Dec-2026	129,297	11,303	140,600	23,902	21,090	99,193

The non-permanence risk analysis was applied to the Project, based on the *AFOLU No- Permanence Risk Tool v4.2*, resulting in a score of 17 points. Therefore, the Caiarara REDD+ Project is considered to have an 17% non-permanence risk.

Indicate the risk classification of non-permanence (%)	17%
Has the non-permanence risk report been attached as an appendix or a separate document?	☒ Yes ☐ No

The number of credits to be held in the AFOLU pooled buffer account is determined as a percentage of the total carbon stock benefits. For avoiding unplanned deforestation project activities, this is calculated using the equation below, using the risk classification of non-permanence. Leakage emissions do not factor into the buffer calculations.

$$Buffer_{Total,t} = (\Delta C_{BSL,PA-UDef,t} - \Delta C_{MP,PA-UDef,t}) \times Buffer\%$$

Where:

$Buffer_{Total,t}$ = Cumulative total permanence risk buffer withholding in year t (t CO2e)

$\Delta C_{BSL,PA-UDef,t}$ = Cumulative net GHG emissions in the baseline within the UDef PA in year t (t CO2e)

$\Delta C_{MP,PA-UDef,t}$ = Total emissions from carbon stock change in all pools in the UDef PA in year t (t CO2e)

$Buffer\%$ = Buffer withholding percentage, from VCS AFOLU Non-Permanence Risk Tool (%) t = 1, 2, 3, …, t

* years elapsed since the start of the project

The following equation calculates the number of potential Verified Carbon Units (VCUs) for years in the crediting period. The calculation results are presented in Table 27. Where this equation results in a decimal number, the value was rounded down.

$VCU_{AUDef,t} = (NER_{UDef,t} - NER_{UDef,t-1}) - (Buffer_{Total,t} - Buffer_{Total,t-1})$

Where:

$VCU_{AUDef,t}$ = Number of potential Verified Carbon Units generated in year t (VCU)

$NER_{UDef,t}$ = Cumulative total net GHG emission reductions of the project activity in year t (t CO2e)

$NER_{UDef,t-1}$ = Cumulative total net GHG emission reductions of the project activity in year t – 1 (t CO2e)

$Buffer_{Total,t}$ = Cumulative total permanence risk buffer withholding in year t (t CO2e)

$Buffer_{Total,t-1}$= Cumulative total permanence risk buffer withholding in year t – 1 (t CO2e)

t = 1, 2, 3, …, t

* years elapsed since the start of the project

Table 27: Estimated VCU reduction

Period		Estimated baseline emissions (tCO2e)	Estimated project emissions (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated buffer pool allocation (tCO2e)	Estimated VCU reduction (tCO2e)
Start	End					
21-Jan-2024	31-Dec-2024	132,838	22,583	19,926	18,743	74,974
1-Jan-2025	31-Dec-2025	136,719	23,242	20,508	19,291	77,165
1-Jan-2026	31-Dec-2026	140,600	23,902	21,090	19,839	79,354

3.3 Monitoring

3.3.1 Data and parameters available for validation (VCS, 3.16)

Data/parameter	$A_{PA\text{-}UDef}$
Data unit	ha
Description	Area of project where activities aimed at avoiding unplanned deforestation will take place
Data source	Calculated within a GIS

Applied value	32,625
Justification for the choice of data or description of the measurement methods and procedures applied	The Project Area was delineated in accordance with the requirements set forth in both the VM0048 methodology and the VMD0055 module, ensuring compliance with eligibility criteria, forest integrity, and spatial boundary definition.
Purpose of the data	• Calculation of project emissions
Comments	None

Data/parameter	DLF
Data unit	%
Description	Displacement leakage factor
Data source	Determined by project proponent
Applied value	15
Justification for the choice of data or description of the measurement methods and procedures applied	According to Section 3.15.14 (VCS Standard v4.7), for validation, projects may apply the optional default activity-shifting leakage deduction of 15 percent to the gross GHG emission reductions and/or removals
Purpose of the data	• Calculation of leakage
Comments	None

Data/parameter	$EA_{EF,t}$
Data unit	%
Description	Ex ante effectiveness of halting baseline emissions in year t i
Data source	Determined by project proponent

Applied value	83
Justification for the choice of data or description of the measurement methods and procedures applied	Project decided to apply an 17% conservative discount to account for the risk of the Project's ineffectiveness in preventing deforestation. This value was based on the Project's permanence risk.
Purpose of the data	• Calculation of project emissions
Comments	None

3.3.2 Monitored Data and Parameters (VCS, 3.16)

Data/parameter	$AD_{BSL,PA\text{-}UDef}$
Data unit	ha/yr
Description	Unplanned deforestation activity data allocated to the UDef PA
Data source	Verra (through the AD Baseline Allocation Report and VT0007)
Description of the measurement methods and procedures to be applied	Approach determined and applied by CTress
Monitoring/recording frequency	To be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	215.68
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	• Calculation of baseline emissions
Calculation method	Application of Appendix 1 (VMD0055) by Verra

Comments	None

Data/parameter	$AD_{BSL,LB-UDef}$
Data unit	ha/yr
Description	Unplanned deforestation activity data allocated to the UDef LB
Data source	Verra (VT0007 and Appendix 1 in VMD0055)
Description of the measurement methods and procedures to be applied	VT0007 and Appendix 1 (VMD0055)
Monitoring/recording frequency	To be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	480.30
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	• Calculation of leakage emissions
Calculation method	Application of Appendix 1 (VMD0055) and VT0007 by Verra
Comments	None

Data/parameter	$C_{AB\_tree,i}$
Data unit	tCO_{2e}/ha
Description	Forest carbon stock in aboveground pool in stratum i
Data source	The mean carbon stock in aboveground tree biomass per unit area is estimated based on field measurements

Description of the measurement methods and procedures to be applied	Carbon stock in the aboveground pool was estimated using field data collected through a forest inventory, allometric equations and carbon conversion factor, in accordance with the criteria of module VMD0001 and VMD0055. All procedures used are described in the Carbon Stock Report (to be provided as evidence)
Monitoring/recording frequency	Every ten years, in accordance with Section 4.1 of Module VMD0001
Applied value	559.86
Monitoring equipment	See Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001
Purpose of the data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Calculation method	See VMD0001
Comments	None

Data/parameter	$C_{BB\_tree,i}$
Data unit	tCO_{2e}/ha
Description	Forest carbon stock in belowground pool in stratum i
Data source	The mean carbon stock in aboveground tree biomass per unit area is estimated based on field measurements
Description of the measurement methods and procedures to be applied	Carbon stock in the belowground pool was estimated using field data collected through a forest inventory, allometric equations and carbon conversion factor, in accordance with the criteria of module VMD0001 and VMD0055. All procedures used are described in the Carbon Stock Report (to be provided as evidence)
Monitoring/recording frequency	Every ten years, in accordance with Section 4.1 of Module VMD0001

Applied value	134.37
Monitoring equipment	See Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001
Purpose of the data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Calculation method	See VMD0001
Comments	None

Data/parameter	$C_{DW,i}$
Data unit	tCO_{2e}/ha
Description	Forest carbon stock in the dead wood pool in stratum i
Data source	The mean carbon stock in dead wood biomass per unit area is estimated based on field measurements
Description of the measurement methods and procedures to be applied	Carbon stock in the dead wood pool was estimated using field data collected through a forest inventory, allometric equations and carbon conversion factor, in accordance with the criteria of module VMD0002 and VMD0055. All procedures used are described in the Carbon Stock Report (to be provided as evidence)
Monitoring/recording frequency	Every ten years, in accordance with Section 4.1 of Module VMD0002
Applied value	67.74
Monitoring equipment	See Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0002
Purpose of the data	• Calculation of baseline emissions • Calculation of project emissions

	● Calculation of leakage
Calculation method	See VMD0002
Comments	None

Data/parameter	$\Delta Cc_{AB-LI,i}$
Data unit	tCO_{2e}/ha
Description	Conservatively estimated emissions from carbon stock change in the UDef AP in aboveground and litter pools in forest stratum i
Data source	Estimated based on field measurements
Description of the measurement methods and procedures to be applied	Emissions from carbon stock change in the aboveground pool were estimated using field data collected through a forest inventory and, when necessary, applying a discount factor, in accordance with the criteria of module VMD0001 and VMD0055. All procedures used are described in the Carbon Stock Report (to be provided as evidence)
Monitoring/recording frequency	Every ten years, in accordance with Section 4.1 of Module VMD0001
Applied value	545.51
Monitoring equipment	See Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001
Purpose of the data	● Calculation of baseline emissions ● Calculation of project emissions
Calculation method	See VMD0001
Comments	- Since the discount factor is 0, the applied value is the same for UDef-LB - Carbon stock in the litter pool was not accounted for in the Project

Data/parameter	$\Delta Cc_{BB-DW,i}$
Data unit	tCO_{2e}/ha
Description	Conservatively estimated emissions from carbon stock change over a 10-year period in the UDef PA in belowground biomass and dead wood pools in forest stratum i
Data source	The mean carbon stock in belowground pool per unit area is estimated based on field measurements
Description of the measurement methods and procedures to be applied	Emissions from carbon stock change in the belowground and dead wood pools were estimated using field data collected through a forest inventory and, when necessary, applying a discount factor, in accordance with the criteria of module VMD0001, VMD0002 and VMD0055. All procedures used are described in the Carbon Stock Report (to be provided as evidence)
Monitoring/recording frequency	Every ten years, in accordance with Section 4.1 of Module VMD0001
Applied value	179.94
Monitoring equipment	See Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001 and VMD0002
Purpose of the data	• Calculation of baseline emissions • Calculation of project emissions
Calculation method	See VMD0001 and VMD0002
Comments	- Since the discount factor is 0, the applied value is the same for UDef-LB

Data/parameter	$\Delta Cc_{SOC-WP,i}$
Data unit	tCO_{2e}/ha

Description	Conservatively estimated emissions from carbon stock change over a 20-year period in the UDef PA in the soil organic carbon and wood product pools in forest stratum i
Data source	N/A
Description of the measurement methods and procedures to be applied	Carbon stock in the soil organic carbon and wood product pools was not accounted for in the Project
Monitoring/recording frequency	N/A
Applied value	0
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	• Calculation of baseline emissions • Calculation of project emissions
Calculation method	N/A
Comments	None

Data/parameter	$C_{p,post,i}$
Data unit	tCO_{2e}/ha
Description	Estimated carbon stock in post-deforestation pool p in forest stratum i
Data source	Published and/or peer-reviewed study: Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference Report: Land Use Sector, Land Use Change and Forests, 2020

Description of the measurement methods and procedures to be applied	Estimated carbon stock in post-deforestation were estimated in accordance with the criteria of module VMD0055.
Monitoring/recording frequency	Every ten years, in accordance with Section 4.1 of Module VMD0001
Applied value	179.94
Monitoring equipment	See Carbon Stock Report (to be provided as evidence)
QA/QC procedures to be applied	See VMD0001 and VMD0002
Purpose of the data	• Calculation of baseline emissions • Calculation of project emissions
Calculation method	See VMD0001 and VMD0002
Comments	- Since the discount factor is 0, the applied value is the same for UDef-LB

Data/parameter	$\Delta C_{MP,PA-UDef}$
Data unit	tCO_{2e}
Description	Cumulative net GHG emissions in the UDef PA
Data source	Calculated by the project proponent
Description of the measurement methods and procedures to be applied	Estimated by the project proponent using spreadsheet
Monitoring/recording frequency	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
Applied value	To be calculated at each verification event

Monitoring equipment	N/A
QA/QC procedures to be applied	Described in the topic "Monitoring of project implementation"
Purpose of the data	• Calculation of project emissions
Calculation method	Equation 39 of the Module VMD0055
Comments	None

Data/parameter	$\Delta C_{LK\text{-}ME,t}$
Data unit	tCO_{2e}
Description	Cumulative net GHG emissions due to market-effects leakage in year t
Data source	Calculated by the project proponent using national databases on fuelwood production
Description of the measurement methods and procedures to be applied	Emissions due to market-effects leakage were estimated using fuelwood production data collected through a national database and applying a leakage factor default of 0.4, in accordance with the criteria of module VMD0011
Monitoring/recording frequency	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
Applied value	To be calculated at each verification event
Monitoring equipment	N/A
QA/QC procedures to be applied	See VMD0011
Purpose of the data	• Calculation of leakage
Calculation method	See VMD0011
Comments	None

Data/parameter	$PROP_{MIG}$
Data unit	Proportion
Description	Proportion of households living in the project activities region that are recent migrants and are engaging in land use activities identified as a baseline driver of unplanned deforestation
Data source	N/A
Description of the measurement methods and procedures to be applied	$PROP_{MIG}$ has been assigned a conservative value according to VMD0055
Monitoring/recording frequency	To be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	1.0
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	• Calculation of leakage
Calculation method	N/A
Comments	None

Data/parameter	$\Delta C_{OLB,t}$
Data unit	tCO_{2e}/ha
Description	Emissions from carbon stock change due to land cover transition in areas available for activity shifting outside the UDef LB, as calculated for year t
Data source	Verra

Description of the measurement methods and procedures to be applied	See Appendix 2 A2.2 in Module VMD0055
Monitoring/recording frequency	To be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	424.99
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	• Calculation of leakage
Calculation method	See Appendix 2 A2.2 in Module VMD0055
Comments	None

Data/parameter	ΔCLK-AS,t
Data unit	tCO2e
Description	Cumulative net GHG leakage emissions due displacement of unplanned deforestation activities in year t
Data source	Calculated by the project proponent
Description of the measurement methods and procedures to be applied	Estimated using spreadsheet
Monitoring/recording frequency	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
Applied value	To be calculated at each verification event
Monitoring equipment	N/A

QA/QC procedures to be applied	Described in the topic “Monitoring of project implementation”
Purpose of the data	• Calculation of leakage
Calculation method	Equation 47 of the Module VMD0055
Comments	None

Data/parameter	$E_{BSL,BiomassBurn,i,t}$
Data unit	tCO_{2e}
Description	Non-CO2 emissions due to biomass burning as part of project activities in forest stratum i in year t
Data source	Calculated by the project proponent using the parameters and criteria defined in Module VMD0013
Description of the measurement methods and procedures to be applied	See VMD0013
Monitoring/recording frequency	To be renewed in 2027. Subsequent baselines will be renewed at least every six years
Applied value	<table><tr><td>Start</td><td>End</td><td>$E_{BSL,BiomassBurn,i,t}$</td></tr><tr><td>1/21/2024</td><td>12/31/2024</td><td>1,434</td></tr><tr><td>1/1/2025</td><td>12/31/2025</td><td>1,434</td></tr><tr><td>1/1/2026</td><td>12/31/2026</td><td>1,434</td></tr></table>
Monitoring equipment	N/A
QA/QC procedures to be applied	See VMD0013

Purpose of the data	• Calculation of baseline emissions
Calculation method	Application of Equation 1 presented in Module VMD0013 for the calculation of non-CO_2 emissions. In the baseline scenario, the burned area ($A_{burn,i,t}$) is considered equal to $AD_{BSL,PA-UDEF,i,t}$.
Comments	None

Data/parameter	$E_{MP,BiomassBurn,i,t}$
Data unit	tCO_{2e}
Description	Non-CO2 emissions due to biomass burning as part of project activities in forest stratum i in year t
Data source	Calculated by the project proponent using the parameters and criteria defined in Module VMD0013
Description of the measurement methods and procedures to be applied	See VMD0013
Monitoring/recording frequency	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently.
Applied value	To be calculated at each verification event
Monitoring equipment	N/A
QA/QC procedures to be applied	See VMD0013
Purpose of the data	• Calculation of project emissions
Calculation method	Application of Equation 1 presented in Module VMD0013 for the calculation of non-CO_2 emissions
Comments	None

Data/parameter	$AD_{MP,PA-UDEF,i,t}$
Data unit	ha
Description	Deforestation in the UDef PA in forest stratum i in year t
Data source	Calculated by the project proponent using the parameters and criteria defined in Section 5.3.3.2 of the Module VMD0055
Description of the measurement methods and procedures to be applied	See Section 5.3.3.2 of the Module VMD0055 and the Standard operating procedures (SOPs) described in the Monitoring Plan
Monitoring/recording frequency	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently.
Applied value	To be calculated at each verification event
Monitoring equipment	N/A
QA/QC procedures to be applied	See Section 5.3.3.2 of the Module VMD0055 and the Standard operating procedures (SOPs) described in the Monitoring Plan
Purpose of the data	• Calculation of project emissions
Calculation method	See Section 5.3.3.2 of the Module VMD0055 and the Standard operating procedures (SOPs) described in the Monitoring Plan
Comments	None

Data/parameter	$AD_{MP,LB-UDEF,i,t}$
Data unit	ha
Description	Deforestation in the UDef LB in forest stratum i in year t
Data source	Calculated by the project proponent using the parameters and criteria defined in Section 5.3.3.2 of the Module VMD0055

Description of the measurement methods and procedures to be applied	See Section 5.3.3.2 of the Module VMD0055 and the Standard operating procedures (SOPs) described in the Monitoring Plan
Monitoring/recording frequency	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently.
Applied value	To be calculated at each verification event
Monitoring equipment	N/A
QA/QC procedures to be applied	See Section 5.3.3.2 of the Module VMD0055 and the Standard operating procedures (SOPs) described in the Monitoring Plan
Purpose of the data	• Calculation of project emissions
Calculation method	See Section 5.3.3.2 of the Module VMD0055 and the Standard operating procedures (SOPs) described in the Monitoring Plan
Comments	None

Data/parameter	Buffer%
Data unit	%
Description	Buffer withholding percentage
Data source	Calculated by the project proponent using the VCS AFOLU Non-Permanence Risk Tool
Description of the measurement methods and procedures to be applied	See VCS AFOLU Non-Permanence Risk Tool
Monitoring/recording frequency	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently.
Applied value	To be calculated at each verification event

Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	• Calculation of VCUs
Calculation method	See VCS AFOLU Non-Permanence Risk Tool
Comments	None

Data/parameter	$NER_{UDef,t}$
Data unit	tCO_{2e}
Description	Cumulative total ex post net GHG emission reductions of the project activity in year t
Data source	Calculated by the project proponent using Equation 52 of the Module VMD0055
Description of the measurement methods and procedures to be applied	N/A
Monitoring/recording frequency	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently.
Applied value	To be calculated at each verification event
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	• Calculation of VCUs
Calculation method	See Equation 52 of the Module VMD0055
Comments	None

3.3.3 Monitoring Plan (VCS, 3.16, 3.20)

The Climate Impact Monitoring Plan will encompass key issues for demonstrating the reduction of emissions from deforestation and degradation due to avoided unplanned deforestation within the Project Area and in the Leakage Belt, according to the methodology VM0048, Module VMD0055.

The monitoring plan consists of two main parts:

i) Monitoring changes in forest cover, carbon stocks, GHG emissions and spatial variable datasets used in modeling considering a fixed baseline period (TASK 1);

ii) Baseline reassessment at the close of a fixed baseline period (TASK 2).

TASK 1 - Monitoring of changes in forest cover, carbon stocks, GHG emissions and spatial variable datasets used in modeling considering a fixed baseline period

1. Monitoring of project implementation

a. Technical description

The monitoring of project implementation aims to verify that key steps and procedures for reporting GHG emission reductions are being carried out, in full compliance with the technical and methodological requirements defined in the applicable modules of methodology VM0048. This includes the establishment and documentation of the project boundary, stratification zones, leakage belt, and leakage management zones, as well as the monitoring of ex-post parameters

b. Data to be collected

The data to be collected include:

- Project area for unplanned deforestation reduction (UDef PA);
- Leakage belt (UDef LB);
- Leakage management zone;
- Project activity region.
- Data to be collected for monitoring of actual carbon stock changes and greenhouse gas emissions (described in item 2)
- Data to be collected for monitoring of leakage carbon stock changes and greenhouse gas emissions (described in item 3)
- Data to be collected for estimation of ex post net carbon stock changes and greenhouse gas emissions (described in item 4)

c. Overview of data collection procedures

Data collection will be conducted by trained technical teams. Procedures will include geospatial analysis, field verification, and integration of remote sensing data.

d. Quality control and quality assurance procedures

Quality control and assurance (QC/QA) procedures will include cross-verification of geographic coordinates using geoprocessing tools, validation of spatial layers, and consistency checks between field data and remote sensing outputs. Standard Operating Procedures (SOPs) and specific protocols will be applied to address any discrepancies, measurement errors, or data gaps, ensuring reliability and traceability. All rules as well as the calculation and data collection procedures established in the modules and documents referenced in Table 11 in Section 3.2.2.3, will be applied and validated.

e. Data archiving

All collected data will be archived in secure digital systems, with regular backups and access control. The files will include detailed metadata, field records, georeferenced images, thematic maps, and technical reports. Archiving will be carried out in a way that ensures data traceability and availability for verification, auditing, and periodic reporting purposes.

f. Organization and responsibilities of the parties involved in all of the above.

Responsibilities will be allocated according to technical expertise:

- The geoprocessing team will manage spatial analysis and boundary delineation;
- Field technicians will conduct on-site data collection;
- The quality assurance team will oversee QC/QA protocols;
- Project coordination will be responsible for data integration, double checking of the results, reporting, and communication with Verra and other stakeholders.

2. Monitoring of actual carbon stock changes and greenhouse gas emissions

a. Technical description

The monitoring of actual changes in carbon stocks and greenhouse gas emissions aims to assess the project's impacts on forest resources by identifying any changes that may occur throughout the project's lifecycle.

b. Data to be collected

The data to be collected and monitored include parameters related to carbon stocks in aboveground and belowground reservoirs and deadwood. Data for estimating post-deforestation stocks will also be updated, if necessary. In addition, activity data allocated to the UDef PA, with annual deforestation estimates, must be periodically updated in accordance with the requirements of VM0048.

Parameter	Description	Unit	Source	Frequency

$C_{AB\_tree,i}$	Forest carbon stock in aboveground tree biomass in stratum i	t CO_2e/ha	Forest inventory	Every ten years
$C_{AB\_tree,post,}$	Post-land use transition carbon stock in aboveground tree biomass in stratum i	t CO_2e/ha	Published and/or peer-reviewed	Every ten years
$C_{BB\_tree,i}$	Forest carbon stock in belowground tree biomass in stratum i	t CO_2e/ha	Forest inventory	Every ten years
$C_{BB\_tree,post,i}$	Post land use transition carbon stock in belowground tree biomass in stratum i	t CO_2e/ha	Published and/or peer-reviewed	Every ten years
$C_{DW,i}$	Forest carbon stock in the dead wood pool in stratum i	t CO_2e/ha	Forest inventory	Every ten years
$\Delta Cc_{AB-LI,i}$	Conservatively estimated emissions from carbon stock change in the UDef AP in aboveground and litter pools in forest stratum i	t CO_2e/ha	Calculated	Every ten years
$\Delta Cc_{BB-DW,i}$	Conservatively estimated emissions from carbon stock change over a 10-year period in the UDef PA in belowground biomass and dead wood pools in forest stratum i	t CO_2e/ha	Calculated	Every ten years
$\Delta Cc_{SOC-WP,i}$	Conservatively estimated emissions from carbon stock	t CO_2e/ha	Calculated	Every ten years

	change over a 20-year period in the UDef PA in the soil organic carbon and wood product pools in forest stratum i			
$AD_{BSL,PA-UDef}$	Unplanned deforestation activity data allocated to the UDef PA	ha/year	Verra	At least every six years
$E_{BSL,BiomassBurn,i,t}$	Non-CO2 emissions due to biomass burning as part of project activities in forest stratum i in year t	t CO_2e	Calculated	At least every six years

c. Overview of data collection procedures

The data for monitoring carbon stocks will be obtained through field forest inventories (measurement of diameter at breast height in tree individuals), the use of allometric equations for biomass estimation, and the application of conversion factors for carbon and carbon equivalent. Satellite imagery and public databases may be used to support the detection of changes in forest cover and to validate field data.

UDef PA activity data will be requested via the AD Allocation Request Form submitted to Verra and received through the AD Baseline Allocation Report.

Any other GHG emissions occurring in the baseline scenario within the UDef PA will be estimated. In the first baseline, only emissions from biomass burning will be accounted for, in accordance with the calculation assumptions of Module VMD0013. Emissions from fossil fuel use and nitrogen fertilizer will not be included.

d. Quality control and quality assurance procedures

The monitoring of actual carbon stocks will be carried out periodically, in accordance with the applicability conditions and rules defined in module VMD0055, as well as the calculation and data collection procedures established in modules VMD0001, VMD0002, and VMD0016. Quality control and assurance procedures include:

- Designation of a specialized team to perform field data measurements and the necessary calculations;
- Ensuring that the responsible team understands the minimum requirements of precision, accuracy, and conservativeness to guarantee the quality of the data and information generated, with training provided if necessary;

- Data tabulation, presentation of statistical descriptors, and reporting of all materials and methods used in carbon stock data collection and calculation, thus ensuring verification and replicability of the information;
- Reporting of stock change and emission information through spreadsheets structured by theme, with traceable equations and values

e. Data archiving

All collected data will be archived in secure digital systems, with regular backups and access control. The files will include detailed metadata, field records, georeferenced images, thematic maps, and technical reports. Archiving will be carried out in a way that ensures data traceability and availability for verification, auditing, and periodic reporting purposes.

f. Organization and responsibilities of the parties involved in all of the above

The organization and responsibilities will be distributed among the project proponents and other parties involved, according to their respective competencies. A specialized team, to be designated, will be responsible for primary data collection, stock estimation, and information reporting. The project's technical coordination will be responsible for requesting activity data from Verra, applying QC/QA procedures, and consolidating and storing the data and information.

3. Monitoring of leakage carbon stock changes and greenhouse gas emissions

a. Technical description

The monitoring of changes in carbon stock leakage and greenhouse gas emissions aims to predict and quantify emissions resulting from the displacement of deforestation activities as a potential impact of the Project.

b. Data to be collected

The data to be collected and monitored include parameters related to carbon stocks in the Leakage Belt, as mentioned and listed in item 2-b. In addition, the activity data allocated to the UDef LB, with annual deforestation estimates, must be periodically updated in accordance with the requirements of VM0048.

Parameter	Description	Unit	Source	Frequency
$AD_{BSL,LB\text{-}UDef}$	Unplanned deforestation activity data allocated to the UDef LB	ha/year	Verra	At least every six years

$\Delta C_{OLB,t}$	Emissions from carbon stock change due to land cover transition in areas available for activity shifting outside the UDef LB, as calculated for year t	t CO_2e/ha	Verra	At least every six years
$PROP_{MIG}$	Proportion of households living in the project activities region that are recent migrants and are engaging in land use activities identified as a baseline driver of unplanned deforestation	proportion	Calculated	At least every six years

c. Overview of data collection procedures

The data for monitoring carbon stocks will be obtained through field forest inventories (measurement of diameter at breast height in tree individuals), the use of allometric equations for biomass estimation, and the application of conversion factors for carbon and carbon equivalent. Satellite imagery and public databases may be used to support the detection of changes in forest cover and to validate field data.

UDef LB activity data will be requested via the AD Allocation Request Form submitted to Verra and received through the AD Baseline Allocation Report.

d. Quality control and quality assurance procedures

The same procedure applied in item 2-d.

e. Data archiving

All collected data will be archived in secure digital systems, with regular backups and access control. The files will include detailed metadata, field records, georeferenced images, thematic maps, and technical reports. Archiving will be carried out in a way that ensures data traceability and availability for verification, auditing, and periodic reporting purposes.

f. Organization and responsibilities of the parties involved in all of the above

The organization and responsibilities will be distributed among the project proponents and other parties involved, according to their respective competencies. A specialized team, to be designated, will be responsible for primary data collection, stock estimation, and information reporting. The project's technical coordination will be responsible for requesting activity data from Verra, applying QC/QA procedures, and consolidating and storing the data and information

4. Estimation of ex post net carbon stock changes and greenhouse gas emissions

a. Technical description

The monitoring and analysis of potential changes that may occur in the UDef PA and UDef LB, such as deforestation and degradation. The monitoring period AD will be estimated using a sample-based approach. The AD will be conservatively adjusted to account for estimated statistical uncertainty. In addition to emissions from unplanned deforestation, emissions from biomass burning and potential leakage will also be monitored

b. Data to be collected

Parameter	Description	Unit	Source	Frequency
$AD_{MP,PA\text{-}UDEF,i,t}$	Deforestation in the UDef PA in forest stratum i in year t	ha	Calculated	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
$E_{MP,BiomassBurn,i,t}$	Non-CO_2 emissions due to biomass burning as part of project activities in forest stratum i in year t	t CO_2e	Calculated	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
$\Delta C_{MP,PA\text{-}UDef}$	Cumulative net GHG emissions in the UDef PA in year t	t CO_2e	Calculated	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
$AD_{MP,LB\text{-}UDEF,i,t}$	Deforestation in the UDef LB in forest stratum i in year t	ha	Calculated	Monitoring must be conducted at least every five years, or

				prior to each verification event where verification occurs more frequently
$\Delta C_{LK-AS.t}$	Cumulative net GHG leakage emissions due to the displacement of unplanned deforestation activities in year t	t CO_2e	Calculated	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
$\Delta C_{LK-ME.t}$	Cumulative net GHG emissions due to market-effects leakage in year t	t CO_2e	Calculated	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
$\Delta C_{LK-UDef,t}$	Cumulative net GHG emissions due to leakage from the project activity in year t	t CO_2e	Calculated	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
Buffer%	Buffer withholding percentage	%	Calculated	Monitoring must be conducted at least every five years, or prior to each verification event where verification occurs more frequently
$NER_{UDef,t}$	Cumulative total ex post net GHG emission reductions of the project activity in year t	t CO_2e	Calculated	Monitoring must be conducted at least every five years, or prior to each verification event

				where verification occurs more frequently

c. Overview of data collection procedures

UDef PA and UDef LB Deforestation Data for the Monitoring Period.

Step 1. Analyze changes in land use types- The first step aims to identify changes in land use (Change Category, CHC) as a way to estimate the area of unplanned deforestation at the beginning and end of the monitoring period within the UDef PA and UDef LB. Step 1 in Section 5.3.3.2 of VMD0055 guides this analysis, and the results must be presented according to Table 8 of the module, page 38.

Table 8: Format for recording estimated sampling frame proportions, with reference to associated parameter names

Sampling Stratum (ss)	Change Category (CHC)			Stratum Total
	Deforestation (UDef)	Stable Forest (SF)	Change Category	
Stratum 1	$Prop_{UDef,ss=1}$	$Prop_{SF,ss=1}$	...	$W_{Sss=1}$
Stratum 2	$Prop_{UDef,ss=2}$	$Prop_{SF,ss=2}$	...	$W_{Sss=2}$
Stratum SS	$Prop_{UDef,ss=SS}$	$Prop_{SF,ss=SS}$	...	$W_{Sss=3}$
***CHC* total**	***Prop*$_{UDef}$**	***Prop*$_{SF}$**	**...**	**1**

A sample-based approach will be used, supported by satellite imagery of the areas of interest. This approach generates counts of sample units within each defined sampling stratum (ss). These counts of sample units are referred to as CountCHC,ss for the CHC categories. The total count of sample units in the sampling stratum ss (Countss) will be summarized following the format of Table 6 in VMD0055.

Table 6: Format for recording plot observations, with reference to associated parameter names

Sampling Stratum (ss)	Change Categories (CHC)				Stratum Total
	Unplanned Deforestation (UDef)	Stable Forest (SF)	*CHC*	*CHC*	
ss = 1	$Count_{UDef,ss=1}$	$Count_{SF,ss=1}$	...	...	$Count_{ss=1}$
ss = 2	$Count_{UDef,ss=2}$	$Count_{SF,ss=2}$	...	...	$Count_{ss=2}$
ss = 3	$Count_{UDef,ss=3}$	$Count_{SF,ss=3}$	...	...	$Count_{ss=SS}$

The weights of each sampling stratum (wsss) must be calculated for sampling strata 1, 2, 3, ..., SS by dividing the mapped area of each stratum (Ass) by the total sampling area. Areas that are excluded are not considered part of the project sampling; therefore, Ass should not include any portion of a sampling stratum that would otherwise extend into the identified excluded areas. The data will be presented as shown in Table 7 of VMD0055.

Table 7: Calculation of sample stratum weights

Sampling Stratum (ss)	Sampling Stratum Area (ha)	Sampling Stratum Weight (ratio)
ss = 1	$A_{ss=1}$	$ws_{ss=1} = A_{ss=1} / A_{PSF}$
ss = 2	$A_{ss=2}$	$ws_{ss=2} = A_{ss=2} / A_{PSF}$
ss = SS	$A_{ss=SS}$	$ws_{ss=SS} = A_{ss=SS} / A_{PSF}$
Sampling strata total (SS)	A_{PSF}	1

Where:

Ass = Area of sampling stratum ss within the project sampling framework (ha);

APSF = Project sampling area (ha);

ss = 1, 2, 3, ..., Sampling stratum ss

The counts of sample units for the CHC change classes (unplanned deforestation (UDef), stable forest (SF), stable non-forest (SNF), and regeneration (Reg)) observed within each sampling stratum ss (CountCHC,ss), along with the stratum weights (wsss), are used to estimate the proportion of the area in each CHC change category (PropCHC,ss). For each sampling stratum, the weighted proportion (PropCHC,ss) of the total area (APSF) that falls within sampling stratum ss and is classified as a CHC change category (unitless) must be calculated by multiplying the stratum weight (wsss) by the number of CHC observations in the sampling stratum and dividing by the total observation count for the respective stratum. This will be done using Equation 23 in VMD0055:

$$Prop_{CHC,ss} = ws_{ss} \times \frac{Count_{CHC,ss}}{Count_{ss}}$$

Equation 23 of VMD0055

Step 2. Calculate the Total Area of Each AD Category – This analysis aims to estimate the area of each CHC category, taking into account any land use changes that may have occurred. These areas are referred to as: $A_{PA,CHC,i}$ (the area of UDef PA in stratum i classified as CHC change category during monitoring period i) and $A_{LB,CHC,i}$ (the area of UDef LB in stratum i classified as CHC change category during monitoring period i). Step 2 in Section 5.3.3.2 of VMD0055 guides the calculation of these areas using Equations 24 and 25.

$$A_{PA,CHC,i} = A_{PSF} \times \sum_{ss=1}^{SS} Prop_{PA,CHC,i,ss}$$

Equation 24 df VMD0055 (UDef PA)

$$A_{LB,CHC,i} = A_{PSF} \times \sum_{ss=1}^{SS} Prop_{LB,CHC,i,ss}$$

Equation 25 of VMD0055 (UDef LB)

Step 3. Calculate the Uncertainty of the Estimated Areas for Each Change Category – These calculations take into account the standard error of the CHC proportion $S_{PropCHC}$, the standard error of the areas S_{ACHC}, and the percentage uncertainty of the estimated area of the CHC change category within the project area during the monitoring period $U\%_{ACHC}$. Step 3 in Section 5.3.3.2 of VMD0055 guides these calculations using Equations 26, 27, and 28.

$$S(Prop_{CHC}) = \sqrt{\sum_{ss=1}^{SS} \frac{ws_{ss}^{2} \times \frac{Count_{CHC,ss}}{Count_{ss}} \times \left(1 - \frac{Count_{CHC,ss}}{Count_{ss}}\right)}{Count_{ss} - 1}}$$

Equation 26 of VMD0055

$$S(A_{CHC}) = S(Prop_{CHC}) \times A_{PSF}$$

Equation 27 of VMD0055

$$U\%(A_{CHC}) = t_{alpha=10\%} \times \frac{S(A_{CHC})}{\sum_{i=1}^{M} A_{PA,CHC,i} + \sum_{i=1}^{M} A_{LB,CHC,i}} \times 100$$

Equation 28 of VMD0055

Step 4. Conservatively Inflate the Estimated Area of Unplanned Deforestation – The total estimated area of unplanned deforestation (CHC = UDef) within the project area is inflated based on its level of uncertainty. Where the percentage uncertainty of the estimated unplanned deforestation area is less than or equal to 10%, the estimate may be used without modification, and the inflation factor is zero. Where the percentage uncertainty is greater than 10%[72], the area estimate must be increased using the inflation factor IF_{UDef}. The inflation factor, IF_{UDef}, is calculated using Equation 29 in VMD0055.

$$IF_{UDef} = \frac{U\%(A_{UDef})}{100 \times t_{alpha=10\%}} \times t_{alpha=66.67\%}$$

Equation 29 of VMD0055

Where:

IF_{UDef} = Inflation factor for the area of unplanned deforestation (%)

U%(AUDef) = Percentage uncertainty of the estimated area of unplanned deforestation (where CHC = UDef) within the project area during the monitoring period (%)

[72] The uncertainty cannot exceed the maximum acceptable uncertainty established in the VCS Methodology Requirements.

$t_{alpha=10\%}$ = t-distribution value for a two-sided 90% confidence interval (a value of 1.6449 can be used for DA analyses)

$t_{alpha=66.67\%}$ = t-distribution value for a one-sided 66.67% confidence interval (a value of 0.4307 can be used for AD analyses)

Using the estimated area and the inflation factor, calculate the final AD values for UDef PA and UDef LB, according to Equations 30 and 31 presented in VMD0055.

$$A_{PA,inflated,UDef,i} = A_{PA,UDef,i} \times \left(1 + IF_{UDef}\right)$$

Equation 30 of VMD0055 (UDef-PA)

$$A_{LB,inflated,UDef,i} = A_{LB,UDef,i} \times \left(1 + IF_{UDef}\right)$$

Equation 31 of VMD0055 (UDef-LB)

$A_{PA,inflated,UDef,i}$ and $A_{LB,inflated,UDef,i}$ are the areas of unplanned deforestation within forest stratum i of the UDef PA and UDef LB, respectively, during the monitoring period, conservatively inflated to account for uncertainty (in ha)

Step 5. Determine the AD for Unplanned Deforestation – Annual emissions per stratum in the UDef PA and UDef LB are equal to the AD calculated for the monitoring period divided by the number of years in the monitoring period, according to Equations 32 and 33.

$$AD_{MP,PA-UDef,i,t} = \frac{A_{PA,inflated,UDef,i}}{MPL}$$

Equation 32 of VMD0055 (UDef-PA)

$$AD_{MP,LB-UDef,i,t} = \frac{A_{LB,inflated,UDef,i}}{MPL}$$

Equation 33 of VMD0055 (UDef-LB)

The total emissions from carbon stock changes in all reservoirs in forest stratum i, in year t, in the UDef PA and UDef LB, respectively, will be calculated using Equations 34 and 35 in VMD0055.

$$\begin{aligned}\Delta C_{MP,PA-UDef,i,t} &\\ = \ & (AD_{MP,PA-UDef,i,t} \times \Delta Cc_{MP,AB-LI,i}) \\ & + \frac{1}{10} \times \sum_{t-10}^{t} (AD_{MP,PA-UDef,i,t} \times \Delta Cc_{MP,BB-DW,i,t}) \\ & + \frac{1}{20} \times \sum_{t-20}^{t} (AD_{MP,PA-UDef,i,t} \times \Delta Cc_{MP,SOC-WP,i,t})\end{aligned}$$

Equation 34 of VMD0055 (UDef-PA)

$$\Delta C_{MP,LB-UDef,i,t} = (AD_{MP,LB-UDef,i,t} \times \Delta C_{MP,LB,AB-LI,i}) + \frac{1}{10} \times \sum_{t-10}^{t} (AD_{MP,LB-UDef,i,t} \times \Delta C_{MP,LB,BB-DW,i,t}) + \frac{1}{20} \times \sum_{t-20}^{t} (AD_{MP,LB-UDef,i,t} \times \Delta C_{MP,LB,SOC-WP,i,t})$$

Equation 35 of VMD0055 (UDef-LB)

Similarly, the aggregation of all strata i, at time t, will be calculated using Equations 36 and 37 in VMD0055.

$$\Delta C_{MP,PA-UDef,t} = \sum_{i=1}^{M} \Delta C_{MP,PA-UDef,i,t}$$

Equation 36 of VMD0055 (UDef-PA)

$$\Delta C_{MP,LB-UDef,t} = \sum_{i=1}^{M} \Delta C_{MP,LB-UDef,i,t}$$

Equation 37 of VMD0055 (UDef-LB)

Ex-post Estimation of Other GHG Emissions in the UDef PA

Equation 1 of module VMD0013 (E-BPB), previously calculated using $A_{burn,i,t}$ (burned area in stratum i in year t, in ha) to generate $E_{BSL,BiomassBurn,i,t}$ (non-CO_2 emissions from biomass burning in the baseline scenario), will, in the ex-post analysis, use the parameter $E_{MP,BiomassBurn,i,t}$ instead.

Thus, the ex-ante Equations 20 and 21 will be adapted to ex-post Equations 38 and 39.

$$GHG_{MP_{PA}-UDef,E,t} = \sum_{t=1}^{t*} \sum_{i=1}^{M} E_{MP,BiomassBurn,i,t}$$

Equation 38 of VMD0055 (adapted)

Estimation of Net Emissions Caused by Unplanned Deforestation and Degradation

The net emissions, which were previously calculated using Equations 21 and 22, will now be calculated using Equations 39 and 40 in VMD0055:

$$\Delta C_{MP,PA-UDef,t} = \sum_{t=1}^{t*} \sum_{i=1}^{M} (\Delta C_{MP,PA-UDef,i,t}) + GHG_{MP_{PA}-UDef,E,t}$$

Equation 39 of VMD0055 (UDef PA)

$$\Delta C_{MP,LB-UDef,t} = \sum_{t=1}^{t*} \sum_{i=1}^{M} (\Delta C_{MP,LB-UDef,i,t})$$

Equation 40 of VMD0055 (UDef LB)

Ex-post Estimation of Emissions Due to Leakage

Ex-post monitoring of leakage is essential to accurately quantify the unintended displacement of deforestation and the associated greenhouse gas emissions outside the project area. Four types of leakage must be assessed:

(i) Activity shifting by geographically constrained agents, monitored within the Leakage Belt (LB);

The difference in carbon stocks between the baseline scenario and the monitoring period within the UDef LB, from the project start date to year t, will be calculated as:

$$\Delta C_{LK-net-LB,t} = \Delta C_{BSL,LB-UDef,t} - \Delta C_{MP,LB-UDef,t}$$

Equation 41 of VMD0055

If ΔCLK-net-LB,t < 0: This means that the actual emissions were lower than projected — that is, there was less deforestation than expected.

These negative values will not be attributed as positive leakage to the project. They are handled in subsequent steps to ensure that the project is not unfairly penalized.

Where significant (as determined following Appendix 1 of VM0048), other GHG emissions occurring in the UDef LB will be evaluated with Equation 42 and 43 of the Module VMD0055:

$$GHG_{LB,E,i,t} = \frac{\sum_{t=1}^{t*}(E_{BSL,FC,i,t} + E_{BSL,BiomassBurn,i,t} + N2O_{BSL,direct-N,i,t})}{\sum_{t=1}^{t*} AD_{BSL,PA-UDef,i,t}}$$

Equation 42 of VMD0055

$$GHG_{MP,LK-UDef,E,t} = \sum_{t=1}^{t*}\sum_{i=1}^{M}\left(\left(AD_{BSL,LB-UDef,i,t} - AD_{MP,LB-UDef,i,t}\right) \times GHG_{LB,E,i,t}\right)$$

Equation 43 of VMD0055

The net GHG emissions are summed for the UDef LB as:

$$\Delta C_{LK,LB,t} = \Delta C_{LK-net-LB,t} + GHG_{MP,LK-UDef,E,t}$$

Equation 44 of VMD0055

(ii) Activity shifting by geographically mobile agents is estimated on the basis of:

a) The net reduction (displacement) of deforestation from the baseline to the end of the monitoring period within the combined area of the UDef PA and UDef LB;

b) The proportion of geographically mobile deforestation agents projected in the baseline. This is calculated considering the Proportion of Migrated Land Cover Transition Agents in the Baseline (PROPMIG). For the first baseline, a conservative value of 1.0 will be assigned, in accordance with Section 5.3.4.4 of Module VMD0055.

c) Area-weighted emission factors for areas accessible to geographically mobile deforestation agents outside the UDef LB, calculated and provided by Verra. For the first baseline, the value will be 424.99 tCO2e.

It is conservatively assumed that geographically mobile agents of unplanned deforestation recently settled in the project activities region are primarily driven by a need to secure agricultural land. The amount of leakage to areas outside the UDef LB is taken as the total area of avoided land cover transition in the UDef PA, scaled by the proportion of recent migration (PROPMIG)

$$AD_{AS-OLB,t} = PROP_{MIG,t} \times \sum_{t=1}^{t*} \sum_{i=1}^{M} (AD_{BSL,PA-UDef,i,t} - AD_{MP,PA-UDef,i,t})$$

Equation 45 of VMD0055

The area of deforestation that is displaced from the UDef PA ($AD_{AS\text{-}OLB,t}$) is assumed to result in land cover conversion of an equal extent outside the UDef LB. Cumulative total emissions from carbon stock change due to activity shifting to areas available outside the UDef LB is calculated by the equation below:

$$\Delta C_{LK,OLB,t} = AD_{AS-OLB,t} \times \Delta C_{OLB,t}$$

Equation 46 of VMD0055

Total activity-shifting leakage emissions are the sum of leakage from within and outside the UDef LB. Where total leakage is calculated to be less than zero, ΔCLK-AS,t is assigned a value of zero.

$$\Delta C_{LK-AS,t} = \Delta C_{LK,LB,t} + \Delta C_{LK,OLB,t}$$

Equation 47 of VMD0055

iii) Market-effects leakage: occurs where in the baseline case commodities were produced for regional, national or international markets. Since the collection of native forest wood for charcoal production has been identified through socioeconomic studies conducted in the project activity zone, these emissions will be accounted for using VMD0011: Estimation of Emissions from Market-effects (LK-ME), Section 5.2.

iv) Emissions due to leakage mitigation measures: They correspond to non-CO2 GHG emissions from biomass burning or fertilizer use, and to tCO2e emissions from carbon stock changes in above- and belowground tree biomass in the leakage management zone. However, activities proposed by Caiarara REDD+ Project are not expected to result in significant emission increases; therefore, they will not be accounted for.

Finally, total leakage emissions are equal to the sum of emissions from activity shifting, market-effects, and GHG emissions associated with leakage mitigation measures, and will be calculated using the equation below.

$$\Delta C_{LK-UDef,t} = \Delta C_{LK-AS,t} + \Delta C_{LK-ME,t} + GHG_{LK,E,t}$$

Equation 49 of VMD0055

Net GHG Emission Reductions

The total net GHG emissions reductions of the avoiding unplanned deforestation project activity are calculated as follows:

$$NER_{UDef,t} = \Delta C_{BSL,PA-UDef,t} - \Delta C_{MP,PA-UDef,t} - \Delta C_{LK-UDef,t}$$

Equation 50 of VMD0055

Calculation of AFOLU Pooled Buffer Account Contribution

The number of credits to be held in the AFOLU pooled buffer account is determined as a percentage of the total carbon stock benefits. For avoiding unplanned deforestation project activities, this is calculated using the equation below. Leakage emissions do not factor into the buffer calculations.

$$Buffer_{Total,t} = \left(\Delta C_{BSL,PA-UDef,t} - \Delta C_{MP,PA-UDef,t}\right) \times Buffer\%$$

Equation 51 of VMD0055

Calculation of Verified Carbon Units

The number of potential Verified Carbon Units (VCUs) for years in which monitoring has been conducted and submitted for verification is calculated as follows:

$$VCU_{AUDef,t} = \left(NER_{UDef,t} - NER_{UDef,t-1}\right) - \left(Buffer_{Total,t} - Buffer_{Total,t-1}\right)$$

Equation 52 of VMD0055

Where this equation results in a decimal number, the number will be rounded down.

d. Quality control and quality assurance procedures

Standard Operating Procedures (SOPs) and specific protocols will be applied to address any discrepancies, measurement errors, or data gaps, ensuring reliability and traceability. These documents will be available as evidence.

e. Data archiving

All collected data will be archived in secure digital systems, with regular backups and access control. The files will include detailed metadata, field records, georeferenced images, thematic maps, and technical reports. Archiving will be carried out in a way that ensures data traceability and availability for verification, auditing, and periodic reporting purposes.

f. Organization and responsibilities of the parties involved in all of the above

Responsibilities will be allocated according to technical expertise:

- The geoprocessing team will manage spatial analysis according to the SOP;
- Field technicians will conduct on-site data collection;
- The quality assurance team will oversee QC/QA protocols;
- Project coordination will be responsible for data integration, reporting, and communication with Verra and other stakeholders.

TASK 2 - Baseline reassessment at the close of a fixed baseline period

The methodological procedure to update the baseline will be the same as in the first baseline estimation and will meet all the requirements described in Section 3.2.6 of the VCS Standard v4.

3.3.4 Disclosure of the Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

The Climate Monitoring Plan and all its results will be publicly disclosed on the VERRA registration page and on the Biofilica Ambipar Environmental Investments S/A website. Validated and verified monitoring results will be available in the public version of the report on the Verra website, ensuring free access to any interested party.

Summaries of the main results will be prepared in accessible language and presented in person to the communities through adapted printed and digital materials. Copies will also be made available at strategic locations (farm headquarters and community centers). The results will be disclosed after each official project verification, with regular updates following the monitoring cycles defined in the PDD.

Information about the Project will also be disseminated via email and WhatsApp (with a pre-established list of contacts of interested parties), and on social media, such as Instagram and LinkedIn, of the project proponent.

In this way, it is expected that all necessary efforts to provide full transparency and distribute key Project information and documents will be achieved.

3.4 Optional Criterion: Benefits of Adaptation to Climate Change

3.4.1 Regional Climate Change Scenarios (CCB, GL1.1)

Climate change represents one of the greatest global challenges of the 21st century. One of the main indicators of this phenomenon is the increase in global average temperature, which has already risen by approximately 1.1°C since pre-industrial levels, according to the Intergovernmental Panel on Climate Change (IPCC)[73] . This warming has been driven mainly by the increase in the concentration of greenhouse

[73] IPCC. Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, 2022. Available at: https://www.ipcc.ch/report/ar6/wg2/

gases (GHG) in the atmosphere, with direct impacts on precipitation patterns, the frequency and intensity of extreme events, and the availability of essential natural resources, such as water and fertile soil.

In Brazil, the effects of climate change are noticeable and are likely to intensify in the coming years. Studies indicate that the country may face more frequent prolonged droughts, reduced water availability in several regions, and changes in rainfall patterns, with a direct impact on agriculture, food security, and biodiversity conservation (Brazilian Panel on Climate Change, 2016)[74] (World Bank, 2024)[75] .

The country's size and geographical diversity result in a wide range of climatic conditions, from equatorial and tropical climates in the north and center to humid subtropical climates in the south and semi-arid conditions in the northeast. The following data are from the Brazil Climate Risk Profile Report, prepared by the World Bank (2025)[76] :

- Historical warming: Surface air temperatures have been rising steadily, with accelerating trends over time—0.17 °C per decade from 1951 to 2020, increasing to 0.29 °C per decade from 1991 to 2020.
- Future projections: According to the high emissions SSP3-7.0 scenario, warming is projected to intensify – increasing by 0.34 °C per decade from 2000 to 2050 and 0.51 °C per decade from 2050 to 2100.
- Precipitation trends: Historically, precipitation has decreased by 33.4 mm per decade (1.9% per decade), with the largest declines in the northeast and south. Projections indicate further national declines, with an average reduction of 63 mm (3.6%) by mid-century, especially in the northwest.
- Intensification of the dry season: Rainy seasons are projected to decrease, while dry seasons increase, exacerbating drought risks.
- Heat extremes: Maximum temperatures are rising faster than minimum temperatures, increasing the frequency of extremely hot days. Future warming will increase daytime and nighttime temperatures, amplifying risks to health, agriculture, and ecosystems.
- Extreme events: Heavy rainfall events are expected to occur more frequently and with shorter return periods, posing greater risks of flooding and landslides.

The Amazon, in particular, is one of the most vulnerable regions due to its essential role in regulating the climate and water cycle of the entire South American continent. Reduced forest cover could compromise the transport of moisture to other regions of the country, affecting areas such as the Midwest and Southeast, which are responsible for a large part of the country's agricultural production (Nobre, 2021)[77] .

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[74] PBMC – Brazilian Panel on Climate Change. Impacts, Vulnerabilities and Adaptation to Climate Change. Executive Summary, 2016. Available at: https://www.pbmc.coppe.ufrj.br/

[75] WORLD BANK. Climate Change Knowledge Portal. World Bank Group, 2024. Available at: https://climateknowledgeportal.worldbank.org/

[76] WORLD BANK. Climate Risk Country Profile. Brazil. 2025. Available at:https://climateknowledgeportal.worldbank.org/sites/default/files/country-profiles/17303-WB_Brazil%20Country%20Profile-WEB.pdf

[77] NOBRE, Carlos A.; SATORI, H.; SCARANO, F. R. Amazonia against the clock: A regional pact to keep the Amazon biome forests standing. One Earth, Volume 4, Issue 7, 2021, Pages 939–945. Available at: https://www.sciencedirect.com/science/article/pii/S2590332221001883

Climate change also has a significant influence on land use patterns. Rising temperatures and irregular rainfall can lead to a loss of productivity in natural and agroforestry systems, favoring the advance of unsustainable activities such as the expansion of extensive livestock farming and illegal deforestation. Environmental degradation, coupled with economic pressures, tends to encourage the conversion of forest areas to alternative uses of lower ecological value, resulting in the fragmentation of ecosystems and a reduction in ecosystem services (Marengo, 2022)[78] .

In the Amazon context, this process can be even more critical. The reduction in atmospheric humidity, aggravated by the loss of vegetation cover, compromises the forest's natural regeneration capacity and alters the dynamics of local production systems. Studies indicate that, without mitigation measures, part of the Amazon rainforest may enter an irreversible process of savannization, and that savanna ecosystems are expanding in the center of the Amazon (Flores & Holmgren, 2021)[79] .

Projections from the Climate Change Knowledge portal[80] , a platform developed by the World Bank to provide information, data, and tools related to climate change and development, show that the average surface air temperature in the state of Pará has already increased since 1950. According to the most optimistic scenario (SSP1), warming will be limited to about 1°C additional by the end of the century. In the absence of climate action, as presented in the SSP5 scenario, Pará could face warming of up to 6-10°C (depending on model uncertainty) above the historical average, which would have serious impacts on health, biodiversity, agriculture, and water availability.

[78] MARENGO, J. A. et al. Climate change and health in cities in Brazil. The Lancet Regional Health – Americas, v. 5, 2022. Available at: https://www.sciencedirect.com/science/article/pii/S2667193X2100188X.

[79] Flores, B.M., Holmgren, M. White-Sand Savannas Expand at the Core of the Amazon After Forest Wildfires. *Ecosystems* **24,** 1624–1637 (2021). https://doi.org/10.1007/s10021-021-00607-x

[80] The data presented are from CMIP6, derived from the sixth phase of CMIPs and available at: https://climateknowledgeportal.worldbank.org/country/brazil/climate-data-projections . CMIPs form the database for the IPCC Assessment Reports. CMIP6 is the most recent phase of a collaborative research project that aims to improve understanding of past, present, and future climate change by providing comprehensive and reliable climate projections to inform adaptation policies and strategies.

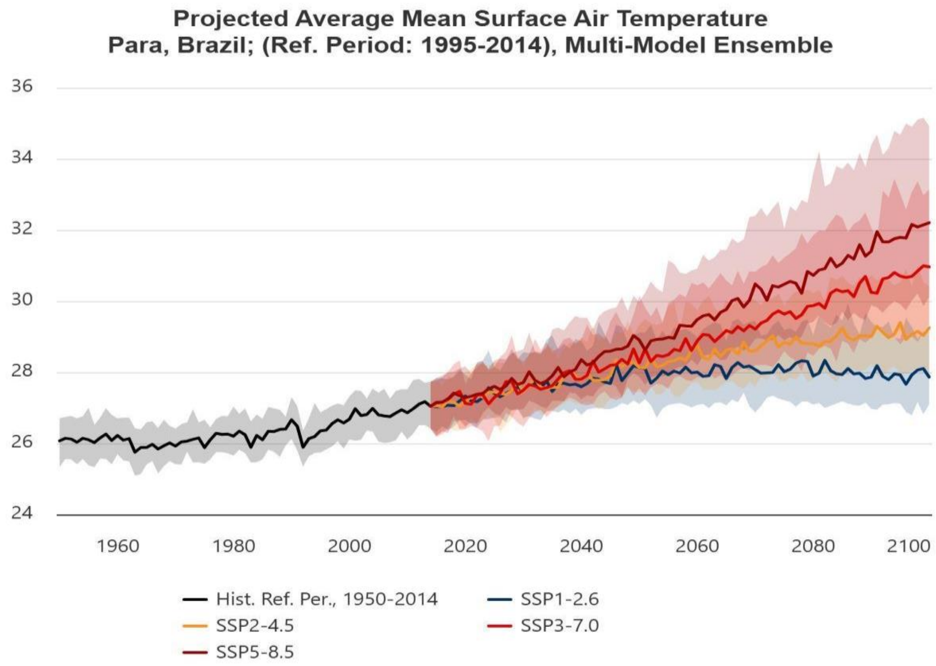


Figure 30. Average surface air temperature, state of Pará, Brazil.

Regarding precipitation projections until 2100, high emission scenarios show a significant drop in precipitation by the end of the century.

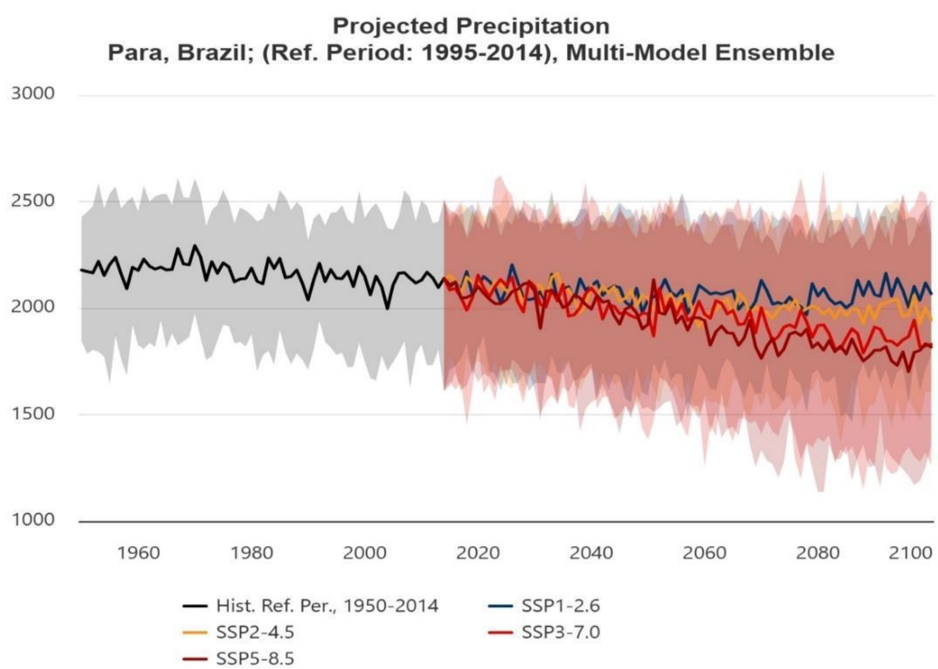


Figure 31. Precipitation projection, state of Pará, Brazil.

More precisely in the municipality of Breu Branco, where the Caiarara REDD+ Project is located, there is already evidence of worsening droughts and reduced precipitation, reflecting trends identified by regional studies (INPE, 2021) and by community reports during the socioeconomic assessment conducted in the area. This scenario directly threatens the integrity of local forests and could trigger processes of forest degradation, biodiversity loss, water scarcity, and increased vulnerability of communities that depend directly on natural resources. The intensification of droughts and reduced water availability affect the ecological balance of the region, hindering the natural regeneration of vegetation and impacting the life cycles of local fauna and flora.

A study conducted by Silva et al. (2021)[81] analyzed the impacts of extreme drought in southeastern Pará between 2015 and 2020 and identified a significant reduction in the forest leaf area index, indicating prolonged water stress and compromised photosynthetic capacity of native vegetation. The authors also reported increased mortality of drought-sensitive tree species, as well as changes in the phenological patterns of regional flora, which may have cascading effects on local fauna and ecological cycles. The study concludes that prolonged drought events have the potential to profoundly alter the structure and functionality of forest ecosystems in the region, reinforcing the urgency of mitigation and conservation strategies.

In a scenario without the project, these trends would intensify. The absence of concrete conservation and mitigation actions would lead to further deforestation and environmental degradation, contributing even more to GHG emissions and aggravating the effects of climate change. On the other hand, with the implementation of the Caiarara REDD+ Project in Breu Branco, it is possible to promote the conservation of native forests, significantly reducing carbon emissions and maintaining essential ecosystem services such as climate regulation, the water cycle, and biodiversity protection.

In addition, the preservation of vegetation directly contributes to the maintenance of local water resources, which are fundamental for both ecosystems and the livelihoods of vulnerable communities. The project also has the potential to generate a multiplier effect: by establishing a conservation model based on incentives and the valuation of standing forests, it can positively influence other rural landowners and companies in the region to adopt similar practices, fostering a network of REDD+ projects that strengthen regional climate resilience. The project's actions also mobilize and engage local stakeholders, promoting a culture of care and appreciation for natural resources.

3.4.2 Climate Change Impacts (CCB, GL1.2)

Climate change has had growing impacts on Amazonian ecosystems and the human populations that depend on them. Rising average temperatures, more extreme events, and changes in precipitation patterns

[81] SILVA, R. P.; OLIVEIRA, A. A.; COSTA, F. R. C. Effects of extreme droughts on forest structure and functioning in southeastern Pará. Revista Brasileira de Climatologia, v. 29, p. 112-129, 2021. Available at: https://revistas.ufpr.br/revistaclima/article/view/88043

have compromised the ecological integrity of the Amazon, affecting biodiversity, the hydrological cycle, and regional climate stability (Marengo, 2020)[82] .

In the state of Pará, these effects are increasingly noticeable, with more frequent and intense droughts, more forest fires, and less water available. These things have had a direct impact on local natural and socioeconomic systems. The region's biodiversity faces increasing risks due to habitat degradation, forest fragmentation, and water stress, factors that compromise the resilience of ecosystems and the regenerative capacity of native species (Silva, 2021)[83] .

A study conducted by Silva et al. (2021)[84] showed that extreme droughts in southeastern Pará resulted in significant loss of vegetation cover and reduced photosynthetic capacity of the forest, with direct impacts on the ecological structure and survival of species sensitive to water stress. The same study pointed out that these effects are reflected in the availability of natural resources for local communities, also affecting their quality of life and food security.

The effects of climate change on the well-being of communities are evident. Local communities in the rural area of the municipality of Breu Branco, where the Caiarara REDD+ Project is located, mainly practice subsistence agriculture, without access to modern technologies and with low capacity to adapt to climate variations. The socioeconomic assessment carried out in the area (STA, 2024) revealed that drought has drastically impacted agricultural production, reducing the availability of food for family consumption and decreasing income from the sale of surplus crops. Food insecurity therefore becomes a real risk for these families, especially during prolonged periods of drought. In addition, water scarcity directly affects the daily lives of communities, hindering access to drinking water and the development of productive activities. The situation worsens during periods of intense drought, when local water sources drastically reduce their volume, compromising the health and well-being of residents.

In the context of the project, forest conservation promoted by REDD+ Caiarara plays a crucial role in mitigating these impacts. Maintaining forest cover contributes to microclimate regulation, water resource protection, and the preservation of natural habitats for local fauna. Forest fragmentation, aggravated by deforestation, compromises the survival of species that need large continuous areas to maintain their reproductive and feeding cycles; reduced relative humidity and increased incidence of forest fires also directly affect these animals.

Field studies in the project area have identified the presence of endangered species such as the Macaco Cairara (*Cebus kaapori*) and the Tiriba-pérola (*Pyrrhura coerulescens*), whose survival depends directly on the preservation of forest ecosystems. Climate change intensifies the risks faced by these species by reducing their habitats and increasing their ecological vulnerability to environmental disturbances.

[82] MARENGO, J. A. et al. Climate Change in Brazil: Vulnerabilities and Adaptation. Advanced Studies, v. 34, n. 99, 2020. Available at: https://www.scielo.br/j/ea/a/RF7ym5qTQbGvZT3Th3nLz5y/?lang=pt

[83] SILVA, R. P.; OLIVEIRA, A. A.; COSTA, F. R. C. Effects of extreme droughts on the structure and functioning of forests in southeastern Pará. Brazilian Journal of Climatology, v. 29, p. 112-129, 2021. Available at: https://revistas.ufpr.br/revistaclima/article/view/88043

[84] Ibid

Thus, by preserving the forest and its associated resources, the Caiarara REDD+ Project contributes significantly to strengthening the ecological and social resilience of the region. The implementation of conservation actions promotes the maintenance of essential ecosystem services, reducing the adverse impacts of climate change. In addition, the project acts as an important instrument for local adaptation, ensuring better conditions for biodiversity conservation, water resource protection, and improved well-being for local communities. By ensuring that forest areas remain standing, the project actively contributes to halting environmental degradation and creating a more balanced environment that is resistant to future climate change.

3.4.3 Necessary and designed adaptation measures (CCB, GL1.3)

Based on the Theory of Change, the table below presents the activities designed to assist communities and biodiversity in adapting to the impacts of climate change.

Table 28. Project activities to assist communities and biodiversity in adapting to climate change.

Activity	Climate change risks and issues addressed	Project activities for adaptation to climate change	Indicators[85]
Strengthening asset surveillance and monitoring of deforestation and hot spots	* Increased deforestation. * Damage caused by forest fires. * Increased CO_2 emissions. * Loss of biodiversity.	* Monitoring and surveillance in the project area to prevent the area from being invaded for deforestation or illegal hunting, avoiding greenhouse gas emissions and biodiversity loss. * Satellite monitoring allows for the immediate identification of possible fire spots, enabling a rapid response in the field to verify and mitigate possible damage. * The preservation of these areas helps maintain ecosystem resilience, especially for biodiversity, in relation to the challenges posed by climate change.	Deforested area. Available in section 3.3.2.
Biodiversity conservation and monitoring	* Population decline. * Species extinction. * Loss of biodiversity.	* Monitoring of endangered and endemic species, enabling the implementation of a species conservation plan.	Macaco Cairara (*Cebus kaapori*). Available in section 5.5.2.

[85] The indicators used already exist and are described in the joint plans for the Climate, Community, and Biodiversity scopes.

Development and strengthening of value chains	* Food insecurity. * Decrease in income generation. * Decreased productivity due to longer periods of drought.	* Through technical assistance and rural extension, it is possible to provide knowledge of technical and sustainable practices to address the consequences of climate change, such as the lengthening and intensification of droughts already observed by community members.	Number of education/training activities carried out. Available in section 4.4.1.
Promotion of education actions	* Environmental degradation. * Forest fires. * Loss of biodiversity due to illegal hunting.	* Raising awareness of the importance of wild flora and fauna, reducing illegal deforestation, hunting, and fishing practices. * Community members trained to take action on various environmental fronts, such as techniques to prevent the spread of forest fires and how to fight them.	Number of education/training activities carried out. Available in section 4.4.1.

4 COMMUNITY

4.1 Community scenario without project

4.1.1 Descriptions of Communities at Project Start (CCB, CM1.1)

To describe the communities at the start of the project, a socioeconomic assessment was carried out by a specialized team from Soluções Técnicas e Serviços de Engenharia Ambiental Ltda (STA) between October 2023 and April 2024. The method used to characterize and identify the population was based on secondary and primary data obtained through field surveys, combined with the systematization of previous studies and surveys, as well as available information. The analysis of the social, economic, environmental, and cultural dimensions includes elements such as historical and current occupation, population density, number of social groups and actors, levels of education, and access to education and health services. The identification and characterization in the field was carried out with communities in the region and workers from the proponent's farms.

4.1.1.1 Surrounding communities

4.1.1.1.1 History of community occupation

The most recent history of occupation in the region can be traced back to the government strategy for the Amazon in the 1960s and 1970s, which intensified the migratory process to the region from the 1970s onwards, almost tripling its population (CAVALCANTE, 2005). Large projects such as the Tucuruí Hydroelectric Plant caused countless people to move to the region with a common goal: to find work on the construction sites. This transformed the region into a significant population attraction between the 1970s and 1980s (CAVALCANTE, 2005).

The municipality of Breu Branco, where the Project is located, developed to meet the demand for immigration from these people. With the emancipation of the village of Breu Branco in the 1990s and its rise to municipality, it began to experience moments of development with the exploitation of silicon ore available in the region, as well as timber exploitation.

However, not all of the people who moved to the region in search of opportunities in large projects were able to find work, as the number of people seeking employment opportunities was much higher than the number of jobs available. Thus, individuals and families had to find other ways to support themselves in the region, one of which was migration to rural areas and subsistence farming.

This new working relationship in the countryside ended up influencing the emergence and expansion of new rural communities, as is the case in the municipality of Breu Branco, where there are several recent communities. The table below shows the year each community was founded, showing that most of them are recent.

Table 29. Year of origin of communities around the Caiarara REDD+ Project. Source: Field research, 2023/2024. Prepared by: STA, 2024.

Communities	Year of origin of the community	Years of existence
Branquelândia	2003	21
Café Brasil	1986	38
California	1998	26
Jutaí II	1990	34
Mamorana	2008	16
Roça Comprida	1991	33
Vila dos Remédios	1991	33
Vila Monte Alegre	1986	38
Vila Carol (1)	2004	20
Assentamento Irmã Dorothy	2009	15

PA Boa Esperança	1997	27

Note: (1) Data collection was carried out using a simplified questionnaire with key informants.

Therefore, we can consider that these are not traditional communities, since they are composed of families who came from other states and cities in search of quality of life and employment opportunities.

Currently, the total population of the communities identified for this project is 5,505 inhabitants, grouped into 1,294 families. The table below shows the number of people and families in each community identified.

Table 30. Number of families and residents per community. Source: Field research, 2023/2024. Prepared by: STA, 2024.

Communities	No. of residents	No. of families
Branquelândia	500	120
Café Brasil	149	80
California (1)	352	80
Jutaí II	122	30
Mamorana	1200	240
Roça Comprida	2,300	460
Vila dos Remédios	248	100
Vila Monte Alegre	84	30
Vila Carol (2)	200	34
Assentamento Irmã Dorothy	150	30
PA Boa Esperança	200	90
Total	5,505	1,294

Notes: (1) The California Community estimated the number of families; (2) Data collection was carried out using a simplified questionnaire with key informant.

With regard to the migratory flow of people leaving the communities, a classification was made during the socioeconomic diagnosis. Based on a matrix identifying the main reasons for people leaving their communities, it was possible to map reasons according to age group and gender, according to six categories: work, financial, education, professional qualification, health, and marriage.

The main reasons found for people leaving their communities are the search for education, job opportunities, and better health conditions. Most people who leave their communities to study return after completing their studies or training. Displacement for health reasons is common, mainly involving elderly people who move in search of medical care and usually return shortly afterwards.

4.1.1.1.2 Diversity within the community

- **Age of the population in the communities**

According to data collected during the socioeconomic assessment, adult men are the largest population group in the area, totaling 1,837 people, followed by adult women, totaling 1,498. Children are the third largest group, with 867 people. Next comes the group of adolescents, aged 13 to 17, totaling 694 people. The elderly are the smallest group, with 409 people. The table below shows the number of people by gender and community.

Table 31. Number of people by age group and gender in communities. Source: Field research, 2023/2024. Prepared by: STA, 2024.

Communities	Adult men	Adult women	Elderly	Children	Adolescents	Total
Branquelândia	135	125	53	120	67	500
Café Brasil	55	41	25	20	8	149
California	90	100	50	66	46	352
Jutaí II	30	27	12	25	28	122
Mamorana	510	270	100	165	155	1200
Roça Comprida	802	741	113	323	321	2300
Vila dos Remédios	76	60	22	70	20	248
Vila Monte Alegre	16	17	10	33	8	84
Assentamento Irmã Dorothy	65	47	11	23	4	150
PA Boa Esperança	58	70	13	22	37	200
Total	1,837	1,498	409	867	694	5,305

- **Ethnicity of the population in the communities**

The majority of the population in the communities located around the project area is of mixed race, accounting for between 90% and 65% of the total population. The table below shows the ethnic identification gathered during the survey, based on the population's self-identification.

Table 32. Percentage of each ethnic group in the population of the communities surveyed. Source: Field research, 2023/2024. Prepared by: STA, 2024.

Community	Identified ethnicity (%)			
	White	Black	Mixed	Indigenous
Branquelândia	2%	8%	90%	0%
Café Brasil	5%	25%	70%	0%
California	0%	30%	70%	0%
Jutaí II	10%	20%	70%	0%
Mamorana	2%	18%	80%	0%
Roça Comprida	4%	16%	80%	0%
Vila dos Remédios	5%	30%	65%	0%
Vila Monte Alegre	0%	30%	70%	0%
Assentamento Irmã Dorothy	2%	8%	90%	0%
PA Boa Esperança	No information.			

4.1.1.1.3 Education

Some communities have primary schools structures (up to 4th grade), such as Branquelândia, Mamorana, and Vila dos Remédios. In the communities of Califórnia and Roça Comprida, there are schools offering complete primary education.

In order for students to attend secondary school, they must travel to the city of Breu Branco using school transportation provided as a public service, which, according to reports, serves all identified communities. There are no daycare centers in the communities.

All communities have access to regular school meals, with the exception of the Assentamento Irmã Dorothy and the Boa Esperança PA, which reported not having access.

The school enrollment rate was not collected, as families did not have this information. During field activities, the team tried to interact with schools to verify this data, but without success. Thus, we present below, according to the IBGE, data by color/race and age groups in relation to the literacy rate by municipality. Considering the overall total, it can be said that younger people have higher literacy rates in the municipalities in the Project region.

Table 33. Literacy rate of people aged 15 years or older by color or race and age groups. Source: 2022 Demographic Census, IBGE. Prepared by: STA, 2024.

Municipality	Age group	Color or race					
		Total	White	Black	Yellow	Mixed	Indigenous
Breu Branco (PA)	15 to 19	97.28	97.23	96	100	97.51	75
	20 to 24	96.94	96.66	97.22	100	96.93	100
	25 to 34	95.69	96.46	95.14	100	95.6	100
	35 to 44	89.33	93.99	87.72	100	88.52	82.35
	45 to 54	79.83	83.83	71.74	100	80.44	75
	55 to 64	65.17	72.67	56.75	50	64.81	84.62
	65 years or older	49.57	58.01	41.57	33.33	48.86	50
Goianésia do Pará (PA)	15 to 19	97.99	98.04	98.8	100	97.84	100
	20 to 24	96.8	96.16	96.22	-	97.01	100
	25 to 34	95.2	96.27	92.34	100	95.41	95
	35 to 44	89.26	91.01	85.09	100	89.58	70
	45 to 54	77.4	83.1	65.13	100	78.27	76.92
	55 to 64	65.48	74.82	55.43	100	64.62	60
	65 years or older	49.92	61.9	36.6	100	48.41	25
Moju (PA)	15 to 19	96.95	96.79	96.76	100	97.03	90

	20 to 24	96.53	96.72	96.6	88.89	96.52	90
	25 to 34	95.26	95.18	94.36	100	95.45	95.24
	35 to 44	89.67	91.33	88.05	10	89.7	91.67
	45 to 54	80.64	85.34	78.59	100	80.33	70.59
	55 to 64	70.69	76.38	65.28	83.33	70.88	57.14
	65 years or older	56.7	60.88	52.83	85.71	56.88	30

4.1.1.1.4 Health

All communities have some access to public health services. Some of them have health centers, basic health units, or are visited by community health workers on a regular basis. The table below shows infrastructure and visit information by community.

Table 34. Public health services available. Source: Field research, 2023/2024. Prepared by: STA, 2024.

Communities	Health Center	Basic Health Unit	Visits by Community Health Agents (ACS)	Frequency of ACS visits	Other infrastructure/services, etc.
Branquelândia	-	-	Yes	Twice a week	Visit from the Family Health Doctor (every 2 months)
Café Brasil	-	-	Yes	Once a month	-
California	-	-	Yes	Did not know how often	Visit from the Family Health Doctor (2 to 3 times a year)
Jutaí II	-	-	Yes	Once every two months	-
Mamorana	-	Yes	Yes	Once a month	Visit to the family doctor (once a month)

					PCCU exam (once a month)
Roça Comprida	Yes	-	Yes	Once a month	Doctor's appointment (3 times a week)
Vila dos Remédios	-	Yes	Yes	Once a month	Ambulance available Visit from family doctor (once a month)
Vila Monte Alegre	-	-	Yes	Once a month	-
Assentamento Irmã Dorothy	-	-	Yes	Once a month	Vaccination campaign (periodic)
PA Boa Esperança	Yes	-	Yes	1 to 2 times a month	Visit from the Family Health Doctor (once a month)

In more serious health situations, such as fractures, heart attacks, cancer, diabetes, snake bites, need for specialized consultations, among others, families seek care in the city of Breu Branco, mainly at the Health Unit or the Breu Branco Hospital.

Although families have access to some health services and support for transportation to hospitals in the urban center of the city, they frequently use their knowledge of medicinal plants to treat everyday illnesses in the communities.

Information was collected on the main causes of death in the communities over the last 20 years (since 2004). The data show that there were deaths related to age, when referring to the term "old age" to describe the causes of death of elderly people in the community. There is also a group of diseases related to health care, such as diabetes, heart attack, and stroke. Deaths due to cancer and accident-related diseases (drowning and "lightning") were also recorded. It is important to note that there were reports associated with knives (PA Boa Esperança) and alcohol (Café Brasil).

With regard to the most frequent diseases in the communities by age group, the following table lists the diseases that most affect children, young people, adult women, elderly women, adult men, and elderly men. Children and young people are the most affected by viruses, diarrhea, and respiratory infections in general, such as influenza.

Table 35. Most frequent diseases in communities by age group and sex. Source: Field research, 2023/2024. Prepared by: STA, 2024.

Communities	Children	Young	Adult women	Elderly women	Adult men	Elderly men
Branquelândia	Influenza, diarrhea, pneumonia	Flu	Hypertension, diabetes	Hypertension, diabetes	Spine, hypertension	Spine, hypertension
Café Brasil	Flu	Flu	Flu	Flu, high blood pressure, diabetes	Flu	Flu, high blood pressure, diabetes
California	Flu, diarrhea	Flu, fatty liver	High blood pressure, fatty liver	Osteoarthritis, high blood pressure, diabetes	High blood pressure and diabetes	Diabetes, high blood pressure
Jutaí II	Flu, diarrhea	Flu	Flu	Flu, high blood pressure	Flu	Flu, high blood pressure
Mamorana	Diarrhea, vomiting, flu	Headache, anxiety	Hypertension, diabetes, spine	Hypertension, diabetes, spine	Spine	Spine, hypertension, osteoarthritis
Roça Comprida	Flu, vomiting, diarrhea	Flu, diarrhea, appendicitis	Spine, hypertension, diabetes	Spine, hypertension, diabetes	Spine, prostate, hypertension	Spine, prostate, hypertension
Vila dos Remédio	Flu	Flu	Flu, high blood pressure	Flu, diabetes	Flu, high blood pressure	Flu, diabetes
Vila Monte Alegre	Stomach ache, flu	Stomach ache, flu	Stomach ache, flu	Diabetes, headache	Diabetes, headache	Bone pain, stomach pain, diabetes

Communities	Children	Young	Adult women	Elderly women	Adult men	Elderly men
Assentamento Irmã Dorothy	Flu, pneumonia	Flu	Gastritis, gynecological	Diabetes, hypertension	Spine	Diabetes, hypertension, prostate
PA Boa Esperança	Flu, worms, diarrhea	Flu, kidneys	Flu, breast cancer	Diabetes, hypertension, cholesterol	Spine	Vision, hypertension, diabetes

4.1.1.1.5 Basic sanitation

Basic sanitation data for the communities also reflect the reality of the municipalities in the region, with low rates of service coverage and access to services. Regarding access to drinking water, the communities reported that four have access to the general distribution network (with a water tank installed in the community), namely Mamorana, Roça Comprida, Vila dos Remédios, and PA Boa Esperança. However, it is noted that all communities, even these four, still access water for consumption from (individual) wells, springs, and cisterns.

With regard to sanitary sewage in the communities, there is no public sewage system identified. The survey found that, for most of the communities studied (80%), less than half of the houses have a bathroom or toilet. And 20% believe that more than half of the houses have a bathroom or toilet. Expanding the analysis, when diagnosing where sewage from bathrooms or toilets (if existing) is discharged, the communities reported that sewage is discharged into masonry pits on solid ground or into rudimentary pits (cesspools) also on solid ground (built by families for individual use).

With regard to the disposal of domestic waste, it was found that two communities have a household solid waste collection service provided by the municipality of Breu Branco (Roça Comprida and Vila dos Remédios). However, all communities report that one method of disposal practiced by families is the burning of approximately 61% of domestic waste on their plots.

4.1.1.1.6 Productive economic activities

Within the productive structure of the communities surrounding the project properties, subsistence agriculture is the basis of productive and economic activities, associated with other diverse activities such as extractivism, fishing, livestock, and charcoal production.

Within family farming, the main product reported was cassava for flour. However, other products also appeared, such as corn, beans, rice, watermelon, cocoa, cashews, and pitaya.

With regard to family income, most families earn a minimum wage per month. It appears that part of the families' fixed income comes from social benefits, retirement, or pensions. The communities of California, Mamorana, Vila dos Remédios, and PA Boa Esperança reported that family income is between one and two minimum wages per month.

The trend for family farming is to continue with the same pattern, with traditional slash-and-burn techniques for planting cassava for flour production, the main source of income for families, progressively wearing down the land, with consequent future depletion of arable areas. This causes encroachment on surrounding forest areas, increasing the areas of young secondary forest and reducing the areas of native forest. Thus, with the reduction of arable land, there is a tendency for families to leave the communities, abandoning the land and selling their plots.

4.1.1.2 Farm Workers Universe

For the survey of farm workers, direct workers employed by Dow Brasil and by subcontractors were considered. The total universe of the survey comprised 57% of the total number of employees, totaling 207 people (43 direct and 164 subcontracted).

4.1.1.2.1 Socioeconomic data

- **Gender and age diversity**

Of the workers interviewed, 90% were male, while 10% were female. These data show a male predominance among the interviewees, although there is also female representation in the sample.

The analysis of the average age of respondents reveals a diverse distribution: the most representative age group was 31 to 40 years old, totaling 41.46% of respondents; followed by the 41 to 59 age group, which corresponds to 30.49% of the sample; respondents between 19 and 30 years old represent 27.44% of the total; and there was one respondent aged 60 or over, totaling 0.61% of the survey. These data show a varied age distribution among respondents.

- **Education**

In terms of education, most respondents have completed or incomplete high school, representing 48% of the sample. Next, those with complete or incomplete elementary education total 31% of respondents. A portion has complete or incomplete higher education, representing 18% of the sample. In addition, there is a small proportion of illiterate or uneducated respondents, corresponding to 3% of the sample. This educational diversity among respondents reflects a breadth of experience and knowledge in the sample analyzed.

- **Income**

Most respondents (40.24%) reported that their family income in the last month was between 1 and 2 minimum wages. A similar proportion of respondents (39.63%) indicated that their family income was between 2 and 4 minimum wages. These two groups represent the majority of the sample and suggest a predominance of families with low to medium income. A smaller portion of respondents (8.54%) reported a family income between 4 and 6 minimum wages. On the other hand, 6.71% of respondents reported a household income of less than 1 minimum wage, suggesting the existence of some families in a more precarious financial situation. Only 4.88% of respondents reported a household income of more than 6 minimum wages, indicating a minority with a higher income level.

When asked about the adequacy of their family income, respondents presented a variety of perceptions, as shown in the table below.

Table 36. Classification of workers interviewed regarding the adequacy of their income. Source: Field research, 2023/2024. Prepared by: STA, 2024.

Classification of income for workers	%
More than enough, even enough to save	32%
It is sufficient, enough to pay all bills and not be in debt	41%
Not enough, we always have small debts	23%
It's insufficient to support my family	4%

These responses reflect the diverse economic realities faced by respondents, highlighting both those who feel comfortable with their financial situation and those who struggle to cover all family expenses.

4.1.1.2.2 Places of residence and origin

The survey conducted with the company's workers revealed that the majority reside in the municipality of Breu Branco, representing 46% of the total. Next, a significant portion reside in surrounding communities, totaling 45%. A small proportion of respondents reside in other locations, as categorized under the option "Other," which corresponds to 9% of the total.

Regarding the origin of the workers before they started working on the farm, it was possible to identify that the majority lived in surrounding communities, representing 41% of the total. Next, 40% of respondents came from municipalities close to the farm, such as Breu Branco, Moju, Tucuruí, and Goianésia do Pará. A portion equivalent to 11% came from other states, indicating considerable geographical mobility among workers. In addition, 8% of workers came from other municipalities in Pará.

4.1.1.2.3 Worker training and occupational safety

The survey revealed that the vast majority of workers (84%) said they had received some type of training or qualification for the job they perform on the farm. On the other hand, a portion (16%) reported not having received specific training or qualification. These results suggest that most workers received some type of preparation for their duties, which may contribute to the effective performance of their activities. Of those interviewed who stated that they had received training or capacity building for their roles (84%), the majority participated in various courses and training programs, the most frequently cited of which were: Firefighter

- 27 citations; Brush Cutter Course and Introductory Training - 14 citations each; NR35 course - 11 workers; Platform Operator Course (5 workers); NR33, Machine Operator and First Aid (4 mentions for each course).

When asked about their perception of safety at work, the vast majority of respondents (98%) said they felt safe in their job. Only a small proportion of respondents (2%) indicated that they felt unsafe. These results suggest a high level of confidence among workers in the safety of their work environment, possibly reflecting the effectiveness of the safety measures implemented by the company.

When asked about job satisfaction, the survey revealed that most workers (90%) expressed a high level of satisfaction with their work, rating it as "Very" satisfactory. A small portion (9%) rated their level of satisfaction as "Average." A minority reported "Somewhat" dissatisfied (1%). These results indicate a predominance of high levels of satisfaction among the company's workers.

4.1.2 Interactions between Communities and Community Groups (VCS, 3.19; CCB, CM1.1)

The communities identified in the socioeconomic assessment interact well with no apparent conflicts. Community residents interact with each other in various ways, such as in schools, where students from different communities attend the same schools, in sports, where there are women's and men's soccer championships between communities, and in situations of health need, since they use the same facilities for treatment and consultations. Workers also appear to have friendly interactions with each other, interacting during breaks and lunch.

4.1.3 High Conservation Values (CCB, CM1.2)

In the socioeconomic assessment, High Conservation Value (HCV) attributes 4, 5, and 6 were identified in the project area for the surrounding communities: HCV 4 refers to ecological services such as water supply, since the communities near the Moju River (Vila Monte Alegre, PA Esperança, and Irmã Dorothy) use it to collect water; HCV 5 refers to resources that are fundamental to basic needs, as communities use the forests as a source of food through hunting and fishing and as a source of wood for construction, utensils, and charcoal; and HCV 6 refers to cultural reproduction, due to the use of medicinal plants, fruit trees, and vines.

Although some communities in the project area use Dow's forest areas for water supply, hunting and fishing, and extraction of timber and non-timber forest products, in general the communities have their own forest area and also have access to surrounding forest areas, so they are not exclusively dependent on the forests in the Project Area. Given this, the HCVs identified in the assessment will not be considered by the project, as the communities do not use only the Project Area for subsistence, but also surrounding forest areas and their own community areas to meet their food, water, and cultural reproduction needs. In addition, they have access to basic infrastructure and are close to urban areas, which facilitates access to markets and economic opportunities and reduces the communities' dependence on the forest for food and income generation.

However, with the successful implementation of the Project, the ecosystem services critical to the survival and reproduction of families in the surrounding communities will be maintained and protected through the preservation of the forest cover in the Project Area, thereby benefiting local communities.

4.1.4 Scenario without the project: Community (CCB, CM1.3)

In the scenario without the Caiarara REDD+ Project, the socioeconomic trend of communities in the region faces considerable challenges, especially related to agriculture, rural exodus, and climate change. Although family farming remains the main source of income, families have found it difficult to improve their production practices due to a lack of technical assistance. Without adequate support, farmers are unable to adapt to increasingly adverse conditions, especially in the face of the effects of climate change.

Drought, mentioned by most communities, has intensified in recent years, with reduced rainfall directly affecting agricultural productivity. In addition, this water scarcity increases the risk of fires in crops and pastures, further compromising families' livelihoods. Traditional agricultural techniques, such as slash-and-burn, are depleting available land, leading to encroachment on native forest areas and, consequently, increased deforestation.

The rural exodus is also intensifying. The expansion of soybean cultivation in the surrounding area has put pressure on small family farmers, who, unable to compete or improve their practices, end up being persuaded to sell their land. This movement not only reduces the presence of families in communities, but also contributes directly to increased deforestation, as large areas of forest are converted into monocultures by large producers.

Another factor that aggravates deforestation in the absence of the project is charcoal production, which uses wood illegally extracted from the forest and intensifies environmental degradation, in addition to the growing practice of illegal hunting and fishing by communities, which threatens local biodiversity. These unregulated activities, together with unsustainable agricultural expansion, accelerate forest destruction and compromise the ecological balance of the region.

Therefore, it is concluded that the most likely scenario for communities in the absence of the Caiarara REDD+ Project is the continuation of a series of factors that contribute to deforestation, including low production diversification, limited agricultural productivity, lack of technical assistance for adaptation to climate change, and growing pressure from soybean expansion. These factors, combined with socioeconomic vulnerability, encourage rural exodus and environmental degradation. In the scenario with the project, a significant improvement in the socioeconomic conditions and well-being of communities is expected, with the strengthening of sustainable production, increased access to public policies, and the promotion of climate-resilient agricultural practices. The project will also foster the creation of socio-environmental impact strategies, generating a favorable and sustainable environment that strengthens the permanence of families in the region and protects forests, in addition to raising awareness about illegal and predatory hunting and fishing of endangered wildlife.

4.2 Net positive impacts on the community

4.2.1 Expected impacts on the community (CCB, CM2.1)

The expected impacts were identified through a Socioeconomic Diagnosis conducted in the project region, involving local communities and farm workers. The social impact assessment was carried out using the methodology developed by Richards and Panfil (2011), Richards (2011) and Pitman (2011), which presents a matrix with a descriptive summary of the impacts mapped during field research in the communities and with other agents in the project's surroundings. The matrix shows the group directly involved, the impact, the benefits, the costs and risks associated with each impact, and the likely changes in the group's well-being. The results of the mapping are presented below.

Community group	Communities located near the Project Area that use the forest area.
Impact(s)	Positive - Development and strengthening of value chains through training and coordination to obtain technical assistance to encourage environmental practices and sustainable and adaptive agricultural production models. - Environmental awareness and permanence of families on their land. - Strengthening of historically marginalized groups. - Improved well-being of stakeholders. Negative - Loss of access to subsistence hunting and fishing areas. - Loss of access to areas for extractive products in general.
Type of benefit/cost/risk	• Expected or existing impacts: - All impacts are considered expected, as they may occur as project activities are implemented. • Direct and/or indirect: - All impacts are considered direct because they directly influence the knowledge and practices of community members for the maintenance of their families in their communities, protection of biodiversity, and application of productive techniques and social organization. • Benefits: - Increased technical capacity of community members to develop sustainable and adaptive practices that add value and diversity to their production and income generation. - Improved living conditions for communities.

	- Greater environmental and social awareness of the environment in which they live. - Maintenance of families in the territory through the use of sustainable practices, access to technical training, social organization, and knowledge of biodiversity protection. • Costs: - Human and financial resources related to training, capacity building, and technical assistance. • Risks: - Low participation of community groups in training processes. - Few families practicing sustainable production techniques. - Low engagement of families in collective and community activities in the pursuit of public policies and services. - Possible illegal and uncontrolled entry into forest areas of the Caiarara REDD+ Project to search for water, extractive products in general, and to hunt and fish.
Change in well-being	- Families remaining in the communities. - Gradual and/or sustainable reduction in the use of the project area. - Improvement in family income. - Negative impact on the maintenance and subsistence system due to the use of some forest and non-forest products by families.

Community group	Workers at the Água Azul I and II Reforestation Farms (RAAI and RAAII).
Impact	Positive - Strengthening of human capital through access to professional training and skills development. - Greater efficiency in field surveillance operations.
Type of benefit/cost/risk	• Expected or existing impacts: - All impacts are considered expected, as they will occur as project activities are implemented.

	• Direct and/or indirect: - All impacts are positive and direct on workers, based on the implementation of training and capacity building that directly influence their conduct, dynamics, and quality of work. • Benefits: - Increased effectiveness and productivity of farm activities related to the protection and monitoring of forest areas. - Training and capacity building to improve the daily activities of workers. • Costs: - Human and financial resources related to training and capacity building. • Risks: - Low worker engagement during project activities.
Change in well-being	- Enhance the professional quality of workers. - Improve the quality and working conditions of employees.

4.2.2 Mitigation of Negative Impact on the Community (VCS, 3.19; CCB, CM2.2)

The Socioeconomic Diagnosis conducted in the communities surrounding the Project identified that some families accessed the project area for hunting, fishing, and extraction of forest products such as wood, vines, and medicinal plants. Restricted access resulting from the protection of the area may impact on the way of life and livelihoods of these families. To mitigate these impacts, the Caiarara REDD+ project will implement sustainable productive alternatives through coordination with public institutions and local organizations, promoting the development of value chains, technical assistance, and adaptive agricultural practices, with the aim of reducing dependence on natural resources in the project area.

In addition, to ensure a transparent and collaborative relationship between the project proponent and the communities, a communication and engagement plan has been implemented that includes channels for receiving feedback and conflict resolution mechanisms, with special attention to the safety and equitable participation of women and vulnerable groups. These measures follow the principles of equity and respect for local diversity, ensuring safe and accessible environments for all, in order to strengthen dialogue and engagement among stakeholders, as well as encouraging community involvement in both the activities carried out and the preservation of forests.

The project does not employ or release hazardous substances, does not use pesticides or chemical fertilizers, and does not generate air pollutants, solid waste, or industrial effluents that affect air, water, or soil quality. If any environmental risk is identified, preventive actions will be taken in accordance with the precautionary principle, even in the face of scientific uncertainty.

The protection and sustainable use of areas recognized as High Conservation Value (HCV) – including natural resources essential for subsistence and sites of cultural importance – are ensured through monitoring, dialogue with communities, and continuous adaptation of management strategies.

In addition, mitigation measures will be periodically evaluated. If necessary, they will be adjusted and adapted according to new needs and conditions identified, thus ensuring the continued effectiveness of these actions on community well-being and environmental preservation.

4.2.3 Net positive community well-being (VCS, 3.19; CCB, CM2.3, GL1.4)

The activities proposed by the Caiarara REDD+ Project encourage socioeconomic and sustainable development for the stakeholders involved, focusing on education and assistance to promote more sustainable agricultural practices and habits, to develop and strengthen value chains, and to engage through mechanisms of transparency, communication, and incentives for collective participation in the actions developed by the Project.

In the scenario without the Project, as described in Section 4.1.4, the context of low income and lack of technical assistance and rural extension services causes families to seek alternatives to increase their income from economic and subsistence activities practiced in an unsustainable and unplanned manner that can promote deforestation and forest degradation in the region.

The project coordinates the technical assistance necessary for community members to adapt their agricultural and productive practices to climatic conditions, with an emphasis on sustainable production techniques that allow for greater efficiency in the use of water resources and adaptation to more severe periods of drought.

Value chain development and strengthening activities enable communities to diversify their sources of income, reducing their dependence on a single economic activity and thus increasing their capacity to adapt. Strengthening local governance ensures communication between stakeholders and proponents, promoting actions that bring effective solutions and meet the needs of all stakeholders.

Environmental education and actions to strengthen inclusion, diversity, and equity are also essential to ensure that all segments of the community have the knowledge and tools to adapt to the impacts of climate change. These activities promote greater awareness of the impacts of climate change and empower vulnerable groups to contribute meaningfully to decision-making and the implementation of adaptive practices.

The project's educational actions will cover a range of topics, depending on the demands and needs identified by stakeholders. These actions may range from technical training to workshops on specific topics that help community development, according to the reality of each group. In addition, there will be an emphasis on environmental education, addressing issues such as climate change, the importance of forest conservation, prevention of deforestation and predatory hunting, among others. These activities are essential to raise awareness among communities about environmental challenges and provide the necessary tools for them to adopt more sustainable and adaptive practices, which directly contributes to the well-being of stakeholders by ensuring the protection of the natural resources on which they depend.

Regarding actions to improve the well-being of stakeholders, the project intends to promote lectures, workshops, and seminars on health and well-being topics. The topics will depend on the demand of each stakeholder, always consulting on the results of socioeconomic assessments and the opinion of each group. With this, the project seeks to improve the living conditions of communities and workers in relation to hygiene and basic sanitation habits, health awareness, vaccines, and services offered by the city government, among others.

With regard to the empowerment of women and young people, the project seeks to promote the inclusion of these groups in economic and community activities. Through specific training and opportunities to participate in local governance processes, women and young people will be able to acquire new skills and knowledge that will help them access the labor market or strengthen their positions within the community. This empowerment provides greater autonomy and increases the capacity of these groups to actively contribute to the sustainable development of the region, resulting in positive impacts on both individual and collective well-being.

In summary, the project ensures that all community groups benefit positively, promoting improvements in their well-being and strengthening their resilience in the face of climate challenges. With activities covering education, technical assistance, and value chain strengthening, communities will be empowered to adapt their productive practices to new climate conditions. In addition, the empowerment of vulnerable groups ensures that everyone can contribute to the sustainable development of the region, ensuring that the project brings lasting and equitable impacts for the well-being of all.

4.2.4 High Conservation Values Protected (CCB, CM2.4)

As described in Section 4.1.3, the High Conservation Value Attributes identified in the Project Area will not be considered by the project since the communities do not rely solely on Dow's forests for their livelihoods, but also on their own community forests and those surrounding the Project Area, which ensures that they have sufficient areas to meet their basic needs.

According to the Theory of Change developed (Section 2.1.17), the Project provides for the implementation of activities to change cultural habits and create sustainable production alternatives that avoid deforestation and hunting, integrating new environmentally responsible practices into the daily lives of communities, ensuring greater resilience and long-term sustainability and promoting adaptation to climate change, since sustainable and diversified practices enable communities to better face environmental challenges.

The project's actions are focused on forest conservation, community strengthening, environmental education, and sustainable production practices in order to reduce communities' dependence on deforestation and predatory hunting and fishing.

The project maintains channels for ongoing dialogue with communities and participatory monitoring mechanisms, ensuring that any potential HCV is promptly identified and protected, in accordance with CCB principles.

4.3 Other impacts on stakeholders

4.3.1 Impacts on other stakeholders (VCS, 3.18, 3.19; CCB, CM3.1)

Positive impacts of the project can be observed, which may bring well-being to other actors, such as:

- All local communities, as well as other actors residing in the project region, whether or not they participate in project activities, will benefit from all positive impacts related to the conservation and protection of forest cover and biodiversity;

- All communities and other local actors will benefit from sustainable development, as well as from the opportunities generated by the project activities, improving their quality of life and well-being;

- All stakeholders in the region will benefit not only from project activities, but also from greater access to public policies;

Negative impacts on other stakeholders are not anticipated or are unlikely in the Caiarara REDD+ Project, and may include:

- Lack of engagement of communities and other actors in Project activities and other articulations;

- Failure to communicate Project actions and the establishment of possible conflicts arising from the implementation and conduct of activities.

4.3.2 Mitigation of negative impacts on other stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

As mentioned above, no negative impacts on other actors are expected in the context of the Project, a premise supported by socio-environmental safeguards already in place and the direct involvement of stakeholders in the project design. To reinforce this premise, participatory strategies have been implemented to ensure the inclusion of stakeholders in the design of activities and decision-making. In addition, the project has a communication plan that includes clear procedures for conflict resolution. If this plan proves ineffective, adjustments will be made to the forms of communication and conflict referral processes, always seeking the best solution.

If any unforeseen adverse effects are identified, the project provides for the implementation of corrective measures based on participatory monitoring mechanisms. The project continuously monitors social impacts, focusing on the early identification of potential risks to the well-being of stakeholders. The communication plan includes formal protocols for receiving complaints, handling grievances, and mediating conflicts, with periodic reviews to ensure effectiveness.

4.3.3 Net impacts on other stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

As described and detailed in Section 4.3.1, negative impacts on the well-being of other local stakeholder groups are not anticipated, since the activities to be carried out in relation to the surrounding communities are based mainly on coordination with government agencies and other local institutions precisely to promote improvements in living conditions, greater access to public policies, and activities related to improving existing practices. The activities outlined and proposed for the Project seek only impacts that promote the development, inclusion, and well-being of communities and other stakeholders.

4.4 Community impact monitoring

4.4.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

This section presents the variables to be monitored related to the community, considering the scope of the Project. The monitoring variables are directly linked to the Project's objectives for the surrounding communities and property workers and to the expected products, results, and impacts identified in the project's causal model and related to community well-being. Monitoring the Project's impacts on communities and other stakeholders is an essential management tool, allowing for the evaluation of the effectiveness of activities in achieving the proposed objectives. The development of a monitoring system will be based on the planned goals, supported by the construction of indicators to be collected, verification tools, and procedures for analyzing and evaluating results. When necessary, the system will indicate corrective measures or improvements to achieve the desired progress.

The Community Impact Monitoring Plan will be based on process and outcome indicators, described below. In addition, this plan will be continuously reviewed, with evaluation and validation by stakeholders, ensuring its adaptability to the dynamics and needs of the communities involved.

Data / Parameter	Number of meetings/gatherings held
Data unit	Number
Description	Measurement of the number of meetings/gatherings held to contribute to the development and improvement of actions and activities related to the social activities of the Project.
Description of measurement methods and procedures to be applied	All meetings or gatherings held will be identified and recorded, including minutes, reports, and audiovisual recordings. In addition, stakeholder participation will be recorded through signed lists and registration forms. The data and reports produced will be stored in digital files throughout the Project's life cycle. Original (physical) reports, meeting minutes, and field notes produced during the execution of social activities will also be stored.
Monitoring frequency	Annual
Purpose of data	The indicator provides data on the relationship between stakeholders and project activities. In addition, the data allows for the analysis and identification of trends, such as the most popular topics, the most participatory groups, etc. This information can be used to adjust the strategy to improve future activities.

Comments	-

Data / Parameter	Number of education/training activities carried out
Data unit	Number
Description	Measurement of the number of actions carried out by the Project, contributing to the technical training and environmental awareness of the Project's stakeholders.
Description of measurement methods and procedures to be applied	All education and training actions carried out will be identified and recorded, including detailed documentation of these actions, such as activity reports, materials distributed, audiovisual records, and the topics covered. In addition, participants' interactions during the activities will be recorded and feedback on the content and dynamics of the actions will be collected using signed lists and registration forms. The data and reports produced will be stored in digital files throughout the project's life cycle. Original (physical) reports, meeting minutes, and field notes produced during the implementation of social activities will also be stored.
Monitoring frequency	Annual
Purpose of the data	To count how many actions were taken and what their themes were, in order to evaluate what is being done by the project. This information can be used to adjust the strategy to improve future activities.
Comments	-

Data / Parameter	Number of people benefiting from the project
Data unit	Number
Description	Counting of any person who has benefited during the implementation and monitoring of the Project, through the planned actions and activities related to the social scope

Description of the measurement methods and procedures to be applied	The people who benefited during the project actions will be identified. This will include members of local communities, workers involved in the activities, and other impacted groups, such as training and workshop participants. The record will be made through attendance lists, registration forms, and reports on participation in socio-productive and training activities. This will make it possible to account for the people involved in the project activities. The data and reports produced will be stored in digital files throughout the project life cycle. Original (physical) reports, meeting minutes, and field notes produced during the implementation of social activities will also be stored.
Monitoring frequency	Annual
Purpose of data	To monitor the evolution of participation and benefits generated by project activities that reach stakeholders.
Comments	-

4.4.2 Disclosure of the Monitoring Plan (CCB, CM4.3)

The Community Monitoring Plan and all its results will be publicly disclosed on the VERRA registration page and on the Biofilica Ambipar Environmental Investments S/A website. The validated and verified monitoring results will be available in the public version of the report on the Verra website, ensuring free access to any interested party.

Summaries of the main results will be prepared in accessible language and presented in person to the communities through adapted printed and digital materials. Copies will also be made available at strategic locations (farm headquarters and community centers). The results will be disclosed after each official project verification, with regular updates following the monitoring cycles defined in the PDD.

Information about the Project will also be disseminated via email and WhatsApp (with a pre-established list of contacts of interested parties), and on social media, such as Instagram and LinkedIn, of the project proponent.

In this way, it is expected that all necessary efforts to provide full transparency and distribute key Project information and documents will be achieved.

4.5 Optional Criterion: Exceptional Community Benefits

Not applicable. The Caiarara REDD+ Project is not intended to be validated for gold level in this section.

4.5.1 Exceptional Community Criteria (CCB, GL2.1)

Not applicable.

4.5.2 Short- and long-term community benefits (CCB, GL2.2)

Not applicable.

4.5.3 Community participation risks (CCB, GL2.3)

Not applicable.

4.5.4 Marginalized and/or vulnerable community groups (CCB, GL2.4)

Not applicable.

4.5.5 Net impacts on women (CCB, GL2.5)

Not applicable.

4.5.6 Benefit Sharing Mechanisms (CCB, GL2.6)

Not applicable.

4.5.7 Communication of Benefits, Costs, and Risks (VCS, 3.18; CCB, GL2.7)

Not applicable.

4.5.8 Governance and Implementation Structures (CCB, GL2.8)

Not applicable.

4.5.9 Capacity Development of Small Producers/Community Members (CCB, GL2.9)

Not applicable.

5 BIODIVERSITY

5.1 Biodiversity scenario without project

5.1.1 Existing conditions (VCS, 3.19; CCB, B1.1)

The biodiversity assessments for the Caiarara REDD+ Project were carried out by Ambiens Resolve Inovação Sustentabilidade e Meio Ambiente LTDA for fauna groups and by STA Soluções Técnicas e Serviços de Engenharia Ambiental LTDA for flora group. The study was conducted in January 2024, based on primary and secondary data collection to support project activities and understand the dynamics of the region, as well as to map existing conditions. The list of species identified in the fauna and flora studies can be seen in Appendix 1: SPECIES LIST.

5.1.1.1 Fauna

Each fauna group was identified through available bibliography and field identification, which was carried out using different methodologies for each class.

For the data analysis stage, the species identified in primary and secondary data were organized into lists containing crucial information, such as family, scientific name, common name, preferred habitat, whether the species is endemic or exotic, and whether its occurrence is probable or confirmed, based on the available scientific literature. To assess the level of threat of extinction, it was verified whether the species were listed in the List of Threatened Fauna of the State of Pará (Decree 802/2008), the National List of Endangered Fauna (MMA[86]) and the Global List of Endangered Fauna (IUCN[87]).

5.1.1.1.1 Avifauna

For the **avifauna** group, field sampling was carried out using MacKinnon's quantitative list technique, which consists of compiling successive lists of species during walks along predefined routes. This makes it possible to inventory species and compare spatial differences. During sampling, routes were traveled at a relatively constant speed of 2 km/h. Each time ten different species were detected, a new list was started, with a given species being marked only once on each list, even if it was recorded several times in a row—such species would only be marked again if it was recorded after a new list was started (RIBON, 2010)[88] .

Sampling times included the morning (30 minutes before sunrise until five hours after), afternoon (between three and four hours before sunset), and night (from sunset until three hours after), totaling 8 to 12 hours of sampling per day. The species were recorded through visual or auditory contact with the aid of Bushnell 10 x 42 binoculars and a Tascam digital recorder coupled with a Yoga directional microphone. Whenever possible, the species detected were photographed and/or their sounds were recorded. As a result, bibliographic data analysis revealed 387 bird species distributed across 64 families. Among these, 19 species are listed in at least one endangered fauna list: 5 species on the Pará state list (Decree 802/2008), 15 on the national list, and 8 on the global list (IUCN). Five species (1%) are endemic to the Belém Endemism Center, 11 species (2.8%) are endemic to the Amazon, and one species is considered exotic, the pardal (*Passer domesticus)*.

In the field study, **124 bird species** were recorded, distributed across 42 families. Of these, 7 species recorded in the field had not been previously identified in the bibliographic data. During field sampling, two species recorded are classified as endemic of the Belém Endemism Center: the araçari-de-pescoço-vermelho (*Pteroglossus bitorquatus)* and the tiriba-pérola (*Pyrrhura lepida*), while other two identified are classified as endemic of the Amazon biome, the marianinha-de-cabeça-amarela (*Pionites leucogaster*) and the bacacu-preto (*Xipholena lamellipennis*). In addition, eight species found are cited in at least one list of

[86] MMA ORDINANCE No. 148, OF JUNE 7, 2022. National List of Endangered Species. MINISTRY OF THE ENVIRONMENT
[87] IUCN. 2023. The IUCN Red List of Threatened Species. Version 2023-1. <www.iucnredlist.org>.
[88] RIBON, R. 2010. Bird sampling using Mackinnon's list method. In: Ornitol. e Conserv. Ciência Apl. Técnicas Pesqui. e Levant. (eds. Von Matter, S., Straube, F., Accordi, I., Piacentini, V. & Cândido-Junior, J.). Technical Books, Rio de Janeiro, pp. 33–44

endangered fauna, with three in the IUCN threat categories: the tiriba-pérola (*Pyrrhura lepida)*, the ararajuba (*Guaruba guarouba)* and the marianinha-de-cabeça-amarela (*Pionites leucogaster)*, all classified as "Vulnerable" (VU).

Figure 32. Araçari-de-pescoço-vermelho (*Pteroglossus bitorquatus*) on the left and tiriba-pérola (*Pyrrhura lepida*) on the right. Photos: Ambiens.

To assess the adequacy of the cumulative sampling effort, collector curves were constructed based on 100 randomizations of the data, using the number of contacts as the sampling unit. The collector curve is a procedure for assessing how close an inventory is to recording all species in the study site. However, due to the high richness found in tropical ecosystems, collector curves rarely reach stabilization (SANTOS, 2006). Therefore, a total richness estimator was used, based on extrapolations from the collector curve obtained in the samples, with the aim of estimating the total richness of the sampled community (first-order Jackknife estimator). To construct the collector curves, estimated richness, abundance profiles, diversity indices, and equitability, the R program with the "vegan" package was used (Oksanen et al. 2022).

As a result of the assessment of the adequacy of the sampling effort, the collector curve did not reach a horizontal plateau, indicating that more bird species should be found with increased sampling effort. In fact, the number of species recorded in the field (124) represented 66% of the number of species estimated by the first-order Jackknife estimator (187 species). The richness recorded during the sampling represented 32% of the total richness found in the bibliographic data (387 species).

5.1.1.1.2 Mastofauna

For **mastofauna** (medium and large mammals), a literature review and field sampling were carried out through active search. The active search method using trails was used to obtain direct visual contact with individuals or to view traces. Visual records were made for species with diurnal habits, such as arboreal species. For species with nocturnal habits, indirect sampling techniques are appropriate, such as identification through footprints, feces, burrows, and rooting[89] . In addition, nine camera traps were installed to record species present at the site.

The work was carried out from 6 a.m. to 11 a.m. and from 3:30 p.m. to 9 p.m. Trails were established in proportion to the types of environments present in the study areas, taking advantage, as far as possible, of roads, stream banks, and pre-existing trails. The routes were traveled at a slow pace at a constant speed of approximately 2 km/h, totaling 41.47 km of distance traveled.

As a result, 40 species of medium and large mammals were identified through bibliographic data analysis. Among these, 14 are listed in at least one list of endangered fauna and 9 are endemic to the Amazon biome. In the field study, **24 species** were recorded**,** among which six are endemic to the Amazon biome and 8 are cited in at least one list of endangered fauna, with six in the threat categories of the global list (IUCN), namely: tamanduá-bandeira (*Myrmecophaga tridactyla)*, bugio-de-mãos-ruivas (*Alouatta belzebul)*, anta (*Tapirus terrestris*), sagui-preto (*Saguinus niger*) and queixada (*Tayassu pecari)*, classified as "Vulnerable" (VU), and the macaco-caiara (*Cebus kaapori),* classified as "Critically Endangered" (CR).

Figure 33. Tamanduá-bandeira (*Myrmecophaga tridactyla*) on the left and Macaco-caiara (*Cebus kaapori*) on the right. Photos: Ambiens.

The collector's curve for medium and large mammal species did not reach a horizontal plateau, indicating that it is very likely that more species will be found in the study areas if more hours of sampling effort are spent. In fact, the observed number of medium and large mammal species (24) was close to the estimated number according to the estimator (35.2), representing 68% of the estimated number of species according to the first-order Jackknife estimator.

5.1.1.1.3 Herpetofauna

For the **herpetofauna** group, active searching was also used as a methodology for identification, conducting visual inspections in locations with the greatest potential to harbor species representative of these groups,

[89] PARDINI, R., DITT, E.H., CULLEN, Jr. L., BASSI, C., RUDRAN, R., 2003. Rapid survey of medium and large mammals. p. 181-201. In: Cullen Jr., Laury, Rudran, Rudy, Valladares-Padua, Cláudio (Eds.). Methods of study in conservation biology and wildlife management. Ed. UFPR. 665 p.

covering pre-existing trails and water bodies. In addition to visual inspections, auditory detections of anuran amphibians were also performed.

The active search carried out during the daytime (6:00 a.m. to 11:00 a.m.) is aimed at recording reptiles, which occurred in forests and their surroundings, cultivated areas, in leaf litter, under fallen logs, on the branches of shrubs and trees, in tree hollows, in bromeliads, under rocks, and in burrows, as well as possible traces such as feces and tracks. Afternoon searches were conducted from 5:00 p.m. to 7:00 p.m., and night searches from 7:00 p.m. to 10:00 p.m., to record amphibians and reptiles. These searches were focused on nocturnal reptiles and amphibians near watercourses, rivers, lakes, and wetlands.

As a result of the bibliographic analysis, 172 species representing the herpetofauna were listed, including 85 species of amphibians distributed across 13 families and 87 species of reptiles distributed across 21 families.

In the field study, **23 species** of herpetofauna were recorded, with 15 amphibians and 8 reptiles. Among them, two species are endemic, the sapo-de-seta (*Adelphobates galactonotus)* and the sapo-do-Pará (*Leptodactylus paraensis)*, and five are listed in at least one list of endangered fauna, one of which is listed in an IUCN threat category: the jabuti-tinga (*Chelonoidis denticulata)*, classified as "Vulnerable" (VU) because it is targeted by hunting and illegal trade.

Figure 34. Sapo-de-seta (*Adelphobates galactonotus*) on the left and Jabuti-tinga (*Chelonoidis denticulata*) on the right.

Considering all sampling areas, the collector curve did not show a tendency to stabilize, indicating that the number of species observed may increase with the addition of samples. In fact, the recorded value (23) represents 82% of the value of the first-order Jackknife estimator (28 species).

5.1.1.1.4 Ichthyofauna

For the ichthyofauna, only bibliographic data were analyzed.

For the **ichthyofauna** group, 170 species were found. Of these, 99 are endemic to the Amazon biome river basins, 65 species are endemic to South America, one species is endemic to Brazilian territory, and five species were not evaluated.

Among the species recorded, three are listed as Data Deficient (DD), 150 species are classified as Least Concern (LC), and 17 species were not evaluated, according to the IUCN international list. On the Brazilian list of endangered fauna, two species are classified as Critically Endangered (CR), the cascudo (*Baryancistrus niveatus)* and the arraia-aramaçá (*Paratrygon aiereba)*, and one as Endangered (EN), the pacu (*Mylesinus paucisquamatus)*.

5.1.1.1.5 Insects

For insect groups, only bibliographic data were analyzed.

The **insects** were divided into four groups: aquatic insects (orders Ephemeroptera, Odonata, Plecoptera, Hemiptera), bees (superfamily Apoidea), flies (order Diptera), and ants (family Formicidae). Of the species recorded in the bibliographic data for the region studied, 154 are aquatic insects, 55 are bees (16 of which are considered rare), 26 are flies, and 118 are ant species.

More details and information on the survey and analysis of fauna species can be found in the Final Fauna Diagnosis Report, presented to the auditor.

5.1.1.2 Flora

Floral identification was carried out through an inventory of the project area and a survey of bibliographic data. The inventory was carried out in conjunction with the carbon stock study, where random plots were set up to identify individuals at the site. The species found in primary and secondary data were listed with their scientific name, common name, and family. In addition, we sought to analyze the existence of endangered, rare, and/or endemic species, considering global (IUCN) and national interests (MMA; CNCFLORA); the existence of Genetically Modified Organisms (GMOs); and the existence of invasive species.

The vegetation cover of the properties is classified into two phytophysiognomies: Dense Ombrophilous Forest (DOF) and Campinarana. The DOF is characterized by tropical climatic factors of high temperatures (averages of 25ºC) and high precipitation, well distributed throughout the year (from 0 to 60 dry days), which determines a bioecological situation with virtually no dry season. The term Campinarana is a regional Amazonian name that means "false field," usually presenting low floristic diversity, with a predominance of generalist species from open areas, generally widely distributed.

The phytosociological survey was carried out on plots of 1 hectare each (100 x 100 m), with five plots in the dense ombrophilous forest area chosen at random from among the 17 plots established for the carbon stock inventory, and three plots established at random in the Campinarana area.

The plots allocated in the DOF were divided into five 20 x 100m transects, and all trees and palms with DBH > 10 cm were counted. In Campinarana, due to the low floristic richness and, according to Ferreira et al. (2013 and 2014), which show the need to inventory small areas, the plots were not inventoried in their entirety: in each one, five 10 x 10 m subplots were established and the number of woody individuals with DBH > 1.0 cm was counted.

5.1.1.2.1 Dense Ombrophilous Forest

As a general result, regarding floristic characterization, the list of species likely to occur, resulting from bibliographic research, listed 418 species of flora in the region, and the list of confirmed species, resulting from field collection, identified 173 species of flora belonging to 83 botanical families in the study site (see Appendix 1: SPECIES LIST).

In the Project Area (Dense Ombrophilous Forest), 2,508 individuals belonging to 149 species were inventoried. Considering the lists of endangered species, 13 species of potential occurrence and 14 species of confirmed occurrence at some degree of threat were found in the project area. Of these, seven are classified in the threatened categories of the global list (IUCN), namely: samaúma (*Eriotheca longipedicellata)*, cedro (*Cedrela fissilis), guajará-cinza (Pouteria oppositifolia)* and itauba (*Mezilaurus itauba)* classified as "Vulnerable" (VU), louro-canela (*Ocotea fragrantissima)* and maçaranduba (*Manilkara elata)* classified as "Endangered" (EN) and acapu (*Vouacapoua americana)* classified as "Critically Endangered" (CR).

Most of these species are of high commercial value and have suffered or continue to suffer from exploitation, resulting in a sharp decline in their populations. No local endemic species were found, but species restricted to the state of Pará were identified, indicating endemism on a regional geographical scale. No invasive species were found in the project area.

5.1.1.2.2 Campinarana

Campinarana, as it is not a forest formation, is not within the Caiarara REDD+ Project Area. However, the Campinarana areas located within Dow's properties in the Project Zone are considered High Conservation Value Attributes because they are a rare ecosystem (see Section 5.1.2).

In Campinarana, 117 individuals belonging to 33 species were inventoried. These values reveal a density of individuals and species richness compatible with the results obtained in other studies in nearby locations.

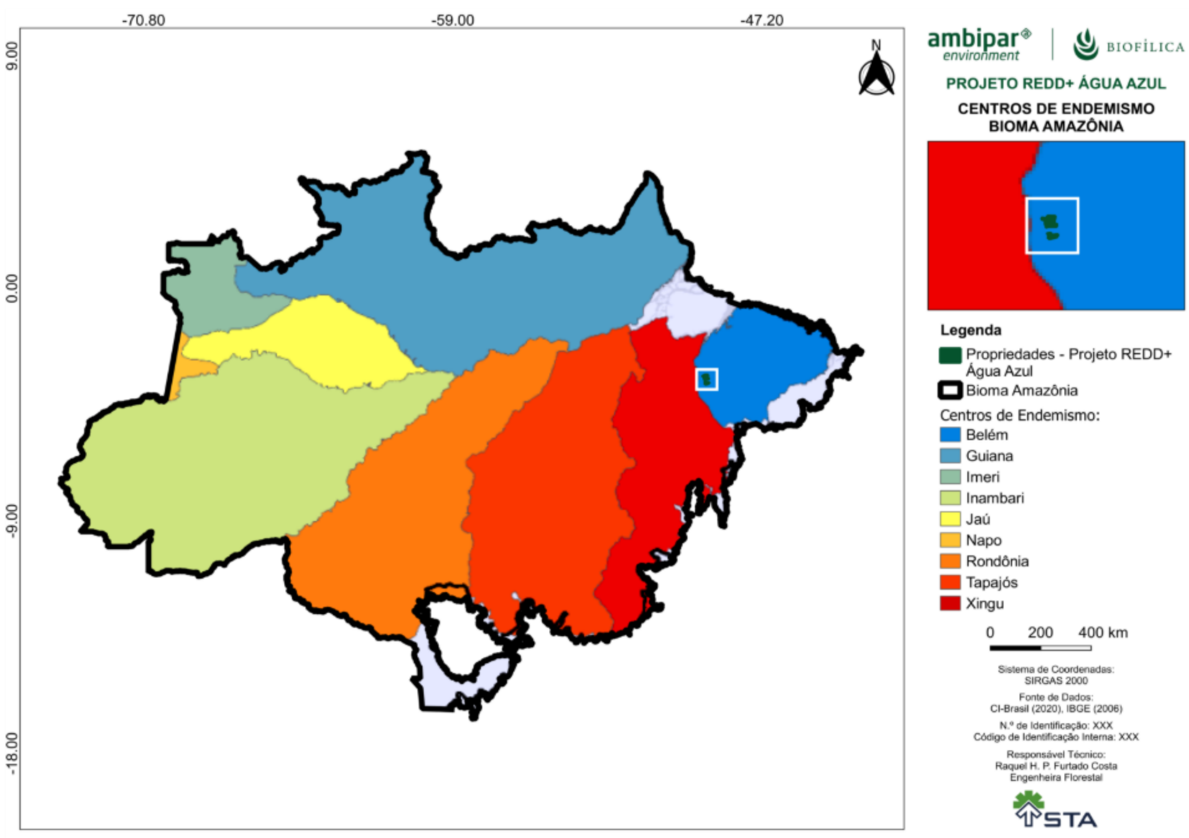
The low floristic richness found in the Amazonian Campinaranas is justified by the fact that the species present there are highly specialized and adapted to critical environmental factors, such as high temperatures, nutrient-poor soils, high acidity, and poor drainage (Mardegan et al., 2008). White sand soils are limiting to plant growth, especially due to their very low nitrogen concentration (Mardegan et al., 2008). Thus, the number of species is reduced in comparison to other types of vegetation in the Amazon, such as the ombrophilous forest (Anderson, 1975).

Ferreira (2009) and Ferreira et al. (2014) highlighted the low shared species richness between areas, showing that each area has many unique species. According to the authors, the low similarity of species between the Campinaranas areas of the eastern Amazon shows that the number of rare species is high, which is one of the most important criteria in choosing new areas for the creation of conservation units. These authors advocated the creation of new conservation units in part of the eight fragments of Campinaranas in the lower Tocantins River region due to the large number of species with low individual density and some species unique to the fragments. According to Ferreira et al. (2014), this strategy would fill a major gap in the protection of the Campinaranas flora, whose preservation will contribute to the maintenance of natural ecosystems of regional or local importance and help regulate the economic use of these areas, so as to make it compatible with conservation objectives.

5.1.1.3 Conclusion

The Caiarara REDD+ Project area is located in the eastern part of the Amazon, specifically in one of the eight Amazonian Centers of Endemism, the Belém Center of Endemism (Figure 35). This region is of extreme biological importance due to its high rate of endemism, where many species of flora and fauna are found exclusively in this area. However, the Belém Endemism Center is also the region with the highest rates of deforestation in the entire Amazon, with about 76% of its area already deforested[90] . This high rate of deforestation threatens not only endemic species, but also the ecosystems and environmental services that sustain local biodiversity and other species that inhabit the region.

Figure 35. Map showing the location of the Caiarara REDD+ Project in relation to the Centers of Endemism in the Brazilian Amazon.



[90] ALMEIDA AS, VIEIRA ICG. Belém Endemism Center: status of remaining vegetation and challenges for biodiversity conservation and ecological restoration. Rev Estud Univ. 2010; 36: 95–111. https://doi.org/10.1007/s13398-014-0173-7.2

The threat of extinction of the species found in the assessments is directly related to their exploitation, deforestation of their habitats, and climate change. Forest communities may experience sharp declines in species in the near future due to climate change (Gomes et al., 2019)[91] . Species adapted to drought may be expanding their geographic range and increasing in abundance, as the dry season has become longer and more frequent in the Amazon (Esquivel-Muelbert et al., 2019)[92] .

The decline in the forest community directly impacts the abundance of fauna species, since with the decline of the forest, the supply of food, protection, and shelter for these species is also reduced, creating a scenario of greater vulnerability and, consequently, a greater risk of extinction for the species identified in the project area.

The fauna and flora species identified through local surveys in the project area with some degree of threat of extinction and/or endemic species and the demonstration of the absence of negative impacts on them are presented in the following table.

Table 37. Endemic or endangered species identified in the project area

Endemic and/or endangered fauna species: - Avifauna: Marianinha-de-cabeça-amarela (*Pionites leucogaster*); Araçari-de-pescoço-vermelho (*Pteroglossus bitorquatus*); Tiriba-pérola (*Pyrrhura lepida*); Bacacu-preto (*Xipholena lamellipennis*); Ararajuba (*Guaruba guarouba*); Arapaçu-galinha-do-pará (*Dendrexetastes paraensis*). - Mastofauna: Onça-pintada (*Panthera onca*); Tamanduá-bandeira (*Myrmecophaga tridactyla*); Macaco-caiarara (*Cebus kaapori*); Bugio-de-mãos-ruivas (*Alouatta belzebul);*	The fauna species listed here are classified as threatened according to the International Union for Conservation of Nature (IUCN), MMA (2022), and Pará (2008), or are endemic to the region. The Caiarara REDD+ Project will not negatively affect the habitats of endemic, rare, threatened, or endangered species, as the activities planned for this REDD+ project are focused on research, conservation, and monitoring of forest habitats in the Amazon biome. Positive impacts are expected from the project due to the proposed activities, generating net benefits to biodiversity that would not likely occur in the absence of the Project.

[91] Gomes, V.H.; Vieira, I.C.; Salomão, R.P.; Ter Steege, H. (2019). Amazonian tree species threatened by deforestation and climate change. Nature Climate Change, 9(7): 547-553.

[92] Esquivel-Muelbert, A.; Baker, T.R.; Dexter, K.G.; Lewis, S.L.; Brienen, R.J.; Feldpausch, T.R., et al. (2019). Compositional response of Amazon forests to climate change. Global Change Biology, 25: 39–56.

Gato-mourisco (*Herpailurus yagouroundi*); Onça-parda (*Puma concolor*); Anta (*Tapirus terrestres*); Veado-roxo (*Mazama nemorivaga*); Catipuru (*Microsciurus flaviventer*); Sagui-preto (*Saguinus niger*); Mico-de-cheiro (*Saimiri sciureus* (collinsi)); Anta (*tapirus terrestris*); Queixada (*Tayassu pecari*). - Herpetofauna: Sapo-de-seta (*Adelphobates galactonotus*); Sapo-do-Pará (*Leptodactylus paraenses*); Jabuti-tinga (*Chelonoidis denticulata*).	
Endemic and/or endangered plant species: Cedro (*Cedrela fissilis*); Sumauma-da-terra-firme (*Eriotheca longipedicellata*); Maçaranduba (*Manilkara elata*); Itauba (*Mezilaurus itauba*); Louro-canela (*Ocotea fragrantissima*); Acapu (*Vouacapoua americana*); Guajará-cinza (*Pouteria oppositifolia*); Breu Branco (*Protium heptaphyllum*); Araracanga (*Aspidosperma album*).	The plant species listed here are classified as threatened according to the IUCN, MMA (2022), CNCFLORA (2024) and Pará (2008), or are endemic to the region. The Caiarara REDD+ Project will not negatively affect endemic, rare, threatened, or endangered species, as the planned activities aim to maintain the forest stand. Thus, project activities aimed at combating fire and deforestation can be highlighted. Positive impacts are expected from the project due to the proposed activities, generating net benefits for biodiversity that would probably not occur in the absence of the Project.

5.1.2 High Conservation Values (CCB, B1.2)

High conservation value	HCV 1: Concentrations of biological diversity, including endemic, rare, threatened, or endangered species, significant at the global, regional, or national level.
Qualifying attribute	The Caiarara REDD+ Project area is located in the eastern portion of the Amazon, specifically in one of the eight Amazonian Centers of Endemism, the Belém Center of Endemism. This region is of extreme biological importance due to its high rate of endemism, where many species of flora and fauna are found exclusively in this area. With regard to birds, according to BirdLife DataZone, there are two Important Bird Areas (IBAs) in the project region: Caxiuanã/Portel (PA05)[93] and Rio Capim (PA06)[94] . In addition, the results of the bird survey indicate that thirteen bird taxa recorded in bibliographic data or field samples are considered endemic to the Amazon biome. Among these, five are endemic to the Belém Endemism Center, meaning they are restricted to the region: the jacamin-de-costas-escuras (*Psophia obscura), the* tiriba-pérola (*Pyrrhura lepida)*, the araçari-de-pescoço-vermelho (*Pteroglossus bitorquatus)*, the pica-pau-dourado-de-Belém (*Piculus paraensis)* and the mãe-de-taoca (*Phlegopsis nigromaculata)*. Regarding species at risk of extinction according to the IUCN, three bird species were recorded in the PA: the tiriba-pérola (*Pyrrhura coerulescens)*, the ararajuba (*Guaruba guarouba)* and the marianinha-de-cabeça-amarela (*Pionites leucogaster),* classified as "Vulnerable" (VU). With regard to mammals, five species identified are threatened with extinction according to the IUCN, namely: tamanduá-bandeira (*Myrmecophaga tridactyla)*, bugio-de-mãos-ruivas (*Alouatta belzebul*), anta (*Tapirus terrestris*) e queixada (*Tayassu pecari*) in the "Vulnerable" (VU) class, and the macaco-caiarara (*Cebus kaapori)* in the "Critically Endangered" (CR) class. In addition, nine mammal species recorded in the Caiarara REDD+ project region according to bibliographic data are endemic to the Amazon. Two primates stand out: the macaco-caiarara (*Cebus kaapori*) and the cuxiú-preto (*Chiropotes satanas*). The cuxiú-preto, classified as "Endangered" (EN) according to the global list, occurs in

[93] BirdLife International (2025) Site factsheet: Caxiuanã / Portel. Downloaded from <https://datazone.birdlife.org/site/factsheet/caxiuan%C3%A3-portel> 25/07/2025.

[94] BirdLife International (2025) Site factsheet: Rio Capim. Downloaded from <https://datazone.birdlife.org/site/factsheet/rio-capim> 25/07/2025.

	eastern Amazonia and the remaining populations of the species are mainly threatened by deforestation and fragmentation of their habitat, in addition to suffering from hunting pressure. The macaco-caiarara, classified as "Critically Endangered" (CR) on the global list, is an extremely rare primate with an even smaller distribution than the cuxiú-preto and suffers from the same threats. Unfortunately, the area where these primates occur coincides with the area of greatest human occupation in the Amazon, which has seen a reduction of approximately 76% of its original forest cover (Almeida & Vieira, 2010). It is believed that both primates have seen a reduction of at least 80% of their original population in the last 30 years alone ([95]). The macaco-caiarara was recorded on the RAA I farm during data collection for the diagnosis, a very important record as it is a rare primate to be observed in the wild. Regarding flora species, seven species threatened with extinction according to the IUCN have confirmed occurrence in the Project area, namely: *Cedrela fissilis* (cedro), *Eriotheca longipedicellata* (sumauma-da-terra-firme), *Pouteria oppositifolia* (Guajará-cinza) and *Mezilaurus itauba* (itauba), classified as "Vulnerable" (VU), *Manilkara elata* (Maçaranduba) and *Ocotea fragrantissima* (Louro-canela), classified as "Endangered" (EN); and *Vouacapoua americana* (Acapu), classified as "Critically Endangered" (CR). Nevertheless, according to the IPAM 2024 report, which classified areas based on conservation priority, the project area is classified as extremely important, as it is home to numerous species of fauna and flora and is under intense pressure from human activities. Therefore, the project area as a whole has significant concentrations of endemic, rare, and endangered species, meeting the criteria for High Conservation Value 1 (HCV 1).
Focal area	The project area as a whole should be the focus area for the preservation of endemic, rare, and endangered species and needs to be managed and monitored to maintain or enhance this High Conservation Value Attribute.

[95] Chico Mendes Institute for Biodiversity Conservation. 2018. Red Book of Brazilian Fauna Threatened with Extinction. Brasília: ICMBio. 4162 p.

High conservation value	HCV 3: Areas containing rare or threatened ecosystems
Qualifying attribute	For an area to be characterized as HCV3, it must include ecosystems, habitats, and refugees of special importance due to their rarity or the level of threat they face, their rare or unique species composition.[96] These characteristics are relevant to the vegetation of Campinarana: it constitutes an ecological disjunction of a type of field vegetation with characteristic species, whose core area is located in the northwestern portion of the Amazon, distributed irregularly throughout the territory of Pará, occupying 0.3% of the state (Ferreira et al. 2010)[97] . The term Campinarana is a regional Amazonian name that means "false field," but it encompasses different phytophysiognomies, interconnected by edafoclimatic, physiognomic, and floristic gradients. Small areas of disjointed Campinara can occur throughout the Amazon Biome, including in the state of Pará, which has more than a dozen small 'campinas islands' (Ferreira, 2009)[98] . There is a specific threat to this vegetation: Ferreira et al. (2010; 2013, 2014)[99] report the destruction of Campinaranas due to illegal sand removal for construction, which results in huge holes in the exploited sites, since the soil of the campinas has a sandy texture, high leaching rates, and low fertility indices (Barbosa & Ferreira, 2004)[100] , and are unable to return to their original condition due to their low resilience (Costa & Pina-Rodrigues, 1996)[101] .

[96] Brown, E.; Dudley, N.; Lindhe, A.; Muhtaman, D.R.; Stewart, C.; Synnott, T. 2013. General guide for identifying High Conservation Values. HCV Resource Network (Portuguese version). Disponível em: https://global-uploads.webflow.com/624493bb51507d22cf218d50/628687247a00da00330a681e_HCVCommonGuide_portuguese_IFL2018.pdf Acesso em 28/03/2024.

[97] Ferreira, L.V., Thales, M.C., Pereira, J.L.G., Fernandes, J.A. Marin, Furtado, C. da S. & Chaves, P.P. 2010. Biodiversity. In: Monteiro, M.A., Menezes, C.R.C. & Galvão, I.M.F. (Eds.). Ecological-Economic Zoning of the Eastern Zone and Calha Norte of the State of Pará: Diagnosis of the Physical-Biotic Environment. Pará Rural Program Management Center, 2, Belém, p. 25-102. Available at: https://www.amazonia.cnptia.embrapa.br/publicacoes_estados/Para/.

[98] Ferreira, C. A. C. (2009). Comparative analysis of woody vegetation in the grassland ecosystem of the Brazilian Amazon. Doctoral thesis (Biological Sciences, Botany). National Institute for Amazonian Research, Manaus, 300p. Available at: https://repositorio.inpa.gov.br/handle/1/12804 Accessed on: 01/26/2024.

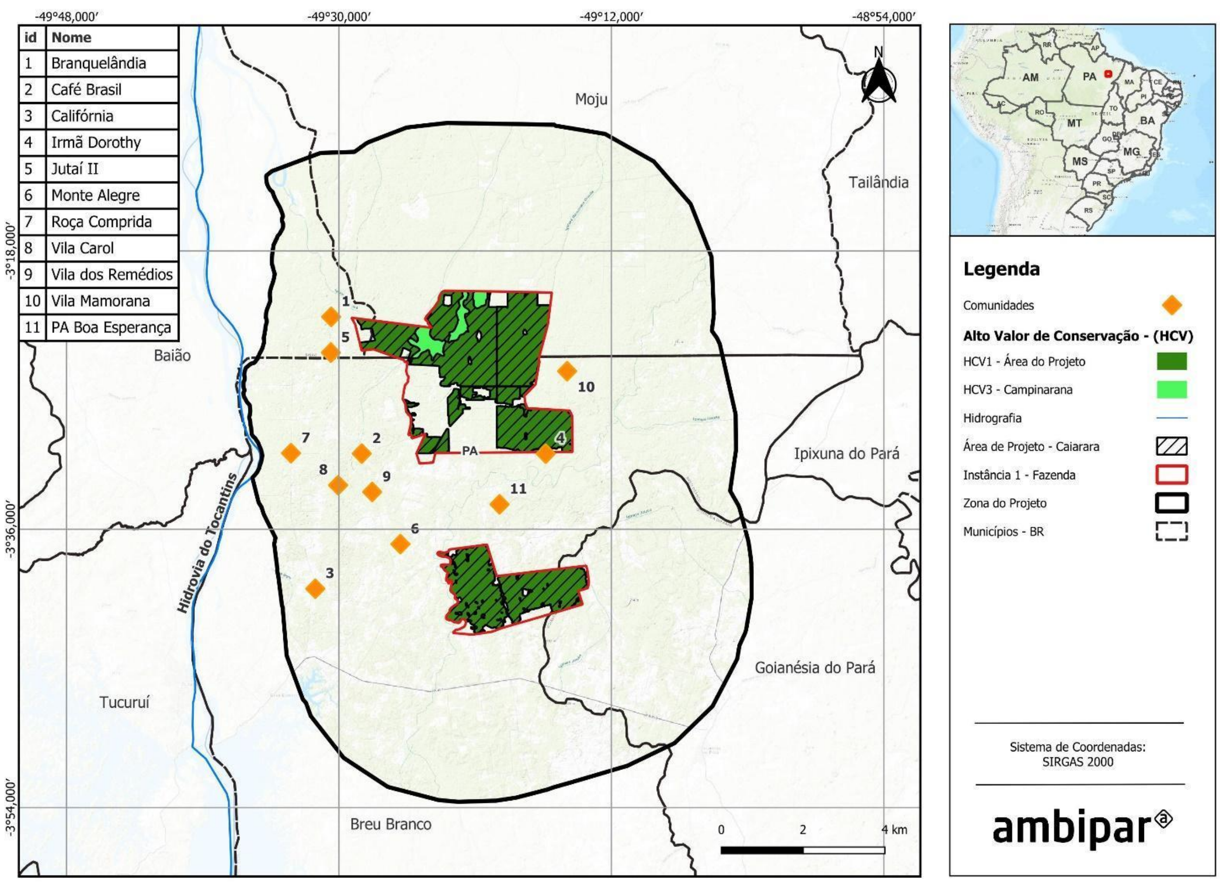
[99] Ferreira, L. V., Chaves, P. P., Cunha, D. D. A., & Parolin, P. (2014). Floristics and structure of the campinaranas of the lower Tocantins River as a basis for the creation of new conservation units in the state of Pará. Pesquisas Botânica, 65, 169-182.

[100] Barbosa, R. I., & Ferreira, C. A. C. (2004). Above-ground biomass of a "campina" ecosystem in Roraima, northern Brazilian Amazon. Acta Amazonica, 34, 577-586.

[101] Costa, L. G. S., & Piña-Rodrigues, F. C. M. (1996). Technical feasibility of recovering degraded areas. Belém: FCAP. Documentation and Information Service.

Focal area	Although the campinaranas areas are not within the Caiarara REDD+ Project area, they will be indirectly impacted by the project's activities to conserve vegetation cover and protect and monitor biodiversity. The areas are located in the Project Zone, within the properties where the Project was implemented, and cover 1,921 hectares, representing 4.2% of the farm area.

Figure 36. Map of the focal area of High Conservation Value Attributes identified for the Caiarara REDD+ Project.



5.1.3 Scenario without project: Biodiversity (CCB, B1.3)

If the Caiarara REDD+ Project is not implemented, the area will continue to be subject to various threats, including: (I) illegal logging, both for timber and charcoal; (II) deforestation accompanied by land use change, mainly for pasture; (III) forest fires intensified by land use change; (IV) illegal extraction of white sand from the Campinarana areas; and (V) global and local climate change caused by deforestation (Flora Diagnosis – STA, 2024).

Future scenarios for the Amazon[102] indicate that by 2050, around 40% of forests could be lost, threatening 51% of common species and 43% of rare species. In the improved governance scenario, 21% of forests would be lost, threatening 16% of common species and 25% of rare species. In both scenarios, forest loss in the Eastern Amazon would be more severe, with forest losses ranging from 42% to 76%.

The project area is located in the eastern Amazon, in one of the eight Amazonian Centers of Endemism, the Belém Center of Endemism. The future scenario of deforestation in the region would strongly impact endemic and endangered species, with more than 86% experiencing population losses of more than 80% of their individuals.

Habitat loss resulting from deforestation in the project area would decrease the size of local populations, reducing the diversity of flora and, consequently, the fauna that depend on the forest. The main question is 'how much could this influence population size at the regional level?'. Gomes et al (2019)[103] estimated the proportion of more than 5,000 tree species in different regions of the Amazon and projected how these species would decline in scenarios of deforestation and climate change. Based on this information, we can predict strong impacts on the Project's species if deforestation continues at the current rate, especially considering the importance of threatened and endemic species.

Therefore, the implementation of the Caiarara REDD+ Project is crucial for the forest to be preserved, maintaining carbon stocks and influencing other areas and landowners in the surrounding area to do the same. In addition, the project's activities aim to preserve existing biodiversity and monitor endemic and endangered species so that they can thrive.

5.2 Positive net impacts on biodiversity

5.2.1 Expected changes in biodiversity (VCS, 3.19; CCB, B2.1)

There has been no deforestation, conversion, or drainage of natural ecosystems in the last 10 years in the Project Area, according to satellite image analysis using MapBiomas and PRODES data. It can be verified that the Project Area has been entirely classified as forest since 2014.

Expectations for changes in biodiversity as a result of the Caiarara REDD+ Project were estimated using the Theory of Change method, described in Section 2.1.17. With this definition in mind, the cause-and-effect link between the activities planned for the project, the actions taken, and the expected short-, medium-, and long-term results become clearer.

All activities were designed with the aim of promoting the long-term preservation of forest cover in the Project Area and, consequently, protecting the existing fauna and flora in the area. With regard to biodiversity, it is understood that the most significant changes compared to the scenario without the

[102] Ter Steege, H.; Pitman, N.C.; Sabatier, D.; Baraloto, C.; Salomão, R.P.; Guevara, J.E., et al. (2013). Hyperdominance in the Amazonian tree flora. Science, 342, 1243092. DOI: 10. 1126/science.1243092

[103] Gomes, V.H.; Vieira, I.C.; Salomão, R.P.; Ter Steege, H. (2019). Amazonian tree species threatened by deforestation and climate change. Nature Climate Change, 9(7): 547-553.

implementation of the project are linked to the conservation of natural habitats. This allows for the preservation or growth of species that would otherwise be at risk due to land use change.

Therefore, it is clear that safeguarding the forest will result in beneficial changes to biodiversity as anticipated through the Project's actions and the REDD+ mechanism. In addition, the biodiversity monitoring plan, through the monitoring of fauna and flora indicators, will allow for the verification and investigation of possible changes that could adversely affect biological diversity. This will provide a clearer understanding of the population dynamics of species, including endemic and endangered species, as well as conflicts resulting from human interaction with nature.

In summary, the expected changes in biodiversity in the Project are:

- Maintenance of species identified in the scenario without the project;
- Ensuring the conservation of habitats and species in the Brazilian Amazon;
- Reduce illegal activities through the Project's activities;
- Raise awareness and sensitivity among local workers and communities regarding environmental issues, reducing pressure from hunting;
- Contribute to the conservation of endangered and/or endemic species;
- Promote an ecologically balanced environment;
- Maintain landscape connectivity.

Changes in the main elements of biodiversity are described in the following tables.

Biodiversity element	Flora
Estimated change	Positive change due to activities to reduce deforestation and forest degradation, resulting in the maintenance and/or increase of flora species and populations.
Justification for change	The activities that will lead to the change were developed using the Theory of Change method, based on the flora diagnosis carried out in the Project Area. The indicators related to flora will be monitored throughout the project to validate the expected results. The Caiarara REDD+ Project will promote the maintenance of native forests through activities aimed at monitoring forests to contain the expansion of deforestation. To this end, the project area will be monitored via satellite to identify deforestation points (if any) and hot spots to prevent possible fires in the area. In addition, activities will be carried out locally, such as strengthening and increasing the intensity of property surveillance on the two farms in the Project to prevent third

	parties from invading the area and damaging the flora through deforestation and fauna through illegal hunting.

Biodiversity Element	Fauna
Estimated Change	Positive change due to activities to reduce deforestation and forest degradation, resulting in the conservation of habitats and, consequently, the maintenance and/or increase of local fauna species and populations.
Justification for Change	The activities that will lead to change were developed using the Theory of Change method, based on the fauna diagnosis carried out in the Project Area. The indicators related to fauna will be monitored throughout the project to validate the expected results. The activities designed for the Project are focused on the conservation of wildlife habitats, such as remote monitoring, asset surveillance, firefighting and mitigation, as well as environmental education to raise awareness about the consequences and damage caused by illegal hunting and fishing. These actions were designed based on the socio-environmental diagnosis carried out previously and will be applied in the Project to ensure the success of the conservation objectives throughout the project's lifetime.

5.2.2 Mitigation measures (VCS, 3.19; CCB, B2.3)

The data collected for biodiversity-related studies were satisfactory in terms of assessing the current biodiversity conservation context in the Project Area. However, longer-term studies are needed to understand the variations that occur in biodiversity over time as the forest changes.

According to the Brazilian Diagnosis of Biodiversity & Ecosystem Services[104] , the main threats to biodiversity are climate change and land use change, i.e., deforestation or any activity involving the conversion of native vegetation areas to other uses, such as agriculture, livestock, or mining.

In order to improve the population conditions of the species and mitigate the negative impacts caused by the internal and external factors mentioned above, the activities of the Caiarara REDD+ Project were

[104] BPBES (2018): Summary for decision makers of the assessment report of the Brazilian Platform on Biodiversity and Ecosystem Services. Campinas, SP. 24 pages.

designed and structured to act as mitigating measures for the main threats to biodiversity, such as continuous monitoring of fauna and flora and monitoring of endangered species classified as High Conservation Value Attributes (HCVAs), with the aim of understanding ecosystem dynamics and detecting any significant changes.

Furthermore, through environmental education activities planned for stakeholders and the promotion of sustainable practices, it is hoped to achieve community involvement and awareness of the importance of these species in maintaining ecological balance, reducing potential threats. Finally, increased asset surveillance will prevent third-party activities such as illegal hunting and forest degradation, contributing to the effective protection of key species. In this sense, no significant negative impacts on biodiversity are expected as a result of project activities. All activities have been designed and structured to act as mitigating measures for the main internal and external impacts and threats to biodiversity identified in the studies carried out in Socioeconomic and Biodiversity Diagnostics (STA, 2024 and Ambiens, 2024). Thus, the proposals presented will be more effective and significant, resulting in relevant and beneficial changes for the region, its surroundings, and biodiversity. The activities were structured in accordance with the precautionary principle, adopting preventive actions even in the face of scientific uncertainties, with the aim of avoiding potential damage to biodiversity and High Conservation Value Attributes.

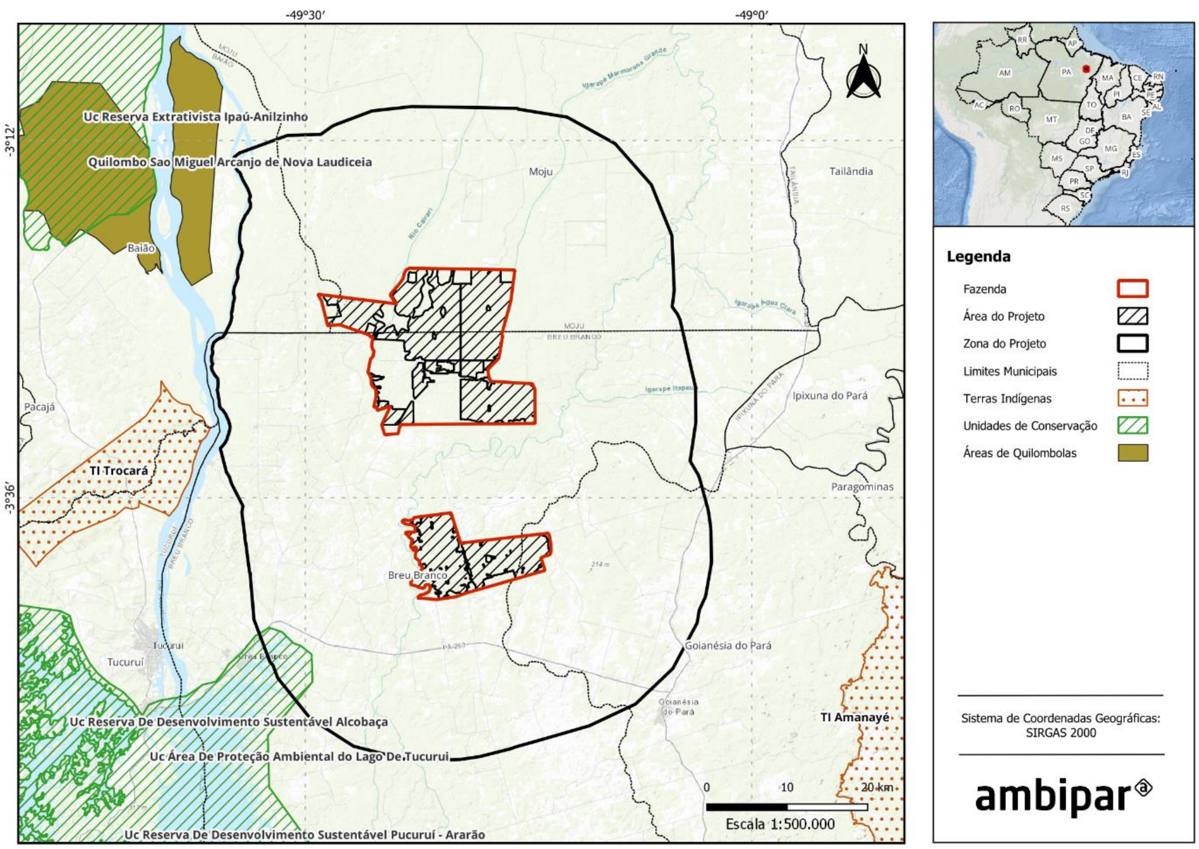
5.2.3 Net positive impacts on biodiversity (CCB, B2.2, GL1.4)

The expectations of net positive impact on biodiversity as a result of the Caiarara REDD+ Project were estimated using the Theory of Change method described in Section 2.1.17. Project activities were designed to contribute directly to reducing deforestation and degradation, maintaining forest cover, and conserving the diversity of fauna and flora species and their habitats. As a result, the net impacts on biodiversity in the project area are expected to be positive compared to the conditions of the land use scenario without the project, since the latter suffers from deforestation and forest cover degradation and its future scenario projection is the intensification of biodiversity loss of fauna and flora (Section 5.1.3).

In this context, the project will implement measures related to the study, conservation, and monitoring of existing flora and fauna species in the area, in addition to monitoring the HCVs listed in section 5.1.2, to contribute positively to local biodiversity. For species identified as being under some degree of threat, the project will contribute to the protection of their habitats by preserving and maintaining the forest cover of the Project Area, in addition to implementing perimeter monitoring and surveillance activities to prevent trespassing by third parties, thereby contributing to reducing hunting, fishing, and illegal deforestation.

The existence of few Conservation Units, Indigenous Lands, and Quilombola Areas in the region surrounding the farms makes the Project Area an important connector between these protected areas. The protection of forest massifs creates climate refuges for biodiversity, assisting in adaptation to climate change, which reinforces the expected positive impact of the project. The following map shows the protected areas surrounding the Project.

Figure 37. Map of Quilombola Areas, Conservation Units, and Indigenous Lands around the Caiarara REDD+ Project Zone.



In addition, the project can contribute to maintaining the local climate, minimizing global climate change by maintaining habitat connectivity, especially considering a larger spatial scale such as in the Project Area; in addition to protecting ecosystem services such as pollination, seed dispersal, gene flow, soil protection, hydrographic basins, among others.

5.2.4 High Conservation Values Protected (CCB, B2.4)

The project identified the presence of HCV 1, which represents the existence of concentrations of biological diversity including endemic, rare, threatened, or endangered species that are significant at the global, regional, or national level; and HCV 3, which represents the existence of areas containing rare or threatened ecosystems.

The activities developed by the Caiarara REDD+ Project throughout its duration will generate positive impacts on high conservation value attributes, as they include measures to maintain biodiversity-related attributes, such as the conservation and monitoring of fauna and flora species, including those threatened with extinction, in the case of HCV 1, and in addition, the project anticipates positive impacts on the campinaranas areas, considered rare and threatened ecosystems, in the case of HCV 3.

Forest conservation and protection, provided for by the REDD+ Project through activities developed using the Theory of Change (see Section 2.1.17) of strengthening asset surveillance, remote monitoring of deforestation and hot spots, continuous monitoring of biodiversity, among others, will directly contribute to the preservation of these vulnerable attributes, ensuring the maintenance of biodiversity and mitigating the risks that threaten them.

Thus, no negative effects related to biodiversity are expected from the activities proposed by the project on High Conservation Value attributes. However, the project also has adaptive response mechanisms in place in case unforeseen negative impacts are identified during the project's life cycle, such as reviewing project activities and strengthening mitigation actions to prevent, detect, and respond to potential risks to biodiversity, ensuring an adaptive management structure.

5.2.5 Species used (VCS, 3.19; CCB, B2.5, B2.6)

No fauna or flora species will be introduced as part of the Caiarara REDD+ Project activities.

5.2.6 Invasive species (VCS, 3.19; CCB, B2.5)

There are no invasive species in the Caiarara REDD+ Project area.

5.2.7 Exclusion of GMOs (CCB, B2.7)

No GMO species will be used to generate GHG emission reductions or carbon dioxide removal in the Caiarara REDD+ Project.

5.2.8 Justification of Inputs (VCS, 3.19; CCB, B2.8)

The Caiarara REDD+ Project will not use any fertilizers, chemical pesticides, biological control agents, or other inputs.

5.2.9 Residual products (VCS, 3.19; CCB, B2.9)

Dow Brazil has a Waste Management Plan in place, which aims to minimize waste generation, involving the stages of identification/classification, packaging, temporary storage, and transportation to the appropriate destination, mitigating the effects of pollution caused by waste generated in the environment.

For the activities proposed by the Project, no additional waste generation other than that described above is expected. Any waste produced will be separated for recycling or composting, whenever possible, considering the aforementioned Plan, as well as the environmental laws related to the subject. If separation for recycling or composting is not possible, the waste will be disposed of as regular trash. As new activities are selected to comprise the Project, the identification, classification, and management of their waste will be properly conducted.

5.3 Impacts on external biodiversity

5.3.1 Negative impacts on external biodiversity (CCB, B3.1) and mitigation measures (CCB, B3.2)

In general, a REDD+ Project's main objective is to reduce emissions from unplanned deforestation and forest degradation. Therefore, it is indisputable that the conservation provided by this mechanism benefits the Project Zone as a whole. The Caiarara REDD+ Project aims to achieve greater control over anthropogenic disturbances that would negatively impact biodiversity, mainly habitat loss due to deforestation for the predominant economic activities in the region. However, with the strengthening of measures aimed at conserving the forest and its resources, these disturbances and activities may naturally shift to areas outside the Project Zone.

It is worth noting that all Project activities were discussed and evaluated (Section 2.3.10) with the communities surrounding the Project Area in order to understand the scope of the Project activities and forest conservation. Furthermore, it is important to note that the activities proposed by the Project will extend to stakeholders located within the Project Area, but with potential positive impacts beyond these boundaries.

The measures designed to mitigate negative impacts on external biodiversity are presented in the following table:

Negative offsite impact	Mitigation measure(s)
Relocation of illegal activities outside the Project Area	To prevent the displacement of illegal hunting, fishing, and deforestation activities, the mitigation measure established is to engage rural communities surrounding the Project Area through activities proposed by the Theory of Change (see Section 2.1.17), which are: Strengthening local governance; Developing and strengthening value chains; Improving the well-being of stakeholders; Promoting educational activities; and Strengthening inclusion, diversity, and equity. Through these activities, the Project seeks to offer sustainable production alternatives, encourage new environmentally responsible practices in the daily lives of communities, raise awareness about the protection of biodiversity and ecosystem services, provide training and improve the well-being of surrounding residents, seeking greater engagement with the Caiarara REDD+ Project. The long-term goal is to promote changes in cultural habits, integrate sustainable practices, and prevent illegal deforestation and hunting, thereby strengthening environmental preservation and fostering a greater understanding of the importance of standing forests, protected local biodiversity, and the benefits of a balanced environment.

The REDD+ Caiarara Project has adaptive response mechanisms, such as reviewing activities and reinforcing mitigation actions, should an unforeseen negative impact on external biodiversity be identified during the Project's life cycle, with a view to preventing, detecting, and responding to any risks to biodiversity, ensuring an adaptive management structure.

5.3.2 Net external biodiversity benefits (VCS, 3.19; CCB, B3.3)

The main positive impact expected outside the Project Zone is the promotion of biodiversity through the maintenance of large forest remnants that may serve as refuge and protection for threatened species and ecosystems. Thus, despite the possible displacement of activities and disturbances outside the Project Zone, which may cause occasional negative impacts, the connectivity of the landscape, the promotion of gene flow between forest areas, the protection of ecosystem services such as pollination, seed dispersal, and protection of soils and hydrographic basins, the maintenance of the local climate, the maintenance of natural habitats, and the *in situ* monitoring of endangered and endemic species are benefits that justify the presence of the Caiarara REDD+ Project. If any negative impacts on external or internal biodiversity are identified during the Project's life cycle, the Caiarara REDD+ Project is committed to proposing measures to mitigate the impact in the best possible way.

5.4 Biodiversity Impact Monitoring

5.4.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

This section presents the biodiversity-related variables to be monitored, the types of measurement, and the sampling methods related to the activities within the biodiversity scope of the Project. The monitoring variables are directly linked to the Project's biodiversity conservation objectives and to the expected outputs, outcomes, and impacts identified in the Project's causal model related to biodiversity conservation.

The Project seeks to protect High Conservation Value (HCV) attributes through socio-environmental activities that aim to contain deforestation and degradation, in accordance with the Theory of Change developed (see Section 2.1.17). As described in Section 5.1.2, the identified High Conservation Value Attributes are related to HCV 1: Concentrations of biological diversity including endemic, rare, threatened or endangered species that are significant at the global, regional or national level, and HCV 3: Areas containing rare or threatened ecosystems. The Monitoring Plan includes the assessment of the effectiveness of the actions taken to maintain and improve the aforementioned HCVs, which will be carried out based on the analysis of the monitored biodiversity indicators presented below.

Flora

Data / Parameter	Species richness with DBH > 10 cm and Shannon-Weaver indices
Area	Project area and Campinarana
Data unit	Number

Description	Number of species with DBH > 10 cm in the REDD+ Caiarara project area
Description of measurement methods and procedures to be applied	To monitor the number of species with DBH > 10 cm in the project area, it is necessary to identify all trees present in the selected transects in the field.
Monitoring frequency	At each verification
Purpose of the data	The data allows for analysis and identification of the richness of plant species in the area, as well as comparison with each data collection, enabling the identification of decreases or increases in biodiversity.
Comments	-

Data / Parameter	Number of plant species with some degree of threat or endemic to State of Pará
Area	Project area and Campinarana
Data	Number
Description	Estimate the number of species with some degree of threat or endemic to the State of Pará based on species identification carried out in the field during the campaign.
Description of measurement methods and procedures to be applied	Identification will be carried out by cross-referencing species information obtained in the field with the IUCN and Pará lists. This will make it possible to verify which species identified in the project area are classified as threatened or endemic to the region. The information obtained will be transferred to an Excel document for storage and comparison.
Monitoring frequency	At each verification
Purpose of the data	The data allows the identification of the number of species with some degree of threat or endemic to the project area.
Comments	-

Data / Parameter	Density of individuals of *Vouacapoua americana* (Acapu)
Area	Project area
Data unit	Number
Description	Density of individuals of the species *Vouacapoua americana* (Acapu) identified in each monitoring campaign in the Project Area
Description of measurement methods and procedures to be applied	To monitor the density of the species, the density of individuals of *Vouacapoua americana* (Acapu) identified in the respective monitoring campaign will be highlighted from the results of the flora monitoring. Other data collection methodologies may be established jointly between proponents and strategic partners defined throughout the Project and detailed prior to each monitoring campaign.
Monitoring frequency	At each verification
Purpose of the data	The data allow the density of individuals of the species *Vouacapoua americana* (Acapu) to be identified, assisting in the analysis of population trends in the Project Area.
Comments	-

Data / Parameter	Density of individuals of *Ocotea fragrantissima* (Louro-canela).
Area	Project area
Data unit	Number
Description	Density of individuals of the species *Ocotea fragrantissima* (Louro-canela) identified in each monitoring campaign in the Project Area
Description of measurement methods and procedures to be applied	To monitor the density of the species, the density of individuals of *Ocotea fragrantissima* (Louro-canela) identified in the respective monitoring campaign will be highlighted from the flora monitoring results. Other data collection methodologies may be established jointly between proponents and strategic partners defined throughout the Project and detailed prior to each monitoring campaign.
Monitoring frequency	At each verification.

Purpose of the data	The data allows for the identification of the density of individuals of the species *Ocotea fragrantissima* (Louro-canela), assisting in the analysis of population trends in the Project Area.
Comments	-

Data / Parameter	Density of individuals of Maçaranduba (*Manilkara elata*)
Area	Project area
Data unit	Number
Description	Density of individuals of the species Maçaranduba (*Manilkara elata*) identified in each monitoring campaign in the Project Area
Description of measurement methods and procedures to be applied	To monitor the density of the species, the density of individuals of Maçaranduba (*Manilkara elata*) identified in the respective monitoring campaign will be highlighted from the results of the flora monitoring. Other data collection methodologies may be established jointly between proponents and strategic partners defined throughout the Project and detailed prior to each monitoring campaign.
Monitoring frequency	At each verification.
Purpose of the data	The data allow the identification of the density of individuals of the Maçaranduba (*Manilkara elata*) species, assisting in the analysis of population trends in the Project Area.
Comments	-

Fauna

Data / Parameter	Fauna species richness
Data unit	Number
Description	Number of species per fauna group (avifauna, mastofauna, and herpetofauna)
Data source	Data collection in field campaigns

Description of measurement methods and procedures to be applied	Fauna species richness has different methodologies for each group, which are: - Avifauna: data collection can be done using MacKinnon's quantitative list technique. This technique was developed for rapid surveys in tropical environments rich in bird species. - Mastofauna: data collection can be done through camera trap records, sightings, and identification of tracks and droppings. - Herpetofauna: data collection can be done by active search, i.e., visual and auditory contacts.
Monitoring frequency	At each verification
Purpose of data	The data allows for the identification of the diversity and abundance of species existing in the project area.
Comments	-

Data / Parameter	Number of fauna species with some degree of threat or endemic to the Belém Endemism Center
Area	Project area
Data unit	Number
Description	Estimate the number of species with some degree of threat or endemic to the Belém Endemism Center based on species identification carried out in the field during the campaign.
Description of measurement methods and procedures to be applied	Identification will be carried out by cross-referencing species information obtained in the field, and the IUCN and Pará lists. This will make it possible to verify which species identified in the project area are classified as threatened or endemic to the region. The information obtained will be transferred to an Excel document for storage and comparison.
Monitoring frequency	At each verification
Purpose of the data	The data allows the identification of the number of species with some degree of threat or endemic to the project area.

Comments	-

Data / Parameter	Number of records of Macaco-cairara (*Cebus kaapori*)
Area	Project area
Data unit	Number
Description	Number of records of the species Macaco-cairara (*Cebus kaapori*) identified in each monitoring campaign in the Project Area
Description of measurement methods and procedures to be applied	To monitor the species indicator, the number of records of Macaco-cairara (*Cebus kaapori*) individuals identified in the respective monitoring campaign will be highlighted from the fauna monitoring results. Other data collection methodologies may be established jointly between proponents and strategic partners defined throughout the Project and detailed prior to each monitoring campaign.
Monitoring frequency	At each verification
Purpose of the data	The data allows the number of individuals of the Macaco-cairara (*Cebus kaapori*) species to be identified, assisting in the analysis of population trends in the Project Area.
Comments	-

5.4.2 Disclosure of the Biodiversity Monitoring Plan (CCB, B4.3)

The Biodiversity Monitoring Plan and all its results will be publicly disclosed on the VERRA registration page and on the Biofilica Ambipar Environmental Investments S/A website. The validated and verified monitoring results will be available in the public version of the report on the Verra website, ensuring free access to any interested party.

Summaries of the main results will be prepared in accessible language and presented in person to the communities through adapted printed and digital materials. Copies will also be made available at strategic locations (farm headquarters and community centers). The results will be disclosed after each official project verification, with regular updates following the monitoring cycles defined in the PDD.

Information about the Project will also be disseminated via email and WhatsApp (with a pre-established contact list of stakeholders), and on social media, such as Instagram and LinkedIn, of the project proponent.

In this way, it is expected that all necessary efforts to provide full transparency and distribute key Project information and documents will be achieved.

5.5 Optional Criterion: Exceptional Biodiversity Benefits

5.5.1 High Biodiversity Conservation Priority Status (CCB, GL3.1)

The Caiarara REDD+ Project Area is considered a high priority area for biodiversity conservation, as it meets the "vulnerability" criterion of Key Biodiversity Areas (KBA) (Langhammer et al., 2007)[105] , as they were identified in the data collection for the fauna and flora diagnosis as endangered species in the categories "Critically Endangered" (CR) and "Endangered" (EN), according to the IUCN Red List[106] .

In addition to harboring a large number of fauna and flora species and species with some degree of threat and/or endemic, it also has a rare and threatened ecosystem, the campinarana areas, located in the Project Zone within the farm properties, which need to be protected urgently (see Section 5.1.2 for more details). In the Project region, the presence of endangered flora and fauna species was verified according to the List of Endangered Species of Pará (Decree 802/2008), the National List of Endangered Species (MMA[107]) and the Global List of Endangered Species (IUCN), as described in section 5.1.1.

Considering the criteria of the IUCN Red List of Threatened Species and meeting the requirements of *the CCB Standard v3.1* Gold Label for Exceptional Biodiversity Benefits, the species classified as "Critically Endangered" and "Endangered" are:

- Fauna:
    - Macaco-cairara (*Cebus kaapori*) – "Critically Endangered" (CR);

- Flora:
    - Acapu (*Vouacapoua americana*) – "Critically Endangered" (CR);
    - Louro canela (*Ocotea fragrantissima*) – "Endangered" (EN);
    - Maçaranduba *(Manilkara elata)* "Endangered" (EN).

5.5.2 Population trends of key species (CCB, GL3.2, GL3.3)

The key species and their respective population trends for the Caiarara REDD+ Project can be found in the tables below.

[105] Langhammer, P.F., Bakarr, M.I., Bennun, L.A., Brooks, T.M., Clay, R.P., Darwall, W., De Silva, N., Edgar, G.J., Eken, G., Fishpool, L.D.C., Fonseca, G.A.B. da, Foster, M.N., Knox, D.H., Matiku, P., Radford, E.A., Rodrigues, A.S.L., Salaman, P., Sechrest, W., and Tordoff, A.W. 2007, "Identification and gap analysis of Key Biodiversity Areas: Targets for comprehensive protected area systems, Best Practice Protected Areas Guidelines Series No. 15. IUCN, Gland, Switzerland.

[106] IUCN 2025. The IUCN Red List of Threatened Species. Version 2025-1. https://www.iucnredlist.org.

[107] MMA DECREE No. 148, OF JUNE 7, 2022. National List of Endangered Species. MINISTRY OF THE ENVIRONMENT

Key species	Macaco-cairara (*Cebus kaapori*)
Population trend at the start of the project	The species is considered "critically endangered" (CR) by the IUCN (2023) and is in continuous decline in terms of mature individuals[108] . The number of individuals in the project area is unknown.
Scenario without project	Without the Caiarara REDD+ Project, the project area is vulnerable to illegal logging and hunting. As a result, the population of this species is declining and its threat status is worsening, mainly due to habitat loss and degradation caused and exacerbated by deforestation and forest degradation, as well as predatory hunting.
Scenario with project	With the REDD+ Caiarara project, *Cebus kaapori* will be able to maintain or improve the population status of its species, since its habitat will be preserved and the risk of forest degradation and illegal predatory hunting will be mitigated through project activities. In addition, the species will be continuously monitored to establish action plans if necessary.

Key species	Acapu (*Vouacapoua americana*)
Population trend at the start of the project	The species is considered "critically endangered" (CR) by the IUCN[109] and the population trend is declining. The number of individuals in the project area is unknown.
Scenario without the project	Without the Caiarara REDD+ Project, the project area is vulnerable to deforestation and illegal logging. The species in question is critically endangered due to overexploitation for the construction of fences and furniture. Without forest protection, the remaining individuals in the project area are likely to be affected.

[108] Fialho, M.S., Jerusalinsky, L., Moura, E.F., Ravetta, A.L., Laroque, P.O., de Queiroz, H.L., Boubli, J.P. & Lynch Alfaro, J.W. 2021. *Cebus kaapori* (amended version of 2020 assessment). *The IUCN Red List of Threatened Species* 2021: e.T40019A191704766. https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T40019A191704766.en.

[109] Varty, N. & Guadagnin, D.L. 1998. *Vouacapoua americana. The IUCN Red List of Threatened Species* 1998: e.T33918A9820054. https://dx.doi.org/10.2305/IUCN.UK.1998.RLTS.T33918A9820054.en.

Scenario with project	With the REDD+ Caiarara Project, *Vouacapoua americana* will be able to maintain or improve the population status of its species, since the project's activities aim to preserve the forest by preventing forest degradation by third parties, thus preventing the species in question from being exploited.

Key species	Louro-canela (*Ocotea fragrantissima* Ducke)
Population trend at the start of the project	The species is considered "endangered" (EN) by the IUCN[110] and the population trend is declining. The number of individuals in the project area is unknown.
Scenario without the project	Without the Caiarara REDD+ Project, the project area is vulnerable to deforestation practices. The species in question is endangered due to illegal logging. Without forest protection, the remaining individuals in the project area are likely to be affected.
Scenario with project	With the REDD+ Caiarara Project, the Louro-canela (*Ocotea fragrantissima* Ducke) will be able to maintain or improve the population status of its species, since the project's activities aim to preserve the forest by preventing forest degradation by third parties, thus preventing the species in question from being exploited.

Key species	Maçaranduba (*Manilkara elata*)
Population trend at the start of the project	The species is considered "endangered" (EN) by the IUCN[111] and the population trend is declining. The number of individuals in the project area is unknown.
Scenario without the project	Without the Caiarara REDD+ Project, the project area is vulnerable to deforestation practices. The species in question is endangered

[110] Moraes, M., Fernandez, E., Martinelli, G. & Quinet, A. 2021. *Ocotea fragrantissima. The IUCN Red List of Threatened Species* 2021: e.T189605533A189605535. https://dx.doi.org/10.2305/IUCN.UK.2021-2.RLTS.T189605533A189605535.pt.

[111] Pires O'Brien, J. 1998. *Manilkara elata. The IUCN Red List of Threatened Species* 1998: e.T35609A9943876. https://dx.doi.org/10.2305/IUCN.UK.1998.RLTS.T35609A9943876.en.

	due to illegal logging. Without forest protection, the remaining individuals in the project area are likely to be affected.
Scenario with project	With the REDD+ Caiarara Project, the Maçaranduba *(Manilkara elata)* will be able to maintain or improve the population status of its species, since it is the objective of the project activities to preserve the forest by preventing forest degradation by third parties, thus preventing the species in question from being exploited.

APPENDIX 1: SPECIES LIST

Table 38. List of species recorded in the Project Area in fauna and flora surveys. END= endemism; BR= Brazilian list of endangered species (MMA, 2022); PA= list of endangered species of the Pará State (PA, 2008); IUCN= IUCN Red List of endangered species.

Group	Cientific Name	Common Name	END	BR	PA	IUCN
avifauna	*Amazona amazonica*	curica				LC
avifauna	*Amazona farinosa*	papagaio-moleiro				LC
avifauna	*Amazona farinosa*	papagaio-moleiro				LC
avifauna	*Ara chloropterus*	arara-vermelha				LC
avifauna	*Ara macao*	araracanga				LC
avifauna	*Ardea alba*	garça-branca				LC
avifauna	*Attila cinnamomeus*	tinguaçu-ferrugem				LC
avifauna	*Attila spadiceus*	capitão-de-saíra-amarelo				LC
avifauna	*Automolus paraensis*	barranqueiro-do-pará				LC
avifauna	*Brotogeris chrysoptera*	periquito-de-asa-dourada				LC
avifauna	*Brotogeris chrysoptera*	periquito-de-asa-dourada				LC
avifauna	*Buteo nitidus*	gavião-pedrês				LC
avifauna	*Butorides striata*	socozinho				LC
avifauna	*Cacicus haemorrhous*	guaxe				LC
avifauna	*Cairina moschata*	pato-do-mato				LC
avifauna	*Campephilus rubricollis*	pica-pau-de-barriga-vermelha				LC

avifauna	*Campylopterus largipennis*	asa-de-sabre-cinza				LC
avifauna	*Cantorchilus leucotis*	garrinchão-de-barriga-vermelha				LC
avifauna	*Cathartes melambrotus*	urubu-da-mata				LC
avifauna	*Celeus undatus*	pica-pau-barrado				LC
avifauna	*Celeus undatus*	pica-pau-barrado				LC
avifauna	*Ceratopipra rubrocapilla*	cabeça-encarnada				LC
avifauna	*Cercomacra cinerascens*	chororó-pocuá				LC
avifauna	*Cercomacroides laeta*	chororó-didi				LC
avifauna	*Cercomacroides laeta*	chororó-didi				LC
avifauna	*Chaetura chapmani*	andorinhão-de-chapman				LC
avifauna	*Chaetura spinicaudus*	andorinhão-de-sobre-branco				LC
avifauna	*Chelidoptera tenebrosa*	urubuzinho				LC
avifauna	*Colonia colonus*	viuvinha				LC
avifauna	*Coragyps atratus*	urubu				LC
avifauna	*Crotophaga ani*	anu-preto				LC
avifauna	*Crypturellus cinereus*	inambu-pixuna				LC
avifauna	*Crypturellus soui*	tururim				LC
avifauna	*Crypturellus strigulosus*	inambu-relógio				LC
avifauna	*Crypturellus variegatus*	inambu-anhangá				LC
avifauna	*Cyanerpes caeruleus*	saí-de-perna-amarela				LC
avifauna	*Cyanoloxia rothschildii*	azulão-da-amazônia				LC

avifauna	*Cyclarhis gujanensis*	pitiguari				LC
avifauna	*Dacnis cayana*	saí-azul				LC
avifauna	*Dendrexetastes paraensis*	arapaçu-galinha-do-pará	BR	VU		-
avifauna	*Dromococcyx phasianellus*	peixe-frito				LC
avifauna	*Elanoides forficatus*	gavião-tesoura				LC
avifauna	*Falco rufigularis*	cauré				LC
avifauna	*Formicarius analis*	pinto-do-mato-de-cara-preta				LC
avifauna	*Formicarius colma*	galinha-do-mato				LC
avifauna	*Galbula dea*	ariramba-do-paraíso				LC
avifauna	*Grallaria varia*	tovacuçu				LC
avifauna	*Guaruba guarouba*	ararajuba		VU	VU	VU
avifauna	*Harpagus bidentatus*	gavião-ripina				LC
avifauna	*Harpagus bidentatus*	gavião-ripina				LC
avifauna	*Heliothryx auritus*	beija-flor-de-bochecha-azul				LC
avifauna	*Herpetotheres cachinnans*	acauã				LC
avifauna	*Ibycter americanus*	cancão				LC
avifauna	*Icterus pyrrhopterus*	graúna				LC
avifauna	*Ictinia plumbea*	sovi				LC
avifauna	*Jacamerops aureus*	jacamaraçu				LC
avifauna	*Jacana jacana*	jaçanã				LC
avifauna	*Lamprospiza melanoleuca*	pipira-de-bico-vermelho				LC

avifauna	*Leptotila rufaxilla*	juriti-de-testa-branca				LC
avifauna	*Leptotila rufaxilla*	juriti-de-testa-branca				LC
avifauna	*Lipaugus vociferans*	cricrió				LC
avifauna	*Lipaugus vociferans*	cricrió				LC
avifauna	*Lophotriccus galeatus*	caga-sebinho-de-penacho				LC
avifauna	*Loriotus cristatus*	tiê-galo				-
avifauna	*Melanerpes cruentatus*	benedito-de-testa-vermelha				LC
avifauna	*Micrastur semitorquatus*	falcão-relógio				LC
avifauna	*Microcerculus marginatus*	uirapuru-veado				LC
avifauna	*Momotus momota*	udu				LC
avifauna	*Monasa morphoeus*	chora-chuva-de-cara-branca				LC
avifauna	*Monasa morphoeus*	chora-chuva-de-cara-branca				LC
avifauna	*Monasa nigrifrons*	chora-chuva-preto				LC
avifauna	*Myiozetetes cayanensis*	bentevizinho-de-asa-ferrugínea				LC
avifauna	*Myrmotherula axillaris*	choquinha-de-flanco-branco				LC
avifauna	*Myrmotherula longipennis*	choquinha-de-asa-comprida				LC
avifauna	*Nannopterum brasilianum*	biguá				-
avifauna	*Notharchus hyperrhynchus*	macuru-de-testa-branca				LC
avifauna	*Notharchus tectus*	macuru-pintado				LC
avifauna	*Nyctidromus albicollis*	bacurau				LC
avifauna	*Nystalus torridus*	rapazinho-estriado-do-leste				-

avifauna	*Ortalis superciliaris*	aracuã-de-sobrancelhas				LC
avifauna	*Patagioenas cayennensis*	pomba-galega				LC
avifauna	*Patagioenas speciosa*	pomba-trocal				LC
avifauna	*Patagioenas subvinacea*	pomba-botafogo				LC
avifauna	*Pauxi tuberosa*	mutum-cavalo				-
avifauna	*Phaethornis ruber*	rabo-branco-rubro				LC
avifauna	*Pheugopedius genibarbis*	garrinchão-pai-avô				LC
avifauna	*Piaya cayana*	alma-de-gato				LC
avifauna	*Pionites leucogaster*	marianinha-de-cabeça-amarela	Ama			VU
avifauna	*Pionus menstruus*	maitaca-de-cabeça-azul				LC
avifauna	*Piprites chloris*	papinho-amarelo				LC
avifauna	*Pitangus sulphuratus*	bem-te-vi				LC
avifauna	*Podilymbus podiceps*	mergulhão-caçador				LC
avifauna	*Polioptila plumbea*	balança-rabo-de-chapéu-preto				LC
avifauna	*Progne chalybea*	andorinha-grande				LC
avifauna	*Psarocolius bifasciatus*	japuguaçu				LC
avifauna	*Psarocolius decumanus*	japu				LC
avifauna	*Psarocolius viridis*	japu-verde				LC
avifauna	*Pseudastur albicollis*	gavião-branco				LC
avifauna	*Pteroglossus aracari*	araçari-de-bico-branco				LC
avifauna	*Pteroglossus bitorquatus*	araçari-de-pescoço-vermelho	Bel		VU	NT

avifauna	*Pteroglossus inscriptus*	araçari-de-bico-riscado				LC
avifauna	*Pygiptila stellaris*	choca-cantadora				LC
avifauna	*Pyrrhura coerulescens (Pyrrhura lepida)*	tiriba-pérola	Bel	VU	EN	VU
avifauna	*Querula purpurata*	anambé-una				LC
avifauna	*Ramphastos tucanus*	tucano-de-papo-branco				LC
avifauna	*Ramphastos vitellinus*	tucano-de-bico-preto				LC
avifauna	*Ramphocaenus melanurus*	chirito				LC
avifauna	*Ramphocelus carbo*	pipira-vermelha				LC
avifauna	*Rhytipterna simplex*	vissiá				LC
avifauna	*Rupornis magnirostris*	gavião-carijó				LC
avifauna	*Sarcoramphus papa*	urubu-rei				LC
avifauna	*Sclerurus macconnelli*	vira-folha-de-peito-vermelho				-
avifauna	*Spizaetus melanoleucus*	gavião-pato				LC
avifauna	*Stelgidopteryx ruficollis*	andorinha-serradora				LC
avifauna	*Tachornis squamata*	andorinhão-do-buriti				LC
avifauna	*Tachyphonus rufus*	pipira-preta				LC
avifauna	*Tamatia tamatia*	rapazinho-carijó				-
avifauna	*Tangara gyrola*	saíra-de-cabeça-castanha				LC
avifauna	*Thamnomanes caesius*	ipecuá				LC
avifauna	*Thamnomanes caesius*	ipecuá				LC
avifauna	*Thraupis episcopus*	sanhaço-da-amazônia				LC

avifauna	*Thraupis palmarum*	sanhaço-do-coqueiro				-
avifauna	*Tinamus guttatus*	inambu-galinha				NT
avifauna	*Tityra semifasciata*	anambé-branco-de-máscara-negra				LC
avifauna	*Tolmomyias flaviventris*	bico-chato-amarelo				LC
avifauna	*Trogon melanurus*	surucuá-de-cauda-preta				LC
avifauna	*Tyranneutes stolzmanni*	uirapuruzinho				LC
avifauna	*Tyrannus melancholicus*	suiriri				LC
avifauna	*Veniliornis affinis*	picapauzinho-avermelhado				LC
avifauna	*Willisornis vidua*	rendadinho-do-xingu				LC
avifauna	*Xipholena lamellipennis*	bacacu-preto	Ama	VU		NT
avifauna	*Xiphorhynchus guttatus*	arapaçu-de-garganta-amarela				LC
mastofauna	*Alouatta belzebul*	bugio-de-mãos-ruivas	X	VU		VU
mastofauna	*Cebus kaapori*	macaco-cairara	X	CR		CR
mastofauna	*Cerdocyon thous*	cachorro-do-mato				LC
mastofauna	*Cuniculus paca*	paca				LC
mastofauna	*Dasyprocta leporina*	cutia				LC
mastofauna	*Dasypus novemcinctus*	tatu-galinha				LC
mastofauna	*Didelphis marsupialis*	gambá				LC
mastofauna	*Herpailurus yagouaroundi*	gato-mourisco		VU		LC
mastofauna	*Hydrochoerus hydrochaeris*	capivara				LC
mastofauna	*Leopardus pardalis*	jaguatirica				LC

mastofauna	*Mazama nemorivaga*	veado-roxo	X			LC
mastofauna	*Microsciurus flaviventer*	catipuru	X			LC
mastofauna	*Myrmecophaga tridactyla*	tamanduá-bandeira		VU	VU	VU
mastofauna	*Nasua nasua*	quati				LC
mastofauna	*Panthera onca*	onça-pintada		VU	VU	NT
mastofauna	*Pecari tajacu*	cateto				LC
mastofauna	*Procyon cancrivorus*	mão-pelada				LC
mastofauna	*Puma concolor*	onça-parda		VU	VU	LC
mastofauna	*Saguinus niger*	sagui-preto	X			VU
mastofauna	*Saimiri sciureus (collinsi)*	mico-de-cheiro	X			LC
mastofauna	*Sapajus apella*	macaco-prego				LC
mastofauna	*Tamandua tetradactyla*	tamandua-mirim				LC
mastofauna	*Tapirus terrestris*	anta		VU		VU
mastofauna	*Tayassu pecari*	queixada		VU		VU
hepetofauna	*Allobates sp.*	-				LC
hepetofauna	*Adelphobates galactonotus*	Sapo-de-seta	X			LC
hepetofauna	*Adenomera andreae*	Sapinho-do-folhiço				LC
hepetofauna	*Boa constrictor*	Jiboia				LC
hepetofauna	*Boana geographica*	Perereca				LC
hepetofauna	*Callimedusa tomopterna*	Rã-lêmure-laranja				LC
hepetofauna	*Chelonoidis carbonarius*	Jabuti-piranga				LC

hepetofauna	*Chelonoidis denticulata*	Jabuti-tinga				VU
hepetofauna	*Chironius multiventris*	Cobra-cipó				LC
hepetofauna	*Corallus hortulanus*	Suaçubóia				LC
hepetofauna	*Elachistocleis bicolor*	Rãzinha-apito-de-guarda				LC
hepetofauna	*Erythrolamprus reginae*	Jabutibóia				LC
hepetofauna	*Leptodactylus fuscus*	Rãzinha-assobiadora				LC
hepetofauna	*Leptodactylus knudseni*	Rã-pimenta				LC
hepetofauna	*Leptodactylus mystaceus*	Rã-piadeira				LC
hepetofauna	*Leptodactylus paraensis*	Sapo-do-pará	X			LC
hepetofauna	*Osteocephalus taurinus*	Perereca				LC
hepetofauna	*Physalaemus ephippifer*	Rã-do-folhiço				LC
hepetofauna	*Pithecopus hypochondrialis*	Perereca-macaco				LC
hepetofauna	*Rhaebo guttatus*	Sapo-dorado				LC
hepetofauna	*Rhinella icterica*	Sapo-cururu				LC
hepetofauna	*Thecadactylus rapicauda*	Lagartixa-cauda-de-nabo				LC
hepetofauna	*Tropidurus oreadicus*	Calango				LC
flora	*Anacardium giganteum W.Hancock ex Engl.*	Cajuaçu				LC
flora	*Anacardium occidentale L.*	-				LC
flora	*Anacardium spruceanum Benth. ex Engl.*	-				-
flora	*Astronium graveolens Jacq.*	Muiracatiara				LC
flora	*Astronium lecointei Ducke*	Muiracatiara				LC

flora	*Tapirira guianensis Aubl.*	Tatapiririca				LC
flora	*Annona exsucca DC.*	Envira branca				LC
flora	*Duguetia echinophora R.E. Fr.*	Envira surucucu				LC
flora	*Duguetia flagellaris Huber.*	Ameju amarelo /Catinga de cutia				LC
flora	*Guatteria blepharophylla Mart.*	-				LC
flora	*Guatteria punctata (Aubl.) R.A.Howard*	Envira preta				LC
flora	*Guatteria schomburgkiana Mart.*	Envira vermelha / Embira preta				LC
flora	*Xylopia cayennensis Maas*	Inbiriba / pau santo/ Envira branca				LC
flora	*Aspidosperma album (Vahl) Benoist ex Pichon*	Araracanga	X		VU	LC
flora	*Aspidosperma carapanauba Pichon*	Carapanauba				LC
flora	*Geissospermum vellosii Allemão*	Quinarana				-
flora	*Himatanthus articulatus (Vahl) Woodson*	Sucuúba				LC
flora	*Secondatia floribunda A.DC*	Catuaba / Limorana				-
flora	*Didymopanax morototoni (Aubl.) Decne. & Planch.*	Morototó				LC
flora	*Syagrus inajai (Spruce) Becc.*	Jarana				LC
flora	*Handroanthus impetiginosus (Mart. ex DC.) Mattos*	-				NT
flora	*Jacaranda copaia (Aubl.) D.Don*	Parapará				LC
flora	*Protium altissimum (Aubl.) Marchand*	Breu-amescla / Breu barrote/Almesclão				LC
flora	*Protium altsonii Sandwith*	Breu manga/Breu branco				LC
flora	*Protium apiculatum Swart*	Breu vermelho				LC
flora	*Protium heptaphyllum (Aubl.) Marchand*	Breu branco			VU	LC

flora	*Protium stevensonii (*Standl.) Daly	Breu barrote/ Breu de palma				LC
flora	*Protium trifoliolatum* Engl.	Breu				LC
flora	*Trattinnickia burserifolia* Mart.	Breu sucuruba				LC
flora	*Caraipa grandifolia* Mart.	Tamaquaré				LC
flora	*Caryocar glabrum (*Aubl.) Pers.	Piquiarana				LC
flora	*Couepia robusta* Huber	Coco-pau/Pajurá				LC
flora	*Gaulettia canomensis (*Mart.) Sothers & Prance	-				LC
flora	*Hymenopus heteromorphus (*Benth.) Sothers & Prance	Macucu				LC
flora	*Hymenopus macrophyllus (*Benth.) Sothers & Prance	Anoerá				LC
flora	*Licania coriacea* Benth.	Casca seca / Pajurá				LC
flora	*Platonia insignis* Mart.	Bacuri				-
flora	*Symphonia globulifera* L. f.	Anani				LC
flora	*Combretum rotundifolium* Rich.	-				LC
flora	*Terminalia amazonia (*J.F.Gmel.) Exell	Tanimbuca/meringiba-de-mata/Cuiarana				LC
flora	*Terminalia dichotoma* G. Mey.	Tanimbuca				LC
flora	*Cordia bicolor* A.DC.	Freijó branco				LC
flora	*Cordia goeldiana* Huber	Freijó cinza				-
flora	*Minquartia guianensis* Aubl.	Acariquara				LC
flora	*Heisteria acuminata (*Bonpl.) Engl.	Itaubarana				LC
flora	*Heisteria scandens* Ducke	-				LC
flora	*Conceveiba guianensis* Aubl.	Urucurana				LC

flora	*Glycydendron amazonicum Ducke*	Castanha-de-porco				LC
flora	*Hevea brasiliensis (Willd. ex A. Juss.) Müll. Arg.*	Seringueira				LC
flora	*Hevea guianensis Aubl.*	Seringarana /Seringa vermelha				LC
flora	*Abarema cochleata (Willd.) Barneby & J.W.Grimes*	-				LC
flora	*Abarema jupunba (Willd.) Britton & Killip*	Ingarana/Saboeiro				LC
flora	*Alexa grandiflora Ducke*	Melancieiro				-
flora	*Amphiodon effusus Huber*	Gema de ovo				LC
flora	*Bowdichia nitida Spruce ex Benth.*	Sucupira preta				LC
flora	*Bowdichia virgilioides Kunth*	-				LC
flora	*Copaifera duckei Dwyer*	Copaiba				-
flora	*Copaifera martii Hayne*	Copaiba				LC
flora	*Cynometra marginata Benth.*	Iperana				LC
flora	*Dimorphandra sp.*	Louro-Tamaquaré				-
flora	*Dinizia excelsa Ducke*	-				LC
flora	*Diplotropis brasiliensis (Tul.) Benth*	-				LC
flora	*Diplotropis purpurea (Rich.) Amshoff*	Sucupira preta / Sucupira amarela				LC
flora	*Dipteryx odorata (Aubl.) Forsyth f.*	Cumaru				DD
flora	*Enterolobium maximum Ducke*	Fava tamboril				LC
flora	*Enterolobium schomburgkii (Benth.) Benth.*	Orelha de macaco				LC
flora	*Hydrochorea pedicellaris (DC.)*	Esponjeira				-
flora	*Hymenaea courbaril L.*	Jatobá				LC

flora	*Hymenolobium modestum Ducke*	Angelim da mata				LC
flora	*Inga brachystachys Ducke*	-				LC
flora	*Inga edulis Mart.*	Ingá timbó				LC
flora	*Inga heterophylla Willd.*	Ingá vermelho				LC
flora	*Inga thibaudiana DC.*	Ingá vermelho / Ingá xixica				LC
flora	*Inga umbellifera (Vahl) DC.*	Ingá xixi branco				LC
flora	*Mora paraensis (Ducke) Ducke*	Pracuúba branca				NT
flora	*Ormosia nobilis Tul.*	-				LC
flora	*Parkia gigantocarpa Ducke*	Fava atanã / Visgueiro				-
flora	*Parkia nitida Miq.*	Fava vermelha				LC
flora	*Parkia paraensis Ducke*	Fava branca/Visgueiro				NT
flora	*Parkia pendula (Willd.) Benth. ex Walp.*	Fava bolota				LC
flora	*Peltogyne campestris Huber ex Ducke*	-				LC
flora	*Peltogyne venosa (Vahl) Benth.*	Roxinho				-
flora	*Pseudopiptadenia psilostachya (DC.)*	-				LC
flora	*Pseudopiptadenia suaveolens (Miq.) J.W.Grimes*	Timborana				LC
flora	*Pterocarpus santalinoides L'Hér. ex DC.*	Mututi da mata				LC
flora	*Schizolobium parahyba var. amazonicum Barneby*	Fava paricá/ paricá				LC
flora	*Swartzia grandifolia Bong. e1 Benth.*	Coração de Negro				LC
flora	*Tachigali glauca Tul. (Ducke)*	Tachi preto				-
flora	*Tachigali guianensis (Benth.) Zarucchi & Herend.*	Tachi branco /Tachi vermelho				-

flora	*Tachigali paraensis (Huber) Barneby*	Tachi pitomba/Tachi branco				LC
flora	*Tachigalia paniculata Aubl.*	Tachi / Tachi preto				LC
flora	*Vatairea guianensis Aubl.*	Sucupira amarela				LC
flora	*Vatairea paraensis Ducke*	Angelim Amargoso / Fava amargosa				LC
flora	*Vouacapoua americana Aubl.*	Acapu		EN		CR
flora	*Zygia racemosa (Ducke) Barneby & J.W.Grimes*	Angelim rajado				LC
flora	*Goupia glabra Aubl.*	Cupiúba				LC
flora	*Endopleura uchi (Huber) Cuatrec.*	Uchi				LC
flora	*Humiria balsamifera (Aubl.) A.St.-Hil.*	-				LC
flora	*Sacoglottis guianensis Benth.*	Uxirana				LC
flora	*Vantanea guianensis Aubl.*	-				LC
flora	*Vismia guianensis (Aubl.) Pers.*	Lacre				LC
flora	*Licaria puchury-major (Mart.) Kosterm.*	-				-
flora	*Licaria crassifolia (Poir.) P.L.R.Moraes*	Louro pimenta				LC
flora	*Mezilaurus itauba (Meisn.) Taub. ex Mez*	Itauba		VU		VU
flora	*Nectandra amazonicum Ness*	Louro				-
flora	*Ocotea aciphylla (Nees & Mart.) Mez*	Louro rosa				LC
flora	*Ocotea fragrantissima Ducke*	Louro canela		EN		EN
flora	*Ocotea leucoxylon (Sw.) Laness.*	Louro abacate				LC
flora	*Couratari guianensis Aubl.*	Tauari				LC
flora	*Couratari oblongifolia Ducke & Kunth*	Tauari branco				-

flora	*Eschweilera apiculata (Miers) A.C. Sm.*	Matamatá/Ripeiro/Ripeiro amarelo				LC
flora	*Eschweilera coriacea (DC.) S.A. Mori*	Matamata branco				LC
flora	*Eschweilera grandiflora (Aubl.) Sandwith*	Matamatá -preto				-
flora	*Eschweilera pedicellata (Rich.) S.A. Mori*	Buranji / Jatereu				LC
flora	*Lecythis idatimon Aubl.*	Ripeiro / Matamatá vermelho				LC
flora	*Lecythis lurida (Miers) S.A. Mori*	-				LC
flora	*Lecythis pisonis Cambess.*	Sapucaia				-
flora	*Byrsonima crassifolia (L.) Kunth*	-				LC
flora	*Apeiba glabra Aubl.*	Pente de macaco				LC
flora	*Eriotheca globosa (Aubl.) A. Robyns*	-				LC
flora	*Eriotheca longipedicellata (Ducke) A. Robyns*	Sumauma da terra-firme		VU		VU
flora	*Luehea speciosa Willd.*	Açoita cavalo				LC
flora	*Sterculia excelsa Mart.*	Axixá				LC
flora	*Sterculia pruriens (Aubl.) K.Schum.*	Envira quiabo /Axixá / Axixá branco				LC
flora	*Theobroma grandiflorum (Willd. ex Spreng.) K.Schum.*	Cupuaçu				LC
flora	*Theobroma speciosum Willd. ex Spreng.*	Cacau da mata				-
flora	*Theobroma subincanum Mart.*	Cupuí				LC
flora	*Bellucia grossularioides (L.) Triana*	Goiaba da Mata / Goiabinha / Muuba				LC
flora	*Mouriri grandiflora DC.*	Merauba				LC
flora	*Carapa guianensis Aubl.*	Andiroba				LC
flora	*Cedrela fissilis Vell.*	Cedro		VU		VU

flora	*Emmotum nitens (Benth.) Miers*	-				LC
flora	*Dendrobangia boliviana Rusby*	Caferana / Pau de cubiú				LC
flora	*Bagassa guianensis Aubl.*	Tatajuba				LC
flora	*Brosimum acutifolium Huber*	-				LC
flora	*Brosimum lactescens (S. Moore) C. C. Berg.*	Amapaí				LC
flora	*Brosimum parinarioides Ducke*	-				LC
flora	*Clarisia racemosa Ruiz & Pav.*	Guariúba				LC
flora	*Ficus duckeana C. C. Berg & Ribeiro*	Pau de ficus				-
flora	*Ficus pertusa L. f.*	Figueira				LC
flora	*Maquira guianensis Aubl.*	-				LC
flora	*Perebea mollis (Poepp. & Endl.) Huber*	Inharé				LC
flora	*Virola elongata (Benth.) Warb.*	Ucuúba				LC
flora	*Virola michelii Heckel*	Casca de vidro/ Ucuuba preta				LC
flora	*Eugenia lambertiana DC.*	Goiabinha				LC
flora	*Guapira opposita (Vell.) Reitz*	-				LC
flora	*Neea macrophylla Poepp. & Endl.*	João mole				LC
flora	*Agonandra brasiliensis Miers ex Benth. & Hook. f.*	Pau marfim				LC
flora	*Euplassa pinnata (Lam.) I.M.Johnst.*	Louro faia				LC
flora	*Esenbeckia sp.*	Guarantã				-
flora	*Laetia procera (Poepp.) Eichler*	Pau jacaré / Muirapuama				LC
flora	*Chrysophyllum lucentifolium Cronquist*	Goiabão				LC

flora	*Manilkara bidentata subsp. surinamensis (Miq.)*	Maparajuba				LC
flora	*Manilkara elata (Allemão ex Miq.) Monach*	Maçaranduba				EN
flora	*Manilkara paraensis (Huber) Standl.*	Maparajuba/Maçarandubinha				-
flora	*Pouteria anomala (Pires) T.D.Penn.*	abiu rosadinho				-
flora	*Pouteria caimito (Ruiz & Pav.) Radlk.*	Abiu seco / Abiu vermelho				LC
flora	*Pouteria gongrijpii Eyma*	Abiurana				LC
flora	*Pouteria macrophylla (Lam.) Eyma*	Abiu-cutite				LC
flora	*Pouteria oppositifolia (Ducke) Baehni*	Guajará-cinza				VU
flora	*Pouteria ramiflora (Mart.) Radlk.*	Guajará-bolacha				LC
flora	*Pouteria venosa (Mart.) Baehni*	Guajará				LC
flora	*Simaba cedron Planch.*	Pau para tudo				LC
flora	*Simarouba amara Aubl.*	Marupá				LC
flora	*Cecropia distachya Huber*	-				LC
flora	*Cecropia obtusa Trécul*	Embaúba branca				LC
flora	*Cecropia palmata Willd.*	Embaúba vermelha				LC
flora	*Cecropia sciadophylla Mart.*	Embaubão/Torém				LC
flora	*Erisma lanceolatum Stafleu*	Quarubarana / Quarubatinga				DD
flora	*Qualea paraensis Ducke*	Mandioqueira				LC
flora	*Vochysia guianensis Aubl.*	Quaruba				LC
flora	*Vochysia vismiifolia Spruce ex Warm.*	Quaruba				LC